Experience-Generating Physical Systems

Designing Data for Embodied Learning

Design & Learning Research Group

Complete textual draft: Chapters 1–16 and Appendices A–C

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Chapter 1 Introduction

Imagine a small motor turning a short lever. We ask it to reach an angle, record the command and encoder readings, and watch the lever settle. Then we attach a load and repeat the command. Suppose the second motion is slower. We now have two traces, but what have we learned? The difference might involve the added inertia, gravitational loading, friction, a current limit or the controller's response. A plotting or timing error might also change the apparent comparison.

This imagined investigation begins with a familiar engineering task and opens a larger question: how should we arrange a physical interaction so that the data it produces helps us understand a system or learn useful behavior? The answer depends on the mechanism, the measurements, the actions we choose and the conclusions we intend to draw. These choices are the subject of this book.

1.1 Physical interaction as a source of data

1.1.1 Encounters, recorded traces and interpreted outcomes

During the joint's motion, electrical power, mechanical loading and feedback control act together. The physical encounter includes processes that our instruments may not observe. An encoder reports a position-related quantity; it does not directly report every deformation, internal force or source of heat. The stored command records what was requested, which may differ from the input actually applied after limits and internal control have acted.

A record is therefore a partial account of an encounter. Its meaning depends on the sensor, units, reference frame, calibration and time convention. Calling a recorded column "torque" does not establish that output torque was measured. It may instead contain a requested effort or an estimate derived from current.

Interpretation is another step. The conclusion that a load caused a slower response requires a defensible comparison and consideration of other changes. The encounter, its recorded traces and our explanation of them are connected, but each can fail in a different way. Keeping them distinct makes a surprising trace a reason to investigate rather than an explanation in itself.

1.1.2 Data for understanding systems and learning behavior

The same joint can support different learning purposes. We might want to estimate its effective inertia, predict its response under a new load, or train a controller to reach a target despite changing conditions. Each purpose asks something different of the data. Estimating inertia requires motion that makes its effect distinguishable from competing effects. Learning reliable reaching requires experience of the conditions in which the controller will operate.

A collection of successful motions might be useful for imitation while revealing little about recovery from a disturbance. Conversely, an informative identification experiment need not resemble an efficient way to perform the eventual task. A model can predict a narrow range well without explaining every physical cause, and a policy can succeed without identifying the mechanism's parameters. We must state what is to be learned before deciding which records would support it.

1.2 Experience-Generating Physical Systems

1.2.1 Bodies, environments, sensing and action as a collection arrangement

In this book, *Experience-Generating Physical Systems*, abbreviated *ExpGen PhySyst*, names an integrative perspective on physical systems: design and operate their interactions so that observations, actions and context can be recorded, interpreted and used to improve understanding or capability. "Experience" here concerns encounters and what can be learned from them; it does not imply conscious experience. The perspective brings established mechanics, control, measurement and learning methods into a common design question.

The relevant system includes more than the robot. In our example it includes the lever and load, the motor and its controller, the sensors and clock, and the procedure that selects and repeats motions. A person may choose the next trial; an autonomous policy may do so instead. In either case, actions influence which physical situations become available to observe.

These components cannot be chosen independently of the intended evidence. A position sensor may be sufficient to describe a motion but insufficient to separate all its causes. Adding a sensor, changing a load or altering a command may make a previously ambiguous question answerable.

1.2.2 Designing collection processes for a stated learning purpose

Designing data means designing the conditions under which records are generated and made meaningful. It includes choosing what to interact with, how to act, what to observe, which conditions to cover and how to retain the resulting records. It also includes deciding how those records will be compared and what would count as a useful learning outcome.

Return to the loaded joint. If the question concerns sensitivity to loading, we can specify the load's placement, initial position, command, controller settings and observation timing. Repeating trials helps reveal variability; changing selected conditions tests how far an interpretation extends. If several conditions change together, the resulting difference may remain ambiguous. If a signal is clipped or unavailable, we must retain that limitation when interpreting the motion.

Collecting for a learned controller raises another question: do the recorded situations represent its intended operating conditions? Repeating one easy motion adds records without necessarily resolving an unobserved failure case. The value of additional data depends on the question, the coverage already obtained and what the learning method can use.

1.3 Connecting design, data and learning

1.3.1 Measured, estimated, simulated and human-supplied information

A joint investigation can combine encoder readings, velocity estimates, simulated trajectories and a person's description of the task. These describe different sources or derivations of information. They are not mutually exclusive sensor categories: one demonstration may contain all four.

An estimate inherits assumptions from its measurements and computation. A simulator's internal state follows from its model and numerical procedure; it does not become a measurement of the physical joint merely because it has the same variable name. Human-supplied instructions and labels express judgments or intentions whose meaning also needs to be specified. Physical measurements themselves depend on instruments and calibration.

Useful records preserve these relationships. To interpret an estimated velocity, a later reader needs to know which position samples and timing convention produced it. To interpret a demonstrated action, the reader needs to distinguish operator intent, transmitted targets and realized motion. Preserving provenance makes it possible to revise an interpretation without silently changing the identity of the evidence.

1.3.2 Collection purpose, coverage and evidence across simulation and reality

Simulation lets us examine the consequences of specified assumptions and generate interactions under controlled model conditions. Physical experiments test correspondence within the conditions actually observed. Neither a close match on one trajectory nor a successful simulated policy establishes general agreement between a model and a physical system.

For the joint, a useful comparison aligns initial conditions, commands, observations and timing as far as the experiment permits. A discrepancy then becomes a question about modeling, measurement, implementation or unaccounted conditions. The Sim2Real gap is a bridge between these questions: it directs us toward the next informative measurement or intervention, rather than only a single aggregate error score.

Coverage and selection matter within each environment too. Keeping only successful trials changes the population represented by a dataset. Collecting many neighboring samples does not make them independent trials. A claim about

learning benefit therefore needs declared conditions and a comparison that accounts for collection effort, variability and omitted cases. These concerns will connect dataset design to the evaluation of learned behavior.

1.4 Reading this book

1.4.1 Scope, prerequisites and mathematical language

The book assumes calculus, linear algebra and programming. Probability, estimation, optimization and learning concepts are developed as the argument requires them. Robotics supplies the main setting, with the physical and statistical foundations presented together so that readers can reason about both a mechanism and the data collected through it.

The mechanical language follows Featherstone (2008): angular-first spatial motion vectors, dual spatial forces, spatial inertia and recursive rigid-body dynamics. Their definitions and derivations begin in Chapters 4–5. The connection to code is part of the explanation: a variable's units, coordinate frame, reference point, ordering and timing must agree with the quantity in the derivation. Probability and learning use their own appropriate notation. Appendix A collects a compact reference rather than replacing these explanations.

The path begins with actuation, mechanics and simulation, then develops measurement, identification and collection design. Exploration and teleoperation provide two routes to generating experience. Imitation, generalist policies and predictive models explain different ways to learn from it. Transfer, system co-design and evaluation bring these foundations together by asking what should change next and what evidence justifies that decision.

1.4.2 Continuing investigations and optional computational companions

A controlled joint, contact-rich manipulation and a human-guided arm will recur as investigations. The joint connects commands and measurements to physical inference. Manipulation introduces changing contacts and policy-generated experience. The arm connects human intent, feasible motion and demonstrations. Their purpose is to carry physical meaning across the mathematics, records and learned behavior, without making the argument depend on a particular product or course timetable.

Appendix B documents collection conventions, sample sources and their limits. Appendix C connects worked investigations to computational companions, including optional tools such as asMagicBrain. The main explanations remain self-contained. Constructed examples, measured results and source-reported findings are identified as such.

The question carried through the book is practical and theoretical at once: given a capability we want to understand or learn, which physical interactions should we arrange, what should we record, and how will we know whether those records helped? ExpGen PhySyst makes that question a starting point for design.

Chapter 2 Historical Notes on Physical AI

How should a machine acquire useful behavior, and what evidence would show that it has done so? The question connects ideas about computation with the practical work of constructing, observing and controlling physical systems. Researchers have approached it through different problems, instruments and standards of evidence. Their choices help explain why the same robot can be understood as a mechanism, a feedback system, a learning agent or an apparatus for collecting data.

These notes follow a selective thread through those choices. The purpose is to place the questions of ExpGen PhySyst in context and identify what earlier work helps us ask. The examples do not establish a single origin for Physical AI or a necessary progression from one approach to another.

2.1 Machines and the question of intelligence

2.1.1 Turing, machine intelligence and learning

Alan Turing's 1950 article was titled *Computing Machinery and Intelligence*. It proposed examining machine performance through an imitation game, making the conditions under which intelligence is judged part of the problem itself (Turing 1950). The conversational setting restricts the evidence available to an interrogator. Physical manipulation skill is therefore outside that particular test; a successful conversation alone would not establish it.

The same article also considers learning. In its final section, Turing distinguishes an initial child-machine programme from the process of educating it, discusses experience and feedback, and considers giving a machine sensory organs. Reading this discussion alongside the imitation game reveals a broader interest than conversational performance alone. For this book, it raises a useful question: which initial capacities and subsequent encounters should a learning process receive? That connection is our present-day interpretation, rather than an attribution of modern dataset design to Turing.

2.1.2 Computational ideas and their physical realization

Turing's wartime work also makes the relation between a procedure and its physical realization concrete. The British Bombe was an electromechanical device used in the attack on Enigma. Its development combined Turing's cryptanalytic approach, Gordon Welchman's diagonal-board improvement, and engineering and manufacture by the British Tabulating Machine Company under Harold Keen. Museum accounts place those contributions within a wider codebreaking operation and acknowledge Polish antecedents (The National Museum of Computing, n.d.; Jackson, n.d.).

The Bombe's function was cryptanalysis. Its relevance here is the work required to make a logical procedure operate through an engineered machine. That work joins choices about representation and computation to choices about components, interfaces and operation. In robotics, an algorithm similarly reaches the world through sensors, actuators and timing. The historical example provides a reason to study these connections while preserving the differences between codebreaking machinery and a robot that learns through physical interaction.

2.2 Learning through embodiment and feedback

2.2.1 Mechanics, control and adaptive behavior

Norbert Wiener's *Cybernetics*, first published in 1948, placed control and communication in animals and machines within a shared inquiry (Wiener 2019). Feedback connects observed consequences to subsequent action. A controller's behavior must consequently be understood together with the system it controls and the information returned to it. This connection predates the present use of Physical AI as a research label.

Feedback and learning are distinct mechanisms. A fixed controller can continually correct an error without changing its control law. A learning process can use records of that behavior to revise a model, policy or parameter. The records then

depend partly on the controller that generated them: successful regulation may suppress variations that an identification experiment would need to observe.

Mechanics gives these statements physical content. Loading, inertia, friction and contact affect what a command produces and what feedback can reveal. Chapters 3–5 develop that connection using actuation and Featherstone spatial mechanics. The historical thread motivates studying learning in a system whose physical quantities and feedback paths are explicit.

2.2.2 Situated and behavior-based approaches

Rodney Brooks's 1991 *Intelligence without Representation* argued for an incremental approach grounded in perception and action in the real world (Brooks 1991). For our purposes, the useful question is how much useful behavior can arise from an agent's continuing interaction with its surroundings, and what an internal representation must contribute. This question leaves room to investigate the conditions under which different representations help.

A physical body changes the question. Reach, compliance, sensing location and contact determine which encounters are possible and which distinctions are observable. A learning method receives information through that arrangement; changing the arrangement can change the learning problem.

A later, explicitly interdisciplinary formulation appears in Aslan Miriyev and Mirko Kovač's 2020 comment *Skills for Physical Artificial Intelligence* (Miriyev and Kovač 2020). They emphasize the combination of skills needed for research and education concerned with physical artificial intelligence. It is one contemporary formulation of the term. Its value for this book is the invitation to connect knowledge of physical systems with computational learning, without assigning the subject exclusively to one discipline.

2.3 Learning skills from interaction

2.3.1 Human demonstrations and autonomous exploration

Teaching by demonstration provides one way to shape a learner's experience. Stefan Schaal's *Learning from Demonstration*, published in the 1997 proceedings of NIPS 1996, investigated how demonstrations could assist reinforcement learning through information about a policy, a value function or task dynamics (Schaal 1997). This example connects human guidance to subsequent learning rather than treating recorded behavior as a complete solution in itself.

Demonstration and autonomous exploration make different collection choices. A person selects motions, responds to difficulties and implicitly favors some situations over others. An exploring agent visits situations according to its policy, objectives and available resets. Both generate selected records of interaction. Neither guarantees that all conditions needed for later use have been encountered.

For ExpGen PhySyst, these are design decisions about how experience becomes available. We need to know what the human or agent could observe, which commands were issued, which interventions occurred and how those events were recorded. Chapters 9–12 develop the collection and learning methods that make these questions operational.

2.3.2 ALOHA, LEAP Hand and UMI as contrasting system designs

Three research systems illustrate how apparatus and learning choices meet. *ALOHA* couples a bimanual robot with leader–follower teleoperation and an imitation-learning method, Action Chunking with Transformers. Its reported records distinguish leader joint positions used as action targets from follower joint positions and images used as observations (Zhao et al. 2023). The collection apparatus therefore helps define both the demonstrated action and the information available to learn it. For this book, ALOHA is a concrete example of designing an interface and data representation together.

LEAP Hand instead foregrounds a dexterous hand's kinematics, actuation and maintainable construction (Shaw et al. 2023). Its paper considers several ways of controlling and learning with the hand, including teleoperation, behavior cloning and simulation-trained manipulation. Their sources must be kept distinct: the behavior-cloning example uses prior

Allegro Hand demonstrations mapped to LEAP, alongside video pretraining. The platform illustrates how embodiment and control interfaces shape the use of experience; it should not be read as a single demonstration-collection procedure.

The *Universal Manipulation Interface* (UMI) collects demonstrations with handheld grippers and cameras, separating the collection tool from the robot that later executes a policy (Chi et al. 2024). Visual-inertial motion estimation recovers end-effector trajectories from these records. Alignment, latency handling and filtering for the deployment robot are part of the interface. Robot-free collection still has to produce information compatible with robot execution.

These examples suggest complementary questions: which apparatus makes useful encounters accessible, which signals describe them, and which action interface allows what was learned to be executed? Their comparison is an interpretation through ExpGen PhySyst, not a claim that the three projects share a single method or a direct historical lineage.

2.4 Data and predictive models

2.4.1 Behavior learning and generalist policies

Learning a behavior means learning how observations and task information should influence action. It can be studied on a specialized apparatus or across a broader collection of tasks and robots. Broadening that collection raises questions about common representations, compatible action meanings and which examples contribute to which capabilities.

OpenVLA provides a 2024 example: its authors combine pretrained vision-language components with robot demonstrations to learn language-conditioned action and study adaptation to new settings (Kim et al. 2024). Such work brings robot interaction records into contact with other training sources. The resulting policy still acts through a particular observation and control interface. A model's breadth of training does not make the units, frames, actuator limits or timing of that interface disappear.

The phrase *large behavior models* appears in the October 2024 research partnership announcement by Boston Dynamics and Toyota Research Institute. The announcement connects TRI's learning work with Atlas, teleoperation, data collection and hardware and simulation evaluation (Boston Dynamics and Toyota Research Institute 2024). It is a dated statement of research direction. Here, behavior learning names the learning of action and control; the phrase alone does not specify a distinct theory of intelligence or demonstrate general physical competence.

2.4.2 Internal models, learned world models and physical tests

A predictive model supports a different question from a policy: what consequences might follow an action? Physical equations provide one route to prediction; models learned from records provide another. The relation between prediction and action has a longer history than any recent paper using the term *world model*.

David Ha and Jürgen Schmidhuber's 2018 work offers a specific modern example. Their recurrent world model learns compressed temporal structure from observations and actions, and a controller uses its learned representation. Their experiments also examine training a controller within a generated environment (Ha and Schmidhuber 2018). The reported settings are game environments. The paper is useful here for the distinction between a learned predictor, a controller and the environment against which it is tested.

Generating plausible observations and predicting the consequences relevant to control are different requirements. A learned model may be inaccurate in conditions poorly represented in its collection data, and a controller may exploit those inaccuracies. Ha and Schmidhuber discuss this problem directly. For physical systems, it motivates examining which interactions a dataset covers and which tests can expose a consequential modeling error.

Chapters 6, 13 and 14 will distinguish physical simulation, learned prediction and evidence for transfer. A useful Sim2Real investigation asks which correspondence matters for the intended behavior and what additional data would test it.

2.5 The questions carried forward

2.5.1 Different approaches to building and using robots

A project may begin by designing a mechanism, improving a controller, creating a collection interface or adapting a pretrained model. Each starting point makes some questions prominent. Mechanism design exposes loading and fabrication choices; policy development exposes representations, training data and learning objectives. As the investigation develops, these concerns often meet at the same sensor and actuator interfaces.

The practical boundary between building and using a robot can therefore move. A user who encounters a sensing limitation may modify the apparatus. A designer who wants a new capability may need demonstrations or autonomous exploration. The cases above make this meeting visible without requiring that every investigation redesign every component.

ExpGen PhySyst organizes the meeting around a collection purpose. What must be learned, which encounters could provide useful evidence, what apparatus makes them possible, and how will their records be interpreted? Established methods address parts of this problem. The book's task is to connect their assumptions and consequences so that a design choice can be evaluated through what it enables us to observe and learn.

2.5.2 Historical sources, terminology and limits of the account

The references in this chapter have different roles. Turing supplies an argument about evaluation and learning; Wiener supplies a feedback and communication perspective; the robotics papers describe particular apparatus and methods. A museum history establishes attribution, while a corporate announcement records an intended research direction. They should be read with those purposes in mind. Publication dates, later editions and retrospective accounts also need to remain distinct.

Terms such as *embodied intelligence*, *physical intelligence*, *Physical AI* and *world model* acquire meaning within particular arguments and experiments. These notes select useful examples rather than establishing their first use or a universal definition. ExpGen PhySyst is the integrative perspective adopted in this book, not a label retrospectively assigned to every cited author.

The next chapter begins the technical development with actuation and feedback. Its starting point is the motor-driven joint from Chapter 1. Understanding how power, loading, control and sensing shape its motion will give precise physical meaning to the records from which later chapters seek to learn.

Chapter 3 Actuation and Feedback Control

The loaded joint in Chapter 1 moves because electrical energy is converted into mechanical work. Yet the quantity sent by a program is often an angle, rather than a voltage or force. Between that request and the resulting motion lie a drive, a motor, a transmission and one or more feedback loops. Their behavior determines both the physical encounter and the information that can be recorded from it.

Consider two investigations with the same joint. In the first, the aim is to learn how a payload affects acceleration. In the second, the aim is to collect demonstrations of reaching. Accurate regulation may help the second task while concealing the variation needed by the first. If an internal controller supplies more current to maintain the same motion, position records alone can make two different loads appear almost identical. The actuation system is therefore part of the collection apparatus, even when a learning algorithm treats it as a ready-made component.

This chapter develops a model of that apparatus. We begin with a simplified electrical motor, connect its effort to a joint through mechanics, then examine feedback and its implementation. The final experiment brings these ideas together: what should be commanded, measured and retained if a joint's records are to support embodied learning? The numerical examples are constructed calculations. They illustrate consequences of stated assumptions and make no claim about the performance of a particular commercial actuator.

3.1 From electrical power to mechanical action

An actuator is an arrangement that can apply mechanical effort through a usable interface. A motor supplies one part of it: an electromagnetic interaction that produces torque. A drive switches and regulates electrical power. A transmission may change speed, torque and mechanical impedance. Sensors and control determine how a request is converted into action. Integration changes packaging and interfaces, but it does not remove these functions.

Our initial model uses one motor shaft and one rotary output. Positive shaft and joint directions will be defined together. Angles are in radians, angular velocities in radians per second, effort in newton metres and current in amperes. A direction convention is necessary even for a scalar quantity: a negative current can be a positive joint effort if wiring or gearing reverses the chosen direction.

3.1.1 Motor equations, back electromotive force and torque

Let θ_m be shaft angle, i winding current and u the effective terminal voltage. For a simplified permanent-magnet motor, the electrical balance and electromagnetic torque are

$$L\dot{i} = u - Ri - k_e\dot{\theta}_m, \tag{3.1}$$

$$\tau_e = k_t i. \tag{3.2}$$

Here R is resistance, L inductance, k_e the back-electromotive-force constant and k_t the torque constant. Back electromotive force opposes the applied voltage in the positive-speed motoring condition. It expresses the electrical consequence of moving the motor through its magnetic field; it is not a friction term. Reversing motion or current changes the associated power flows.

Equations (3.1)–(3.2) describe a useful averaged model, not every switching or magnetic effect. A brushless motor and its commutating drive can sometimes be represented by an equivalent torque-producing current, but phase current, bus current and the drive's reported current need not have the same meaning. Phase-versus-line conventions, peak-versus-root-mean-square values, pole pairs and winding configuration must be reconciled before importing a published torque constant. In the ideal energy-consistent model, $k_t = k_e$ numerically in compatible SI conventions; this equality is not permission to combine constants with different definitions. The equivalent-circuit assumptions are also set out in the official motor technical note (Google DeepMind, n.d.).

The distinction between a motor model and a drive interface matters when designing records. A requested duty ratio for pulse width modulation is not necessarily the terminal voltage u. Supply variation, switching, dead time and limits

intervene. Likewise, a current target is an input to a regulator rather than a measurement of its achieved current. To study the electrical transient, both the bandwidth of the experiment and the available electrical signals must be appropriate. If only slow joint reports are accessible, fitting L from them may be unjustified even when inductance appears in the underlying physics.

For a stationary shaft and fixed u, the current approaches u/R with time constant L/R. At nonzero speed, the voltage needed to hold a current includes $k_e\dot{\theta}_m$. Thus a motor that can supply an effort at rest may be unable to supply the same effort at high speed. Neglecting the inductive term only after stating a quasi-steady assumption gives

$$i \simeq \frac{u - k_e\dot{\theta}_m}{R}. \tag{3.3}$$

This relationship already connects collection choices to physical feasibility: the same requested effort at different speeds need not be realized in the same way.

3.1.2 Speed, work, power and mechanical loading

Mechanical work and instantaneous power for the shaft are

$$W = \int \tau_e \, d\theta_m, \qquad P_m = \tau_e\dot{\theta}_m. \tag{3.4}$$

Large holding torque can require substantial current while mechanical power is zero. Conversely, modest torque at high speed can transfer substantial power. These statements distinguish effort, energy and power; none is a substitute for the others. Multiplying the electrical balance by current gives, for constant inductance and consistent motor constants,

$$ui = Ri^2 + \frac{d}{dt}\left(\frac{1}{2}Li^2\right) + \tau_e\dot{\theta}_m. \tag{3.5}$$

The terms are terminal electrical power, resistive heating, magnetic-energy change and electromagnetic mechanical power. Drive losses and additional motor losses would require further terms. A holding experiment can therefore heat a motor even though its encoder records no movement.

To connect effort to a physical body we use the angular-first spatial language of Featherstone (2008). A spatial motion vector $\mathbf{v}$ and a spatial force $\mathbf{f}$, expressed in the same frame and about the same origin, have the order

$$\mathbf{v} = \begin{bmatrix} \boldsymbol{\omega} \\ \mathbf{v}_O \end{bmatrix}, \qquad \mathbf{f} = \begin{bmatrix} \mathbf{n}_O \\ \mathbf{F} \end{bmatrix}, \qquad P = \mathbf{f}^{\mathsf{T}}\mathbf{v}. \tag{3.6}$$

$\mathbf{n}_O$ is moment about O, $\mathbf{F}$ force, and $\mathbf{v}_O$ the linear velocity associated with that origin. Each three-vector retains its own physical units. The six-vector is not six entries with one common unit. Chapter 4 develops transformations and the general rigid-body balance; here we need only the power relation at a fixed-axis joint.

Choose O on a fixed revolute axis and express its unit direction $\mathbf{a}$ in the same frame. The joint motion subspace and power relation are

$$\mathbf{S} = \begin{bmatrix} \mathbf{a} \\ \mathbf{0} \end{bmatrix}, \qquad \mathbf{v} = \mathbf{S}\dot{q}, \qquad P = \mathbf{f}^{\mathsf{T}}\mathbf{S}\dot{q} = \tau\dot{q}, \qquad \tau = \mathbf{S}^{\mathsf{T}}\mathbf{f}. \tag{3.7}$$

Generalized joint effort is a projection of a wrench, not its arbitrary first entry. If the reference point is moved away from the joint axis, the linear part of the motion subspace generally ceases to be zero. Moving the point for one vector alone breaks the power correspondence.

The same convention determines inertia. For a body of mass m, center of mass $\mathbf{c}$ measured from O, and rotational inertia $\mathbf{I}_C$ about its center of mass, let $\mathbf{c}_\times\mathbf{x} = \mathbf{c} \times \mathbf{x}$. Its spatial inertia and fixed-axis kinetic energy are

$$\mathbf{I}_O = \begin{bmatrix} \mathbf{I}_C + m\mathbf{c}_\times\mathbf{c}_\times^{\mathsf{T}} & m\mathbf{c}_\times \\ m\mathbf{c}_\times^{\mathsf{T}} & m\mathbf{1}_3 \end{bmatrix}, \tag{3.8}$$

$$T = \frac{1}{2}\mathbf{v}^{\mathsf{T}}\mathbf{I}_O\mathbf{v} = \frac{1}{2}J_L\dot{q}^2, \qquad J_L = \mathbf{S}^{\mathsf{T}}\mathbf{I}_O\mathbf{S}. \tag{3.9}$$

$\mathbf{1}_3$ is the 3×3 identity. Thus the familiar scalar inertia about a pivot is a specialization of spatial inertia. For a slender uniform link of length ℓ pivoted at one end about an axis perpendicular to its length, this projection gives $m\ell^2/12 + m(\ell/2)^2 = m\ell^2/3$. A point payload m_p at radius r_p adds $m_p r_p^2$. The geometry of a fixture therefore changes the acceleration produced by an effort even when the motor and controller are unchanged.

For the planar example used below, $q = 0$ means the link hangs vertically downward, positive q is counterclockwise, and gravity acts downward. Taking potential energy as

$$V(q) = -C_g \cos q, \qquad C_g = g\,(m\ell/2 + m_p r_p)\,, \qquad g_q(q) = \frac{\partial V}{\partial q} = C_g \sin q \tag{3.10}$$

makes the sign explicit. The gravitational generalized force is $-g_q$; the positive term g_q appears on the left of the required-effort equation. A reader should be able to recover that sign from the geometry, rather than guess it from a plotted trace.

3.2 Transmission and actuation architecture

The motor and the load rarely share all their preferred operating conditions. A motor may rotate quickly with modest shaft torque, while a joint needs lower speed and greater effort. A transmission changes this relation. It also changes what the joint reveals about internal motion, loss and loading.

3.2.1 Gear ratios, reflected inertia and efficiency

Define a positive reduction ratio N by $\theta_m = Nq$; any reversal has already been absorbed into the coordinate convention. For a rigid, ideal transmission, speed and transmitted effort satisfy

$$\dot\theta_m = N\dot q, \qquad \tau_t = N\tau_s, \qquad \tau_s\dot\theta_m = \tau_t\dot q. \tag{3.11}$$

Here τ_s is torque delivered to the transmission at its input, and τ_t is torque delivered to the load. Neither should be confused with τ_e, because some electromagnetic effort accelerates the motor rotor and overcomes motor-side losses.

Let J_m and b_m be motor rotor inertia and viscous damping, and J_L and b_L their load-side counterparts. Write the motor and load equations before eliminating the internal effort:

$$J_m\ddot\theta_m + b_m\dot\theta_m + \tau_t/N = k_t i, \tag{3.12}$$

$$J_L\ddot q + b_L\dot q + g_q(q) + \tau_f = \tau_t + \tau_{\text{ext}}. \tag{3.13}$$

τ_f denotes additional resisting effort, while τ_{ext} is an externally applied generalized effort that assists positive motion when positive. Substitution gives

$$J_\Sigma\ddot q + b_\Sigma\dot q + g_q(q) + \tau_f = Nk_t i + \tau_{\text{ext}}, \tag{3.14}$$

$$J_\Sigma = J_L + N^2 J_m, \qquad b_\Sigma = b_L + N^2 b_m. \tag{3.15}$$

Here the motor housing is fixed to ground. The N^2 factor follows equally from rotor kinetic energy $\frac{1}{2}J_m(N\dot q)^2$. Increasing reduction therefore changes more than the available effort. It raises the rotor inertia seen at the joint and changes the relationship between contact, acceleration and motor current. These properties influence both physical interaction and the information in a trace. In a multi-joint robot, moving motor housings can introduce additional coupling and gyroscopic terms; the scalar fixed-base result is not a complete rotor model for every articulated mechanism (Featherstone 2008, 186–189).

Efficiency cannot be inserted into every operating condition as a universal constant. In a specified steady motoring regime one may approximate $\tau_t = \eta N\tau_s$, with $0 < \eta \le 1$, from the ratio of delivered to input power. Backdriving requires a different power-flow description, and at zero speed that power ratio is undefined. Use an identified loss model or a documented operating-regime approximation instead of using a motoring efficiency to infer a static holding torque. Equations (3.14) and (3.15) use an ideal gear and retain specified losses explicitly.

Worked example: an effort and energy budget. A uniform $0.25\,\text{kg}$ link is $0.20\,\text{m}$ long and carries a $0.15\,\text{kg}$ point payload at $0.18\,\text{m}$. Take $J_m = 2.0 \times 10^{-5}\,\text{kg m}^2$, $N = 6$, $k_t = k_e = 0.05$ in compatible SI units, $R = 2.0\,\Omega$ and $b_\Sigma = 0.010\,\text{N m s/rad}$. For the separate shaft and load budgets take $b_m = 0$ and $b_L = b_\Sigma$. Ignore Coulomb friction and transmission losses. The projected load inertia, reflected rotor inertia and total inertia are

$$J_L = 0.25(0.20)^2/3 + 0.15(0.18)^2 = 0.0081933\,\text{kg m}^2,$$

$$N^2 J_m = 0.0007200\,\text{kg m}^2, \qquad J_\Sigma = 0.0089133\,\text{kg m}^2. \tag{3.16}$$

At $q = \pi/6$, $\dot{q} = 1.0\,\mathrm{rad/s}$ and $\ddot{q} = 2.0\,\mathrm{rad/s^2}$, gravity contributes $0.25506\,\mathrm{N\,m}$. Equation (3.14) requires

$$Nk_t i = 0.017827 + 0.010000 + 0.255060 = 0.282887\,\mathrm{N\,m}, \qquad i = 0.94296\,\mathrm{A}. \tag{3.17}$$

If $\dot{i} = 0$ at that instant, the required terminal voltage is $2.18591\,\mathrm{V}$. Resistive heating is $1.77833\,\mathrm{W}$ and electromagnetic mechanical power is $0.282887\,\mathrm{W}$, summing to the $2.06122\,\mathrm{W}$ electrical input. That mechanical power includes rotor kinetic-energy change as well as work delivered through the transmission. The load-side effort alone is $0.281447\,\mathrm{N\,m}$.

At rest at the same angle, ideal holding current is $0.85020\,\mathrm{A}$ despite zero mechanical power. Changing only the reduction from 6 to 30 would raise total joint inertia to $0.0261933\,\mathrm{kg\,m^2}$, before considering a different gearbox's losses or mass. These are two distinct design tradeoffs: holding introduces a thermal question, and reduction introduces an interaction dynamics question. Recording angle alone answers neither.

3.2.2 Friction, compliance, backlash and backdrivability

A rigid transmission is an approximation. Away from zero speed a simple resisting-effort model is

$$\tau_f = b_f \dot{q} + \tau_c \,\mathrm{sgn}(\dot{q}). \tag{3.18}$$

If this extra viscous term is used, it must not also be included in b_Σ. The Coulomb term has magnitude τ_c and opposes motion. At rest, static friction is better represented as an interval of possible balancing efforts, rather than defining $\mathrm{sgn}(0) = 0$ and expecting the model to explain sticking. Low-speed reversal data can expose this distinction; fast motion in one direction may not.

To represent compliance, separate the load angle q from the motor-side angle mapped to the output, $q_m = \theta_m/N$. A simple torsional element gives

$$\tau_t = k_s(q_m - q) + d_s(\dot{q}_m - \dot{q}). \tag{3.19}$$

The earlier algebraic constraint $q_m = q$ is now replaced by coupled dynamics. An encoder on the motor measures q_m after scaling; it does not automatically measure load angle q. A tracking plot based on that encoder can look accurate while the load deflects. A second encoder or external measurement can make the deflection observable. The choice of sensor location thus changes the questions a dataset can answer.

Backlash adds a region of relative motion in which opposing transmission surfaces are not both engaged. A dead-zone spring is a rough model, but impacts, preload and direction history can matter. Treating all reversal error as friction discards these competing explanations. A deliberate reversal with independent output observation is more informative than an unlabeled average tracking error.

Backdrivability concerns how external loading can move the mechanism through the transmission and drive. It depends on reflected inertia, friction, compliance, reduction and the energized controller. A freely moving unpowered joint and the same joint under stiff position regulation are different interaction conditions. Likewise, a low reduction does not establish the quality of force sensing. It can reduce some obstacles to inferring external effort, while calibration, internal dynamics and current sensing still matter.

3.2.3 Smart servos, integrated drives and quasi-direct drive

A conventional servo system combines a motor, position feedback, a drive and a controller. A compact smart servo packages much of this arrangement behind a digital interface. Its useful properties may include configurable control modes, internal profiles, feedback registers and protection. The word *smart* describes integration and capabilities, not an assurance that all mechanical states or forces are measured. Specific models can expose different modes and reports even within one product family.

Quasi-direct-drive actuation generally combines a torque-dense motor with a relatively low reduction. It seeks useful joint effort while limiting some transmission effects that obstruct rapid interaction and backdriving. There is no universal ratio at which an actuator becomes quasi-direct drive. The choice must be evaluated through continuous and peak effort, thermal behavior, rotor inertia, losses and task demands. The actuator design study by Wensing et al. (2017) gives a concrete account of these tradeoffs in legged robotics and of the limits of estimating contact force through actuator signals.

Proprioceptive sensing concerns the system's internal mechanical state and interaction-related information. In this actuation setting it often includes position and current used to estimate effort. Current-based effort estimation is valuable, but it does not become an independent output torque measurement. In Equation (3.12), rotor acceleration, losses and transmission dynamics separate electromagnetic torque from delivered torque. Inferring a contact force adds geometry and external dynamics to that chain.

For data design, the useful comparison is therefore between specific architectures and accessible quantities. Can the interface set current or only position? Is position measured at the motor or output? Are limits and profiles known? Can an external observation check the inferred effort? A packaged smart servo can provide a highly useful demonstration interface; a backdrivable drive can provide useful interaction data. Neither architecture removes the need to specify what its records represent.

3.3 Nested feedback loops

Feedback uses a reported or estimated consequence of action to adjust a later input. It can regulate a quantity despite disturbances, but its performance depends on the plant, sensing and timing. A nested architecture often places current regulation inside velocity regulation, with position regulation outside. Different hardware may combine, omit or implement these loops differently. We derive idealized versions to understand their roles before adding implementation limits.

3.3.1 Current and effort regulation

Suppose current i is available, and let i_d be its target. A proportional and integral current regulator with back-electromotive-force feedforward is

$$\begin{aligned} e_i &= i_d - i, \qquad \dot{z}_i = e_i, \\ u_d &= k_e \dot{\theta}_m + K_{pi} e_i + K_{ii} z_i. \end{aligned} \tag{3.20}$$

For constant i_d, exact back-emf compensation and unsaturated voltage, linear perturbations have characteristic polynomial

$$Ls^2 + (R + K_{pi})s + K_{ii}. \tag{3.21}$$

To see the origin of this polynomial, substitute $i = i_d - e_i$ into the motor equation. For constant i_d, $L\dot{e}_i = Ri_d - (R + K_{pi})e_i - K_{ii}z_i$. Differentiating and using $\dot{z}_i = e_i$ gives the corresponding homogeneous second-order equation. The proportional gain augments electrical damping, while integral action can remove a constant current error under these assumptions. Inductance, resistance and gains determine the response. Positive coefficients give stability for this second-order continuous-time model, not a guarantee for every sampled implementation.

If the current loop is sufficiently faster than the outer motion being studied, we may approximate $i \simeq i_d$ and treat $Nk_t i_d$ as an available joint effort in the ideal rigid-transmission model. The approximation has a bandwidth and operating range. Voltage saturation at high speed, current limits, delay, sensor filtering and temperature-dependent resistance can make it inaccurate. A fast desired change of effort also needs inductive voltage; the steady expression (3.3) is insufficient to predict it. For a drive with symmetric terminal-voltage and current limits, feasibility requires

$$|Ri + k_e N\dot{q} + L\dot{i}| \le u_{\max}, \qquad |i| \le i_{\max}. \tag{3.22}$$

These are constraints on combinations of speed, current and current change. They do not define a rectangular joint-effort-versus-speed region. Actual drive and regeneration restrictions may further reduce that region.

An effort controller can use this current loop as an inner element. To interpret its records, retain both the current target and the current report, with their times and limits. Otherwise a learner may be asked to predict the effect of a current that was requested but never achieved. An output torque sensor, when available and calibrated, supplies different evidence and can test the assumptions of current-based estimation.

3.3.2 Velocity and position control

With the ideal inner effort approximation, write a velocity regulator as

$$\tau_d = K_{pv}(\dot{q}_d - \dot{q}) + K_{iv} z_v + \hat{g}_q(q), \qquad \dot{z}_v = \dot{q}_d - \dot{q}. \tag{3.23}$$

The target τ_d is converted to $i_d = \tau_d/(N k_t)$ only within the assumed transmission and calibration model. $\hat{g}_q$ is a gravity compensation estimate. Inertia, damping and gain units now differ from those in the electrical loop: a velocity error in $\mathrm{rad/s}$ must be mapped to joint effort in $\mathrm{N\,m}$.

Position can be regulated by a further outer loop that generates velocity, or by a direct effort law. For the scalar rigid model, a useful direct law is

$$\tau_d = K_p(q_d - q) + K_d(\dot{q}_d - \dot{q}) + \hat{g}_q(q). \tag{3.24}$$

The proportional term creates an effort opposing position error. The derivative term damps relative motion. For smoothly varying references, $\dot{q}_d$ is the reference velocity. A discontinuous position step cannot be differentiated as an ordinary finite velocity; a practical setpoint step may instead use derivative feedback only on measured motion, setting the commanded velocity term to zero. This avoids a derivative kick and is the convention used in the step example below. This is a feedback law with physical dimensions: K_p has units $\mathrm{N\,m/rad}$ and K_d has units $\mathrm{N\,m\,s/rad}$. They cannot be transferred as bare numbers between degree-based motor coordinates, radian-based joint coordinates and differently scaled current interfaces.

For a moving target, optional model feedforward adds $\hat{J}_\Sigma \ddot{q}_d + \hat{b}_\Sigma \dot{q}_d$ to Equation (3.24). Such feedforward can reduce tracking error when the model is useful. It also introduces dependence on the target's derivatives and on estimated physical parameters. The current, velocity and position loops answer different regulation questions; none makes its target identical to the physical consequence by definition.

Closed-loop collection inherits these distinctions. A position-controlled demonstration records targets filtered through internal dynamics. It does not directly provide a torque policy's training actions. Conversely, a collection of current commands is not interchangeable with desired joint positions. The learner's action space must match the semantics of the collected actions, or an explicit conversion with stated assumptions is required.

3.3.3 Closed-loop response, damping and stability

For a constant q_d, exact gravity compensation, no external disturbance and no extra friction, let $e = q - q_d$. Substituting Equation (3.24) into Equation (3.14) gives

$$J_\Sigma \ddot{e} + (b_\Sigma + K_d)\dot{e} + K_p e = 0. \tag{3.25}$$

The model predicts natural frequency and damping ratio

$$\omega_n = \sqrt{K_p/J_\Sigma}, \qquad \zeta = \frac{b_\Sigma + K_d}{2\sqrt{J_\Sigma K_p}}. \tag{3.26}$$

Positive inertia, stiffness and total damping make this ideal homogeneous second-order system asymptotically stable. Critical damping means $\zeta = 1$. These properties describe the stated model and operating conditions. They are not a certificate for a device with unmodeled compliance, saturation or delay.

An energy argument makes the physical connection clear. With $E = \frac{1}{2} J_\Sigma \dot{e}^2 + \frac{1}{2} K_p e^2$, constant inertia gives

$$\dot{E} = -(b_\Sigma + K_d)\dot{e}^2 \le 0. \tag{3.27}$$

The controller adds a virtual potential and damping to the error dynamics. With model error or disturbance, a forcing term enters this balance. For example, without gravity compensation a static equilibrium satisfies $K_p(q_d - q_*) = g_q(q_*)$ in this simplified model. A nonzero steady position error can therefore be physically necessary to supply holding effort, rather than evidence that the encoder has failed.

Worked example: bandwidth is constrained by available effort. For the constructed joint in Equation (3.16), target a $5\,\mathrm{Hz}$ natural frequency and critical damping. With $\omega_n = 2\pi(5)\,\mathrm{rad/s}$, Equation (3.26) gives

$$K_p = 8.7971\,\mathrm{N\,m/rad}, \qquad K_d = 0.55004\,\mathrm{N\,m\,s/rad}. \tag{3.28}$$

Using derivative feedback only on measured velocity, at rest at the downward position a $0.05\,\mathrm{rad}$ target step initially requests $0.43986\,\mathrm{N\,m}$; a $0.35\,\mathrm{rad}$ step requests $3.07899\,\mathrm{N\,m}$. A $2.0\,\mathrm{A}$ current limit permits only $Nk_t i_{\max} = 0.600\,\mathrm{N\,m}$ in our ideal model. The larger request therefore cannot follow the unsaturated linear prediction. No measured settling time is implied by these gain calculations.

Loading consumes the same effort budget. At horizontal rest, gravity alone requires $0.51012\,\mathrm{N\,m}$. The maximum positive acceleration under the same current limit is then approximately $(0.600 - 0.51012)/0.0089133 = 10.084\,\mathrm{rad/s^2}$, before additional losses. A collection protocol that asks for the same aggressive motion at every orientation may systematically encounter different saturation regimes. Those regimes are part of the data-generating conditions, not unexplained noise.

The same gains also need not give the same response after changing reduction or payload, because J_Σ changes. Recording controller gains without recording the mechanism and load is therefore incomplete. Later identification and learning chapters will ask whether these changes should be modeled, estimated online or covered in training data.

3.4 The implemented controller

The continuous equations distinguish physical mechanisms, but a real implementation reads finite-resolution reports and updates commands at particular times. The implemented controller includes the conversion between program variables and those equations. A simulation comparison must make that conversion explicit as well.

3.4.1 Sampling, filtering, saturation and trajectory shaping

Let a controller update at $t_k = kT_s$ and hold its requested input between updates. A zero-order hold is the simplest model of that behavior. The motor and body continue to evolve between samples. Packet arrival, actuator update and logger receipt can occur at different times, so a single nominal frequency does not specify the entire timing chain.

A backward difference estimates velocity from position:

$$\hat{\dot{q}}_k = \frac{q_k - q_{k-1}}{T_s}. \tag{3.29}$$

It estimates an interval's motion, not an exact instantaneous velocity at the new sample. Filtering reduces some fluctuations while adding a temporal response. For example,

$$\hat{v}_k = \alpha \hat{v}_{k-1} + (1-\alpha)\hat{\dot{q}}_k, \qquad 0 \le \alpha < 1 \tag{3.30}$$

specifies a particular discrete filter. Its coefficient and initialization are needed to interpret the estimate. A supplied device velocity report may use a different estimator; naming both columns "velocity" conceals that difference.

Saturation limits an effort request to a feasible range:

$$\tau_a = \mathrm{clip}(\tau_d, -\tau_{\max}, \tau_{\max}). \tag{3.31}$$

τ_a here denotes the limited software request, not independently measured mechanical torque. Current-loop error and physical losses can still separate it from delivered effort. If an integral controller continues accumulating error while its output is limited, it can produce a long recovery transient. One anti-windup approach feeds the difference between limited and unlimited outputs back into the integrator. Another conditionally stops integration. Their details affect the records and must not be omitted from an exact implementation comparison.

Trajectory shaping changes the target before the feedback law sees it. Velocity and acceleration bounds can turn a position step into a ramped or smooth profile. A joint may accurately track the internal profile while lagging behind the original step. Treating that profile as an unexplained actuator delay confuses an intentional command transformation with a plant property. If accessible, record the original request and profile separately.

For a first indication of sampled stability, ideal forward-Euler integration maps a continuous pole λ to $1 + T_s\lambda$. Stability of that numerical approximation requires $|1 + T_s\lambda| < 1$. For a negative real pole $-a$, this becomes $0 < T_s a < 2$. This calculation explains why a stable continuous model does not automatically justify an arbitrary discrete update. It is not the exact stability criterion for every held-input controller or simulator. An actual discrete design must include the plant discretization, feedback timing and delays; later simulation work will make those distinctions explicit.

3.4.2 Requested targets, applied inputs and sensor reports

A useful record distinguishes at least four stages: a task request, the controller's generated target, the drive's realized electrical input and the mechanical response reported by sensors. These are physical and computational relationships, not simply alternative column names. A position request may become an internal profile, then current and voltage, then shaft motion, transmitted motion and an encoder report. Some stages may be inaccessible. Unobserved quantities should remain identified as unobserved rather than filled with their targets.

Official smart-servo documentation makes such distinctions concrete. For example, the DYNAMIXEL XM430's Goal Current is a target in current control and a current limit in current-based position control. Its Present Current register reports an electrical quantity (ROBOTIS, n.d.). This is a model-specific example; it does not imply that every DYNAMIXEL exposes these modes. A data contract should identify the exact model, mode, scale and meaning of each field.

In mechanics, q, $\dot{q}$ and τ refer to a chosen joint coordinate and its power-conjugate effort. In code, matching that meaning requires the joint identifier, encoder zero, direction, reduction, units and time interpretation. If $\theta_m = Nq$, motor-side and joint-side gains are different. If a report is in degrees, converting it to radians before differentiating changes the numerical velocity by the same scale. The order of conversion, filtering and differentiation should be documented.

MuJoCo provides a useful implementation comparison. Its `motor` actuator is an ideal direct-force shortcut, not an electrical motor with Equation (3.1). For a scalar hinge transmission with `gear="1"`, unit gain and no bias, a limited control is a generalized effort request. Other transmissions or actuator definitions have different semantics (DeepMind Technologies Limited, n.d., n.d.). An illustrative definition is

```
<joint name="hinge" type="hinge" axis="0 0 1"
       armature="0.00072" damping="0.01"/>
<actuator>
  <motor name="effort" joint="hinge" gear="1"
         ctrllimited="true" ctrlrange="-0.6 0.6"/>
</actuator>
```

These fragments belong inside an otherwise valid model; they are not a complete robot. The body supplies J_L and gravity. Joint `armature` supplies the additional $N^2 J_m$ in this scalar example, and joint damping supplies b_Σ. Adding the same rotor inertia again to the body would double count it. A direct effort actuator must likewise not receive the ratio N again if the code already converted current to $Nk_t i$.

For this specific ideal model, a controller may assign its limited effort request to `data.ctrl[actuator_id]`. The corresponding `data.qfrc_actuator[dof_id]` is the model's applied generalized actuator force at the simulation instant at which it is evaluated. The actuator index and velocity-degree-of-freedom index need not be equal. After advancing simulation, associate each logged quantity with the time and evaluation stage at which it was computed; do not assume that reading several arrays together guarantees a common physical instant.

Reproducing electrical dynamics would require representing current as a state, updating it from the voltage equation, and applying the resulting electromagnetic effort with consistent timing. A position actuator shortcut would instead introduce a servo law. Neither is inherently wrong: the important question is which processes the chosen model includes and which data interpretation it supports. This mapping preserves Featherstone's joint-effort meaning while making the simulator's additional assumptions visible.

3.5 Collecting data from an actuator

Actuator records become useful when their meaning and collection purpose are specified together. The same motion can support a tracking comparison, a parameter estimate, a forward prediction model or an imitation dataset. Those uses share signals but demand different comparisons. ExpGen PhySyst places the design decision before collection: what uncertainty or capability should these interactions help address?

3.5.1 Electrical, proprioceptive and thermal observations

Electrical observations may include current, bus voltage or drive status. Mechanical observations may include encoder position, derived velocity and external force or motion measurements. Temperature observations describe a thermal state at a particular sensing location. A case-temperature sensor is not necessarily a winding-temperature sensor, and an electrical current report is not necessarily phase-resolved current. The observation chain must be understood before a signal is assigned a physical name.

A measured current can support an effort estimate such as

$$\hat{\tau}_{\mathrm{em}} = N\hat{k}_t(i_{\mathrm{rep}} - \hat{i}_0), \tag{3.32}$$

where the report has first been converted to amperes and $\hat{i}_0$ is an estimated offset. This is an output-equivalent electromagnetic effort estimate. It still includes effort associated with rotor acceleration and internal losses. Comparing it with external load torque requires those terms and their uncertainty. In the constructed $N = 6$, $k_t = 0.05$ example, a $0.10\,\mathrm{A}$ current offset changes this estimate by $0.030\,\mathrm{N\,m}$. That error may be consequential even if the position trace appears smooth.

Temperature can change resistance, available continuous effort and protection behavior. A rough lumped model is

$$C_{\mathrm{th}}\dot{T}_w = i^2R(T_w) - \frac{T_w - T_a}{R_{\mathrm{th}}} + P_{\mathrm{other}}, \tag{3.33}$$

where C_{th} is thermal capacitance, R_{th} thermal resistance, T_w a modeled winding temperature and T_a ambient temperature. This is not a claim that a product's reported temperature equals T_w or that a single thermal state explains every time scale. Its role is to show why repeated trials can depend on recent operation. Trial order, rest periods and available temperature reports help distinguish a load effect from thermal history.

Measurements and estimates also have temporal uncertainty. If independent position errors have standard deviation σ_q, the backward difference in Equation (3.29) has noise standard deviation $\sqrt{2}\sigma_q/T_s$. A second difference has $\sqrt{6}\sigma_q/T_s^2$ under the same independent-noise assumption. For $\sigma_q = 0.001\,\mathrm{rad}$ and $T_s = 0.010\,\mathrm{s}$, these are $0.14142\,\mathrm{rad/s}$ and $24.495\,\mathrm{rad/s^2}$. Multiplying the latter by our joint inertia gives an inertial-effort noise scale of $0.21833\,\mathrm{N\,m}$. Smoother-looking position data therefore need not support reliable raw acceleration inference. Quantization, correlated noise and filtering require their own treatment; the stated values are a constructed independent-noise calculation.

Sampling faster can preserve more temporal detail, but applying the same finite difference to independent sample noise amplifies that noise more. This does not mean faster acquisition is intrinsically worse. It means that acquisition, estimation and the physical bandwidth of interest must be designed together. Chapter 7 develops calibration and state inference beyond these first observations.

3.5.2 A controlled joint experiment and its measurement limits

Suppose our question is whether a changed payload explains a changed response. The physical hypothesis is not simply "the plots differ." Under the rigid model, payload changes both J_L and g_q; controller settings, current limits and transmission conditions may change the observed consequence. A useful experiment must expose enough of this chain to evaluate the hypothesis.

Begin by defining the body, load mass and load placement. Define the coordinate zero and direction, the available control mode and the motion range in which the assumptions are credible. Retain model or serial identity, controller settings, limits, supply conditions and relevant thermal history. These are conditions of the comparison, rather than extra descriptive metadata added after plotting.

Next choose interventions that reveal distinct effects. At rest, the rigid model's inertial term vanishes, so a static record cannot identify inertia. Motions at multiple positions change gravitational loading. Acceleration and deceleration expose inertia, while opposite directions can expose asymmetric losses or friction. Repetitions reveal variability; a changed payload tests whether a fitted description extends to a specified new condition. The available actuator and fixture must determine feasible amplitudes and rates. No particular command sequence is claimed here as a verified hardware procedure.

Table 3.1: Physical meanings and limitations of joint-record fields.

Quantity	Meaning to retain	Important limitation
q_d, internal profile	Requested joint target; profile if accessible	Not an encoder measurement
q_{rep}	Converted encoder report, zero, direction and location	May describe motor-side motion
i_d, i_{rep}	Current target or limit, and electrical report	Neither alone measures load torque
u, supply report	Terminal input if measured; otherwise named supply quantity	Bus voltage alone is not terminal voltage
Status, limits, temperature	Active mode, clipping/protection and sensor location	Available reports may omit internal events
Times and trial identity	Command, sample or receipt times; load and repetition	Shared units do not imply synchronized clocks

When extra friction τ_f is neglected and k_t is known, the reduced equation can be written as a linear parameter relation:

$$N k_t i_k + \tau_{\mathrm{ext},k} = \begin{bmatrix} \ddot{q}_k & \dot{q}_k & \sin q_k \end{bmatrix} \begin{bmatrix} J_\Sigma \\ b_\Sigma \\ C_g \end{bmatrix}. \tag{3.34}$$

This form shows why the motions matter. If every sample is at rest, the first two columns vanish. If the chosen motion makes columns nearly dependent, different parameter combinations can explain much the same record. Estimating all coefficients is then poorly determined even if a curve fit looks good. Unknown k_t, unmodeled friction and noisy derived acceleration introduce further ambiguities. Chapter 8 develops identification and its statistical limitations; here the equation helps design the encounter before fitting.

Worked example: what a contrast can isolate. In an ideal constructed pair of observations, let $q = \pi/6$ and $\ddot{q} = 2.0\,\mathrm{rad/s^2}$ in both cases, while velocities are $+1.0$ and $-1.0\,\mathrm{rad/s}$. Equation (3.14) with the example parameters gives electromagnetic joint efforts 0.282887 and $0.262887\,\mathrm{N\,m}$. Their difference gives

$$b_\Sigma = \frac{\tau_+ - \tau_-}{2(1.0\,\mathrm{rad/s})} = 0.0100\,\mathrm{N\,m\,s/rad}. \tag{3.35}$$

This contrast cancels the equal inertia and gravity contributions. Coulomb friction would add another direction-dependent term; unequal accelerations would stop the inertial terms from canceling. The point of the example is not that two real samples solve identification, but that a contrast is informative only for the effects it actually distinguishes.

Timing can produce an apparent physical discrepancy of comparable importance. If a motion is $q(t) = A\sin(2\pi f t)$ and an otherwise identical trace is shifted by d, the difference has amplitude

$$A_{\mathrm{diff}} = 2A\,|\sin(\pi f d)|\,. \tag{3.36}$$

For $A = 0.30\,\mathrm{rad}$, $f = 2.0\,\mathrm{Hz}$ and $d = 0.020\,\mathrm{s}$, this is $0.07520\,\mathrm{rad}$. That is an amplitude, not a root-mean-square error. A mismatch of timestamps can thus resemble a substantial tracking or Sim2Real error. Correcting time alignment is justified when supported by the acquisition chain; choosing an arbitrary shift merely to minimize error can conceal a real control delay.

A compact record should consequently distinguish the fields in Table 3.1. Unavailable fields should be declared rather than inferred silently. Raw reports and conversions should remain distinguishable, so later recalibration can change an estimate without changing the identity of the original observation.

The same records lead to different learning problems. A forward predictor may map recent observations, realized or requested actions and context to future observations. A behavior-cloning policy may instead map an observation history to the demonstrated target. For the former, missing saturation state can make the same nominal action appear to have inconsistent effects. For the latter, mixing position and current actions without a declared representation makes the target itself ambiguous. If unobserved actuator state matters, a short history can sometimes help a predictor, but it does not establish that the missing state has been physically identified.

Collection selection matters too. Keeping only successful reaches may help describe nominal execution while omitting

saturation, reversal or recovery. Treating neighboring samples as independent trials overstates the variety of experience. A useful split reserves complete trials or operating conditions according to the claim being tested, rather than randomly sharing adjacent points from one trace across training and testing. General dataset design is the subject of Chapter 9; actuator physics supplies the first reason that collection conditions must accompany the samples.

Exercises and further investigation. The following questions extend the derivations without requiring access to a particular motor. Use consistent coordinates, units and explicit assumptions.

1. Starting from Equation (3.8), project the inertia of the example link about its hinge. Add the point payload, then explain why moving its mounting radius changes inertia quadratically and the gravity coefficient linearly. Which geometric observations would be needed to test the calculation?
2. Derive Equation (3.14) from the two separate mechanical balances. Explain why scaling torque by N but forgetting $N^2 J_m$ gives an inconsistent acceleration prediction. Repeat the numerical budget for $N = 30$ while keeping the constructed motor and load parameters.
3. Remove gravity compensation from the position controller. Find the implicit holding equilibrium and discuss how changing K_p changes its error and effort. State where the unsaturated conclusion ceases to apply.
4. For a constant current target, derive the characteristic polynomial in Equation (3.21). Explain which assumptions are lost when voltage clipping or imperfect back-emf compensation occurs.
5. Design a comparison that distinguishes a delayed report from a compliant output. State sensor locations, timestamp meanings and which traces would remain ambiguous if only one encoder were available.
6. Inspect the signal meanings of one documented actuator. Construct a table separating request, internal transformation, measurement and estimate. Identify a learning question that its available reports can support and another that needs an additional observation. Do not assume undocumented modes or sensors.
7. Propose two datasets from the same joint: one for identifying loading effects and one for learning reaching behavior. Specify interventions, retained failures and held-out conditions, then explain why their most informative records need not be the same.

The arithmetic of the constructed examples is reproduced in the companion `tools/check_chapter03.py`. It checks numerical balances and conversions; it is not a hardware test or a simulation-performance result. For deeper mechanics, read Featherstone (2008) on spatial vectors, inertia and joint models. For actuator architecture and interaction-oriented design, Wensing et al. (2017) connects electrical and transmission choices to force-control behavior. Official actuator and simulator documentation should be read alongside these foundations when interpreting a concrete interface.

Actuation closes the first link between physical design and data. Motor, transmission, feedback and acquisition choices determine which actions are possible and which consequences are visible. Chapter 4 now develops the spatial mechanics needed to carry that reasoning beyond a single joint, preserving the same correspondence among physical quantities, mathematical representations and computational variables.

Chapter 4 Spatial Mechanics of Rigid Bodies

Attach a small tracking marker to the lever studied in Chapter 3. The encoder reports a joint angle, a camera reports the marker's pose, and a simulator reports the motion of a modeled body. All three records concern the same intended movement, but they do not contain interchangeable quantities. The marker is displaced from the joint, its axes may be rotated, and the simulator may use a different origin or orientation representation. A large numerical discrepancy can appear before there is any disagreement about the underlying physics.

Spatial mechanics gives us a language for resolving these differences and for describing the forces and inertia behind the motion. Its six-dimensional vectors combine angular and linear quantities while retaining their different units and transformation rules. The value is more than compact notation: a single power relation connects motion, loading and joint effort, and a single body equation combines translation and rotation.

We follow the angular-first convention of Featherstone (2008). The aim is to derive the physical meaning of the representations before assembling a robot's equations in Chapter 5. Every change of coordinates must preserve the physical situation it describes. For ExpGen PhySyst, this is a foundation for designing data: a recorded pose, velocity or wrench is useful only when its body, reference point, axes, units and time are understood.

4.1 Configurations and coordinate frames

The configuration of a rigid body specifies where it is and how it is oriented. Rigid means that distances between its material points are fixed within the model. A real structure can deform, but beginning with a rigid model makes its geometric assumptions explicit. A later discrepancy can then motivate a compliance model rather than being hidden in an undefined coordinate change.

A coordinate frame consists of an origin and three orthonormal axes. It is a representation device, not a new physical body. Two frames can be attached to the same body; one can also express a moving body's quantities in fixed world axes. Distinguishing the body being described from the frame used to describe it prevents a common confusion between physical motion and changing numbers.

4.1.1 Rotations, rigid transformations and joint coordinates

Let ${}^{A}\mathbf{R}_{B}$ map components of a free three-vector from frame B to frame A. Its columns are the B axes expressed in A. A proper rotation satisfies

$$({}^{A}\mathbf{R}_{B})^{\mathsf{T}}\,{}^{A}\mathbf{R}_{B} = \mathbf{1}_{3}, \qquad \det({}^{A}\mathbf{R}_{B}) = 1, \qquad {}^{B}\mathbf{R}_{A} = ({}^{A}\mathbf{R}_{B})^{\mathsf{T}}. \tag{4.1}$$

The determinant condition excludes reflections. A rotation matrix changes coordinates without changing a vector's length. The same numerical matrix can also be used to describe a physical rotation, but these interpretations must not be mixed without identifying what is held fixed.

Let ${}^{A}\mathbf{r}_{AB}$ point from origin O_A to origin O_B, expressed in A. Coordinates of a physical point P obey

$${}^{A}\mathbf{p}_{P} = {}^{A}\mathbf{R}_{B}\,{}^{B}\mathbf{p}_{P} + {}^{A}\mathbf{r}_{AB}. \tag{4.2}$$

Unlike a free vector, point coordinates depend on the origin. The homogeneous matrix combines the rotation and translation:

$${}^{A}\mathbf{T}_{B} = \begin{bmatrix} {}^{A}\mathbf{R}_{B} & {}^{A}\mathbf{r}_{AB} \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix}, \qquad \begin{bmatrix} {}^{A}\mathbf{p}_{P} \\ 1 \end{bmatrix} = {}^{A}\mathbf{T}_{B} \begin{bmatrix} {}^{B}\mathbf{p}_{P} \\ 1 \end{bmatrix}. \tag{4.3}$$

The bottom entry distinguishes a point from a direction, for which that entry would be zero. These 4×4 matrices describe point geometry. They are not the 6×6 transformations used later for spatial motion and force.

Composition follows the frame labels: ${}^{A}\mathbf{T}_{C} = {}^{A}\mathbf{T}_{B}\,{}^{B}\mathbf{T}_{C}$. Matrix order matters. Translating along a body's local axis and then rotating its pose generally places it somewhere different from rotating first and then translating along a fixed world axis. A stored transform therefore needs an explicit mapping direction; a generic filename does not supply that information.

A revolute joint restricts relative configuration to rotation about an axis; a prismatic joint restricts it to translation along an axis. Their scalar coordinates have different units, and their zeros and positive directions are design choices. A free body instead has six independent configuration dimensions. The number of stored orientation parameters can exceed the number of independent dimensions, as the next subsection shows.

Worked example: a point seen from another frame. Suppose O_B lies at ${}^{A}\mathbf{r}_{AB} = (0.20, 0.10, 0)^{\mathsf{T}}$ m, and ${}^{B}\mathbf{R}_{A}$ rotates components by -90° about the common positive z direction. For ${}^{A}\mathbf{p}_{P} = (0.50, 0.40, 0)^{\mathsf{T}}$ m, inversion of Equation (4.2) gives

$$ {}^{B}\mathbf{p}_{P} = {}^{B}\mathbf{R}_{A}({}^{A}\mathbf{p}_{P} - {}^{A}\mathbf{r}_{AB}) = (0.30, -0.30, 0)^{\mathsf{T}}\ \mathrm{m}. \tag{4.4} $$

The point has not moved. Subtracting the origin displacement and then rotating the components changes its description. Rotating the original coordinates without translating would describe a different geometric quantity. A marker pose must undergo the corresponding mounting transformation before it can be compared with a body or joint pose.

4.1.2 Configuration coordinates and tangent velocities

For a hinge, $\dot{q}$ is its scalar angular velocity. For a general orientation, the derivative of its stored parameters is not automatically angular velocity. For instance, a rotation matrix has nine entries subject to orthogonality, and a unit quaternion has four entries subject to a norm constraint. Each represents only three independent rotational dimensions.

Consider a body orientation ${}^{W}\mathbf{R}_{B}(t)$ relative to a fixed world frame W. Let $[\mathbf{x}]_{\times}\mathbf{y} = \mathbf{x} \times \mathbf{y}$ denote the cross-product matrix. Angular velocity can be expressed in either axes:

$$ [\boldsymbol{\omega}^{W}]_{\times} = \dot{\mathbf{R}}\mathbf{R}^{\mathsf{T}}, \qquad [\boldsymbol{\omega}^{B}]_{\times} = \mathbf{R}^{\mathsf{T}}\dot{\mathbf{R}}, \qquad \boldsymbol{\omega}^{W} = \mathbf{R}\boldsymbol{\omega}^{B}, \tag{4.5} $$

where $\mathbf{R} = {}^{W}\mathbf{R}_{B}$. The products are skew-symmetric because the derivative of $\mathbf{R}^{\mathsf{T}}\mathbf{R} = \mathbf{1}_3$ is zero. This gives three independent angular-velocity components rather than nine independent matrix-entry rates.

For a scalar-first Hamilton unit quaternion $Q = (s, \mathbf{u})$ representing the same orientation, define multiplication by

$$ (s_1, \mathbf{u}_1) \otimes (s_2, \mathbf{u}_2) = \left(s_1 s_2 - \mathbf{u}_1^{\mathsf{T}}\mathbf{u}_2,\, s_1\mathbf{u}_2 + s_2\mathbf{u}_1 + \mathbf{u}_1 \times \mathbf{u}_2\right). \tag{4.6} $$

With these conventions,

$$ \dot{Q} = \frac{1}{2}Q \otimes (0, \boldsymbol{\omega}^{B}) = \frac{1}{2}(0, \boldsymbol{\omega}^{W}) \otimes Q, \qquad s^2 + \mathbf{u}^{\mathsf{T}}\mathbf{u} = 1. \tag{4.7} $$

Changing the multiplication order changes which angular-velocity frame the formula accepts. A scalar-last library requires another storage conversion. The choices are valid conventions, but an unmarked mixture is an error.

More generally, distinguish configuration parameters $\mathbf{q}$ from generalized tangent velocity $\boldsymbol{\nu}$:

$$ \dot{\mathbf{q}} = \mathcal{N}(\mathbf{q})\boldsymbol{\nu}. \tag{4.8} $$

The matrix $\mathcal{N}$ depends on the representation. It is an identity for independent scalar joint angles but need not be square for a free body stored with a quaternion. A generalized velocity vector is also not automatically a single spatial motion vector: it may collect scalar joint rates and a free-joint velocity with an implementation-specific ordering.

Quaternions Q and $-Q$ represent the same orientation. Their component distance is 2 for unit Q, even though the physical orientation difference is zero. Consequently, a naive quaternion difference is a poor motion target or data-quality test. A relative rotation can instead be formed from $\delta Q = Q_0^{-1} \otimes Q_1$, with the sign chosen so its scalar part is nonnegative for a shortest rotation. If $\delta Q = (\delta s, \delta\mathbf{u})$, the corresponding rotation-vector magnitude is

$$ \phi = 2\,\mathrm{atan2}(\|\delta\mathbf{u}\|, \delta s), \qquad \delta\boldsymbol{\phi} = \phi\frac{\delta\mathbf{u}}{\|\delta\mathbf{u}\|}. \tag{4.9} $$

The zero-rotation limit is taken continuously; the axis at exactly π has a sign ambiguity. The relative rotation is a finite displacement, not itself an instantaneous velocity. Estimating velocity from it also requires an interval and a frame convention.

For collection design, these facts determine how a pose sequence can be used. A sign change in stored quaternions need not be a physical jump. Dividing raw orientation-parameter differences by elapsed time need not give angular velocity. Retaining the representation and its integration/differentiation rule makes the motion interpretable without forcing every

source to use the same storage layout.

4.2 Spatial motion and spatial force

Rotation and translation are coupled in a rigid body's velocity field. A point away from the rotation axis moves even when the axis is stationary. Forces have the dual property: a force applied away from an origin creates a moment about it. Spatial vectors retain these couplings instead of treating the two three-vectors as unrelated quantities.

4.2.1 Angular-first motion vectors and joint motion subspaces

At a reference point O on a body, express angular velocity and point velocity in common axes and stack them angular-first:

$$\mathbf{v}_O = \begin{bmatrix} \boldsymbol{\omega} \\ \mathbf{v}_{O,\mathrm{lin}} \end{bmatrix}. \tag{4.10}$$

The bold six-vector $\mathbf{v}_O$ describes body motion, while $\mathbf{v}_{O,\mathrm{lin}}$ is its linear part. Angular entries have units $\mathrm{rad/s}$ and linear entries $\mathrm{m/s}$. The mixed units are intentional. A spatial motion vector is neither a pose nor six independent Cartesian position derivatives. An ordinary Euclidean norm of the six entries has no invariant physical meaning: it mixes units and changes when the reference point changes. Learning features may be scaled for a particular objective, but that choice is distinct from a physical invariant such as power or kinetic energy.

For another body-fixed point P, with displacement $\mathbf{r}_{OP}$ in the same axes, rigidity gives

$$\mathbf{v}_{P,\mathrm{lin}} = \mathbf{v}_{O,\mathrm{lin}} + \boldsymbol{\omega} \times \mathbf{r}_{OP}. \tag{4.11}$$

One way to derive this is to write the point trajectory as $\mathbf{p}_P^W = \mathbf{p}_O^W + \mathbf{R}\mathbf{r}_{OP}^B$ and differentiate, using Equation (4.5). A stored linear velocity therefore needs its reference point. World axes alone do not identify that point.

For a revolute joint with unit axis $\mathbf{a}$ through O, relative motion is $\mathbf{v}_J = \mathbf{S}\dot{q}$ with

$$\mathbf{S}_{\mathrm{rev,O}} = \begin{bmatrix} \mathbf{a} \\ \mathbf{0} \end{bmatrix}, \qquad \mathbf{S}_{\mathrm{rev,P}} = \begin{bmatrix} \mathbf{a} \\ \mathbf{a} \times \mathbf{r}_{OP} \end{bmatrix}, \qquad \mathbf{S}_{\mathrm{pris}} = \begin{bmatrix} \mathbf{0} \\ \mathbf{a} \end{bmatrix}. \tag{4.12}$$

The revolute subspace changes its linear block when the reference point moves. The prismatic subspace has no angular block. Their generalized velocities have units $\mathrm{rad/s}$ and $\mathrm{m/s}$ respectively. A joint's motion subspace describes admissible relative motion, not the full absolute velocity of a link in a moving mechanism.

There is another distinction when poses are represented relative to world. For the body pose $(\mathbf{R}, \mathbf{p})$, its motion in body axes at its body origin is

$$\mathbf{v}^B = \begin{bmatrix} \boldsymbol{\omega}^B \\ \mathbf{R}^\mathsf{T}\dot{\mathbf{p}} \end{bmatrix}. \tag{4.13}$$

Using world axes at that same body origin gives $[\boldsymbol{\omega}^W; \dot{\mathbf{p}}]$. Expressing the motion instead about the world origin gives linear part $\dot{\mathbf{p}} - \boldsymbol{\omega}^W \times \mathbf{p}$. All three are legitimate descriptions of the same rigid motion with different axes or reference points. The velocity field can be evaluated at the world origin even when no material point of the body is located there. Calling all three simply "world velocity" would hide a consequential difference.

4.2.2 Wrenches, duality and instantaneous power

Forces applied to a rigid body can be represented at O by a resultant force and its moment:

$$\mathbf{f}_O = \begin{bmatrix} \mathbf{n}_O \\ \mathbf{F} \end{bmatrix}, \qquad \mathbf{n}_O = \sum_j \mathbf{r}_{OP_j} \times \mathbf{F}_j + \sum_j \mathbf{n}_j. \tag{4.14}$$

$\mathbf{n}_j$ denotes any applied pure couple. Moment is in $\mathrm{N\,m}$ and force in N. Translating the reference point changes moment while leaving resultant force unchanged if axes are unchanged. A wrench reports both the body loading and the point about which it is represented.

The mechanical power of these loads follows by substituting Equation (4.11) into the sum of point-force powers:

$$\begin{aligned} P &= \sum_j \mathbf{F}_j^{\mathsf{T}}(\mathbf{v}_{O,\mathrm{lin}} + \boldsymbol{\omega} \times \mathbf{r}_{OP_j}) + \sum_j \mathbf{n}_j^{\mathsf{T}}\boldsymbol{\omega} \\ &= \mathbf{n}_O^{\mathsf{T}}\boldsymbol{\omega} + \mathbf{F}^{\mathsf{T}}\mathbf{v}_{O,\mathrm{lin}} = \mathbf{f}_O^{\mathsf{T}}\mathbf{v}_O. \end{aligned} \tag{4.15}$$

The scalar triple product supplies the moment term. This pairing explains why motion and force transform differently: their transformations must preserve the same physical power.

For a joint with motion subspace $\mathbf{S}$, power gives

$$\mathbf{f}^{\mathsf{T}}\mathbf{S}\dot{q} = \tau\dot{q}, \qquad \tau = \mathbf{S}^{\mathsf{T}}\mathbf{f}. \tag{4.16}$$

For an ideal revolute bearing, constraint loads do no power in the allowed relative motion, so $\mathbf{S}^{\mathsf{T}}\mathbf{f}_c = 0$. They can nevertheless be physically large. A scalar joint-effort measurement cannot recover all bearing forces and moments. For a prismatic joint the same projection produces a force, rather than a torque; generalized effort is defined by its power conjugacy to the chosen coordinate.

Power also guides software checks. Stacking force before moment in one array while keeping angular velocity first in another creates a meaningless dot product. Scaling one linear block from millimetres to metres without scaling the associated transformations can corrupt a calculation even when array dimensions agree. An invariance test is useful, but it cannot replace correct physical labels: two consistently misinterpreted arrays can still pass an algebraic identity.

4.3 Changing coordinates and reference points

We now derive the transformations needed to reconcile records from different frames. This is a change of description of the same motion or loading, not the subtraction of an observer's motion. To form a relative spatial velocity, first express both absolute motions with common axes and reference point, then subtract them. Merely rotating a vector into a camera's axes does not subtract the camera's velocity.

4.3.1 Point transformations and spatial transformations

Use the compact notation $\mathbf{E} = {}^{B}\mathbf{R}_{A}$ and $\mathbf{r} = {}^{A}\mathbf{r}_{AB}$. Thus $\mathbf{E}$ maps A components into B, while $\mathbf{r}$ points from O_A to O_B in A components. Equation (4.11) gives

$$\begin{aligned} \boldsymbol{\omega}^B &= \mathbf{E}\boldsymbol{\omega}^A, \\ \mathbf{v}^B_{O_B,\mathrm{lin}} &= \mathbf{E}(\mathbf{v}^A_{O_A,\mathrm{lin}} - [\mathbf{r}]_\times \boldsymbol{\omega}^A). \end{aligned} \tag{4.17}$$

The minus sign comes from $\boldsymbol{\omega} \times \mathbf{r} = -\mathbf{r} \times \boldsymbol{\omega}$. The resulting motion transformation is

$$\mathbf{v}^B = \mathbf{X}_{BA}\mathbf{v}^A, \qquad \mathbf{X}_{BA} = \begin{bmatrix} \mathbf{E} & \mathbf{0} \\ -\mathbf{E}[\mathbf{r}]_\times & \mathbf{E} \end{bmatrix}. \tag{4.18}$$

Here the superscripts include the respective frame origins as well as their axes. If only the axes rotate, $\mathbf{r} = 0$ and both blocks rotate by $\mathbf{E}$. If only the point changes, $\mathbf{E} = \mathbf{1}_3$ and the off-diagonal block retains the angular-to-linear coupling.

In these same symbols, point coordinates transform by

$$\mathbf{T}_{BA} = \begin{bmatrix} \mathbf{E} & -\mathbf{E}\mathbf{r} \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix}, \qquad \mathbf{X}_{BA}^{-1} = \begin{bmatrix} \mathbf{E}^{\mathsf{T}} & \mathbf{0} \\ [\mathbf{r}]_\times \mathbf{E}^{\mathsf{T}} & \mathbf{E}^{\mathsf{T}} \end{bmatrix}. \tag{4.19}$$

$\mathbf{T}_{BA}$ is the same mapping as ${}^{B}\mathbf{T}_{A}$ in the earlier notation. Its translation column is the location of O_A expressed in B, not the vector $\mathbf{r}$ we started with. Writing out one point mapping is a reliable way to check a convention before building a spatial matrix.

These transformations compose: $\mathbf{X}_{CA} = \mathbf{X}_{CB}\mathbf{X}_{BA}$ for compatible instantaneous frames. A rotation from one source and a translation from another source can be combined only if their definitions and times are compatible. A fixed mounting calibration and a time-varying tracked pose have different roles even though both can be stored as matrices.

Worked example: motion at a displaced origin. Use the origins and rotation from Equation (4.4). Suppose a rigid body's angular and linear components at O_A are

$$\boldsymbol{\omega}^A = (0, 0, 2)^\mathsf{T}\,\mathrm{rad/s}, \qquad \mathbf{v}^A_{O_A,\mathrm{lin}} = (0.10, -0.20, 0.30)^\mathsf{T}\,\mathrm{m/s}. \tag{4.20}$$

The displacement adds $\boldsymbol{\omega}^A \times \mathbf{r} = (-0.20, 0.40, 0)^\mathsf{T}\,\mathrm{m/s}$. After rotation, Equation (4.18) gives

$$\boldsymbol{\omega}^B = (0, 0, 2)^\mathsf{T}\,\mathrm{rad/s}, \qquad \mathbf{v}^B_{O_B,\mathrm{lin}} = (0.20, 0.10, 0.30)^\mathsf{T}\,\mathrm{m/s}. \tag{4.21}$$

Rotating the original linear part alone would give $(-0.20, -0.10, 0.30)^\mathsf{T}\,\mathrm{m/s}$. The difference is a reference-point error, not evidence of a slipping transmission or inaccurate learning model. When a dataset combines tracker and simulator velocities, the mounting displacement can matter as much as the axis convention.

4.3.2 Power-preserving motion and force transformation rules

Physical power must agree in both descriptions. Substituting the motion transformation into that requirement gives

$$(\mathbf{f}^A)^\mathsf{T}\mathbf{v}^A = (\mathbf{f}^B)^\mathsf{T}\mathbf{X}_{BA}\mathbf{v}^A \quad \Longrightarrow \quad \mathbf{f}^A = \mathbf{X}_{BA}^\mathsf{T}\mathbf{f}^B. \tag{4.22}$$

The same-direction force transformation is consequently

$$\mathbf{f}^B = \mathbf{X}_{BA}^{-\mathsf{T}}\mathbf{f}^A, \qquad \mathbf{F}^B = \mathbf{E}\mathbf{F}^A, \qquad \mathbf{n}^B_{O_B} = \mathbf{E}(\mathbf{n}^A_{O_A} - [\mathbf{r}]_\times \mathbf{F}^A). \tag{4.23}$$

The moment term subtracts the moment of the resultant force about the displacement from O_A to O_B. Using $\mathbf{X}$ for both motion and force would fail this duality. Using its transpose in the wrong direction is another common error; the mapping labels resolve which direction is intended.

Worked example: the same power in two descriptions. For the preceding motion, take

$$\mathbf{n}^A_{O_A} = (0.50, -0.30, 0.80)^\mathsf{T}\,\mathrm{N\,m}, \qquad \mathbf{F}^A = (3, 4, 5)^\mathsf{T}\,\mathrm{N}. \tag{4.24}$$

The force and moment about O_B become $\mathbf{F}^B = (4, -3, 5)^\mathsf{T}\,\mathrm{N}$ and $\mathbf{n}^B_{O_B} = (0.70, 0, 0.30)^\mathsf{T}\,\mathrm{N\,m}$. Their powers are

$$\begin{aligned} P_A &= 0.80(2) + 3(0.10) + 4(-0.20) + 5(0.30) = 2.60\,\mathrm{W}, \\ P_B &= 0.30(2) + 4(0.20) - 3(0.10) + 5(0.30) = 2.60\,\mathrm{W}. \end{aligned} \tag{4.25}$$

Moment power and force power separately change with the reference point, but their sum does not. An external wrench sensor and a motion tracker can therefore support a power estimate only after their origins, axes and timing have been reconciled. Agreement of one force component is a weaker check.

The motion subspace transforms with motion, $\mathbf{S}^B = \mathbf{X}_{BA}\mathbf{S}^A$. Combining this with Equation (4.23) preserves the generalized effort $\mathbf{S}^\mathsf{T}\mathbf{f}$. Thus a joint-effort calculation does not depend on choosing the sensor origin or the joint origin, provided both the wrench and motion subspace are represented consistently.

4.4 Inertia and rigid-body dynamics

Geometry determines how motion and force are represented. Inertia determines the momentum and energy associated with motion. The connection is important for collected data: pose and velocity records describe kinematics, but explaining acceleration under loading also requires mass distribution and a dynamic balance.

4.4.1 Mass, center of mass and spatial inertia

Let the body have mass m, center-of-mass displacement $\mathbf{c}$ from a body-fixed origin O, and rotational inertia $\mathbf{I}_C$ about its center of mass, all expressed in the body's axes. For point masses, or the analogous mass integral, the rotational inertia is

$$\mathbf{I}_C = \int \left(\|\boldsymbol{\rho}\|^2 \mathbf{1}_3 - \boldsymbol{\rho}\boldsymbol{\rho}^\mathsf{T}\right) dm, \tag{4.26}$$

where $\boldsymbol{\rho}$ is displacement from the center of mass. This definition explains the units $\mathrm{kg\,m^2}$ and why changing a length scale has a quadratic effect on inertia for fixed mass.

The center-of-mass velocity is $\mathbf{v}_{C,\mathrm{lin}} = \mathbf{v}_{O,\mathrm{lin}} + \boldsymbol{\omega} \times \mathbf{c}$. Rigid-body kinetic energy is translation of the center of mass plus rotation about it:

$$T = \frac{1}{2} m \|\mathbf{v}_{O,\mathrm{lin}} - [\mathbf{c}]_\times \boldsymbol{\omega}\|^2 + \frac{1}{2} \boldsymbol{\omega}^\mathsf{T} \mathbf{I}_C \boldsymbol{\omega} = \frac{1}{2} \mathbf{v}^\mathsf{T} \mathbf{I}_O \mathbf{v},$$
$$\mathbf{I}_O = \begin{bmatrix} \mathbf{I}_C + m[\mathbf{c}]_\times [\mathbf{c}]_\times^\mathsf{T} & m[\mathbf{c}]_\times \\ m[\mathbf{c}]_\times^\mathsf{T} & m\mathbf{1}_3 \end{bmatrix}. \tag{4.27}$$

Expanding the square yields both the parallel-axis term and the off-diagonal coupling. The top-left block is rotational inertia about O, not about C. Ignoring the off-diagonal blocks changes the physical model whenever $\mathbf{c} \neq 0$.

Spatial momentum follows as

$$\mathbf{h}_O = \mathbf{I}_O \mathbf{v}_O = \begin{bmatrix} \mathbf{l}_O \\ \mathbf{p} \end{bmatrix}, \qquad \mathbf{p} = m\mathbf{v}_{C,\mathrm{lin}}, \qquad \mathbf{l}_O = \mathbf{I}_C \boldsymbol{\omega} + \mathbf{c} \times \mathbf{p}. \tag{4.28}$$

The momentum pairing has the same angular/linear order as force, with different units: angular momentum above linear momentum. Spatial inertia maps motion to a dual quantity, rather than mapping motion to another motion vector.

Energy invariance determines the inertia transformation. If $\mathbf{v}^B = \mathbf{X}_{BA} \mathbf{v}^A$, then

$$\mathbf{I}_B = \mathbf{X}_{BA}^{-\mathsf{T}} \mathbf{I}_A \mathbf{X}_{BA}^{-1}, \qquad \mathbf{I}_A = \mathbf{X}_{BA}^\mathsf{T} \mathbf{I}_B \mathbf{X}_{BA}. \tag{4.29}$$

Transforming inertia with $\mathbf{X}\mathbf{I}\mathbf{X}^\mathsf{T}$ without checking its mapping direction does not preserve this energy relation. As with force, the inverse-transpose is fixed by the duality, not chosen by the appearance of the matrix.

Physical mass properties impose constraints. A nondegenerate body has positive mass and positive-definite center-of-mass rotational inertia; idealized point or line masses can produce singular limits. The principal moments must also satisfy triangle inequalities because they arise from nonnegative squared distances in Equation (4.26). An arbitrary symmetric positive-definite 3×3 matrix is therefore not necessarily a realizable rotational inertia. Symmetry and positive energy are useful checks, but not the complete physical-consistency test. A rigid body's spatial inertia has ten independent mass-distribution parameters, rather than the twenty-one entries of an arbitrary symmetric 6×6 matrix. This structure is valuable when storing, checking or estimating mass properties.

Worked example: energy about a displaced origin. Take a constructed body with the following mass properties:

$$m = 2.0\,\mathrm{kg}, \qquad \mathbf{c} = (0.20, -0.10, 0.05)^\mathsf{T}\,\mathrm{m},$$
$$\mathbf{I}_C = \mathrm{diag}(0.030, 0.040, 0.050)\,\mathrm{kg\,m^2}. \tag{4.30}$$

The top-left block of its spatial inertia at O is

$$\mathbf{I}_{O,\mathrm{rot}} = \begin{bmatrix} 0.055 & 0.040 & -0.020 \\ 0.040 & 0.125 & 0.010 \\ -0.020 & 0.010 & 0.150 \end{bmatrix} \mathrm{kg\,m^2}. \tag{4.31}$$

For angular velocity $(0.40, -0.20, 1.10)^\mathsf{T}\,\mathrm{rad/s}$ and origin velocity $(0.30, -0.10, 0.20)^\mathsf{T}\,\mathrm{m/s}$, the center-of-mass velocity is $(0.40, 0.10, 0.20)^\mathsf{T}\,\mathrm{m/s}$. The translational energy is $0.21000\,\mathrm{J}$ and rotational energy about C is $0.03345\,\mathrm{J}$, totaling $0.24345\,\mathrm{J}$. Using the complete 6×6 spatial inertia at O gives the same energy. Using only its rotational block and $m\mathbf{1}_3$ while deleting the coupling blocks does not describe the same body motion.

For a dataset, retaining only total mass loses the distribution needed to interpret rotation. Retaining a CAD inertia without its origin and axes also loses essential meaning. Mass, center of mass and rotational inertia should remain associated with their geometry, units and provenance. A manufacturing change, payload or revised density assumption can change these quantities without changing a mesh's apparent shape.

4.4.2 Momentum, energy and Newton–Euler balance

Derivatives of vector components require care when their axes move. Let $\mathbf{v}$ be a body's absolute spatial velocity. Its spatial acceleration is the derivative of that motion. If components are expressed in a frame B fixed to the same body, the

coordinate derivative equals the spatial acceleration components. At the body origin this gives

$$\mathbf{a}^B = \frac{d\mathbf{v}^B}{dt} = \begin{bmatrix} \boldsymbol{\alpha}^B \\ \mathbf{a}^B_{O,\mathrm{class}} - \boldsymbol{\omega}^B \times \mathbf{v}^B_{O,\mathrm{lin}} \end{bmatrix}. \tag{4.32}$$

$\mathbf{a}_{O,\mathrm{class}}$ is ordinary point acceleration relative to the inertial world. Thus the linear block of spatial acceleration is generally not the ordinary acceleration of the frame origin. Simply concatenating angular acceleration and point acceleration gives a different object.

The transport correction can be expressed compactly with the spatial cross-product operators. For $\mathbf{v} = [\boldsymbol{\omega}; \mathbf{v}_{O,\mathrm{lin}}]$, define

$$\mathbf{v}\times = \begin{bmatrix} [\boldsymbol{\omega}]_\times & \mathbf{0} \\ [\mathbf{v}_{O,\mathrm{lin}}]_\times & [\boldsymbol{\omega}]_\times \end{bmatrix}, \qquad \mathbf{v}\times^* = -(\mathbf{v}\times)^\mathsf{T}. \tag{4.33}$$

The first acts on motion; the second on force or momentum. For arbitrary motion $\mathbf{z}$ and dual vector $\mathbf{f}$, their relation satisfies

$$\mathbf{z}^\mathsf{T}(\mathbf{v} \times^* \mathbf{f}) = -(\mathbf{v} \times \mathbf{z})^\mathsf{T}\mathbf{f}. \tag{4.34}$$

These are six-dimensional operators, not an attempt to use the ordinary three-dimensional cross product on arbitrary six-vectors.

In an arbitrary moving frame F, differentiating components introduces its motion. If $\mathbf{v}_F$ is the absolute spatial velocity of that frame, expressed with the same origin and axes, then

$$\mathbf{a}^F = \frac{d\mathbf{v}^F}{dt} + \mathbf{v}^F_F \times \mathbf{v}^F. \tag{4.35}$$

For a frame attached to the body, $\mathbf{v}^F_F = \mathbf{v}^F$ and $\mathbf{v}^F \times \mathbf{v}^F = 0$, explaining Equation (4.32). For compatible coordinate descriptions of the same spatial acceleration, $\mathbf{a}^B = \mathbf{X}_{BA}\mathbf{a}^A$. The required transport terms must first have been included in any raw component derivatives.

The net external wrench on a rigid body equals the spatial momentum derivative. In body-fixed coordinates the result is

$$\mathbf{f}^B_{\mathrm{net}} = \frac{d\mathbf{h}^B}{dt} + \mathbf{v}^B \times^* \mathbf{h}^B = \mathbf{I}_B\mathbf{a}^B + \mathbf{v}^B \times^* \mathbf{I}_B\mathbf{v}^B. \tag{4.36}$$

$\mathbf{I}_B$ is constant because its axes and origin are fixed in the body. One must not silently assume that the same inertia matrix is constant in fixed world coordinates while the body rotates. The second term in Equation (4.36) accounts for motion-dependent momentum transport. About the center of mass, its angular part includes the familiar $\boldsymbol{\omega} \times \mathbf{I}_C\boldsymbol{\omega}$ gyroscopic term. The ordinary Newton–Euler balances are recovered explicitly at that point:

$$\begin{aligned} \mathbf{F} &= m(\mathbf{a}_{C,\mathrm{class}} - \boldsymbol{\omega} \times \mathbf{v}_{C,\mathrm{lin}}) + \boldsymbol{\omega} \times (m\mathbf{v}_{C,\mathrm{lin}}) = m\mathbf{a}_{C,\mathrm{class}}, \\ \mathbf{n}_C &= \mathbf{I}_C\boldsymbol{\alpha} + \boldsymbol{\omega} \times (\mathbf{I}_C\boldsymbol{\omega}). \end{aligned} \tag{4.37}$$

The linear transport terms cancel. Omitting one of them would make the spatial equation disagree with the familiar center-of-mass force balance.

Gravity, joint loading, contact forces and any other external loads contribute to $\mathbf{f}_{\mathrm{net}}$. Gravity may be represented explicitly by

$$\mathbf{f}_g = \begin{bmatrix} \mathbf{c} \times m\mathbf{g} \\ m\mathbf{g} \end{bmatrix} \tag{4.38}$$

in the chosen axes about O. Here $\mathbf{g}$ is the gravitational acceleration vector, not a positive scalar magnitude with an unstated direction. Alternative algorithms can absorb gravity into an acceleration convention; those conventions must be reconciled before comparing intermediate arrays.

The energy relation follows by pairing Equation (4.36) with velocity. Since $\mathbf{v} \times \mathbf{v} = 0$, the transport term contributes no net power, so in body-fixed coordinates

$$\mathbf{v}^\mathsf{T}\mathbf{f}_{\mathrm{net}} = \mathbf{v}^\mathsf{T}\mathbf{I}_B\mathbf{a} = \frac{d}{dt}\left(\frac{1}{2}\mathbf{v}^\mathsf{T}\mathbf{I}_B\mathbf{v}\right). \tag{4.39}$$

This describes the rigid body's kinetic-energy rate under the net external wrench. If gravity is separated into potential energy, the corresponding total-energy balance must separate its power consistently.

Worked example: nonzero point acceleration with zero spatial acceleration. A body rotates at a constant $2.0\,\mathrm{rad/s}$ about a fixed z axis. A body-fixed point is $0.20\,\mathrm{m}$ from that axis. Choose an instant and body axes in which its radial displacement lies along positive x. The point velocity and ordinary acceleration are

$$\begin{aligned}\mathbf{v}_{P,\mathrm{lin}} &= (0, 0.40, 0)^{\mathsf{T}}\,\mathrm{m/s},\\ \mathbf{a}_{P,\mathrm{class}} &= (-0.80, 0, 0)^{\mathsf{T}}\,\mathrm{m/s^2}.\end{aligned} \tag{4.40}$$

However,

$$\mathbf{a}_{P,\mathrm{class}} - \boldsymbol{\omega} \times \mathbf{v}_{P,\mathrm{lin}} = \mathbf{0}, \qquad \boldsymbol{\alpha} = \mathbf{0}. \tag{4.41}$$

Its spatial acceleration is zero, while centripetal point acceleration is not. This is a difference in represented quantities, not a failure of Newton's law. Substituting the spatial acceleration into Equation (4.36) retains the required momentum-transport term.

An ideal accelerometer at P supplies another quantity: specific force, ${}^{B}\mathbf{R}_{W}(\mathbf{a}^{W}_{P,\mathrm{class}} - \mathbf{g}^{W})$, before bias and noise. It is not directly the linear block of $\mathbf{a}$. A stationary supported sensor can therefore report nonzero specific force. Chapter 7 will develop measurement models; the mechanical distinction already tells us which quantities should not be compared as if they were identical.

4.5 Geometric data in a mechanism

The preceding equations describe one rigid body. A mechanism connects bodies through joints, making each pose depend on a common configuration. Kinematics supplies the first consistency relations among its records, before dynamic models and learning methods are introduced.

4.5.1 Forward kinematics, Jacobians and generalized effort

For a serial mechanism, forward kinematics composes relative transforms from the base to a link or tool. Fixed mounting geometry and joint-dependent transforms both enter that product. The result predicts a pose from configuration. It does not establish that an encoder's zero, an extrinsic calibration or a rigid-body assumption is correct in the physical apparatus.

Differentiating the pose gives a Jacobian mapping generalized velocity to the chosen body's spatial motion at a declared point:

$$\mathbf{v}_P = \mathbf{J}_P(\mathbf{q})\boldsymbol{\nu}, \qquad \mathbf{J}_P = \begin{bmatrix}\mathbf{J}_\omega \\ \mathbf{J}_p\end{bmatrix}. \tag{4.42}$$

Both blocks here use the same axes and point convention as the spatial velocity. $\mathbf{J}_p\boldsymbol{\nu}$ is the ordinary velocity of P; the angular block maps to body angular velocity. For a revolute coordinate in common axes, a geometric column is

$$\mathbf{J}_j = \begin{bmatrix}\mathbf{a}_j \\ \mathbf{a}_j \times (\mathbf{p}_P - \mathbf{p}_{O_j})\end{bmatrix}, \tag{4.43}$$

where $\mathbf{a}_j$ is its current axis direction and O_j a point on the axis. A prismatic column is $[\mathbf{0}; \mathbf{a}_j]$. These expressions apply to the declared scalar joint coordinates; free-base parameterizations require the corresponding generalized-velocity map.

The Jacobian need not be square. At a singular configuration, some desired instantaneous tool motions are unavailable or poorly conditioned with respect to joint rates. This is a geometric restriction on attainable observations and actions. A demonstration collector cannot obtain arbitrary Cartesian motions merely by naming them as targets; the mechanism, joint limits and control must realize them.

Power again determines the dual map. For a wrench applied to the mechanism at P, compatible with the Jacobian's axes and moment reference point,

$$P = \mathbf{f}_P^{\mathsf{T}}\mathbf{J}_P\boldsymbol{\nu} = \boldsymbol{\tau}^{\mathsf{T}}\boldsymbol{\nu}, \qquad \boldsymbol{\tau} = \mathbf{J}_P^{\mathsf{T}}\mathbf{f}_P. \tag{4.44}$$

This is the generalized effort contributed by that applied wrench. It is not automatically the actuator effort required to produce the resulting motion; inertia, gravity and other loads enter the dynamic balance. A balancing actuator effort has

the opposite sign in a static case when no other loads contribute. Nor does a known joint-effort vector uniquely determine an arbitrary six-dimensional wrench. Null spaces and other unknown loads can make that inference ambiguous.

Worked example: a planar arm and a tool wrench. Consider two revolute joints, link lengths $\ell_1 = 0.30\,\mathrm{m}$ and $\ell_2 = 0.20\,\mathrm{m}$, and joint angles $q_1 = \pi/6$, $q_2 = -\pi/3$. Both joints rotate about positive z axes, with q_2 measured relative to the first link. The tool position and orientation angle in the plane are

$$\mathbf{p}_P = \begin{bmatrix} \ell_1 \cos q_1 + \ell_2 \cos(q_1 + q_2) \\ \ell_1 \sin q_1 + \ell_2 \sin(q_1 + q_2) \\ 0 \end{bmatrix}, \qquad \psi = q_1 + q_2. \tag{4.45}$$

In world axes at the tool point, the nonzero Jacobian rows are

$$\mathbf{J}_{p,xy} = \begin{bmatrix} -\ell_1 \sin q_1 - \ell_2 \sin(q_1 + q_2) & -\ell_2 \sin(q_1 + q_2) \\ \ell_1 \cos q_1 + \ell_2 \cos(q_1 + q_2) & \ell_2 \cos(q_1 + q_2) \end{bmatrix}, \qquad \mathbf{J}_{\omega,z} = \begin{bmatrix} 1 & 1 \end{bmatrix}. \tag{4.46}$$

The tool lies at $(0.433013, 0.050000, 0)^{\mathsf{T}}\,\mathrm{m}$, and $\mathbf{J}_{p,xy}$ evaluates to

$$\begin{bmatrix} -0.050000 & 0.100000 \\ 0.433013 & 0.173205 \end{bmatrix}\,\mathrm{m/rad}. \tag{4.47}$$

For joint rates $(0.50, -0.25)^{\mathsf{T}}\,\mathrm{rad/s}$, the tool velocities are

$$\begin{aligned} \mathbf{v}_{P,\mathrm{lin}} &= (-0.050000, 0.173205, 0)^{\mathsf{T}}\,\mathrm{m/s}, \\ \boldsymbol{\omega} &= (0, 0, 0.25)^{\mathsf{T}}\,\mathrm{rad/s}. \end{aligned} \tag{4.48}$$

Apply a tool force $(2, -1, 0)^{\mathsf{T}}\,\mathrm{N}$. Also apply a moment about the tool point of $(0, 0, 0.040)^{\mathsf{T}}\,\mathrm{N\,m}$. The resulting generalized effort is

$$\boldsymbol{\tau} = (-0.493013, 0.066795)^{\mathsf{T}}\,\mathrm{N\,m}. \tag{4.49}$$

The joint and tool power calculations both give $-0.263205\,\mathrm{W}$. Negative power means the applied wrench removes mechanical energy from the instantaneous motion. This calculation does not infer a motor current or predict a trajectory; it relates a declared wrench to generalized effort at one configuration.

The example offers several checks for a collected arm trajectory. Forward kinematics predicts tool pose; the Jacobian predicts its velocity; the wrench projection predicts one load contribution. Agreement among them is stronger than agreement between two plots carrying similar labels. It remains conditional on the geometry, timing and calibration used to produce them.

4.5.2 Recorded poses, reference points and kinematic consistency checks

A geometric record should identify the body and point described, the frame mapping direction, orientation convention, units and sample time. It should also distinguish a measured pose, a pose inferred from joint readings, a commanded target and a simulator state. Two poses with the same matrix shape can have different evidential roles. Preserving this distinction lets a later model use the records without silently treating its own prediction as an independent observation.

Mounting information is especially important when combining modalities. A marker-to-body transform relates a tracked marker to the body's modeled origin. An inertia frame can be displaced and rotated relative to the body frame. A tool site can be displaced from both. Apparent disagreements among their reported positions may be exactly what rigid geometry predicts. Changing only the plotted labels does not reconcile them.

MuJoCo illustrates the implementation boundary. Its stored configuration `qpos` and tangent velocity `qvel` can have different lengths when quaternion joints are present. A free joint stores three translational coordinates and a scalar-first unit quaternion. Its translational velocity is expressed in world axes, while its angular velocity is expressed in the local body axes; this mixed generalized-velocity convention is not the same as directly stacking a Featherstone motion vector (DeepMind Technologies Limited, n.d.).

Use representation-aware operations to compare and update configurations. In MuJoCo, `mj_differentiatePos` computes a tangent difference, and `mj_integratePos` integrates a tangent velocity. Subtracting or adding raw quaternion

entries does not perform the same operations. The API accounts for different configuration and velocity dimensions, while the user must still identify which frames a returned quantity uses (DeepMind Technologies Limited, n.d.).

For an observation point defined as a site, a particularly clear mapping uses the site Jacobians. After updating forward kinematics at a known state, the following illustrative Python fragment constructs an angular-first velocity in world axes at the site origin:

```
import numpy as np
import mujoco

jacp = np.zeros((3, model.nv))
jacr = np.zeros((3, model.nv))
mujoco.mj_forward(model, data)
mujoco.mj_jacSite(model, data, jacp, jacr, site_id)
v_P_W = jacp @ data.qvel
omega_W = jacr @ data.qvel
v_sp_P_W = np.concatenate((omega_W, v_P_W))
```

The linear and rotational Jacobians are returned in world axes and describe that site point and its orientation. The site is fixed to a body. This fragment explains quantity semantics; it is not a complete model or a new simulation experiment. Computing a pose and Jacobian does not validate the geometry against physical measurements.

The reference point remains visible in the variable names. Rotating both blocks into site axes changes the basis at the same point. To express the motion about a different origin requires the additional displacement block of Equation (4.18). It is safer to form this quantity through an explicitly chosen point than to reorder internal arrays whose origins or conventions have not been checked. In particular, MuJoCo's body, inertial and site frames are distinct, and an internal center-of-mass quantity need not refer to the regular body origin. The official API reference and versioned implementation should be checked together when a prose description is ambiguous (DeepMind Technologies Limited, n.d.; Google DeepMind, n.d.).

A kinematic review can proceed through a small number of physical relationships. Test rotation orthogonality and quaternion norm, then the forward/inverse frame maps on known points. Compare transformed velocities at the same reference point. Check the Jacobian prediction against a small configuration displacement using the representation's integration rule. Where compatible force information exists, check power duality and generalized-effort projection. These tests address different failures; one passing identity does not establish the validity of an entire model.

Finite-difference checks should use a range of small displacement scales. A large displacement introduces nonlinear truncation error, while an extremely small one can expose floating-point cancellation. If the configuration is near a limit, a contact change or an orientation branch choice, explain how that affects the comparison. A mathematical check on a smooth kinematic map is different from checking a discontinuous contact event.

Worked example: what a mounting error can imitate. For a body rotating at $2.0\,\mathrm{rad/s}$, an unaccounted perpendicular point displacement of $0.050\,\mathrm{m}$ changes linear velocity by $0.100\,\mathrm{m/s}$. A learner trained on one origin and evaluated on the other can appear to have a systematic prediction error even with correct rigid-body motion. For small errors at a common instant, differentiating Equation (4.11) gives the first-order relation

$$\delta\mathbf{v}_{P,\mathrm{lin}} \simeq \delta\mathbf{v}_{O,\mathrm{lin}} + \delta\boldsymbol{\omega} \times \mathbf{r}_{OP} + \boldsymbol{\omega} \times \delta\mathbf{r}_{OP}. \tag{4.50}$$

Angular-velocity uncertainty is amplified by a long offset, and mounting uncertainty matters more at high angular speed. This is a local sensitivity relation, not a probability model or a measured error distribution. It gives a reason to choose sensor placement and calibration effort according to the motion and learning question.

The same geometry can improve observability. A marker displaced from a rotation axis has a larger positional excursion than one close to it, potentially making rotation easier to observe through position. That benefit depends on visibility, resolution, motion and mounting accuracy. Changing placement changes both the obtainable signal and its interpretation, so it is a physical data-design decision rather than merely a display choice.

Kinematic consistency also has limits. A smooth, self-consistent trajectory does not establish correct mass, friction, contact or actuator behavior. Likewise, two geometric descriptions can agree after applying a calibration fitted on those same records while failing elsewhere. Preserve calibration provenance and reserve separate observations or poses when testing its extension. Measurement and identification chapters will examine those inferences; the present chapter establishes the quantities they concern.

Exercises and further investigation. Use angular-first motion vectors, moment-first wrenches and explicitly stated frames. The numerical companion `tools/check_chapter04.py` reproduces the constructed calculations with standard-library arithmetic.

1. Derive the spatial motion transformation from the velocity of a displaced rigid-body point. Verify its inverse and composition. State why the same matrix is not used to transform homogeneous point coordinates.
2. Recalculate the motion and wrench examples about a third origin of your choice. Check power and joint-effort invariance. Construct a deliberately inconsistent point/axes pairing and identify which physical interpretation fails, even if some component values look plausible.
3. Starting from center-of-mass kinetic energy, derive all four blocks of spatial inertia. Transform the constructed body's inertia to its center of mass and recover the energy $0.24345\,\mathrm{J}$. Explain why retaining only the diagonal blocks changes the model.
4. For the same body's center-of-mass frame, take
$$\begin{aligned}\boldsymbol{\omega} &= (0.40, -0.20, 1.10)^{\mathsf{T}}\ \mathrm{rad/s},\\ \mathbf{v}_{C,\mathrm{lin}} &= (0.30, -0.10, 0.20)^{\mathsf{T}}\ \mathrm{m/s},\\ \boldsymbol{\alpha} &= (0.10, 0.20, -0.30)^{\mathsf{T}}\ \mathrm{rad/s^2},\\ \mathbf{a}_{C,\mathrm{class}} &= (0.50, 0.40, -0.20)^{\mathsf{T}}\ \mathrm{m/s^2}.\end{aligned}$$
Compute spatial acceleration and the net wrench. Verify that the linear force equals $m\mathbf{a}_{C,\mathrm{class}}$, not simply the mass times the spatial-acceleration linear block.
5. Derive the planar arm Jacobian from both position differentiation and joint-axis geometry. Check it by a central finite difference. Explain why a prescribed tool force determines a joint load contribution but measured joint effort need not uniquely determine tool contact force.
6. Construct a quaternion sequence with alternating signs and unchanged physical orientations. Compare component differences with relative-rotation increments. State the integration and frame conventions needed to turn a pose sequence into angular-velocity estimates.
7. Design a record schema for a joint encoder, a tracked marker and a tool wrench sensor. Identify the mounting transformations, point definitions and timestamps needed to compare them. Separate observed, inferred and simulated fields, and name a claim that kinematic consistency alone cannot support.

For further reading, Featherstone (2008, chapters 2–3) develops the spatial-vector foundation and constrained rigid-body models. Its treatment also makes clear why motion and force belong to dual spaces and why their coordinate transformations differ. The official simulator references provide implementation meanings rather than replacing that physical derivation.

The chapter's central result is a correspondence: geometry specifies the quantities, power fixes their dual maps, and inertia connects motion to momentum and loading. For ExpGen PhySyst, that correspondence makes it possible to combine records without erasing their physical meaning. The next chapter assembles these body-level relationships into coupled robot dynamics, where the choice of motion, loading and contact determines which experiences a mechanism can generate.

Chapter 5 Robot Dynamics and Control

Move the elbow of a two-link arm while allowing its shoulder to turn. The shoulder can accelerate even when its actuator applies no additional effort. Now let the tool touch a support. The same actuator requests can produce a different acceleration, because the support supplies another load and restricts the admissible motion. The records depend on the whole mechanical arrangement, rather than on a separate input–output relationship for each motor.

Chapter 4 gave each rigid body a geometry, spatial velocity, inertia and wrench balance. This chapter connects those bodies through joints. The resulting equations explain dynamic coupling, provide the models used by controllers, and establish what changes when manipulation introduces contact. Their computational structure is part of the explanation: recursive algorithms preserve the same spatial meanings while avoiding unnecessary large-matrix calculations.

For ExpGen PhySyst, these mechanics determine which experiences can be generated and what their records can establish. A controller influences the motion available to observe. Contact changes the relationship between effort and acceleration. A successful grasp can conceal very different internal loads. Designing useful data therefore requires identifying the mechanism, the applied actions, the interaction conditions and the intended inference. The calculations below are constructed examples, with a numerical companion; they are not new measured results or a claim that a particular robot has executed the motions.

5.1 Recursive articulated-body mechanics

A tree mechanism has a unique parent for every body except its base. A serial arm is one example; a hand with several fingers is another when each finger branches from a common palm. Closed kinematic loops and environmental contacts introduce additional constraints and will be considered after the free tree. This distinction separates the body's recursive mechanics from the problem of solving all its interactions.

Index moving bodies by $i = 1, \dots, n$, with parent $p(i)$ and a fixed base indexed by 0. Attach a frame to each body. Its spatial inertia $\mathbf{I}_i$ is constant in that frame, and $\mathbf{X}_i(\mathbf{q})$ maps parent-frame motion to the child frame. As in Chapter 4, motion is angular-first and wrenches are moment-first. All quantities in a local body equation use that body's axes and origin.

We first use one scalar revolute or prismatic coordinate per joint, so $\boldsymbol{\nu} = \dot{\mathbf{q}}$. These assumptions let us derive the recursions directly. Multi-coordinate and floating joints use the same spatial principles with their appropriate joint models and tangent velocities; their configuration derivatives need not equal `qvel` componentwise.

5.1.1 Forward velocity and acceleration recursions

Let $\mathbf{S}_i$ be joint i's motion subspace, expressed in the child frame. Its relative motion and the child's absolute motion are

$$\mathbf{v}_{J_i} = \mathbf{S}_i \dot{q}_i, \qquad \mathbf{v}_i = \mathbf{X}_i \mathbf{v}_{p(i)} + \mathbf{v}_{J_i}. \tag{5.1}$$

The parent motion must first be transformed to the child's origin and axes. Only then can relative motion be added. Adding three-dimensional angular and linear entries from different origins would omit the displacement term derived in Chapter 4.

For the ordinary constant-subspace scalar joints used here, the acceleration recursion is

$$\begin{aligned} \mathbf{c}_i &= \mathbf{v}_i \times \mathbf{v}_{J_i}, \\ \mathbf{a}_i &= \mathbf{X}_i \mathbf{a}_{p(i)} + \mathbf{S}_i \ddot{q}_i + \mathbf{c}_i. \end{aligned} \tag{5.2}$$

The cross term appears because the relative joint motion is carried by a moving body frame. It is not an extra empirical force. A more general joint model can add an intrinsic joint bias acceleration $\mathbf{c}_{J_i}$ to $\mathbf{c}_i$ when its motion subspace varies in the chosen representation. Using the scalar formula for every possible joint without that qualification would lose part of the kinematics.

There are two consistent ways to include gravity. One can initialize the fixed base with zero acceleration and explicitly apply gravity wrenches to every body. Alternatively, the recursive algorithms can initialize

$$\mathbf{v}_0 = \mathbf{0}, \qquad \mathbf{a}_0 = -\mathbf{a}_g, \qquad \mathbf{a}_g = \begin{bmatrix} \mathbf{0} \\ \mathbf{g}^W \end{bmatrix}, \tag{5.3}$$

with the base axes aligned to world. Gravity is then omitted from the external-wrench list. We use this second convention below, following Featherstone (2008, 93–97). It produces the required gravity effort without adding the same gravitational load twice.

With this initialization, $\mathbf{a}_i$ is an algorithmic acceleration: physical spatial acceleration minus gravity represented in the body frame. It should not be compared directly with a tracked point's second derivative or assumed to equal an accelerometer reading. Recovering a physical point acceleration also requires the spatial-to-classical conversion from Chapter 4. The resulting required joint efforts and transmitted wrenches remain physical quantities in the stated rigid model.

A forward pass therefore carries information from the base to the leaves: configuration determines transforms, joint rates determine body velocities, and joint accelerations determine the corresponding body accelerations. This is already useful for data interpretation. A body velocity is not generally its own joint rate, and moving a proximal joint changes the acceleration conditions under which distal measurements are collected.

5.1.2 Backward force accumulation and inverse dynamics

Suppose configuration, velocity and desired acceleration are specified. For each body, begin with the effort required to account for its own inertia and its known nongravitational external wrench:

$$\mathbf{f}_i = \mathbf{I}_i \mathbf{a}_i + \mathbf{v}_i \times^* \mathbf{I}_i \mathbf{v}_i - \mathbf{f}_i^{\text{ext}}. \tag{5.4}$$

$\mathbf{f}_i^{\text{ext}}$ is a wrench acting on body i, expressed about that body's origin. It is subtracted because an assisting external load reduces the wrench that must be supplied through the parent connection. Gravity has already entered through Equation (5.3).

Now process bodies from the leaves toward the base. After all children have contributed to $\mathbf{f}_i$, compute

$$\tau_i = \mathbf{S}_i^{\mathsf{T}} \mathbf{f}_i, \qquad \mathbf{f}_{p(i)} \mathrel{+}= \mathbf{X}_i^{\mathsf{T}} \mathbf{f}_i \quad \text{if } p(i) \neq 0. \tag{5.5}$$

The transpose transports a child wrench into parent coordinates, as fixed by power duality. The generalized effort is the projection along admissible joint motion. Components orthogonal to that motion are transmitted constraint loads; they are not recovered by the scalar projection alone. The fixed-base accumulator is omitted here. A base reaction can be obtained separately by initializing its wrench accumulator and including the root's contribution; it is not needed to compute the moving-joint efforts.

The forward motion pass and backward wrench pass form the recursive Newton–Euler algorithm, abbreviated RNEA. Its input is a motion state and acceleration; its output is the generalized effort required by the specified mechanical model. With a bounded joint size and a tree topology, the passes take work proportional to the number of bodies. This computational statement does not include an independent contact solve, actuator command allocation or a search for an appropriate motion.

Inverse dynamics is valuable for feedforward control, but also for examining data. It predicts the effort consistent with a recorded motion under declared inertias, gravity and external loads. A difference from an observed or current-derived effort is a residual to investigate. It does not by itself identify whether an error came from loading, friction, calibration, timing or the motion estimate.

The numerical companion implements these two passes explicitly. In `rnea`, $\mathbf{X}_i$, $\mathbf{S}_i$, $\mathbf{I}_i$ and $\mathbf{f}_i$ correspond to the transform, axis, body-inertia and local-wrench arrays. The code computes the parent contribution with a transpose and uses the same gravity convention as the derivation. This correspondence matters more than matching a variable name in a different library whose point, ordering or sign convention may differ.

5.2 Computing coupled motion

Inverse dynamics answers which effort is needed for a specified acceleration. Forward dynamics asks which acceleration follows from specified effort and loading. The two questions concern the same mechanical balance, but they organize the computation differently. Neither predicts a finite trajectory until an integration and control-update scheme has also been specified; those numerical choices belong to Chapter 6.

5.2.1 Joint-space inertia and forward dynamics

Each body velocity depends linearly on the generalized velocity at a fixed configuration. If $\mathbf{v}_i = \mathbf{J}_i(\mathbf{q})\boldsymbol{\nu}$, summing body kinetic energies gives

$$T = \frac{1}{2}\sum_i \boldsymbol{\nu}^{\mathsf{T}}\mathbf{J}_i^{\mathsf{T}}\mathbf{I}_i\mathbf{J}_i\boldsymbol{\nu} = \frac{1}{2}\boldsymbol{\nu}^{\mathsf{T}}\mathbf{M}(\mathbf{q})\boldsymbol{\nu}, \qquad \mathbf{M} = \sum_i \mathbf{J}_i^{\mathsf{T}}\mathbf{I}_i\mathbf{J}_i. \tag{5.6}$$

$\mathbf{M}$ is symmetric. It is positive definite when every nonzero modeled generalized velocity has positive kinetic energy. Unsupported coordinates or inappropriate massless idealizations can violate that assumption. Off-diagonal entries express dynamic coupling: changing one joint's velocity can change momentum associated with another.

For an independent fixed-base coordinate set, the equations can be written

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{h}(\mathbf{q},\dot{\mathbf{q}}) = \boldsymbol{\tau}_a + \boldsymbol{\tau}_p + \sum_k \mathbf{J}_k^{\mathsf{T}}\mathbf{f}_k^{\text{ext}}, \qquad \mathbf{h} = \boldsymbol{c} + \boldsymbol{g}. \tag{5.7}$$

$\boldsymbol{c}$ is the velocity-dependent rigid-body bias and $\boldsymbol{g}$ the gravity effort. $\boldsymbol{\tau}_a$ is the actuator effort delivered to the modeled mechanism; $\boldsymbol{\tau}_p$ represents separately specified passive effects such as joint damping. A resisting viscous term would appear here as $-\mathbf{B}\dot{\mathbf{q}}$. It must not also be counted inside $\mathbf{h}$ unless the definition is changed explicitly.

For general tangent coordinates replace $\ddot{\mathbf{q}}$ by $\dot{\boldsymbol{\nu}}$ and construct the bias consistently with the joint models. Gravity is then $\boldsymbol{g} = \mathcal{N}(\mathbf{q})^{\mathsf{T}}\partial V/\partial\mathbf{q}$, where $\dot{\mathbf{q}} = \mathcal{N}\boldsymbol{\nu}$; it is simply the potential gradient for the scalar coordinates used here. A configuration vector with quaternions should not be inserted into an ordinary unconstrained Euler–Lagrange formula as if all its entries were independent.

RNEA supplies useful identities. With no external wrenches,

$$\begin{aligned}\mathbf{h} &= \mathrm{ID}(\mathbf{q},\dot{\mathbf{q}},\mathbf{0}),\\ \mathbf{M}_{:j} &= \mathrm{ID}(\mathbf{q},\mathbf{0},\mathbf{e}_j;\mathbf{g}=\mathbf{0}).\end{aligned} \tag{5.8}$$

$\mathbf{e}_j$ selects a unit acceleration of joint j. The mass-matrix identity uses zero velocity and zero gravity so that no bias is added. With these quantities available, forward dynamics solves

$$\mathbf{M}\ddot{\mathbf{q}} = \boldsymbol{\tau}_a + \boldsymbol{\tau}_p + \sum_k \mathbf{J}_k^{\mathsf{T}}\mathbf{f}_k^{\text{ext}} - \mathbf{h}. \tag{5.9}$$

Use a matrix solve rather than explicitly forming $\mathbf{M}^{-1}$ in numerical code. The inverse notation used in later derivations denotes the mathematical action of this solve.

The composite-rigid-body algorithm, CRBA, computes the mass matrix more directly. Define composite inertia by accumulating descendants with their joints instantaneously locked:

$$\mathbf{I}_i^{\text{comp}} = \mathbf{I}_i + \sum_{j:p(j)=i} \mathbf{X}_j^{\mathsf{T}}\mathbf{I}_j^{\text{comp}}\mathbf{X}_j. \tag{5.10}$$

For a joint i, form $\mathbf{F} = \mathbf{I}_i^{\text{comp}}\mathbf{S}_i$. Then $M_{ii} = \mathbf{S}_i^{\mathsf{T}}\mathbf{F}$. Moving upward through the ancestors transports $\mathbf{F}$ with each child's $\mathbf{X}^{\mathsf{T}}$; projection onto an ancestor's motion subspace gives the corresponding off-diagonal entry. Symmetry supplies its transpose. This builds coupling from the same body-level spatial operations (Featherstone 2008, 102–107).

Worked example: a two-link arm. Use the planar geometry of Chapter 4, with uniform slender links of lengths $(0.30, 0.20)\,\mathrm{m}$ and masses $(0.40, 0.25)\,\mathrm{kg}$. Let $c_i = \ell_i/2$ and $I_i = m_i\ell_i^2/12$ be their center distances and planar rotational inertias. Here q_1 is measured counterclockwise from world horizontal, q_2 relative to link 1, and gravity points downward. This angle convention differs from the downward-zero single joint in Chapter 3.

Introduce three scalar combinations, unrelated to the spatial matrices above:

$$A = I_1 + I_2 + m_1 c_1^2 + m_2(\ell_1^2 + c_2^2),$$
$$B = m_2 \ell_1 c_2, \qquad D = I_2 + m_2 c_2^2. \tag{5.11}$$

Summing the link energies or applying CRBA gives

$$\mathbf{M} = \begin{bmatrix} A + 2B\cos q_2 & D + B\cos q_2 \\ D + B\cos q_2 & D \end{bmatrix}. \tag{5.12}$$

The velocity and gravity terms are

$$\boldsymbol{c} = \begin{bmatrix} -B\sin q_2(2\dot{q}_1\dot{q}_2 + \dot{q}_2^2) \\ B\sin q_2 \dot{q}_1^2 \end{bmatrix},$$
$$\boldsymbol{g} = g\begin{bmatrix} (m_1 c_1 + m_2\ell_1)\cos q_1 + m_2 c_2 \cos(q_1 + q_2) \\ m_2 c_2 \cos(q_1 + q_2) \end{bmatrix}. \tag{5.13}$$

The gravity vector follows by differentiating $V = g[(m_1 c_1 + m_2\ell_1)\sin q_1 + m_2 c_2 \sin(q_1 + q_2)]$. The velocity terms follow from differentiating the kinetic energy, or from RNEA with zero acceleration after separating gravity.

At $\mathbf{q} = (\pi/6, -\pi/3)^{\mathsf{T}}$ and $\dot{\mathbf{q}} = (0.50, -0.25)^{\mathsf{T}}$ rad/s, the values are

$$\mathbf{M} = \begin{bmatrix} 0.0453333 & 0.00708333 \\ 0.00708333 & 0.00333333 \end{bmatrix} \text{kg m}^2,$$
$$\boldsymbol{c} = (-0.00121785, -0.00162380)^{\mathsf{T}} \text{ N m},$$
$$\boldsymbol{g} = (1.359313, 0.212393)^{\mathsf{T}} \text{ N m}. \tag{5.14}$$

For acceleration $(1.0, -0.5)^{\mathsf{T}}$ rad/s^2 and no passive or external load, inverse dynamics requires

$$\boldsymbol{\tau}_a = (1.399887, 0.216186)^{\mathsf{T}} \text{ N m}. \tag{5.15}$$

The companion obtains these values independently from the analytic formulas and spatial RNEA, then recovers the acceleration with forward dynamics. Agreement checks a mathematical implementation; it does not verify the assumed link masses against a physical arm.

5.2.2 Articulated-body elimination and computational alternatives

Forming the complete mass matrix is useful when inspecting coupling, but forward dynamics can also eliminate joint accelerations recursively. This is the articulated-body algorithm, ABA. Unlike composite inertia, articulated inertia describes how a subtree responds when its internal joints can move under specified efforts. The distinction is a physical boundary condition, not just a different name for the same matrix.

Initialize each body with

$$\mathbf{I}_i^{\text{art}} = \mathbf{I}_i, \qquad \mathbf{p}_i^{\text{art}} = \mathbf{v}_i \times^* \mathbf{I}_i \mathbf{v}_i - \mathbf{f}_i^{\text{ext}}. \tag{5.16}$$

The superscripts “comp” and “art” identify algorithmic quantities, not coordinate frames. The accumulated relation for a subtree has the form $\mathbf{f}_i = \mathbf{I}_i^{\text{art}}\mathbf{a}_i + \mathbf{p}_i^{\text{art}}$. After its children have been processed, define

$$\mathbf{U}_i = \mathbf{I}_i^{\text{art}}\mathbf{S}_i, \qquad D_i = \mathbf{S}_i^{\mathsf{T}}\mathbf{U}_i, \qquad u_i = \tau_i - \mathbf{S}_i^{\mathsf{T}}\mathbf{p}_i^{\text{art}}. \tag{5.17}$$

For a scalar joint, D_i is a scalar effective inertia and u_i an effort residual. Projecting the subtree relation along $\mathbf{S}_i$ and using $\mathbf{a}_i = \mathbf{X}_i\mathbf{a}_{p(i)} + \mathbf{c}_i + \mathbf{S}_i\ddot{q}_i$ gives

$$\ddot{q}_i = D_i^{-1}[u_i - \mathbf{U}_i^{\mathsf{T}}(\mathbf{X}_i\mathbf{a}_{p(i)} + \mathbf{c}_i)]. \tag{5.18}$$

Substitution eliminates that joint acceleration from the wrench transmitted to the parent. The remaining inertia and bias are

$$\bar{\mathbf{I}}_i = \mathbf{I}_i^{\text{art}} - \mathbf{U}_i D_i^{-1}\mathbf{U}_i^{\mathsf{T}},$$
$$\bar{\mathbf{p}}_i = \mathbf{p}_i^{\text{art}} + \bar{\mathbf{I}}_i\mathbf{c}_i + \mathbf{U}_i D_i^{-1} u_i. \tag{5.19}$$

Both bias terms matter. Omitting the velocity-bias contribution or the specified-effort contribution changes the predicted

parent motion. Accumulate $\mathbf{X}_i^{\mathsf{T}}\bar{\mathbf{I}}_i\mathbf{X}_i$ and $\mathbf{X}_i^{\mathsf{T}}\bar{\mathbf{p}}_i$ into the parent's articulated quantities. A final forward pass uses Equation (5.18) to recover accelerations, starting with the same gravity initialization as RNEA (Featherstone 2008, 126–132).

Elimination removes resistance along an allowed joint motion: $\bar{\mathbf{I}}_i\mathbf{S}_i = \mathbf{0}$. The reduced matrix is therefore generally positive semidefinite rather than positive definite, and it need not have the ten-parameter structure of a rigid-body spatial inertia. It is an articulated response after a degree of freedom has been eliminated. For multi-coordinate joints, D_i is a small matrix and its inverse again denotes a solve. A singular D_i requires examining the model and its constraints rather than dividing by zero.

ABA has linear complexity for an unconstrained tree with bounded joint size. Dense matrix solution, CRBA with structured factorization and repeated inverse-dynamics evaluations offer other computational routes. Contact and closed-loop constraints add their own equations and costs. Choosing an algorithm changes computational organization; it should not silently change the intended mechanical balance.

Worked example: a free elbow changes shoulder response. At the preceding arm configuration, suppose bias is compensated and an additional effort $(0.10, 0)^{\mathsf{T}}\,\mathrm{N\,m}$ is applied. If the elbow is mechanically locked, the first acceleration is $0.10/M_{11} = 2.20588\,\mathrm{rad/s^2}$. With the elbow free and zero additional elbow effort, eliminate its acceleration:

$$\left(M_{11} - \frac{M_{12}^2}{M_{22}}\right)\ddot{q}_1 = 0.10\,\mathrm{N\,m}, \qquad \ddot{q}_2 = -\frac{M_{21}}{M_{22}}\ddot{q}_1. \tag{5.20}$$

The effective shoulder inertia is $0.03028125\,\mathrm{kg\,m^2}$, yielding $\ddot{q}_1 = 3.30237$ and $\ddot{q}_2 = -7.01754\,\mathrm{rad/s^2}$. This is the same elimination principle used by ABA. The locked case needs a constraint reaction at the elbow and is a different physical experiment. Treating a held joint and a freely moving joint as equivalent collection conditions would obscure this coupling.

5.3 Controlling robot motion

A dynamic model provides a relation between effort and motion. A controller chooses effort to pursue a motion or interaction objective. The resulting closed loop becomes part of the apparatus that generates data. Good tracking can make demonstrations repeatable, while also suppressing variations that would help identify loading or coupling.

5.3.1 Gravity compensation, tracking and stability

Consider an ideal fully actuated fixed-base robot without contact or passive losses. For scalar coordinates, write

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \boldsymbol{g} = \boldsymbol{\tau}_a. \tag{5.21}$$

Here $\mathbf{C}\dot{\mathbf{q}} = \boldsymbol{c}$. A compatible Christoffel choice is

$$C_{ij} = \frac{1}{2}\sum_k \left(\frac{\partial M_{ij}}{\partial q_k} + \frac{\partial M_{ik}}{\partial q_j} - \frac{\partial M_{jk}}{\partial q_i}\right)\dot{q}_k. \tag{5.22}$$

For this choice $\dot{\mathbf{M}} - 2\mathbf{C}$ is skew-symmetric. Consequently $\dot{\mathbf{q}}^{\mathsf{T}}(\dot{\mathbf{M}} - 2\mathbf{C})\dot{\mathbf{q}} = 0$. Many matrices can produce the same vector $\boldsymbol{c}$; the skew-symmetry property should not be asserted for an arbitrary factorization.

Gravity compensation $\boldsymbol{\tau}_a = \boldsymbol{g}$ supplies the holding effort but creates no attracting position target by itself. For a constant reference $\mathbf{q}_d$, define $\mathbf{e} = \mathbf{q} - \mathbf{q}_d$ and choose

$$\boldsymbol{\tau}_a = \boldsymbol{g}(\mathbf{q}) - \mathbf{K}_P\mathbf{e} - \mathbf{K}_D\dot{\mathbf{q}}. \tag{5.23}$$

With constant symmetric positive-definite gains, a storage function is

$$H = \frac{1}{2}\dot{\mathbf{q}}^{\mathsf{T}}\mathbf{M}\dot{\mathbf{q}} + \frac{1}{2}\mathbf{e}^{\mathsf{T}}\mathbf{K}_P\mathbf{e}. \tag{5.24}$$

Differentiating and substituting the plant and controller gives

$$\begin{aligned}\dot{H} &= \dot{\mathbf{q}}^{\mathsf{T}}(\boldsymbol{\tau}_a - \mathbf{C}\dot{\mathbf{q}} - \boldsymbol{g}) + \frac{1}{2}\dot{\mathbf{q}}^{\mathsf{T}}\dot{\mathbf{M}}\dot{\mathbf{q}} + \dot{\mathbf{q}}^{\mathsf{T}}\mathbf{K}_P\mathbf{e} \\ &= -\dot{\mathbf{q}}^{\mathsf{T}}\mathbf{K}_D\dot{\mathbf{q}} \le 0.\end{aligned} \tag{5.25}$$

On an invariant compact domain with a valid coordinate chart and regular inertia, the zero-velocity invariant set also requires $\mathbf{e} = 0$. This supplies an asymptotic setpoint argument under the stated ideal assumptions. It is not a global claim for every angle representation, underactuated robot, sampled controller or contact situation.

A moving reference needs a different argument. Exact computed-torque control uses a smooth feasible $\mathbf{q}_d(t)$ and commands

$$\boldsymbol{\tau}_a = \mathbf{M}(\ddot{\mathbf{q}}_d - \mathbf{K}_v\dot{\mathbf{e}} - \mathbf{K}_q\mathbf{e}) + \mathbf{h}. \tag{5.26}$$

With exact model and state, full actuation and realized effort, this yields

$$\ddot{\mathbf{e}} + \mathbf{K}_v\dot{\mathbf{e}} + \mathbf{K}_q\mathbf{e} = \mathbf{0}. \tag{5.27}$$

The gains here have acceleration-law units: $\mathbf{K}_v$ scales by inverse time and $\mathbf{K}_q$ by inverse time squared. They are not the same gains as joint stiffness and damping in Equation (5.23). Constant symmetric positive-definite gains give exponential convergence of this ideal linear error system. A moving-target PD law with gravity compensation alone does not generally cancel the desired inertial and velocity terms and does not inherit this exact error equation (Lynch and Park 2017).

Model error, delay, saturation and unobserved loading change the realized error dynamics. For example, an imperfect gravity model leaves an additional effort $\hat{\boldsymbol{g}} - \boldsymbol{g}$. Good tracking on one trajectory does not show that every parameter is correct; feedback can compensate for several different physical mismatches. An informative collection protocol retains the controller and its limits so that observed motion is interpreted as closed-loop behavior.

5.3.2 Reference trajectories, tracking errors and physical constraints

A reference trajectory specifies more than successive target positions. Its velocity and acceleration must be compatible with the control law and the mechanism. A step in position has no ordinary finite derivative at the jump; unbounded acceleration cannot be produced by finite motor effort. Internal profiles, discussed in Chapter 3, may therefore differ from the original task request.

For a geometric path $\mathbf{q}_d(s)$ with progress parameter $s(t)$,

$$\dot{\mathbf{q}}_d = \mathbf{q}_{d,s}\dot{s}, \qquad \ddot{\mathbf{q}}_d = \mathbf{q}_{d,ss}\dot{s}^2 + \mathbf{q}_{d,s}\ddot{s}. \tag{5.28}$$

Changing traversal speed changes both inertial and velocity-dependent effort. Slowing a motion can reduce these contributions, but cannot remove a static gravity demand. Joint range, velocity, current, voltage and thermal limits also restrict feasible trajectories. A full-actuation tracking formula is only useful where the actuator arrangement can realize its required effort.

Worked example: a payload makes holding infeasible. At the arm configuration used above, ideal unloaded holding efforts are $(1.359313, 0.212393)^{\mathsf{T}}\,\mathrm{N\,m}$. Suppose the constructed joint limits are 1.50 and $0.40\,\mathrm{N\,m}$. Both holding demands fit. Adding a $0.20\,\mathrm{kg}$ point payload at the tool changes gravity effort by

$$\Delta\boldsymbol{g} = m_p g \begin{bmatrix} \ell_1\cos q_1 + \ell_2\cos(q_1+q_2) \\ \ell_2\cos(q_1+q_2) \end{bmatrix} = (0.849571, 0.339828)^{\mathsf{T}}\,\mathrm{N\,m}. \tag{5.29}$$

The total demand is $(2.208884, 0.552221)^{\mathsf{T}}\,\mathrm{N\,m}$, exceeding both limits even at rest. Slowing the same path does not resolve this holding problem. The mechanism, posture, payload, support or actuator budget must change. This example assumes ideal losses and does not qualify the stated limits as continuous thermal ratings of any product.

For data collection, an infeasible reference is a meaningful condition if it is intentionally studied and recorded. If it is silently treated as a successfully applied action, the resulting action–motion pairs have a different meaning. Retain the requested trajectory, available internal profile, observed motion, active limits and payload context separately. Tracking error should also be defined: joint error, tool-position error and orientation error answer different questions and have different units.

Disturbances and failed reaches can be useful learning examples, but their interpretation depends on why they occurred. A failure under overload is not equivalent to an error under a feasible unsaturated reference. The design question is which operating conditions should be represented, and which observations distinguish them sufficiently for the intended learner.

5.4 Contact and physical interaction

Free-motion dynamics allows joint accelerations to follow from effort alone. Contact introduces geometry-dependent loads and restrictions. Touching, sticking, sliding, breaking contact and impact have different physical conditions. An interaction dataset can cross several of them in a short episode, so a single smooth effort-to-acceleration map may be inadequate.

5.4.1 Contact kinematics and generalized contact effort

Begin with a fixed, sustained contact constraint $\boldsymbol{\phi}(\mathbf{q}) = \mathbf{0}$. For scalar joint coordinates,

$$\mathbf{J}_c\dot{\mathbf{q}} = \mathbf{0}, \qquad \mathbf{J}_c\ddot{\mathbf{q}} + \dot{\mathbf{J}}_c\dot{\mathbf{q}} = \mathbf{0}, \qquad \mathbf{J}_c = \frac{\partial \boldsymbol{\phi}}{\partial \mathbf{q}}. \tag{5.30}$$

The acceleration equation contains the changing Jacobian. Dropping that term when the robot is moving generally changes the required constraint force. A moving environment or prescribed contact motion adds corresponding terms to the right-hand side.

Let $\boldsymbol{\lambda}$ be the environment-on-robot contact load, conjugate to the constraint velocity. Power gives generalized effort $\mathbf{J}_c^{\mathsf{T}}\boldsymbol{\lambda}$. After grouping other known efforts into $\mathbf{r} = \boldsymbol{\tau}_a + \boldsymbol{\tau}_p + \boldsymbol{\tau}_{\text{other}} - \mathbf{h}$, the constrained balance is

$$\begin{bmatrix} \mathbf{M} & -\mathbf{J}_c^{\mathsf{T}} \\ \mathbf{J}_c & \mathbf{0} \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{q}} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} \mathbf{r} \\ -\dot{\mathbf{J}}_c\dot{\mathbf{q}} \end{bmatrix}. \tag{5.31}$$

This is an acceleration-level equality model for a known sustained contact mode, with consistent initial position and velocity. It is not yet a rule for deciding whether a unilateral contact exists.

Eliminating acceleration gives the contact-space relation

$$(\mathbf{J}_c\mathbf{M}^{-1}\mathbf{J}_c^{\mathsf{T}})\boldsymbol{\lambda} = -\dot{\mathbf{J}}_c\dot{\mathbf{q}} - \mathbf{J}_c\mathbf{M}^{-1}\mathbf{r}. \tag{5.32}$$

The matrix $\mathbf{W} = \mathbf{J}_c\mathbf{M}^{-1}\mathbf{J}_c^{\mathsf{T}}$ describes how a contact impulse or force changes its associated velocity or acceleration. For positive-definite $\mathbf{M}$ and independent constraint rows it is positive definite. Redundant contacts can make it singular. The factorization or repeated application of unconstrained dynamics can evaluate its action without explicitly storing an inverse.

Worked example: a support changes arm acceleration. At the previous configuration, set joint velocity to zero and let the tool rest on a frictionless horizontal support at its current height. The upward normal row is

$$\mathbf{J}_n = (0.433013, 0.173205)\,\mathrm{m/rad}. \tag{5.33}$$

Suppose gravity is compensated and the remaining applied effort is $\mathbf{r} = (0, -0.020)^{\mathsf{T}}\,\mathrm{N\,m}$. Unconstrained forward dynamics gives $(1.403509, -8.982456)^{\mathsf{T}}\,\mathrm{rad/s^2}$, which accelerates the tool into the support. Here $\dot{\mathbf{J}}_n\dot{\mathbf{q}} = 0$ and $W = 9.139319\,\mathrm{kg}^{-1}$. Equation (5.32) gives an upward force $\lambda_n = 0.103735\,\mathrm{N}$ and accelerations

$$\ddot{\mathbf{q}} = (1.626016, -4.065041)^{\mathsf{T}}\,\mathrm{rad/s^2}, \qquad \mathbf{J}_n\ddot{\mathbf{q}} = 0. \tag{5.34}$$

The joint accelerations remain nonzero while the instantaneous normal tool acceleration is zero. This is the same mechanism and actuator effort as the free calculation, with an additional interaction condition. If the inferred normal force were negative, this unilateral support could not supply it; the assumed mode would need to change.

5.4.2 Friction, unilateral constraints and impacts

For an ideal nonadhesive contact, choose gap ϕ positive when the bodies are separated and normal force λ_n positive when compressive. Rigid unilateral contact imposes

$$\phi \geq 0, \qquad \lambda_n \geq 0, \qquad \phi\lambda_n = 0. \tag{5.35}$$

These position and force conditions state that separated bodies exert no normal contact force and that rigid contact does not penetrate. They are not a complete dynamic law. At a closed sustained contact, velocity and acceleration consistency

must also hold; separating motion is admissible, and an incoming collision requires an impact rule (Lynch and Park 2017, sections 12.1–12.2).

For isotropic Coulomb friction at a point contact, let $\boldsymbol{\lambda}_t$ and $\mathbf{v}_t$ be relative tangential force and velocity, with force acting on the body whose relative velocity is used. Then

$$\begin{aligned} \|\boldsymbol{\lambda}_t\| &\le \mu\lambda_n && \text{during sticking,} \quad \mathbf{v}_t = 0, \\ \boldsymbol{\lambda}_t &= -\mu\lambda_n \frac{\mathbf{v}_t}{\|\mathbf{v}_t\|} && \text{during sliding,} \quad \mathbf{v}_t \neq 0. \end{aligned} \tag{5.36}$$

The cone bound alone does not determine sliding direction. During sticking, the tangential force is selected by the coupled balance subject to its bound; it need not equal $\mu\lambda_n$. The coefficient and point-contact model are approximations. A finite patch, torsional friction, lubrication or velocity-dependent behavior may need a richer model.

An impact is represented differently from sustained force. Over an ideal instantaneous collision, bounded actuator and gravity forces contribute negligible impulse, while the contact impulse $\boldsymbol{\Lambda}$ changes generalized momentum:

$$\mathbf{M}(\dot{\mathbf{q}}^+ - \dot{\mathbf{q}}^-) = \mathbf{J}_c^{\mathsf{T}}\boldsymbol{\Lambda}. \tag{5.37}$$

Configuration is continuous across this idealization; velocity can jump. For one frictionless closing contact against a stationary environment, Newton restitution specifies

$$v_n^+ = -e\, v_n^-, \qquad \Lambda_n = -\frac{(1+e)v_n^-}{\mathbf{J}_n\mathbf{M}^{-1}\mathbf{J}_n^{\mathsf{T}}}, \qquad v_n^- < 0, \quad 0 \le e \le 1. \tag{5.38}$$

Multiple simultaneous impacts and friction introduce further choices and consistency questions. This isolated-contact expression should not be extended to them without specifying the impact model (Tedrake, n.d.).

Worked example: impulse does not determine peak force. A constructed $2.0\,\mathrm{kg}$ effective mass approaches a stationary surface at $-0.50\,\mathrm{m/s}$ with restitution $e = 0.20$. The normal impulse is $1.20\,\mathrm{N\,s}$, the departure velocity is $0.10\,\mathrm{m/s}$, and kinetic energy decreases from 0.25 to $0.01\,\mathrm{J}$. If a real contact delivered that impulse over $0.010\,\mathrm{s}$, its mean force would be $120\,\mathrm{N}$. The ideal impact model supplies neither this duration nor the peak force. An encoder record that resolves only the velocity change cannot recover the force waveform without additional observations and assumptions.

Simulators can choose compliant or regularized contact instead of exact rigid complementarity. MuJoCo's soft contact model deliberately relaxes strict complementarity (DeepMind Technologies Limited, n.d., section "Soft contact model"). Its forces depend on that model and its settings. Numerical contact data should retain those conditions, while physical records retain the sensing and material context. Chapter 6 examines the numerical consequences in detail.

5.4.3 Impedance, energy and interaction stability

In free reaching, one may prioritize accurate motion. During interaction, the relationship between force and motion is often the design objective. Mechanical impedance describes how motion gives rise to resisting effort; admittance describes the converse input–output organization. The distinction concerns dynamic interaction, not merely whether a controller contains a position gain. Hogan (1985) develops this perspective for manipulation.

A local translational or joint-coordinate model of a desired interaction is

$$\mathbf{M}_d\ddot{\mathbf{e}} + \mathbf{D}_d\dot{\mathbf{e}} + \mathbf{K}_d\mathbf{e} = \mathbf{f}_{\text{ext}}. \tag{5.39}$$

Here the error is measured from a fixed reference in compatible coordinates; effort and error velocity are power conjugate. Do not substitute an arbitrary six-component pose error without accounting for its geometric mapping. Desired inertia, damping and stiffness are not automatically the robot's realized interaction properties. Actuation, model compensation and sensing must produce the stated relation over a declared range.

For constant symmetric $\mathbf{M}_d, \mathbf{K}_d \succ 0$ and $\mathbf{D}_d \succeq 0$, define

$$H_d = \frac{1}{2}\dot{\mathbf{e}}^{\mathsf{T}}\mathbf{M}_d\dot{\mathbf{e}} + \frac{1}{2}\mathbf{e}^{\mathsf{T}}\mathbf{K}_d\mathbf{e}, \qquad \dot{H}_d = \dot{\mathbf{e}}^{\mathsf{T}}\mathbf{f}_{\text{ext}} - \dot{\mathbf{e}}^{\mathsf{T}}\mathbf{D}_d\dot{\mathbf{e}}. \tag{5.40}$$

The ideal fixed-reference relation is passive at this effort–velocity port: stored energy can increase only through supplied external work, apart from its initial storage, while damping dissipates energy. This is a derived property of the stated equation, not a guarantee for an arbitrary controller called impedance control.

Worked example: an interaction energy budget. For one translational coordinate, take $M_d = 2.0\,\mathrm{kg}$, $D_d = 4.0\,\mathrm{N\,s/m}$ and $K_d = 100\,\mathrm{N/m}$. At $e = 0.010\,\mathrm{m}$ and $\dot{e} = 0.10\,\mathrm{m/s}$, stored energy is $0.015\,\mathrm{J}$. An external force of $0.30\,\mathrm{N}$ supplies $0.030\,\mathrm{W}$, while damping dissipates $0.040\,\mathrm{W}$. Thus $\dot{H}_d = -0.010\,\mathrm{W}$. The same balance can permit stored energy to increase if external work exceeds dissipation. Passivity constrains the energy exchange; it does not mean that motion or stored energy must always decrease regardless of external input.

The PD-plus-gravity law has a related mechanical interpretation. With an additional external generalized effort, its storage derivative becomes $\dot{H} = \dot{\mathbf{q}}^{\mathsf{T}}\boldsymbol{\tau}_{\mathrm{ext}} - \dot{\mathbf{q}}^{\mathsf{T}}\mathbf{K}_D\dot{\mathbf{q}}$ under the earlier ideal assumptions. It combines the physical inertia with a virtual spring and damping. Interconnecting compatible passive components gives a useful way to reason about energy exchange, but asymptotic convergence requires additional conditions on the interconnected system.

A moving reference can supply energy. Time-varying gains introduce additional storage-rate terms. Delay, sampling, clipping, flexible transmission and force-estimation error can change the realized port behavior. Consequently, interaction stability should be tested or analyzed for the actual implementation and environment, not inferred from a positive stiffness number alone.

This perspective also changes collection design. Stiff tracking can create large loads against a constrained environment; a compliant controller can permit informative motion while limiting some loads. Which observations are useful depends on whether the learning objective concerns force regulation, motion adaptation or contact-mode selection. A record of the reference alone does not describe the resulting mechanical interaction.

5.5 Manipulation as coupled dynamics

A hand grasping an object and an arm pushing it combine the preceding relationships. The object has its own inertia, contact loads couple it to the robot, and friction restricts feasible forces. This is why manipulation data needs the object and contact context as well as robot joint records.

5.5.1 Grasp forces, object wrenches and hand–object interaction

Let $\boldsymbol{\lambda}$ stack hand-on-object point forces in local contact axes. For contact k, let $\mathbf{R}_k$ map those axes into an object frame whose origin is the object center of mass, and let $\mathbf{r}_k$ locate the contact there. The moment-first grasp map is

$$\mathbf{G}_k = \begin{bmatrix} [\mathbf{r}_k]_\times \mathbf{R}_k \\ \mathbf{R}_k \end{bmatrix}, \qquad \mathbf{G} = [\mathbf{G}_1 \ \cdots \ \mathbf{G}_m], \qquad \mathbf{f}_{\mathrm{grasp}} = \mathbf{G}\boldsymbol{\lambda}. \tag{5.41}$$

It superposes the moments and forces at one common object origin. A point contact supplies three force components subject to its contact law; it does not supply an arbitrary independent six-dimensional wrench. Other contact models require other map columns (Murray et al. 1994).

Using the object's body coordinates, its dynamics is

$$\mathbf{I}_o\mathbf{a}_o + \mathbf{v}_o \times^* \mathbf{I}_o\mathbf{v}_o = \mathbf{G}\boldsymbol{\lambda} + \mathbf{f}_{o,\mathrm{other}}. \tag{5.42}$$

Other loads include gravity and environmental contact when present. Define $\mathbf{J}_h\dot{\mathbf{q}}_h$ to be the hand contact-point velocities expressed in the same contact axes. The hand receives the opposite forces, so its generalized contact load is $-\mathbf{J}_h^{\mathsf{T}}\boldsymbol{\lambda}$. This sign follows from defining $\boldsymbol{\lambda}$ as hand-on-object force.

For sticking point contacts, velocity compatibility and the resulting power cancellation are

$$\begin{aligned} \mathbf{J}_h\dot{\mathbf{q}}_h &= \mathbf{G}^{\mathsf{T}}\mathbf{v}_o, \\ \mathbf{v}_o^{\mathsf{T}}\mathbf{G}\boldsymbol{\lambda} - \dot{\mathbf{q}}_h^{\mathsf{T}}\mathbf{J}_h^{\mathsf{T}}\boldsymbol{\lambda} &= 0. \end{aligned} \tag{5.43}$$

Object and hand exchange power through the same contacts. Internal hand control, object inertia and contact feasibility must be considered together. A desired object wrench may require forces outside the friction cones or efforts beyond the hand's available actuation.

The map can have a nontrivial kernel. If $\mathbf{G}\boldsymbol{\lambda}_{\mathrm{int}} = \mathbf{0}$, adding an internal-force component leaves the net object wrench unchanged. It can still load the fingers and affect friction margin, deformation or heating. Not every kernel vector is physically feasible: the total forces must preserve compressive normal loads, friction bounds and actuator constraints.

Worked example: identical object motion, different grip preload. Consider a rigid $0.10\,\mathrm{kg}$ object held between two opposed point contacts at $x = \pm 0.020\,\mathrm{m}$. Take upward y, gravity downward, and equal vertical tangential forces of $0.4905\,\mathrm{N}$ per contact. Opposing normal forces of $5.0\,\mathrm{N}$ cancel horizontally; their moments and the two tangential-force moments also cancel. The net grasp wrench is an upward $0.9810\,\mathrm{N}$ force with zero moment, balancing gravity.

With $\mu = 0.30$, each tangential force is below its $1.50\,\mathrm{N}$ friction bound. Raising both normal forces to $10.0\,\mathrm{N}$ leaves the same net object wrench and the same static object pose in this ideal rigid model. It increases internal hand loading and the friction bound. Object motion alone cannot distinguish these preloads. Current, force or deformation observations may help, with their own calibration and mechanical assumptions.

For in-hand learning, this example separates task outcome from interaction state. A dataset of successful object rotations can omit how the hand distributed its contact forces. That omission may be acceptable for one policy interface and consequential for another. Changing the object's mass, surface or geometry changes the conditions under which the recorded behavior remains feasible, even if the target orientation is unchanged.

5.5.2 Arm pushing and changes of contact mode

Pushing makes contact changes especially visible. A pusher can approach, touch, stick against a face, slide along it or separate. The object's table contact introduces additional friction and possible rotation. A pusher trajectory is therefore not automatically an object trajectory.

For a planar object with center velocity (v_x, v_y), angular velocity ω_z and contact displacement (r_x, r_y) in common world axes, the material point instantaneously at contact has velocity

$$\mathbf{v}_{o,P} = \begin{bmatrix} -r_y & 1 & 0 \\ r_x & 0 & 1 \end{bmatrix} \begin{bmatrix} \omega_z \\ v_x \\ v_y \end{bmatrix}. \tag{5.44}$$

Let $\mathbf{v}_{h,P}$ be the robot pusher-point velocity. Choose normal $\mathbf{n}$ from pusher toward object and tangent $\mathbf{t}$ along the contact. The instantaneous relative components are

$$v_n = \mathbf{n}^{\mathsf{T}}(\mathbf{v}_{o,P} - \mathbf{v}_{h,P}), \qquad v_t = \mathbf{t}^{\mathsf{T}}(\mathbf{v}_{o,P} - \mathbf{v}_{h,P}). \tag{5.45}$$

At a closed smooth face contact, positive v_n corresponds to separation. Sticking requires compatible point velocities and feasible friction; sliding permits nonzero v_t with the associated friction direction. The contact location and normal can change during the episode.

The object's translational and planar rotational balances are

$$m_o \dot{\mathbf{v}}_o = \mathbf{F}_c + \mathbf{F}_{\text{surface}}, \qquad I_{o,z}\dot{\omega}_z = (\mathbf{r} \times \mathbf{F}_c)_z + n_{\text{surface},z}. \tag{5.46}$$

These horizontal-plane equations presume the supporting vertical balance has been handled separately. An off-center push changes angular acceleration through its moment arm. Surface friction contributes its own distributed wrench, so a contact force inferred by ignoring it has a different meaning. A quasi-static approximation neglects inertia only in a regime where the remaining wrench balance is adequate; it is not implied by the word pushing.

For demonstrations, record the pusher's reference and realized motion, object pose, contact geometry where available, and environment conditions. Contact-mode labels can be observations or inferences and should be marked accordingly. Keeping only the final cube position discards the transitions that a learned controller may need to handle. Conversely, recording many samples of one steady push does not necessarily cover sliding or recovery.

This example connects a concrete collection task to a general principle. Actions change not only a continuous state but also which physical relations are active. Data for embodied learning should preserve the circumstances of those transitions rather than forcing every sample into one undifferentiated action–response description.

5.6 Interpreting dynamic interaction data

The models connect commands, mechanical effort, motion and contact under stated assumptions. Dynamic records must preserve these assumptions and distinguish inputs, observations and model-derived estimates.

5.6.1 Commanded effort, transmitted effort and measured loading

The effort in Equation (5.7) is delivered to the modeled mechanism. A motor voltage, current target, estimated electromagnetic torque and measured joint load are different quantities, as established in Chapter 3. If rotor inertia or transmission compliance is explicitly modeled, its dynamics lies inside the chosen system boundary. If it is approximated by reflected armature or a loss term, that approximation must be counted once and documented.

MuJoCo's generalized arrays illustrate the required distinctions. For an ordinary continuous-time interpretation, its balance separates mass and bias from passive, actuator, user-applied and constraint effort:

$$\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{h} = \boldsymbol{\tau}_{\text{passive}} + \boldsymbol{\tau}_{\text{actuator}} + \boldsymbol{\tau}_{\text{applied}} + \boldsymbol{\tau}_{\text{constraint}}. \tag{5.47}$$

Bias is stored in `qfrc_bias`. Passive and actuator efforts are `qfrc_passive` and `qfrc_actuator`. User generalized effort is `qfrc_applied`; constraint effort is `qfrc_constraint`. Applied Cartesian body loads also contribute after mapping into generalized coordinates. Constraint effort includes joint limits and equality constraints as well as contact; it should not be renamed contact force without separating those contributions (Google DeepMind, n.d.).

This mapping is version-sensitive. In the inspected MuJoCo 3.16.0 interface, call `mj_fullM` with the model, current data and destination array to obtain a dense joint-space matrix. Do not assume that the legacy `qM` storage still exists. The versioned data field is `mjData.M`, with sparse storage. A dense matrix's shape follows the velocity dimension, not the number of quaternion configuration entries (DeepMind Technologies Limited, n.d.).

Applied Cartesian loads provide another ordering warning: `xfrc_applied` stores force followed by torque, in world components at the body center of mass in this interface. That is not our moment-first spatial wrench at an arbitrary body origin. Use the documented mapping and avoid adding the same Cartesian load again as a generalized load.

An inverse-dynamics call is also not an actuator controller. `mj_inverse` takes the desired generalized acceleration in `qacc` and writes `qfrc_inverse`. In the stated continuous convention, that result has the structure

$$\boldsymbol{\tau}_{\text{inverse}} = \mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{h} - \boldsymbol{\tau}_{\text{passive}} - \boldsymbol{\tau}_{\text{constraint}}. \tag{5.48}$$

It is the required combination of actuator and user-applied effort, not necessarily actuator effort alone. The call neither integrates a trajectory nor finds a unique feasible `ctrl`. Discrete inverse/effective-metric options require additional qualification, and contacts follow the simulator's configured model. These are interface meanings checked against source, not results of a newly executed simulator experiment.

The companion `tools/check_chapter05.py` instead makes the chapter's core computational correspondence directly inspectable. Its `rnea`, `crba` and `aba` routines share the declared body frames and model but compute effort, mass and acceleration through different routes. The analytic two-link equations provide an additional check, followed by contact, payload, impact and grasp calculations. Agreement among these calculations tests their consistency without equating simulated values with physical measurements.

5.6.2 What motion alone can and cannot reveal about contact

Suppose motion, mass properties and known noncontact efforts are available. An inverse residual can be formed as

$$\mathbf{r}_{\text{dyn}} = \mathbf{M}\hat{\ddot{\mathbf{q}}} + \mathbf{h} - \hat{\boldsymbol{\tau}}_a - \hat{\boldsymbol{\tau}}_p - \hat{\boldsymbol{\tau}}_{\text{other}}. \tag{5.49}$$

Under an exact model and observations, it equals $\mathbf{J}_c^{\mathsf{T}}\boldsymbol{\lambda}$. In practice it also contains errors in acceleration, actuator effort, friction, inertia and timing. A residual is therefore an estimated unexplained generalized load, not an independent measurement of contact force.

Even a perfect generalized residual need not identify every contact load. Recovering $\boldsymbol{\lambda}$ requires a known contact map and sufficient rank, as well as exclusion or modeling of other loads. The grasp preload example gives a physical ambiguity: internal forces can change while net object motion and wrench remain the same. During impact, motion may constrain an impulse while leaving its force-time history unknown.

These limits suggest useful interventions. Independent force sensing can check current-derived effort. Controlled changes of payload or contact location can distinguish loading effects. Holding one joint versus releasing it tests a different dynamic boundary condition. Contact-free episodes help separate some internal effects before interpreting contact

residuals, though their validity must extend to the conditions being compared. The informative intervention depends on the specific ambiguity rather than on collecting more repetitions indiscriminately.

For a forward learning model, retain action semantics, mechanism and object context, relevant histories and available contact observations. For imitation, retain the demonstrated reference and the controller interface that realizes it. For evaluation, distinguish complete episodes and held-out conditions from neighboring samples within one trajectory. A learner's accurate prediction on one contact mode does not establish that it has represented the mechanism's behavior during separation or impact.

This is where dynamic theory becomes a collection-design tool. It identifies which quantities should covary, which can remain hidden, and which changes would test an explanation. ExpGen PhySyst treats the robot, object, environment, controller and recording process as the arrangement that makes those encounters possible. The theory guides what to record without pretending that every unobserved force has become identifiable.

Exercises and further investigation. Use the stated angular-first convention and identify every body origin, effort sign and gravity treatment. The numerical companion provides constructed reference calculations, not a hardware operating procedure.

1. Derive the RNEA forward and backward passes from the body balance. Repeat a static case with explicit gravity wrenches and with fictitious base acceleration. Explain which intermediate acceleration changes and why the required joint effort agrees.
2. Obtain the two-link mass matrix by summing kinetic energies, by CRBA and by inverse-dynamics unit columns. Check symmetry and positive kinetic energy. Explain why off-diagonal entries cannot generally be deleted when interpreting multi-joint records.
3. Reproduce the locked-versus-free elbow calculation through a 2×2 solve and Schur elimination. Derive the ABA reduced bias and explain what goes wrong if either of its two added terms is omitted.
4. Verify the Christoffel energy identity for the example arm. Derive the PD-plus-gravity storage rate for a constant reference, then identify the additional terms when the reference moves or gravity compensation is imperfect. State the assumptions required by the convergence claim.
5. Recalculate the arm support force and acceleration. Choose an effort that would produce a negative force under the equality assumption, then explain the physically admissible unilateral mode. Include the changing Jacobian term when extending the example to nonzero velocity.
6. Reproduce the impact impulse and energy change. Explain what additional evidence is needed to estimate duration and peak force, and why a finite-time force sample cannot be substituted directly for impulse.
7. Build the opposed-contact grasp map and identify its internal-force direction. Check friction feasibility for two preloads. Propose a sensor or intervention that could distinguish them when the object remains still.
8. For a pushing episode, distinguish approach, sticking, sliding and separation using explicitly defined relative velocities. State which observations support the labels and how table friction changes the interpretation of an inverse residual.
9. Design one collection to investigate dynamic coupling and another to learn contact-aware motion. Specify mechanism/object conditions, action interface, retained failures and held-out regimes. Explain which conclusions remain unsupported without an independent load observation.

For further reading, Featherstone (2008, chapters 5–7) develops the recursive algorithms in depth. Lynch and Park (2017) connects dynamics, control and contact geometry; Murray et al. (1994) develops grasp and manipulation mechanics. Hogan (1985) motivates treating interaction dynamics as a control objective. These sources use conventions that should be checked before transferring individual matrices or signs.

The next chapter turns the mechanical equations into an executable simulation. Integration, control timing and contact regularization will then determine how the modeled interactions are computed and recorded. The physical quantities and algorithmic balances established here provide the reference against which those numerical choices can be examined.

Chapter 6 Physical Simulation

An arm follows a commanded motion, its tool meets an object, and a simulated camera records the result. It is tempting to regard these records as the direct consequences of the mechanical equations. Yet two implementations with the same masses, geometry and controller can produce different trajectories. They may update the controller at different rates, integrate motion differently, or assign different meanings to contact. Even within one implementation, a position read after a step can refer to a different time from a derived sensor value read beside it.

Chapter 5 established the mechanical balance. A simulator turns that balance into a sequence of computed states and observations. Understanding this transformation is necessary both for using simulation as an experimental instrument and for generating learning data. The numerical method becomes part of the arrangement that produces experience. It can introduce a useful approximation, an identifiable limitation or a misleading shortcut.

This chapter develops the connection from spatial mechanics to discrete motion, then examines control timing, contact, observation and numerical checks. MuJoCo provides concrete implementation correspondences. The worked calculations are deliberately small enough to inspect independently of an engine. Their numerical companion verifies the declared ideal examples; it does not establish the fidelity of a particular robot model.

6.1 From physical equations to an executable model

A physical model specifies what is being represented. A numerical model specifies how its evolution will be computed. An observation model specifies what information will be exposed. These are related choices, but agreement at one level does not establish agreement at the others. A detailed mesh does not guarantee correct inertia, and a plausible rendered image does not verify either the contact force or the time assigned to that image.

6.1.1 Topology, geometry and inertial representation

Begin with the system boundary. Identify the modeled moving bodies, their joints, the environment and the objects that can exchange loads with them. Decide which components are rigid, which flexibility is represented, and which effects are absorbed into a reduced parameter. A motor rotor modeled explicitly must not also be counted as the same reflected armature inertia. A rigidly fixed base and a floating base require different generalized velocities and admit different reactions.

For each rigid body, the spatial balance remains

$$\mathbf{I}_i\mathbf{a}_i + \mathbf{v}_i \times^* \mathbf{I}_i\mathbf{v}_i = \mathbf{f}_i^{\text{net}}. \tag{6.1}$$

Here motion is angular-first, wrench is moment-first, and inertia, motion and wrench use one declared body origin and coordinate frame. Equation (6.1) uses physical spatial acceleration and net external wrench, including gravity when it acts on the body. The fictitious-base gravity convention in RNEA is an alternative computational arrangement, not an additional physical force. The transformations and recursions in Chapters 4 and 5 connect the local balances into the articulated mechanism (Featherstone 2008).

The executable model needs several kinds of geometry. A visual surface specifies appearance. Collision geometry determines where interactions are detected and what contact locations and normals can be produced. Mass properties determine momentum and acceleration. These representations can share a source without being interchangeable. A hollow cover can look solid; a complicated visual surface can use simpler collision pieces.

For a massive rigid body, the center-of-mass inertia must be symmetric and physically realizable. Its principal moments are nonnegative and obey the triangle inequalities. For example, $I_1 + I_2 \geq I_3$ follows because $I_1 + I_2 - I_3$ is twice an integral of a squared coordinate against mass. An arbitrary positive-definite 3×3 matrix need not satisfy this condition. Consistent geometry, center of mass and spatial inertia are therefore stronger requirements than accepting six positive-looking numbers.

Scaling also changes more than the picture. If all lengths of a constant-density body are multiplied by s, then

$$m' = s^3 m, \qquad \mathbf{I}'_C = s^5 \mathbf{I}_C. \tag{6.2}$$

The second relation contains both the changed mass and squared length. Resizing a mesh while retaining its old mass and inertia describes a different construction. Either choice can be intentional, but the recorded model must make it clear.

Collision modeling adds another approximation. A convex collision representation can fill a concavity that remains visible in a rendering. Simplifying contact geometry can change grasp support, moment arms or whether a pusher reaches an object. Contact margins and filtering rules also affect which pairs are considered. A detected candidate contact is not automatically a measured physical touch or a positive compressive load. MuJoCo documents these choices separately from its visual rendering (DeepMind Technologies Limited, n.d., section Collision detection).

For data design, ask which geometric details affect the intended inference. A texture change can test visual robustness without changing the modeled mechanics. A change in edge radius can alter both appearance and contact. Keeping these interventions distinguishable helps explain why a learner's performance changed.

6.1.2 State coordinates and actuator input maps

Let $\mathbf{q}$ denote configuration, $\boldsymbol{\nu}$ tangent velocity and $\mathbf{z}$ internal actuator state. A continuous model can be organized as

$$\begin{aligned} \dot{\mathbf{q}} &= \mathbf{N}(\mathbf{q})\boldsymbol{\nu}, \\ \mathbf{M}(\mathbf{q})\dot{\boldsymbol{\nu}} + \mathbf{h}(\mathbf{q}, \boldsymbol{\nu}) &= \boldsymbol{\tau}_p + \boldsymbol{\tau}_a + \boldsymbol{\tau}_{\text{ext}} + \mathbf{J}_c^{\mathsf{T}} \boldsymbol{\lambda}, \\ \dot{\mathbf{z}} &= \boldsymbol{\phi}(\mathbf{q}, \boldsymbol{\nu}, \mathbf{z}, \mathbf{u}). \end{aligned} \tag{6.3}$$

The mass and bias have the meanings derived in Chapter 5. Known applied effort, passive effort and contact effort remain distinct. For scalar joints $\mathbf{N}$ can be the identity; quaternion configuration storage requires the appropriate tangent map.

The actuator contribution can be written schematically as

$$\boldsymbol{\tau}_a = \mathbf{B}(\mathbf{q})\, \mathbf{p}(\mathbf{q}, \boldsymbol{\nu}, \mathbf{z}, \mathbf{u}). \tag{6.4}$$

$\mathbf{p}$ contains actuator-output efforts and $\mathbf{B}$ maps them into generalized effort through the transmissions. A control input can be a reference position, a voltage-like quantity or a force request. Its meaning depends on this map and any internal dynamics, limits and feedback. Setting $\mathbf{u}$ equal to a desired joint torque is justified only for an appropriately defined torque interface.

MuJoCo makes this separation explicit through configuration and velocity storage, actuator activation, controls, output forces and generalized actuator effort. In the inspected interface, `qpos` and `qvel` need not have the same dimension. Actuation need not be a one-control, one-output, one-joint correspondence. The configured transmission and actuator type determine the relationship (section Actuation model).

Physical state alone may not determine the implemented next step. A controller can retain an integral state; a solver can retain a warm start; a sensor can have filter history; a plugin can maintain its own state. Denote the complete state needed for the selected computational experiment by $\boldsymbol{\chi}$. With physical parameters $\boldsymbol{\theta}$ and numerical settings $\boldsymbol{\eta}$, one step is

$$\boldsymbol{\chi}_{k+1} = \Psi_{h,\boldsymbol{\eta}}(\boldsymbol{\chi}_k, \mathbf{u}_j; \boldsymbol{\theta}). \tag{6.5}$$

The physics step is h; control index j can change less frequently than step index k. This discrete map includes the computation's retained state and specified update schedule. It is not simply a list of independent joint equations.

For ExpGen PhySyst, this distinction determines what an experience record must preserve. Two episodes with equal positions and velocities can evolve differently if their actuator or controller states differ. Omitting that history does not make the system memoryless; it makes part of its cause unobserved.

6.2 Computing motion in time

Computing an acceleration is not yet computing a trajectory. Integration approximates the accumulation of motion over a finite interval. The approximation must respect the configuration representation, and its errors must be distinguished from errors in the assumed physics.

6.2.1 Configuration integration, step size and numerical error

For ordinary scalar coordinates, write $\mathbf{a}_k$ for the generalized acceleration evaluated at the start of a step. Forward Euler uses

$$\boldsymbol{\nu}_{k+1} = \boldsymbol{\nu}_k + h\mathbf{a}_k, \qquad \mathbf{q}_{k+1} = \mathbf{q}_k + h\boldsymbol{\nu}_k. \tag{6.6}$$

Velocity-first semi-implicit Euler changes the configuration update:

$$\boldsymbol{\nu}_{k+1} = \boldsymbol{\nu}_k + h\mathbf{a}_k, \qquad \mathbf{q}_{k+1} = \mathbf{q}_k + h\boldsymbol{\nu}_{k+1}. \tag{6.7}$$

The ordering matters even when both methods use the same acceleration calculation. Neither is the exact integral of an arbitrary mechanical system.

For general configuration storage, replace addition by an integration operation $\mathbf{q} \oplus \Delta\mathbf{s}$, where $\Delta\mathbf{s}$ is a tangent displacement. For the body-to-world, scalar-first Hamilton quaternion Q used in Chapter 4, a constant body angular velocity gives

$$\begin{aligned} \Delta Q &= \left(\cos\frac{h\|\boldsymbol{\omega}^B\|}{2}, \frac{\sin(h\|\boldsymbol{\omega}^B\|/2)}{\|\boldsymbol{\omega}^B\|}\boldsymbol{\omega}^B \right), \\ Q_{k+1} &= Q_k \otimes \Delta Q. \end{aligned} \tag{6.8}$$

At zero rate, use the continuous limit of the expression. Body-frame angular velocity multiplies on the right; world-frame angular velocity multiplies on the left. This solves the constant-generator quaternion equation, consistent with Featherstone (2008, 86). It is exact for the held angular velocity, not for a complete rigid-body motion whose angular velocity changes during the step.

MuJoCo's `mj_integratePos` performs the representation-aware configuration operation; `mj_differentiatePos` performs its tangent-valued difference. They avoid treating four quaternion entries as four independent angular coordinates (DeepMind Technologies Limited, n.d.).

Worked example: unit norm is not angular accuracy. Start from identity with $\boldsymbol{\omega}^B = (0, 0, \pi)^\mathsf{T}$ rad/s for $h = 0.1$ s. Equation (6.8) gives

$$Q_{k+1} = (0.98768834, 0, 0, 0.15643447), \qquad \mathbf{R}_{k+1}\mathbf{e}_x = (0.95105652, 0.30901699, 0)^\mathsf{T}. \tag{6.9}$$

The rotation is 18°. A forward-Euler update of the quaternion derivative followed by normalization also has unit norm, but its angle is $2\arctan(h\pi/2)$ rather than $h\pi$. Its signed angular error is approximately -0.00254627 rad. A norm check alone therefore misses a geometrically meaningful integration error.

To study step size, first consider a smooth linear oscillator with mass m, stiffness k_s and $\omega = \sqrt{k_s/m}$:

$$\dot{q} = v, \qquad \dot{v} = -\omega^2 q, \qquad H = \tfrac{1}{2}m(v^2 + \omega^2 q^2). \tag{6.10}$$

Its exact unforced motion conserves H. For forward Euler, expansion of the two squared updated terms gives

$$H_{k+1} = (1 + h^2\omega^2)H_k. \tag{6.11}$$

Energy increases at every nonzero step. Decreasing h reduces the finite-horizon error, but at fixed positive h the growth persists over sufficiently long time.

Velocity-first semi-implicit Euler instead has transition matrix

$$\begin{bmatrix} q_{k+1} \\ v_{k+1} \end{bmatrix} = \begin{bmatrix} 1 - h^2\omega^2 & h \\ -h\omega^2 & 1 \end{bmatrix} \begin{bmatrix} q_k \\ v_k \end{bmatrix}. \tag{6.12}$$

Its determinant is one and its trace is $2 - h^2\omega^2$. For $0 < h\omega < 2$, its distinct eigenvalues lie on the unit circle and its trajectories remain bounded. At $h\omega = 2$, the repeated eigenvalue does not generally give bounded powers; the strict inequality matters. Direct substitution also shows that it conserves the modified quadratic

$$\widetilde{H} = \tfrac{1}{2}m\left(v^2 + \omega^2 q^2 - h\omega^2 qv\right). \tag{6.13}$$

This is positive definite within the same range. Physical energy generally oscillates rather than remaining exactly constant. The term *symplectic Euler* is appropriate for this separable Hamiltonian example; one should not infer a symplectic property for an arbitrary configuration-dependent mechanical implementation merely from the words semi-implicit.

Fully implicit, or backward, Euler evaluates both derivatives at the new state. For this oscillator its matrix is the forward-Euler matrix divided by $1 + h^2\omega^2$, and

$$H_{k+1} = \frac{H_k}{1 + h^2\omega^2}. \tag{6.14}$$

The method is stable at arbitrarily large steps for this linear example, but it removes energy from a conservative system. Stability and physical fidelity answer different questions.

Worked example: three records of the same oscillator. Take $m = 1$ kg, $\omega = 10$ rad/s, $q(0) = 0.1$ m and $v(0) = 0$. The initial energy is 0.5 J. After one second, the exact state is approximately $(-0.08390715\ \text{m}, 0.54402111\ \text{m/s})$. At $h = 0.01$ s, the computed energies are

Method	Energy at one second (J)
Forward Euler	1.352407
Velocity-first semi-implicit Euler	0.477815
Backward Euler	0.184856
Exact solution	0.500000

The companion reproduces these values and tests refinement. For the semi-implicit method at this step, the modified invariant bounds physical energy between approximately 0.476190 and 0.526316 J for this initial condition. At $h = 0.25$ s, the same method is unstable. The comparison concerns integrators, not uncertain material damping. A learner trained on the growing or decaying trajectories could otherwise learn a numerical energy source or loss as if it belonged to the mechanism.

For a smooth Lipschitz ordinary differential equation, these Euler methods have one-step error of order h^2 and global error of order h over a fixed finite interval, under the relevant regularity and solvability conditions. This convergence statement differs from fixed-step, long-time stability. Discontinuities and contact-mode changes can invalidate a smooth error-order argument (Driscoll and Braun, n.d., sections 6.2 and 11.3).

The word *implicit* also needs an implementation meaning. Write the smooth, unconstrained balance as $\mathbf{M}(\mathbf{q})\dot{\boldsymbol{\nu}} = \mathbf{F}(\mathbf{q}, \boldsymbol{\nu})$, including the bias terms in the net right-hand side. At fixed configuration, a velocity linearization gives

$$\left(\mathbf{M} - h\frac{\partial \mathbf{F}}{\partial \boldsymbol{\nu}}\right)(\boldsymbol{\nu}_{k+1} - \boldsymbol{\nu}_k) = h\mathbf{F}(\mathbf{q}_k, \boldsymbol{\nu}_k). \tag{6.15}$$

With viscous force $-\mathbf{D}\boldsymbol{\nu}$, the effective matrix is $\mathbf{M} + h\mathbf{D}$. This can stabilize a fast velocity-dependent decay without solving the full nonlinear end-of-step configuration problem. For $\dot{v} = -\alpha v$, explicit damping gives factor $1 - h\alpha$, whereas backward Euler gives $1/(1 + h\alpha)$. The former decays only for $0 < h\alpha < 2$; the latter is stable for every positive step but can still be inaccurate.

For constant $\mathbf{M}$, $\mathbf{D}$, $\mathbf{K}$ and applied $\mathbf{f}_0$, let $\mathbf{F} = \mathbf{f}_0 - \mathbf{D}\boldsymbol{\nu} - \mathbf{K}\mathbf{q}$. If the linear stiffness is also treated implicitly, substituting $\mathbf{q}_{k+1} = \mathbf{q}_k + h\boldsymbol{\nu}_{k+1}$ gives

$$(\mathbf{M} + h\mathbf{D} + h^2\mathbf{K})\mathbf{a}_{\text{step}} = \mathbf{F}(\mathbf{q}_k, \boldsymbol{\nu}_k) - h\mathbf{K}\boldsymbol{\nu}_k, \qquad \mathbf{a}_{\text{step}} = \frac{\boldsymbol{\nu}_{k+1} - \boldsymbol{\nu}_k}{h}. \tag{6.16}$$

This derivation assumes the stated constant linear model. It explains why an effective integration matrix can differ from physical inertia.

MuJoCo's versioned integrator options select particular approximations and force derivatives. Its Euler option is not the fully explicit teaching method above, and its implicit-in-velocity methods are not automatically the fully implicit oscillator calculation. Version 3.16.0 also provides an opt-in `discrete` mode, which incorporates selected stiffness/damping terms into an effective metric shared with its constraint solve. In that mode, `qacc` describes a step acceleration rather than an unqualified continuous-time acceleration. Record the integrator and its version before assigning physical meaning to such arrays (Google DeepMind, n.d.).

6.2.2 Control updates, sample-and-hold and solver timing

A controller usually acts at its own period T_c. If $T_c = dh$ for integer d, a zero-order hold applies

$$\mathbf{u}_j = \pi(\mathbf{o}_j), \qquad \mathbf{u}(t) = \mathbf{u}_j, \quad jT_c \le t < (j+1)T_c. \tag{6.17}$$

The input request is held. An actuator's internal activation, servo effort or transmission moment arm can continue changing within that interval. Holding a position reference is therefore different from holding the resulting joint torque.

With a fixed held input, the control-interval transition composes d physics steps:

$$\boldsymbol{\chi}_{(j+1)d} = \underbrace{\Psi_h(\cdots \Psi_h(\boldsymbol{\chi}_{jd}, \mathbf{u}_j) \cdots, \mathbf{u}_j)}_{d \text{ physics steps}}. \tag{6.18}$$

Parameters and numerical settings are suppressed here, not removed. Reducing h while keeping T_c fixed changes integration resolution. Reducing both changes the closed-loop experiment.

Worked example: a held torque and a sampled controller. For a constructed scalar joint with $I = 0.02$ kg m^2, no gravity or losses, and a held 1 N m torque, acceleration is 50 rad/s^2. Starting from rest over $T_c = 0.02$ s gives the exact values $v(T_c) = 1$ rad/s and $q(T_c) = 0.01$ rad. Velocity-first semi-implicit Euler gives

$$v_N = aT_c, \qquad q_N = \tfrac{1}{2}a(T_c^2 + T_c h), \qquad N = T_c/h. \tag{6.19}$$

At $h = 0.005$ s, the result is $q_N = 0.0125$ rad. Halving the step twice gives 0.01125 and 0.010625 rad. Velocity is exact in this special constant-acceleration example, while position has a first-order error. Accurate velocity alone does not certify the whole trajectory.

Controller sampling introduces another effect even with exact integration. Consider the scalar plant $\dot{x} = u$ and feedback $u_j = -k_c x_j$ held for T_c. Exact integration gives

$$x_{j+1} = (1 - k_c T_c) x_j. \tag{6.20}$$

Asymptotic decay requires $0 < k_c T_c < 2$. With $k_c = 50$ s^{-1}, $T_c = 0.02$ s gives a zero next sample for this ideal plant, while $T_c = 0.05$ s gives factor -1.5 and divergence. Refining the physics step cannot repair an unstable exact sampled control law. The zero sample at the first setting is a special discrete result, not finite-time convergence of the continuous feedback law.

An engine can evaluate dynamics several times within one nominal step. For example, Runge–Kutta trial stages can reevaluate a state-feedback callback at intermediate states. That represents a different controller from one sampled once per T_c and held. A callback that advances its own memory on every invocation may change its behavior merely because the integration method changed.

In MuJoCo, `mj_step` advances time and integrated state. Derived quantities are computed within its pipeline and need not all refer to the newly returned endpoint. A position-dependent sensor can still describe the earlier position; Runge–Kutta introduces trial-stage timing as well. Refresh the required computations at the declared recording time. `mj_forward` evaluates the forward pipeline without integrating time, but invokes applicable callbacks and sensors; it is not universally a side-effect-free logging operation (DeepMind Technologies Limited, n.d.).

The following illustrative pattern separates a held request, a start-of-step observation and the next integrated state. It assumes ordinary continuous Euler stepping, no callbacks and no stateful or stochastic observation side effects. The copying and storage functions are application-specific.

```
data.ctrl[:] = held_request
for substep in range(decimation):
    mujoco.mj_forward(model, data)
    before = copy_observation(data)
    effort = data.qfrc_actuator.copy()
    mujoco.mj_step(model, data)
    after = (data.time, data.qpos.copy(), data.qvel.copy())
    save_transition(before, held_request.copy(), effort, after)
mujoco.mj_forward(model, data)
next_observation = copy_observation(data)
```

The effort belongs to the evaluated start state; it is not asserted to be an interval average. An endpoint force recomputed afterward describes that endpoint under the specified input, not automatically the force history that generated the preceding motion. A production collector must also handle callbacks, sensor histories and termination explicitly.

Split stepping can place a controller between selected pipeline stages, but it is not interchangeable with every integrator. In the inspected version, `mj_step1`/`mj_step2` does not preserve RK4 integration. Use a version-checked interface rather than inferring the schedule from a function's name.

Finally, simulation time, rendering rate and wall-clock throughput are different clocks. Accelerating execution need not change the simulated control period. Rendering fewer frames need not reduce the number of mechanical steps. A learning record needs the simulated action interval and observation time, not merely the time at which a file was written.

6.3 Computing constraints and contact

Chapter 5 described ideal mechanical constraints and contact laws. Their numerical realization adds decisions about detection, compliance, regularization and solution accuracy. These decisions affect the computed trajectory and the interpretation of a reported contact label or force.

6.3.1 Rigid constraints, compliant contact and modeling choices

For a smooth fixed equality constraint, a local velocity correction can be written as

$$\boldsymbol{\nu}^{+} = \boldsymbol{\nu}^{\mathrm{free}} + \mathbf{M}^{-1}\mathbf{J}^{\mathsf{T}}\boldsymbol{\Lambda}, \qquad \mathbf{J}\boldsymbol{\nu}^{+} = \mathbf{v}_{\mathrm{target}}. \tag{6.21}$$

$\boldsymbol{\Lambda}$ is an impulse, not a continuous force. Substitution gives

$$\mathbf{W}\boldsymbol{\Lambda} = \mathbf{v}_{\mathrm{target}} - \mathbf{J}\boldsymbol{\nu}^{\mathrm{free}}, \qquad \mathbf{W} = \mathbf{J}\mathbf{M}^{-1}\mathbf{J}^{\mathsf{T}}. \tag{6.22}$$

This local construction holds geometry fixed while solving the velocity correction. Position drift, changing contact geometry and the impact law still need treatment. Dividing impulse by h gives a step-average force for that construction; it does not identify a physical peak force.

A unilateral frictionless contact also requires nonnegative compression and a rule for separation. A sustained compliant alternative permits a small deformation. Let $\delta = \max(0, -\varphi)$ be penetration relative to a chosen gap function. A simple active-contact model is

$$\lambda_n = \max(0, k_s\delta + c_s\dot{\delta}). \tag{6.23}$$

The outer bound prevents tensile normal force. This spring–damper choice is an illustrative material approximation, not a universal contact law. Its release rule, force regularity and behavior during impact matter.

For one active contact with effective mass $m_{\mathrm{eff}} = (\mathbf{J}_n\mathbf{M}^{-1}\mathbf{J}_n^{\mathsf{T}})^{-1}$, the local contact scale is

$$\omega_n = \sqrt{k_s/m_{\mathrm{eff}}}, \qquad \zeta = \frac{c_s}{2\sqrt{k_s m_{\mathrm{eff}}}}. \tag{6.24}$$

Other active constraints can change the effective mobility. High stiffness introduces fast motion even if the observed object moves slowly. Large damping can introduce a still faster decay mode; the roots of $m_{\mathrm{eff}}s^2 + c_s s + k_s = 0$ then provide more information than spring frequency alone.

Worked example: resolving a stiff support. Let $m_{\mathrm{eff}} = 0.5$ kg and $k_s = 20\,000$ N/m. Under a static 4.905 N load, the ideal indentation is

$$\delta_{\mathrm{eq}} = 0.00024525\ \mathrm{m}, \qquad \omega_n = 200\ \mathrm{rad/s}, \qquad T_n = \frac{2\pi}{\omega_n} = 0.03141593\ \mathrm{s}. \tag{6.25}$$

The undamped semi-implicit oscillator bound would require $h < 0.01$ s. Operating just below that boundary would still resolve the oscillation poorly. A step of 0.001 s gives about thirty-one steps per natural period; whether this is adequate depends on the quantity being studied and a refinement test. The stability bound is not a fidelity prescription for every damped, switching contact system.

MuJoCo's soft-contact formulation is more structured than substituting Equation (6.23) for every interaction. It combines constraint-space dynamics with parameterized softness and friction. Its `solref` and `solimp` settings influence reference constraint behavior and impedance. They should not be read as a uniquely identified material stiffness or sensor calibration without checking the model and parameterization. Hard and soft contact are alternative approximations to

physical interaction, with different computational and interpretive consequences (DeepMind Technologies Limited, n.d., sections Soft contact model and Constraint solver).

Friction adds further choices. A friction cone, a pyramidal approximation, torsional resistance and a finite contact patch represent different possibilities. A stable grasp can depend on which are present. Reporting only one coefficient of friction omits part of the contact model.

6.3.2 Regularization, solver settings and numerical stability

Regularization changes a mathematical problem; a solver tolerance controls how accurately that selected problem is solved. The following scalar example makes the distinction explicit without attempting to reproduce MuJoCo's full contact law.

Consider a normal impulse $\Lambda \geq 0$, a free closing velocity $b < 0$, and inverse effective mass $W > 0$. A regularized velocity-step problem is

$$\min_{\Lambda \geq 0} \tfrac{1}{2}(W + R)\Lambda^2 + b\Lambda, \qquad R \geq 0. \tag{6.26}$$

Both W and R have inverse-mass units; the objective has energy units. For positive $W+R$, the minimizer and its optimality conditions are

$$\begin{aligned} \Lambda^* &= \max\left(0, \frac{-b}{W + R}\right), \\ s &= (W + R)\Lambda + b \geq 0, \qquad \Lambda \geq 0, \qquad \Lambda s = 0. \end{aligned} \tag{6.27}$$

The physical velocity after the impulse is instead

$$v_n^+ = b + W\Lambda. \tag{6.28}$$

For $R > 0$ and active contact, exact optimality gives $v_n^+ = -R\Lambda < 0$. The regularized slack s is not the physical separation velocity. Zero optimization residual therefore does not certify exact rigid nonpenetration.

Worked example: an exact solver can preserve a model discrepancy. Use $W = 2\,\mathrm{kg}^{-1}$, $R = 0.5\,\mathrm{kg}^{-1}$ and $b = -0.4$ m/s. The exact softened solution is

$$\Lambda^* = 0.16\,\mathrm{N\,s}, \qquad s = 0, \qquad v_n^+ = -0.08\,\mathrm{m/s}. \tag{6.29}$$

With $R = 0$, the hard inelastic scalar problem gives $\Lambda = 0.2$ N s and $v_n^+ = 0$. Solving the $R = 0.5$ problem more accurately cannot turn it into the $R = 0$ problem. Conversely, an iterate $\Lambda = 0.15$ N s in the softened problem has $s = -0.025$ m/s and violates its own optimality conditions. These are two different errors.

For several contacts, matrices, friction constraints and reference terms replace the scalars. A positive-definite regularizer can make a positive-semidefinite quadratic strictly convex, but the chosen objective and admissible contact set still define the model. An arbitrary norm of mixed translational and rotational residuals is also not a unit-independent accuracy measure. Tolerances require a declared scaling.

Iteration limits, stopping criteria and warm starts affect the computed answer. A warm start is a numerical memory, not a measured force. At finite iteration count, two different warm starts can produce slightly different next states. The difference can later grow around contact-mode changes. Changing contact parameters, time step and iteration budget simultaneously makes the cause difficult to identify.

A useful examination therefore varies one layer at a time: hold the contact model fixed while checking solver convergence; hold the control schedule fixed while refining time; then test whether the chosen contact model is adequate for independent physical evidence. Stable execution, small optimization residual and realistic interaction are separate achievements.

6.4 Generating simulated observations and trajectories

A collector turns trajectories into learning records by choosing observations, their timing, reset conditions and retained episodes. Simulation makes these choices repeatable; their consequences remain part of the experiment.

6.4.1 Virtual sensors, rendered images and privileged states

Write a simulated observation at declared time t_j as

$$\mathbf{o}_j = \mathcal{H}(\boldsymbol{\chi}(t_j); \boldsymbol{\theta}_s, \boldsymbol{\eta}_o). \tag{6.30}$$

The observation map includes selected coordinates, frame transformations, rendering and any deliberately modeled acquisition effects. $\boldsymbol{\theta}_s$ describes sensor geometry and calibration assumptions; $\boldsymbol{\eta}_o$ describes computational choices such as image resolution. An added noise process has its own distribution and random state.

Separate at least three kinds of information. A selected state variable is available because the simulator stores it. A virtual sensor computes an output under its declared semantics. An image is a rendered view with camera, lighting, material and occlusion assumptions. An object's exact pose can be privileged simulator information even when its image does not uniquely reveal that pose.

The distinction affects learning. A policy trained with contact force or object pose that will be unavailable at deployment needs an explicit way to replace that information. Privileged variables can be useful for labels, diagnostics or a training-only critic, but those uses should be separated from the policy's executable observation interface. Otherwise an apparently successful simulation experiment can rely on information the intended system will never receive.

Temporal semantics matter as much as variable names. An instantaneous rendering at t_j is not an exposure over a moving interval. An idealized exposure-like image can be represented schematically by

$$\mathbf{I}_j = \frac{1}{T_e} \int_{t_j - T_e}^{t_j} \mathcal{R}(\boldsymbol{\chi}(t), \boldsymbol{\theta}_s)\, dt. \tag{6.31}$$

This simple integral explains temporal averaging; it is not a complete physical camera model. A delayed pose or filtered encoder similarly has a history dependence. Adding independent noise to an instantaneous perfect state does not reproduce every bandwidth, delay or occlusion effect. Chapter 7 develops acquisition and estimation in detail.

Frames and reference points remain explicit. A virtual force output can refer to an internal connection rather than the fingertip contact being studied. An acceleration array can represent an algorithmic or discrete quantity rather than a physical point's second derivative. A rendered marker pose can differ from a body origin by a fixed transform. Use the spatial maps already established before comparing values.

A practical consequence is to keep learning observations and diagnostic records distinct in the same episode. The learner can receive only its intended observations, while the diagnostic record retains simulator state, contacts and effort for explanation. Both must preserve their time and derivation identity. A diagnostic variable should not silently become an input feature.

Appearance and mechanics can also be varied independently. Relabeling a cube's faces changes image evidence while preserving physics if its geometry, mass properties and contact model are held fixed. Changing surface friction or edge shape is a different intervention. These comparisons help establish whether a learned behavior depends on visual recognition, physical interaction or both; no particular performance is implied by making the modifications.

6.4.2 Resets, randomized conditions and episode identity

A reset defines an initial condition, not merely a new file name. It can set joint and object states, actuator memory, controller memory, contact settling conditions and the randomization context. Starting inside an unintended overlap can create an artificial initial impulse that becomes part of every collected episode.

Worked example: identical visible resets can hide different dynamics. Consider the scalar joint with $I = 0.02\ \mathrm{kg\ m^2}$ and delivered actuator effort z, with

$$\dot{z} = (u - z)/\tau_z, \qquad I\dot{v} = z, \qquad \dot{q} = v. \tag{6.32}$$

Let $\tau_z = 0.02$ s, $u = 1$ N m and $q(0) = v(0) = 0$. Two episodes differ only in $z(0)$, equal to zero or 1 N m. Their initial accelerations are zero and 50 rad/s^2. Solving the held-input filter and integrating its output gives

$$\begin{aligned} z(T) &= u + (z_0 - u)e^{-T/\tau_z}, \\ v(T) &= \frac{uT + (z_0 - u)\tau_z(1 - e^{-T/\tau_z})}{I}, \\ q(T) &= \frac{uT^2/2 + (z_0 - u)\tau_z[T - \tau_z(1 - e^{-T/\tau_z})]}{I}. \end{aligned} \tag{6.33}$$

At $T = 0.02$ s, the first episode has $q \approx 0.00264241$ rad and $v \approx 0.36787944$ rad/s. The second has $q = 0.01$ rad and $v = 1$ rad/s. Equal positions, velocities and control requests did not define equal initial states. A training record that omits activation or sufficient history can make this difference appear unexplained.

Engine reset and checkpoint operations therefore need explicit scope. MuJoCo's ordinary reset restores model defaults for selected fields; it does not mean that every field becomes zero. Keyframes restore a defined subset of state and user inputs. Neither operation automatically resets the application's random generator or controller memory. State save/restore APIs expose integration-state groups, but optional features can require additional feature-specific state. For example, sleeping changes the requirements for exact replay. Saving only `qpos` and `qvel` is not a general complete checkpoint (DeepMind Technologies Limited, n.d.).

To generate a distribution of episodes, write

$$\boldsymbol{\theta}_e \sim p_{\text{train}}(\boldsymbol{\theta}), \qquad \boldsymbol{\chi}_{e,0} \sim p_{\text{reset}}(\boldsymbol{\chi}_0 \mid \boldsymbol{\theta}_e). \tag{6.34}$$

The physical parameters and reset distribution answer different questions. Object mass and friction can be randomized while initial placement stays fixed, or the reverse. Independent uniform draws are not automatically a physically plausible population. Geometry, mass, inertia and actuator limits can be coupled.

The sampled values, random-generator algorithm and required state belong to episode identity. A seed is helpful but does not alone specify the sequence if the draw order, generator or parallel execution changes. Likewise, bitwise replay on one software version and architecture does not establish identical results across engines, devices or later versions. The relevant reproducibility claim must be bounded.

Each retained transition should state its relation to the control interval. A simple record is

$$\mathcal{T}_{e,j} = (e, j, t_j, \mathbf{o}_j, \mathbf{u}_j, t_{j+1}, \mathbf{o}_{j+1}, \sigma_{e,j}), \tag{6.35}$$

where $\sigma_{e,j}$ records termination or continuation information. The episode context separately identifies the model, numerical settings, reset and observation definitions. A time-limit truncation is different from a physical failure, success or numerical reset. Treating all as the same terminal event changes the meaning of the data.

The companion constructs two toy episodes with distinct reset seeds, two held actions per episode and four physics substeps per action. It checks timestamp increments, action identity, state continuity and episode boundaries. These are checks on bookkeeping, not evidence of a robot's task success.

For embodied learning, retain relevant unsuccessful and transition cases when they belong to the collection's intended scope. Keeping only completed motions can hide loss of contact, occlusion or failed recovery. Conversely, an automatic numerical reset must not silently become a valid recovery demonstration. Dataset selection and independent evaluation are developed further in Chapters 9 and 16.

6.5 Checking numerical fidelity

Numerical fidelity concerns the computation's relation to the selected mathematical model. Physical validity concerns that model's adequacy for the intended real system and inference. Both matter for simulated data, and neither is established by a plausible animation.

6.5.1 Conservation, limiting cases and step-refinement tests

Begin with checks whose expected behavior is independently known. Remove gravity and losses from a free body, test constant velocity without forces, or lock a joint and compare its reduced dynamics. Check frame changes through

power invariance and compare recursive inverse/forward calculations as in Chapter 5. These tests expose representation and sign errors before complicated contact makes their consequences harder to locate.

Energy provides a further diagnostic when its boundary is defined. For a mechanical system with known actuation, external loads and dissipative power,

$$\dot{H} = \boldsymbol{\nu}^{\mathsf{T}} \boldsymbol{\tau}_a + P_{\text{ext}} - P_{\text{loss}}. \tag{6.36}$$

Spring potential must be included if spring forces are part of the system. Constraint work vanishes only for the corresponding ideal stationary workless constraints; moving supports, friction and impacts require their own terms.

A discrete work residual can be formed as

$$r_H = H_N - H_0 - \sum_{k=0}^{N-1} h \left(\boldsymbol{\nu}_k^{\mathsf{T}} \boldsymbol{\tau}_{a,k} + P_{\text{ext},k} - P_{\text{loss},k} \right). \tag{6.37}$$

This particular residual also contains the chosen quadrature error. A nonzero value is a discrepancy to explain, not automatically physical heat or an unknown actuator loss. The oscillator example shows that a stable integrator can produce systematic energy decay without any physical damping.

Step refinement compares solutions at the same physical times with the same controller period, initial conditions and observation definition. For a smooth quantity of interest s with asymptotic error $s_h = s_* + Ch^p + o(h^p)$,

$$\frac{|s_h - s_{h/2}|}{|s_{h/2} - s_{h/4}|} \longrightarrow 2^p, \qquad |s_{h/2} - s_*| \approx \frac{|s_h - s_{h/2}|}{2^p - 1}. \tag{6.38}$$

The denominator must not vanish, and the approximation assumes that the same leading-order regime has been reached. For poses, compare relative rotations and translations rather than raw quaternion subtraction. For several differently scaled variables, define meaningful units or normalization before combining errors.

Worked example: a refinement trend with a known answer. For the held-torque position, the errors at $h = 0.005$, 0.0025 and 0.00125 s are respectively 0.0025, 0.00125 and 0.000625 rad. The consecutive-difference ratio is exactly two, and Equation (6.38) with $p = 1$ estimates the fine-grid error exactly in this constructed case. The oscillator companion also checks first-order convergence against its analytic endpoint, using

$$E_h = \sqrt{(q_h - q_*)^2 + \left(\frac{v_h - v_*}{\omega} \right)^2}. \tag{6.39}$$

Both terms have length units. The error ratios approach two as the step shrinks, even for forward Euler, whose fixed-step long-time energy behavior is unstable. The two observations are consistent because they concern different limits.

Contact tests need quantities suited to nonsmooth behavior. Peak force can change strongly with step and regularization, while total impulse is more interpretable for an unresolved collision. Compare event timing, impulse, settled indentation, slip distance or a declared task outcome, as appropriate. Small changes near a grazing or mode-changing event can prevent pointwise trajectory agreement without proving an implementation error. Conversely, a visually similar endpoint can conceal a different contact history.

Refine solver accuracy separately from time. Report whether a changed result follows from more iterations of the same problem, a smaller integration step or a changed contact model. Refinement cannot establish that an omitted effect is physically negligible; it can establish consistency within a selected approximation.

6.5.2 Solver artifacts, physical assumptions and limits of the simulated data

A useful discrepancy analysis separates at least four questions. Are the equations and representations implemented consistently? Is the numerical approximation sufficiently resolved for the quantity of interest? Are the physical and observation models adequate for the intended conditions? Does the collection cover the conditions under which learning or evaluation will be used? These questions interact, but an answer to one is not a substitute for the others.

Examples illustrate the separation. A frame error survives a smaller time step. A coarse integrator can create apparent damping even with correct mass properties. An exact softened solver can permit the closing velocity in Equation (6.29). A perfectly reproducible rendering can omit the occlusion or delay that matters on a physical system. Randomizing a parameter over many episodes does not demonstrate that its chosen range represents actual variation.

Learning can amplify these distinctions. An optimizer can exploit compliant overlap, solver-dependent impulses, unrealistic effort changes or privileged observations if they make the objective easier. A large number of such episodes increases the amount of data without correcting its generating assumptions. Diagnostic variables, perturbation tests and held-out numerical settings can help investigate a suspected dependency, but a change of settings is itself another experimental condition and must be recorded.

Parallel engines can organize the same mechanical problem differently. Throughput, random-draw schedules, precision, solver implementation and supported features can change. Matching XML or a seed does not establish matching transition maps. Align the physical model, actuator and observation interfaces, timing and numerical choices for the intended comparison, then record the remaining differences.

MuJoCo organizes articulated dynamics and contact computation for model-based analysis and optimization (Todorov et al. 2012). Interpret its outputs against the model, documented interface and inspected code version. Engine functions here use source-checked 3.16.0 meanings. The companion independently checks the declared small models; it does not test every engine option or qualify a robot for deployment.

The resulting principle for ExpGen PhySyst is concrete. A simulated experience belongs to a specified mechanism, environment, controller, numerical evolution and observation process. Preserve enough of that arrangement to explain what the record contains and to reconstruct the relevant comparison. Then connect it to independent physical observations where the claim concerns reality. Simulation can design and test informative interactions without making its internal values measurements of the world.

Exercises and further investigation. Use the same angular-first mechanics and explicit timing conventions as the chapter. State which results concern an ideal model, a numerical method or physical evidence.

1. Derive the s^3 mass and s^5 inertia scaling laws. Construct a positive-definite center-of-mass inertia matrix that violates a principal triangle inequality, and explain why positivity alone is insufficient.
2. Derive Equation (6.8) from the Hamilton quaternion differential equation. Compare right and left multiplication for a nonidentity pose and distinguish normalization from angular accuracy.
3. Derive all three oscillator transition matrices and energy statements. Verify the modified semi-implicit invariant and explain the failure at $h\omega = 2$. Reproduce finite-horizon convergence separately from long-time behavior.
4. Derive the implicit velocity and linear stiffness updates. For a scalar damped mode, compare monotone decay, oscillatory decay and instability under explicit treatment. Explain why an integrator's name alone does not identify its treatment of stiffness and damping.
5. Reproduce the held-torque refinement and sampled-control examples. Design two comparisons: one changes integration resolution alone and the other changes the control period. Specify what must remain fixed.
6. Recalculate contact indentation, natural period and the fastest damped mode for a chosen damping coefficient. Propose appropriate refinement quantities for a settled support and for an impact.
7. Solve the scalar regularized contact problem, including an inactive separating case. Compare optimization slack with actual velocity. Explain why decreasing tolerance and decreasing regularization are different interventions.
8. Reproduce the hidden-activation reset example. Design a record that makes the two episodes distinguishable without giving a deployed policy unavailable privileged information. State which histories or observations are needed.
9. Specify a simulated pushing or grasping collection: physical model, input meaning, controller/physics/observation periods, reset distribution, retained failures and numerical checks. Identify which claims require additional evidence outside simulation.

The companion `tools/check_chapter06.py` checks the worked examples, invariants, convergence and timing. Driscoll and Braun (n.d.) introduce consistency, convergence and stability. Hairer et al. (2006) develop structure-preserving integration beyond the elementary derivations here. MuJoCo's computation and simulation documentation connect these foundations to the inspected engine interface.

Chapter 7 examines physical measurement and state estimation, where selected, uncertain and temporally structured observations replace the simulator's readily available variables.

Chapter 7 Measurement and State Estimation

An encoder reports a joint angle. A camera sees a marker on the moving link. An inertial sensor responds to the motion of its own mounting point. The records disagree slightly. Before interpreting that disagreement as a mechanical deficiency, we must ask what each instrument observed, in which frame, over which time interval and with which uncertainty. The signals may be consistent descriptions of different quantities, or inconsistent descriptions of the same one. A state estimator cannot decide between these possibilities without a measurement model.

Chapter 6 separated simulated state from generated observations. A physical acquisition system makes that distinction unavoidable: the state is not directly available simply because our equations contain it. Measurements are selected, transformed, sampled and transmitted. Estimated state combines those observations with assumptions about motion and uncertainty.

The chapter develops this chain from probability and calibration to temporal and geometric interpretation, then to recursive estimation. Featherstone mechanics supplies the meanings of motion, acceleration and loading. Probability describes incomplete information about those quantities; it is not a replacement physical theory. For ExpGen PhySyst, the objective is to design encounters whose observations can support a useful learning or evaluation claim. The worked examples and companion are constructed calculations, not measurements from a particular device.

7.1 Measurement models and uncertainty

A measurand is the quantity intended to be measured. Naming it precisely requires more than choosing a column heading: joint angle relative to which zero, force on which body, or pose of which point? A model connects that quantity to an indication produced by the instrument. It also states what remains uncertain after calibration and processing.

7.1.1 Conditional probability, Bayes' rule and expectation

Let $\mathbf{x}$ contain the unknown quantities of interest, $\mathbf{y}$ an observation, and $\mathcal{C}$ the known experimental context. For example, $\mathcal{C}$ can include the actuator request, mounting geometry and acquisition time. In a local Euclidean representation, Bayes' rule gives

$$p(\mathbf{x} \mid \mathbf{y}, \mathcal{C}) = \frac{p(\mathbf{y} \mid \mathbf{x}, \mathcal{C}) p(\mathbf{x} \mid \mathcal{C})}{\int p(\mathbf{y} \mid \mathbf{x}', \mathcal{C}) p(\mathbf{x}' \mid \mathcal{C}) \, d\mathbf{x}'}. \tag{7.1}$$

The likelihood describes how observations could arise at a given state. The prior describes information available before this observation. The posterior combines them, provided the denominator is nonzero. Conditioning on context matters: an encoder value after a reset and the same value during loaded motion need not carry the same information.

A density is not the probability of a particular exact continuous value. Probabilities belong to regions and are obtained by integration. A conditional mean and covariance are

$$\begin{aligned} \hat{\mathbf{x}} &= \mathbb{E}[\mathbf{x} \mid \mathbf{y}, \mathcal{C}], \\ \mathbf{P} &= \mathbb{E}[(\mathbf{x} - \hat{\mathbf{x}})(\mathbf{x} - \hat{\mathbf{x}})^{\mathsf{T}} \mid \mathbf{y}, \mathcal{C}]. \end{aligned} \tag{7.2}$$

For Euclidean squared-error loss, the conditional mean minimizes expected loss. A maximum-density estimate answers a different question and depends on the chosen parameterization. A small covariance does not establish that the assumed likelihood or prior is correct.

Consider a scalar indication $y = x + \epsilon$, with independent Gaussian prior $x \sim \mathcal{N}(\mu_0, \sigma_0^2)$ and noise $\epsilon \sim \mathcal{N}(0, \sigma_y^2)$. Completing the square in the product of the two densities gives

$$\sigma_+^2 = \left(\sigma_0^{-2} + \sigma_y^{-2}\right)^{-1}, \qquad \mu_+ = \sigma_+^2 \left(\frac{\mu_0}{\sigma_0^2} + \frac{y}{\sigma_y^2}\right). \tag{7.3}$$

Information is weighted by precision, the inverse variance. The equation does not justify combining two nominally different measurements as if they observed the same x. Their frames, timing and calibration must first make that assumption defensible.

Gaussian models are useful because means and covariances propagate tractably through linear relations. They are choices, not guarantees that clipping, occlusion or contact-induced errors are Gaussian. Barfoot (2024) develops the probabilistic foundations and their use in robotics. We will retain the conditions under which the linear-Gaussian calculation is exact and identify approximations elsewhere.

7.1.2 Noise, covariance, bias and calibration

A useful local measurement model is

$$\mathbf{y} = \mathbf{g}(\mathbf{x}, \boldsymbol{\theta}_c) + \mathbf{b} + \boldsymbol{\epsilon}. \tag{7.4}$$

Calibration parameters $\boldsymbol{\theta}_c$ can include scale, zero, cross-axis sensitivity and mounting geometry. The term $\mathbf{b}$ can represent an uncorrected offset; $\boldsymbol{\epsilon}$ describes the remaining modeled variation. A correction removes an estimated systematic effect, while uncertainty in that correction remains.

For a scalar reference input s, a simple calibration is $y = as + b + \epsilon$. Estimating a and b requires reference values and conditions that make them distinguishable. A reading at rest is not a known zero load if gravity or preload is still present. Calibration is a measurement relationship; identifying the mechanism's mass or friction from motion is the broader system-identification problem of Chapter 8.

For a differentiable transformation $\mathbf{z} = \mathbf{g}(\boldsymbol{\xi})$, first-order uncertainty propagation gives

$$\mathbf{P}_z \approx \mathbf{G}\mathbf{P}_\xi\mathbf{G}^\mathsf{T}, \qquad \mathbf{G} = \left.\frac{\partial \mathbf{g}}{\partial \boldsymbol{\xi}}\right|_{\hat{\boldsymbol{\xi}}}. \tag{7.5}$$

The input vector can include state and calibration variables. Off-diagonal covariances matter when those variables are dependent. An independent additive noise covariance is added afterward; correlated noise instead requires the corresponding cross terms.

The approximation concerns both scale and nonlinearity. In general, $\mathbb{E}[\mathbf{g}(\boldsymbol{\xi})] \neq \mathbf{g}(\mathbb{E}[\boldsymbol{\xi}])$. A first-order Jacobian can miss a shift in the mean as well as an incorrect variance. The GUM describes uncertainty evaluation and covariance; its 2026 amendment explicitly addresses substantial nonlinearity (Joint Committee for Guides in Metrology 2008, 2026). Type A and Type B describe methods of evaluating uncertainty, rather than synonyms for random and systematic errors.

Worked example: averaging cannot remove a shared calibration offset. Let $y_i = q + b + \epsilon_i$ measure one constant angle. Model the uncertain shared offset as zero-mean with $\sigma_b = 0.005$ rad, and independent readout noise with $\sigma_\epsilon = 0.002$ rad. Offset and readout noise are independent. Then

$$\mathrm{Var}(\bar{y} - q) = \sigma_b^2 + \frac{\sigma_\epsilon^2}{N}, \qquad \mathrm{Cov}(y_i - q, y_j - q) = \sigma_b^2 \quad (i \neq j). \tag{7.6}$$

At $N = 100$, the standard uncertainty is approximately 0.005004 rad. Ignoring the common offset would suggest 0.0002 rad. The extra samples reduce the independent noise term, not the uncertainty in one calibration shared by the whole series. A particular realized offset need not equal its modeled mean of zero.

This distinction matters for learning labels. Thousands of frames derived from one mounting calibration are not thousands of independent geometric references. Retain calibration identity and uncertainty when interpreting the apparent quantity or quality of the data.

7.2 The acquisition chain

A physical quantity passes through a transducer, electronics, conversion and processing before it becomes a stored number. Each stage can change what variation is preserved. Nominal resolution is only one property of that chain.

7.2.1 Resolution, quantization, clipping and integration windows

For an ideal uniform quantizer with code spacing Δ and no overload, the difference between an input and its nearest code lies within $[-\Delta/2, \Delta/2]$. If that error is reasonably modeled as uniform,

$$\mathbb{E}[e_Q] = 0, \qquad \mathrm{Var}(e_Q) = \frac{\Delta^2}{12}. \tag{7.7}$$

The distribution is an assumption about input uncertainty or variation within a code, not a universal independence property of quantization. A nearly constant or periodic input can produce strongly structured errors. The GUM discusses this resolution model explicitly (Joint Committee for Guides in Metrology 2008, appendix F.2.2).

Resolution, repeatability and accuracy therefore differ. Clipping adds another effect: an output at a limit can mean any input beyond that limit. Treating it as an ordinary Gaussian reading at the boundary discards the censoring information. Preserve saturation flags, ranges and raw codes where they affect interpretation.

Worked example: small angle codes can produce noisy velocities. An ideal 12-bit encoder spanning one revolution has $\Delta = 2\pi/4096 \approx 0.00153398$ rad. The uniform within-code model gives $\sigma_q \approx 0.00044282$ rad. For a finite-difference estimate

$$\hat{v}_k = \frac{q_{k+1} - q_k}{h}, \qquad \mathrm{Var}(\hat{v}_k - v_{\mathrm{secant}}) = \frac{2\sigma_q^2}{h^2} \tag{7.8}$$

under independent endpoint errors. At $h = 0.01$ s, the standard deviation is approximately 0.06262 rad/s. The worst-case quantization contribution is bounded by $\Delta/h \approx 0.15340$ rad/s. Independence can fail at low motion, and the exact secant itself represents an interval rather than an arbitrary instantaneous velocity.

An instrument can also integrate its input over a window. A simple scalar average with duration T_w is

$$y(t) = \frac{1}{T_w} \int_{t-T_w}^{t} s(\tau)\, d\tau. \tag{7.9}$$

It describes an interval ending at t, not a point sample of $s(t)$. For a sinusoid, its frequency response is

$$H_w(\omega) = e^{-i\omega T_w/2} \frac{\sin(\omega T_w/2)}{\omega T_w/2}. \tag{7.10}$$

The averaging attenuates variation and shifts its effective center by $T_w/2$. In a camera the image formation and feature extraction are more complicated: averaging an extracted pose is not generally the same as extracting a pose from an exposure-averaged image.

For data collection, store the integration or exposure interval when known. A faster communication rate does not necessarily provide new independent measurements if the sensor repeatedly reports the same integrated result.

7.2.2 Bandwidth, filtering, sampling and aliasing

Sampling at rate f_s makes continuous signals indistinguishable when their sampled sequences agree. Ideal reconstruction requires a suitable band limit and a rate exceeding twice the highest retained frequency. The sampling theorem concerns those assumptions, not all physical signals (Shannon 1949).

Worked example: two motions with the same samples. At times $t_k = k/50$ s,

$$\cos(2\pi 40 t_k) = \cos(2\pi 10 t_k). \tag{7.11}$$

A 40 Hz component and a 10 Hz component yield the same cosine samples. No later processing of this sequence alone can determine which continuous signal produced it. Mechanical harmonics and vibration above the retained band can similarly appear as low-frequency behavior.

An analog anti-alias filter acts before sampling. A digital low-pass applied after conversion cannot generally undo an alias already introduced. Real filters have finite transition bands; practical rate selection must leave room for them (Kester 2008). A first-order analog model,

$$H_f(i\omega) = \frac{1}{1 + i\omega\tau_f}, \qquad \tau_f = \frac{1}{2\pi f_c}, \tag{7.12}$$

has gain $1/\sqrt{1+(f/f_c)^2}$ and phase $-\arctan(f/f_c)$. At $f_c = 10$ Hz and $f = 40$ Hz, gain is about 0.24254 and phase is about $-75.96°$. The component is attenuated, not removed. A 0.01 s rectangular window at 40 Hz separately has gain about 0.75683; its averaging should not be confused with a sufficient anti-alias design.

Filtering can make a plot look smooth while delaying its motion or suppressing a contact transient. The sensor bandwidth, sample rate, filter and algorithmic latency describe different parts of the record. For embodied learning, choose them in relation to the interaction to be represented: a slow pose trajectory and a short impact need not be observable with the same chain.

7.3 Time and synchronization

Two accurate instruments can disagree because they observed different instants. A measurement model therefore needs a temporal reference as well as a spatial reference. Matching rows by array index or arrival order does not establish simultaneity.

7.3.1 Exposure time, device time, receipt time and recording time

Distinguish the physical acquisition interval, the event represented by a device timestamp, the host's receipt time and the eventual recording time. A camera timestamp might identify exposure start, end or another event. A rolling shutter assigns different intervals to different rows. A device can queue several samples before transmission. The file-writing time can include still more delay.

For a device sample k, a useful abstract record is

$$\mathcal{R}_k = (y_k, \tau_{d,k}, t_{\mathrm{recv},k}, \mathcal{I}_k, \mathcal{C}_k), \tag{7.13}$$

where $\mathcal{I}_k$ describes its acquisition interval or known temporal support. Context $\mathcal{C}_k$ states the timestamp event, clock domain, sequence identity and relevant processing. Unknown fields remain unknown; a guessed exposure time should not be presented as measured timing.

Different tasks need different alignment. Associating a force pulse with a collision requires a tighter temporal relationship than displaying a slow pose trace. The relevant condition is the uncertainty relative to the event being studied, rather than whether two streams display the same nominal rate.

7.3.2 Clock offset, drift, latency and jitter

Over a limited interval, an affine clock relation can be written as

$$\tau_d(t) = at + b. \tag{7.14}$$

Here t is reference time, b an offset and $a - 1$ a fractional rate mismatch, often called skew. A constant rate mismatch produces an accumulating time error. Drift in the stricter frequency sense means that the rate itself changes; a longer model can include quadratic and stochastic terms. The timing annex of the cyber-physical systems framework distinguishes these effects (Cyber-Physical Systems Public Working Group 2017).

If a packet acquired at t_k arrives at $t_{\mathrm{recv},k} = t_k + \ell_k$, the delay ℓ_k is another quantity. A one-way arrival sequence alone does not generally separate clock offset from transmission delay. Shared triggers, suitable clock comparison or a documented synchronization protocol provide additional information. A visual alignment of curves can estimate an effective lag, but that lag may combine clocks, exposure, filtering and transport.

Worked example: timing error looks like angle error. Let $\tau_d = 1.0001t + 0.012$ s. At reference time 10 s, the device reports 10.013 s. Pairing it as if it were reference time introduces a 0.013 s error. During constant 2 rad/s motion, this corresponds to a 0.026 rad angle discrepancy. It need not indicate backlash or an inaccurate encoder. The affine inverse $(\tau_d - b)/a$ corrects this constructed clock only because its two parameters are known.

For a small zero-mean timing perturbation δt, an observation $\mathbf{y} = \mathbf{g}(\mathbf{x}(t))$ changes approximately by

$$\delta\mathbf{y} \approx \dot{\mathbf{g}}\,\delta t, \qquad \mathbf{R}_t \approx \dot{\mathbf{g}}\,\sigma_t^2\dot{\mathbf{g}}^{\mathsf{T}}. \tag{7.15}$$

The covariance expression assumes the timing perturbation is independent of the other errors being combined. At 2 rad/s, a 0.003 s timing standard deviation contributes about 0.006 rad of angle uncertainty. A fixed unknown lag instead creates a systematic, motion-dependent discrepancy. A shared uncertain clock offset correlates multiple samples; it should not be replaced by independent jitter.

These relations suggest informative collections. Stationary records help examine offsets and noise but reveal little about timing error. Motion over several rates exposes a discrepancy proportional to rate. Distinguishing it from mechanical rate-dependent effects still requires additional observations or independent timing information.

7.3.3 Dropped samples, irregular sampling and resampling assumptions

A missing sample is not an observed zero, an unchanged state or a successful repetition of the previous command. Retain sequence gaps and missingness flags. During estimation, a missing measurement can mean prediction without correction; during learning, it may require a mask or a history definition. A held actuator request during a gap is different from knowing that the actuator delivered unchanged effort.

Use actual intervals for differences:

$$\hat{v}_k = \frac{q_{k+1} - q_k}{t_{k+1} - t_k}. \tag{7.16}$$

This is a secant across the stated interval. Calling it an instantaneous velocity requires a further approximation.

Worked example: the index spacing is not the time spacing. For the constructed angle $q(t) = ct^2$ with $c = 1$ rad/s^2, sample at 0, 0.01 and 0.03 s. The endpoint secant is $0.0009/0.03 = 0.03$ rad/s. Dividing by the nominal two-step interval 0.02 s would give 0.045 rad/s. For this quadratic, the correct endpoint secant equals velocity at 0.015 s, not at the middle sample's time 0.01 s. A nonuniform three-point derivative at that middle time instead gives 0.02 rad/s. The companion checks these distinctions.

Interpolation adds assumptions between observed samples. Linear interpolation assumes a straight segment in the chosen coordinates. A quaternion requires a rotation-aware interpolation with sign handling; componentwise interpolation followed by normalization is not a general substitute. Filling a missing contact interval can hide a mode change.

Offline smoothing can also use future observations. If its output is presented as an online state estimate or policy input, it leaks information that was unavailable at action time. Preserve whether each derived record is filtered, predicted, interpolated or smoothed, and retain the raw timing needed to inspect the transformation.

7.4 Geometric and interaction observations

Position, orientation, acceleration and load are not interchangeable descriptions of interaction. The spatial language of Chapter 4 allows us to map their origins and axes. A measurement model adds the instrument's projection, sensitivity and uncertainty to that geometry.

7.4.1 Camera projection, depth and metric calibration

Let a point have optical-frame coordinates $\mathbf{p}^C = (X, Y, Z)^\mathsf{T}$, with positive Z in front of a pinhole camera. Ignoring distortion initially,

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \frac{1}{Z} \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}. \tag{7.17}$$

Focal lengths f_x, f_y and principal point c_x, c_y are in pixels. The world-to-camera transform must be applied first, using the rotation and origin conventions already defined:

$$\mathbf{p}^C = {}^C\mathbf{R}_W \left(\mathbf{p}^W - {}^W\mathbf{r}_{WC} \right). \tag{7.18}$$

Here the vector ${}^{W}\mathbf{r}_{WC}$ points from world origin to camera origin. This translation is not simply added with an arbitrary sign. Official OpenCV documentation uses the corresponding world-to-camera extrinsic convention and develops distortion and calibration (OpenCV, n.d.).

An image point defines a ray, not a unique metric depth. Known geometry, multiple views or another range observation supplies additional information. Calibration must include the image size and any cropping or resizing; otherwise even correct intrinsic values can be applied to a different pixel coordinate system.

For known calibration, the local projection Jacobian is

$$\mathbf{J}_{\pi} = \begin{bmatrix} f_x/Z & 0 & -f_x X/Z^2 \\ 0 & f_y/Z & -f_y Y/Z^2 \end{bmatrix}. \tag{7.19}$$

It maps small point-coordinate uncertainty into pixel uncertainty through Equation (7.5). Uncertain intrinsics or extrinsics add inputs and cross-covariances; they are not automatically independent of the point estimate.

Worked example: depth uncertainty grows with range. With $f_x = f_y = 600$ pixels and principal point $(320, 240)$, the point $(0.1, 0.05, 1)$ m projects to $(380, 270)$ pixels. For a rectified stereo pair with baseline $B = 0.08$ m and common focal length $f = 600$ pixels,

$$Z = \frac{fB}{d}, \qquad \sigma_Z \approx \frac{Z^2}{fB}\sigma_d. \tag{7.20}$$

Disparity d includes correction for any rectified principal-point offset. With $\sigma_d = 0.5$ pixels and known f, B, depth 1 m corresponds to $d = 48$ pixels and $\sigma_Z \approx 0.01042$ m. At 2 m, disparity halves and the approximate uncertainty quadruples to 0.04167 m. A fixed pixel precision is not a fixed metric precision. The linear approximation requires small disparity uncertainty away from zero; near zero disparity, the nonlinear depth distribution requires more care.

7.4.2 Pose reconstruction, ambiguity and occlusion

Pose reconstruction relates observations of known or inferred object features to a rigid transformation. The result depends on the feature correspondences, geometric model and viewing configuration. An object with repeated visual features can admit several plausible orientations. Occlusion can remove precisely the features that distinguish them. A returned pose is then an estimate conditional on those assumptions, rather than a direct observation of an unambiguous state.

A covariance near one solution describes local uncertainty; it does not represent all alternative solutions. For rotation, one useful local representation is

$$\mathbf{R} = \hat{\mathbf{R}} \exp([\delta\boldsymbol{\phi}]_{\times}). \tag{7.21}$$

This is a right, body-local perturbation. Its covariance belongs to the three-component $\delta\boldsymbol{\phi}$, not to four unconstrained quaternion coordinates. A left perturbation uses another frame. A six-component pose error, even when arranged angular-first, is not automatically Featherstone's spatial velocity. Its transformation must follow its declared perturbation definition (Barfoot 2024).

Quaternion signs and angle wraps add representation issues. Q and $-Q$ represent the same attitude, and a wrap at a coordinate boundary is not a physical jump. Comparing rotations through a relative rotation avoids that particular ambiguity. It does not resolve a genuinely ambiguous image, an unknown revolution count or an incorrectly matched marker.

For collection design, vary views and motions deliberately. One static view can be excellent for one translation direction and poor for depth or rotation about a symmetric axis. Retain confidence, visibility and reconstruction failures when available. Hiding low-confidence intervals can remove the very recovery behavior a policy must learn. If a pose label is derived from the same images used as input, its uncertainty and failure modes are inherited from that estimator.

7.4.3 Inertial sensing, gravity and sensor frames

An ideal accelerometer measures specific force at its own origin. Let P be that origin, S its axes and W the world frame. Then

$$\begin{aligned}\mathbf{y}_a^S &= {}^S\mathbf{R}_W\left(\mathbf{a}_{P,\text{class}}^W - \mathbf{g}^W\right) + \mathbf{b}_a + \boldsymbol{\epsilon}_a,\\ \mathbf{y}_g^S &= {}^S\mathbf{R}_W\boldsymbol{\omega}^W + \mathbf{b}_g + \boldsymbol{\epsilon}_g.\end{aligned} \tag{7.22}$$

The second line describes a rate gyro. At rest in aligned axes with gravity $(0,0,-g)^\mathsf{T}$, the ideal accelerometer reports upward g, while in free fall it reports zero. Calling its output ordinary acceleration without removing the appropriate gravity term reverses the interpretation.

Mounting away from a reference point introduces rigid-body acceleration:

$$\mathbf{a}_{P,\text{class}}^W = \mathbf{a}_{O,\text{class}}^W + \boldsymbol{\alpha}^W \times \mathbf{r}_{OP}^W + \boldsymbol{\omega}^W \times (\boldsymbol{\omega}^W \times \mathbf{r}_{OP}^W). \tag{7.23}$$

This is the ordinary point-acceleration relation derived from the spatial kinematics, not an empirical correction. The sensor's linear output should not be substituted directly for the lower block of a spatial acceleration vector. That block has the different meaning explained in Chapter 4. Barfoot (2024) develops inertial measurement models with the corresponding frame and mounting effects.

Worked example: a stationary center does not imply a zero sensor signal. At one instant, a body has $\boldsymbol{\omega}^W = (0,0,2)^\mathsf{T}$ rad/s, zero angular acceleration, and a fixed reference point O. Place an aligned sensor at $\mathbf{r}_{OP}^W = (0.1,0,0)^\mathsf{T}$ m. Its ordinary acceleration is $(-0.4,0,0)^\mathsf{T}$ m/s^2. With gravity $(0,0,-9.81)^\mathsf{T}$ m/s^2, ideal specific force is

$$\mathbf{f}_{\text{spec}}^S = (-0.4,0,9.81)^\mathsf{T}\ \text{m/s}^2. \tag{7.24}$$

The horizontal component is a real centripetal effect, not necessarily bias. The companion rotates the sensor axes as another check; rotating axes must rotate the vector's components without changing the physical motion.

An unknown gyro offset accumulates into orientation error when integrated. An accelerometer cannot independently identify arbitrary translational acceleration and gravity direction from one sample. These limitations motivate complementary observations and an explicit estimator rather than a universal instruction to subtract a constant number from one axis.

7.4.4 Force and tactile calibration, hysteresis and loading estimates

A load sensor produces indications related to elastic deformation or another transduction mechanism. For a multi-axis device, calibration commonly maps several coupled channels into force and moment components. Individual raw channels need not each correspond to a single physical axis. A local calibration can be expressed as

$$\mathbf{f}_S = \mathbf{C}(\mathbf{r} - \mathbf{r}_0), \qquad \mathbf{f}_S = \begin{bmatrix}\mathbf{n}_S\\ \mathbf{F}_S\end{bmatrix}. \tag{7.25}$$

Here $\mathbf{r}$ denotes raw channel indications, not a displacement. The calibration, sign of the load, origin and output ordering are part of its definition. A vendor's force-first output must be permuted before using the book's moment-first convention. ATI's documentation gives concrete examples of channel calibration, crosstalk, hysteresis and tare (ATI Industrial Automation 2020).

For known rotation and origin shift, Chapter 4 gives a linear wrench map. Let $\mathbf{E} = {}^S\mathbf{R}_O$ rotate O components into S, and let $\mathbf{r} = {}^O\mathbf{r}_{OS}$ point from O to S. The angular-first motion transform is $\mathbf{X}_{SO}$; power duality gives the moment-first wrench at O:

$$\mathbf{f}_O = \mathbf{A}\mathbf{f}_S, \qquad \mathbf{A} = \mathbf{X}_{SO}^\mathsf{T} = \begin{bmatrix}\mathbf{E}^\mathsf{T} & [\mathbf{r}]_\times\mathbf{E}^\mathsf{T}\\ \mathbf{0} & \mathbf{E}^\mathsf{T}\end{bmatrix}. \tag{7.26}$$

The lever term couples force to moment. This same matrix maps uncertainty: the covariance obeys

$$\boldsymbol{\Sigma}_{f_O} = \mathbf{A}\boldsymbol{\Sigma}_{f_S}\mathbf{A}^\mathsf{T} \tag{7.27}$$

when the transform is exact. Uncertain mounting adds further terms. Do not rotate force while leaving moment at the old origin.

Worked example: translating a wrench also changes its uncertainty. Let aligned axes at S be displaced by 0.05 m along positive x from reference origin O. A nominal pure force at S has $\mathbf{F}_S = (0, 0, 10)^\mathsf{T}$ N and $\mathbf{n}_S = \mathbf{0}$. The moment about O is

$$\mathbf{n}_O = \mathbf{n}_S + \mathbf{r}_{OS} \times \mathbf{F}_S = (0, -0.5, 0)^\mathsf{T}\ \mathrm{N\,m}. \tag{7.28}$$

Assume independent sensor-axis standard deviations of 0.01 N m for moment and 0.1 N for force, and exact mounting. Then $\sigma_{n_{O,y}} \approx 0.01118$ N m and $\mathrm{Cov}(n_{O,y}, F_z) = -0.0005$ N^2 m. The translated channels are correlated even though the original channels were independent.

A scalar calibration also needs its units. If $r = (200\ \mathrm{counts/N})F + 100\ \mathrm{counts}$, then $r = 2100$ counts indicates 10 N under that model. An unknown 20-count offset changes the inferred force by 0.1 N. A tare subtracts a selected baseline. It does not remove gravity changes after tool reorientation or the inertial load of an accelerating payload.

Hysteresis means that the same load can produce different indications depending on loading history. A single-valued fitted curve cannot remove that dependence without a suitable history or state model. Temperature, creep, cross-axis effects and mounting can also change the relationship. Reported uncertainty must refer to the conditions in which it was evaluated.

A pressure-like tactile array adds spatial information but not arbitrary traction. Consider the wrench acting on the instrumented body, about origin O, with every component expressed in common axes. Taxel j measures compressive pressure p_j over known area A_j; $\mathbf{r}_j$ points from O to that taxel and $\mathbf{n}_j$ is the unit direction of force on the body. For an outward surface normal, $\mathbf{n}_j$ is its negative. A normal-only wrench estimate is

$$\hat{\mathbf{f}} = \sum_j \begin{bmatrix} \mathbf{r}_j \times (p_j A_j \mathbf{n}_j) \\ p_j A_j \mathbf{n}_j \end{bmatrix}. \tag{7.29}$$

This assumes known taxel locations and normals, calibrated pressure and a sufficiently represented contact patch. It omits unmeasured shear and other unresolved effects. A model that adds them produces an estimate, not additional directly measured channels.

A wrist sensor likewise measures the transmitted interaction at its own location, including relevant gravity and dynamic loading. Recovering an external contact contribution requires a body/load model and its uncertainty. A current-derived effort, an inverse-dynamics residual and a calibrated load-cell reading should retain those different derivations in a learning dataset.

7.5 Estimating changing state

A measurement does not usually reveal all of the state needed for control. Velocity, orientation or actuator history may need to be inferred from a sequence. A state estimator combines a motion model with observations, while retaining an explicit account of what information they provide.

7.5.1 Prediction–correction and uncertainty propagation

Consider the discrete linear model

$$\begin{aligned} \mathbf{x}_k &= \mathbf{F}_{k-1}\mathbf{x}_{k-1} + \mathbf{B}_{k-1}\mathbf{u}_{k-1} + \mathbf{w}_{k-1}, \\ \mathbf{y}_k &= \mathbf{H}_k\mathbf{x}_k + \mathbf{r}_k. \end{aligned} \tag{7.30}$$

Let process and measurement noises have zero means and covariances $\mathbf{Q}_{k-1}$ and $\mathbf{R}_k$. Initially assume independence across times, between noise types and from the initial state error. These assumptions exclude several common shared-calibration and processing dependencies unless they are modeled explicitly.

Propagating the previous estimate and its covariance gives

$$\begin{aligned} \hat{\mathbf{x}}_k^- &= \mathbf{F}_{k-1}\hat{\mathbf{x}}_{k-1}^+ + \mathbf{B}_{k-1}\mathbf{u}_{k-1}, \\ \mathbf{P}_k^- &= \mathbf{F}_{k-1}\mathbf{P}_{k-1}^+\mathbf{F}_{k-1}^\mathsf{T} + \mathbf{Q}_{k-1}. \end{aligned} \tag{7.31}$$

The superscripts distinguish prediction before this measurement from the corrected estimate afterward. They are not

coordinate-frame labels. The innovation is the difference between the observation and its prediction:

$$\mathbf{e}_k^y = \mathbf{y}_k - \mathbf{H}_k\hat{\mathbf{x}}_k^-, \qquad \mathbf{S}_k = \mathbf{H}_k\mathbf{P}_k^-\mathbf{H}_k^\mathsf{T} + \mathbf{R}_k. \tag{7.32}$$

A discrepancy is judged relative to its predicted observation uncertainty, not by its raw magnitude alone. Assume $\mathbf{S}_k$ is positive definite on the retained measurement components.

For Gaussian prior and noises, the conditional state and observation have cross-covariance $\mathbf{P}_k^-\mathbf{H}_k^\mathsf{T}$. Conditioning this joint Gaussian, or completing its quadratic exponent, gives

$$\begin{aligned}\mathbf{K}_k &= \mathbf{P}_k^-\mathbf{H}_k^\mathsf{T}\mathbf{S}_k^{-1},\\ \hat{\mathbf{x}}_k^+ &= \hat{\mathbf{x}}_k^- + \mathbf{K}_k\mathbf{e}_k^y.\end{aligned} \tag{7.33}$$

The Kalman gain weights the innovation by the information coupling state and measurement. In code, solve a linear system with $\mathbf{S}_k$ instead of explicitly constructing its inverse.

With prediction error $\mathbf{e}_k^- = \mathbf{x}_k - \hat{\mathbf{x}}_k^-$, the error after a linear correction is $(\mathbf{1} - \mathbf{K}_k\mathbf{H}_k)\mathbf{e}_k^- - \mathbf{K}_k\mathbf{r}_k$. Under the declared independence, its covariance is

$$\mathbf{P}_k^+ = (\mathbf{1} - \mathbf{K}_k\mathbf{H}_k)\mathbf{P}_k^-(\mathbf{1} - \mathbf{K}_k\mathbf{H}_k)^\mathsf{T} + \mathbf{K}_k\mathbf{R}_k\mathbf{K}_k^\mathsf{T}. \tag{7.34}$$

This Joseph form is a sum of positive-semidefinite contributions in exact arithmetic and is useful for numerical implementation. Symmetry, factorization and conditioning still require attention in finite precision. For the optimal linear gain it agrees with the shorter covariance expression $\mathbf{P}_k^- - \mathbf{P}_k^-\mathbf{H}_k^\mathsf{T}\mathbf{S}_k^{-1}\mathbf{H}_k\mathbf{P}_k^-$.

The recursive formulation is central to Kalman (1960); Barfoot (2024) develops its modern robotics interpretation. With linear dynamics and Gaussian initial state and independent Gaussian noises, this is an exact posterior update. With only suitable second moments, the same equations give the optimal linear estimate, not an exact arbitrary non-Gaussian posterior. Correlated process and measurement errors require additional cross terms.

For a constant-velocity translational model, $\mathbf{x} = (p, v)^\mathsf{T}$ and

$$\mathbf{F}(h) = \begin{bmatrix}1 & h\\ 0 & 1\end{bmatrix}, \qquad \mathbf{Q}(h) = q_c\begin{bmatrix}h^3/3 & h^2/2\\ h^2/2 & h\end{bmatrix}. \tag{7.35}$$

The covariance follows by integrating continuous white acceleration of spectral density q_c, with units $\mathrm{m}^2/\mathrm{s}^3$, through the position/velocity dynamics. It is a stochastic idealization, not the variance of one finite acceleration sample. Independent random acceleration held constant over each interval instead gives

$$\mathbf{Q}_{\text{held}}(h) = \sigma_a^2\begin{bmatrix}h^4/4 & h^3/2\\ h^3/2 & h^2\end{bmatrix}, \qquad [\sigma_a^2] = \mathrm{m}^2/\mathrm{s}^4. \tag{7.36}$$

The different powers and units identify different models. Variable sample intervals require the corresponding $\mathbf{F}(h)$ and $\mathbf{Q}(h)$.

Worked example: a position observation also corrects velocity. Take prior mean $(0\ \mathrm{m}, 1\ \mathrm{m/s})^\mathsf{T}$ and covariance $\mathrm{diag}(0.0004, 0.04)$ in the corresponding squared units. Predict for $h = 0.1$ s with $q_c = 0.2\ \mathrm{m}^2/\mathrm{s}^3$. Then

$$\hat{\mathbf{x}}^- = \begin{bmatrix}0.1\\ 1\end{bmatrix}, \qquad \mathbf{P}^- = \begin{bmatrix}0.0008666667 & 0.005\\ 0.005 & 0.06\end{bmatrix}. \tag{7.37}$$

Covariance entries have units m^2, m^2/s and $\mathrm{m}^2/\mathrm{s}^2$ according to their row and column. Observe position $y = 0.12$ m with $\mathbf{H} = [1\ 0]$ and $\sigma_y = 0.01$ m. The innovation is 0.02 m and $S = 0.0009666667\ \mathrm{m}^2$. The correction gives

$$\mathbf{K} \approx \begin{bmatrix}0.89655172\\ 5.17241379\end{bmatrix}, \qquad \hat{\mathbf{x}}^+ \approx \begin{bmatrix}0.11793103\\ 1.10344828\end{bmatrix}. \tag{7.38}$$

The first gain entry is dimensionless; the second has units s^{-1}. Velocity changes because the motion model created position–velocity cross-covariance. The companion checks the Joseph covariance against Gaussian information-form conditioning. The result is not a direct velocity measurement.

For nonlinear dynamics and observations, an extended Kalman filter propagates nonlinear means and uses local Jacobians for covariance and correction. This is a local approximation; multimodal pose hypotheses, large errors and discontinuous contact can violate its assumptions. Rotation estimates should be updated through the chosen tangent perturbation,

with covariance in compatible coordinates, rather than by treating nine rotation entries as independent state variables.

7.5.2 Observability, missing observations and model assumptions

Observability asks whether different initial states can be distinguished from the available observation sequence under the model. For a time-invariant, noise-free linear system, stacking observations gives

$$\mathcal{O} = \begin{bmatrix} \mathbf{H} \\ \mathbf{HF} \\ \vdots \\ \mathbf{HF}^{n-1} \end{bmatrix}. \tag{7.39}$$

Full column rank makes the n-component state distinguishable over that horizon. Rank alone does not measure robustness to noise: nearly dependent columns can make estimation poorly conditioned. A nonlinear system also needs a more local or trajectory-dependent analysis.

For the constant-velocity model and a position observation,

$$\mathcal{O}_2 = \begin{bmatrix} 1 & 0 \\ 1 & h \end{bmatrix}. \tag{7.40}$$

It has full rank for $h \neq 0$. Inferring velocity from repeated positions is therefore possible under this ideal model, but short intervals and noisy positions can make it difficult. This connects observability to the differentiation example without claiming that an algebraically observable state is measured accurately.

Worked example: extra samples do not separate position from bias. Augment the state by an unknown constant measurement bias:

$$\mathbf{x}_a = \begin{bmatrix} p \\ v \\ b \end{bmatrix}, \qquad \mathbf{F}_a = \begin{bmatrix} 1 & h & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{H}_a = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}. \tag{7.41}$$

Successive observation rows are $[1 \; jh \; 1]$. Their rank is two, with indistinguishable direction $(1, 0, -1)^{\mathsf{T}}$. Increasing true position and decreasing bias by the same amount leaves every observation unchanged. A calibrated reference or other independent information is required. A prior can constrain this direction, but supplies information rather than proving observability from these measurements.

During missing observations, the filter predicts without that correction. Uncertainty can grow according to the motion model, even when the displayed mean remains smooth. Occlusion can also create informative missingness: failure to detect an object may depend on its pose or the contact itself. Dropping such intervals changes the population being represented.

Innovations help examine model consistency. The normalized innovation quantity

$$d_k^2 = (\mathbf{e}_k^y)^{\mathsf{T}} \mathbf{S}_k^{-1} \mathbf{e}_k^y \tag{7.42}$$

combines residual scale and predicted uncertainty. Under the exact Gaussian model it has the corresponding chi-squared distribution. Large or serially structured innovations motivate investigation; they do not uniquely identify a sensor fault. Automatically removing all large innovations can hide contact transitions or legitimate failures. Record the selection rule and retained alternatives.

7.5.3 State uncertainty, parameter uncertainty and joint estimation

State uncertainty concerns the system's changing condition. Parameter uncertainty concerns the coefficients and calibration with which that condition is interpreted. Model inadequacy concerns omitted or incorrect relationships. These distinctions determine what a covariance does and does not describe.

For $\mathbf{y} = \mathbf{g}(\mathbf{x}, \boldsymbol{\theta})$, a first-order propagation with uncertain state and parameters gives

$$\begin{aligned} \mathbf{P}_y \approx{}& \mathbf{G}_x \mathbf{P}_x \mathbf{G}_x^{\mathsf{T}} + \mathbf{G}_\theta \mathbf{P}_\theta \mathbf{G}_\theta^{\mathsf{T}} \\ &+ \mathbf{G}_x \mathbf{P}_{x\theta} \mathbf{G}_\theta^{\mathsf{T}} + \mathbf{G}_\theta \mathbf{P}_{\theta x} \mathbf{G}_x^{\mathsf{T}}. \end{aligned} \tag{7.43}$$

Ignoring calibration uncertainty or state–parameter dependence can misrepresent the uncertainty of a derived pose. An omitted independent calibration contribution understates covariance; with dependent inputs, the omitted cross terms can change the result in either direction.

An augmented estimator can include parameters in its state, for example $\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k$. Adding parameter process noise assumes that the parameter can drift; it does not provide evidence of that physical drift. Joint estimation retains cross-covariance, but cannot recover an unobservable combination merely by adding it to the state vector. The bias example makes this failure explicit.

Tuning $\mathbf{Q}$ or $\mathbf{R}$ until residuals look acceptable is not a substitute for identifying timing errors, shared references or missing dynamics. An overconfident model can produce a narrow posterior around a biased answer. Conversely, a large covariance can obscure whether an experiment was informative. Chapter 8 develops identification and experimental excitation; here the measurement arrangement must at least preserve the information needed to distinguish those explanations.

7.6 Designing a measurement arrangement

The preceding mathematics makes sensor selection an information-design problem. Useful observations reveal the quantities and distinctions that matter for the intended task. More channels or a higher nominal rate do not automatically provide complementary information.

7.6.1 Sensor placement, complementary information and temporal uncertainty

Start from the question the record should answer. Does the learner need joint motion, object pose, contact mode, transmitted loading or a history sufficient to infer them? Locate each sensor relative to the mechanism and identify the physical map from the intended quantity to its indication. A camera that sees motion tangent to its image plane and an encoder that measures a joint coordinate can be complementary, but their agreement still depends on geometry and time.

For a fixed unknown vector with linear observations $\mathbf{y}_i = \mathbf{H}_i\mathbf{x} + \mathbf{r}_i$, independent Gaussian noises and a Gaussian prior, accumulated precision is

$$\mathbf{P}_+^{-1} = \mathbf{P}_0^{-1} + \sum_i \mathbf{H}_i^{\mathsf{T}}\mathbf{R}_i^{-1}\mathbf{H}_i. \tag{7.44}$$

This is the vector counterpart of scalar precision weighting. Measurement directions that are weak or missing can remain uncertain despite many repeats. Correlated channels require their joint covariance rather than this independent sum. Here the equation examines measurement information under a specified model; designing excitation to identify physical parameters is developed in Chapter 8.

A marker far from a rotation axis produces more displacement per angular change, which can improve visual angle sensitivity. It can also change visibility, mechanical loading and the effect of mounting uncertainty. A displaced IMU increases rotational acceleration signals as well as mounting sensitivity. A load sensor's placement changes the transmitted wrench and the body loads that must be separated. Sensor position is therefore part of the physical and inferential design.

Temporal uncertainty can dominate a spatial improvement. At a moving configuration, combine the local spatial covariance with the timing contribution only under a stated dependence model. Independent observations that share one uncertain transform or clock remain correlated through that shared quantity. A simulation comparison should reproduce or explicitly exclude those effects rather than treating a perfect state array as a calibrated sensor.

Software must preserve the same semantics. A simulator's virtual sensor array, its direct state and a processed camera pose can have different origins, reference points and update stages. In the inspected MuJoCo 3.16.0 interface, configured sensor histories and delays further affect what a read returns. Its force/torque sensors describe a specified internal interaction at a site, rather than automatically isolating an external fingertip contact (DeepMind Technologies Limited, n.d.). Map axis ordering, action/reaction sign and time before comparing them with physical records.

The companion makes these mathematical maps explicit through routines for projection, inertial motion, wrench covariance and Kalman correction. It checks analytic Jacobians by finite differences and the covariance update through an independent information-form calculation. Those are implementation checks on the examples, not a claim that a particular acquisition pipeline has been calibrated.

7.6.2 A joint experiment with synchronized encoder and optical records

Consider a constructed revolute-joint investigation that brings the chapter together. A marker lies $L = 0.2$ m from the rotation axis and moves in an image-parallel plane at optical depth 1 m. Use calibrated focal length 600 pixels in both directions and known principal point. The analytic motion and marker geometry are

$$q(t) = 0.3\sin(2\pi f t)\ \text{rad}, \quad f = 1\ \text{Hz}, \qquad \mathbf{p}^C(q, L) = \begin{bmatrix} L\cos q \\ L\sin q \\ 1\ \text{m} \end{bmatrix}. \tag{7.45}$$

The motion is defined for all times used by the synthetic acquisition. For real finite records, missing prehistory would need explicit treatment.

The marker's image radius is 120 pixels. Its local angle sensitivity has magnitude 120 pixels/rad. With independent isotropic 0.5-pixel uncertainty and exact geometry, the local optical angle uncertainty is about 0.00417 rad. That statement excludes timing and calibration uncertainty; it should not be interpreted as the accuracy of an actual camera system.

Worked example: an apparent mechanical discrepancy caused by timing. Generate 200 reported times $t_k = k/200$ s over one period. The encoder records $q(t_k)$, but let the optical sample have actual acquisition time $t_k - d$ with $d = 0.02$ s while retaining report label t_k. Without added noise, the continuous maximum paired-angle discrepancy is

$$\max_t |q(t) - q(t-d)| = 2(0.3)\,|\sin(\pi f d)| \approx 0.03767431\ \text{rad}. \tag{7.46}$$

At the marker, this becomes approximately 0.00753442 m or 4.52065 pixels of image displacement. The chosen grid contains a maximizing time. Correct pairing using the known acquisition times removes this constructed residual exactly. The zero residual follows from the analytic generator and supplied timing; it is not an empirical accuracy result.

Changing the assumed lever length to 0.22 m instead produces a 12-pixel radius discrepancy under the same ideal projection. The local geometric directions explain the distinction:

$$\frac{\partial \mathbf{p}}{\partial L} = \begin{bmatrix} \cos q \\ \sin q \\ 0 \end{bmatrix}, \qquad \frac{\partial \mathbf{p}}{\partial q} = \begin{bmatrix} -L\sin q \\ L\cos q \\ 0 \end{bmatrix}. \tag{7.47}$$

They are orthogonal in this particular arrangement. A small timing error acts through $(\partial \mathbf{p}/\partial q)\dot{q}\,\delta t$ in the tangential direction; a length error acts radially. Fitting a mechanical correction before examining these directions could explain the wrong cause. An angle reconstructed by an ideal atan2 direction alone would not reveal the radius error, so retain the observations relevant to the comparison.

The assumptions make this a diagnostic example rather than a general identifiability guarantee. Unknown camera geometry, perspective, occlusion, shared calibration or changing depth can couple the directions. Several motions and independent references may then be needed.

A useful physical collection would retain the raw encoder indications, image observations and their timing; calibration and marker geometry; requested and observed motion; visibility and saturation information; and the processing that produces estimates. Stationary holds examine some noise and offset properties. Moving records reveal rate-dependent timing and mounting effects. Reversals can expose hysteresis or transmission effects, but only after the measurement alternatives are considered. Separate calibration records from independent checks under changed conditions.

The state estimator can then operate on declared observations and actual intervals, with appropriately propagated uncertainty and explicit missing measurements. A derived pose or velocity remains an estimate with provenance. If used as a learning target, retain how it was obtained and which errors can be shared across episodes.

For ExpGen PhySyst, this is the central connection: designing data includes designing the measurement arrangement that makes an experience interpretable. Mechanics defines the quantity; sensing constrains the available evidence; estimation combines it under explicit assumptions. More data becomes useful when those records preserve the distinctions needed for the intended learning or evaluation question.

Exercises and further investigation. Use declared units, origins, axes, acquisition times and uncertainty assumptions. Distinguish constructed calculations from physical evidence.

1. Derive the scalar Gaussian posterior by completing the square. Explain how the result changes when the two inputs share a calibration error rather than being independent.
2. Reproduce the common-offset averaging example. Design an additional reference observation that could reduce the shared uncertainty, and distinguish correcting an offset from eliminating its uncertainty.
3. Derive the quantization and finite-difference uncertainty formulas. Construct a slow-motion case in which independent uniform endpoint errors are a poor approximation. State how angle wrapping is handled.
4. Reproduce the alias example and first-order filter response. Compare adding samples before and after an alias has occurred. Explain the difference between sample rate, bandwidth and integration window.
5. Derive the affine-clock correction and timing-induced covariance. Identify what a one-way packet-arrival record cannot determine by itself. Reproduce the irregular-sample derivatives at their declared times.
6. Derive the camera Jacobian and stereo-depth sensitivity. Include uncertain focal length, baseline and correlated left/right pixel errors in a covariance calculation. State where the linear approximation fails.
7. Derive the IMU lever-arm expression from rigid-body kinematics. Recalculate the worked example after rotating the sensor axes and compare rest, fixed-axis rotation and free fall.
8. Derive the moment-first wrench transform and covariance map. Reproduce the translated covariance and cross-covariance. Explain why tare, current-derived effort and pressure-only taxels are insufficient to identify arbitrary external contact loading.
9. Derive the Kalman update from conditional Gaussian information. Reproduce the numerical posterior and compare Joseph and information covariance forms. Contrast continuous white acceleration with random acceleration held over each interval.
10. Construct the observability matrices with and without sensor bias. Explain why a prior can constrain an unobservable direction without creating a new physical observation. Examine prediction through a gap.
11. Reproduce the encoder/optical example using the companion. Compare timing, radius and axis-convention errors. State which conclusions depend on knowing the generating geometry and acquisition times.
12. Design a measurement arrangement for a grasping or pushing investigation. Specify the measurands, complementary channels, calibration, temporal support, retained failures and estimate provenance. Identify which uncertainties are shared and which observations the deployed learner will actually receive.

Barfoot (2024) provides a broader treatment of probabilistic state estimation and geometric sensing. The GUM and its nonlinearity amendment provide a metrological reference for uncertainty evaluation; Shannon (1949) supplies the classical band-limited sampling foundation. These readings complement, rather than replace, checking the actual instrument and implementation contracts.

Chapter 8 turns to identifying a system and designing experiments that distinguish model explanations. The measurement and estimation foundations here establish what the resulting data can support.

Chapter 8 System Identification and Experimental Design

Two models reproduce a joint's motion almost equally well. One attributes its slow response to a large inertia; the other attributes it to weak delivered effort. Collecting more repetitions of the same motion may reduce random variation without distinguishing those explanations. The next useful experiment might change a known load, rather than increase the number of samples. What must change depends on the model, the measured quantities and the question the model is intended to answer.

Chapter 7 established what an observation means and how uncertainty enters state estimation. System identification asks a related but different question: which relationships and parameters explain the observed response of a defined system? It uses mechanics and probability together. The physical equations restrict possible explanations; experimental design determines which distinctions the data can support.

This chapter develops that connection from the system boundary to reusable identification records. Featherstone's spatial mechanics provides a parameterization of rigid-body dynamics, while estimation and validation expose its evidential limits. For ExpGen PhySyst, an interaction is valuable when its record can change what is known about the system or improve a declared learning task. Every numerical example below is constructed and checked by an arithmetic companion. None is a new hardware experiment, measured dataset or simulator benchmark.

8.1 Define the system to be identified

Begin with the intended use. Predicting the next joint position, estimating link mass, compensating friction and selecting an action are different objectives. They can require different model boundaries and different data. A compact input–output model can predict a particular operating regime without assigning its coefficients a unique mechanical interpretation.

8.1.1 Open-loop boundaries, closed-loop feedback and input–noise dependence

A useful discrete description separates internal state, delivered input and recorded output:

$$\begin{aligned} \mathbf{x}_{k+1} &= \mathbf{F}(\mathbf{x}_k, \mathbf{u}_k, \boldsymbol{\theta}, \mathcal{C}_k) + \mathbf{w}_k, \\ \mathbf{y}_k &= \mathbf{H}(\mathbf{x}_k, \boldsymbol{\theta}_c, \mathcal{C}_k) + \mathbf{e}_k. \end{aligned} \tag{8.1}$$

Here $\boldsymbol{\theta}$ describes the system being identified, $\boldsymbol{\theta}_c$ describes the measurement relationship, and $\mathcal{C}_k$ records relevant conditions. Disturbance and observation noise have different roles. An unobserved actuator state belongs inside the boundary if its dynamics affects the chosen input–output relationship.

The input must name a physical or computational interface. A position reference sent to an embedded servo is not the same input as motor current or delivered joint effort. Identifying reference-to-position behavior includes the embedded controller and its limits. Identifying effort-to-motion dynamics needs an effort observation or a justified actuation model. Changing the interface changes the identification problem.

In an open-loop experiment, an externally specified input may be independent of relevant disturbances by design. Independence still requires an assumption: saturation, human intervention or rejected trials can make the retained input depend on the response. Under feedback, the controller explicitly uses observations, so input and disturbance can become dependent.

Consider the elementary regression $y_k = bu_k + e_k$. Its through-origin least-squares coefficient is

$$\hat{b} = \frac{\sum_k u_k y_k}{\sum_k u_k^2} = b + \frac{\sum_k u_k e_k}{\sum_k u_k^2}. \tag{8.2}$$

With suitable moment limits, the error term tends to $\mathbb{E}[ue]/\mathbb{E}[u^2]$. More samples do not remove a nonzero input–error correlation.

Worked example: feedback changes the regression. Use a constructed static loop, $y = 2u + e$ and $u = r - \frac{1}{2}y$. This is an algebraic illustration, not a dynamic servo model. Solving gives $u = \frac{1}{2}r - \frac{1}{4}e$ and $y = r + \frac{1}{2}e$. Let r, e be independent, zero-mean and unit-variance. Then

$$\mathbb{E}[u^2] = 0.3125, \qquad \mathbb{E}[uy] = 0.375, \qquad \frac{\mathbb{E}[uy]}{\mathbb{E}[u^2]} = 1.2. \tag{8.3}$$

The physical coefficient is 2, but the naive regression approaches 1.2. The companion reproduces the moments exactly with the four equally weighted combinations $r, e \in \{-1, 1\}$.

An independent reference can sometimes provide an instrument: if $\mathbb{E}[re] = 0$ and $\mathbb{E}[ru] \neq 0$, then $\mathbb{E}[ry]/\mathbb{E}[ru] = b$ in this scalar model. Both noise orthogonality and relevance are needed. Merely naming a recorded signal an instrument does not supply either property.

Feedback does not invalidate every estimator. The relevant dependence is between the regressors and the residual at their actual times. A correctly specified causal innovation model can retain the orthogonality required for consistent prediction-error estimation despite feedback. The disturbance model, delays and external excitation matter (Forssell and Ljung 1999). The lesson is to model the closed loop and its noise, rather than declare all closed-loop data unusable or all least-squares fits trustworthy.

8.1.2 Physical parameters, effective terms and available inputs

Featherstone mechanics makes a useful distinction between nonlinearity in motion and linearity in inertial parameters. At a fixed body origin, write the mass m, first moment $\mathbf{h} = m\mathbf{c}$ and rotational inertia $\mathbf{I}_O$. The angular-first spatial inertia is

$$\mathbf{I} = \begin{bmatrix} \mathbf{I}_O & [\mathbf{h}]_\times \\ -[\mathbf{h}]_\times & m\mathbf{1}_3 \end{bmatrix}. \tag{8.4}$$

Its ten independent coefficients can be collected as

$$\boldsymbol{\pi} = (m, h_x, h_y, h_z, I_{xx}, I_{yy}, I_{zz}, I_{xy}, I_{xz}, I_{yz})^\mathsf{T}. \tag{8.5}$$

The off-diagonal entries here are the actual entries of $\mathbf{I}_O$; another source's signed products of inertia may use a different convention. First moments have units kg m, while inertia entries have units kg m^2.

For specified physical spatial motion $\mathbf{v}$ and acceleration $\mathbf{a}$, the required net applied body wrench is

$$\mathbf{f} = \mathbf{I}(\boldsymbol{\pi})\mathbf{a} + \mathbf{v} \times^* \mathbf{I}(\boldsymbol{\pi})\mathbf{v} = \mathbf{Y}_b(\mathbf{v}, \mathbf{a})\boldsymbol{\pi}. \tag{8.6}$$

Motion is angular-first, wrench is moment-first, and both use the same body origin and axes. Because $\mathbf{I}$ is linear in $\boldsymbol{\pi}$, each regressor column is obtained by substituting one unit coefficient. The unit bases are algebraic devices; an individual basis need not describe a physically realizable body. A separately known external wrench contributes to this net load; inverse dynamics subtracts it to find the wrench still required through the support or actuator connection. Inside RNEA, $\mathbf{a}$ follows Chapter 5's gravity-adjusted algorithmic convention; the corresponding gravity wrench is then not added a second time.

Worked example: one spatial balance becomes one regressor evaluation. Take a uniform 2 kg box with side lengths $(0.4, 0.2, 0.1)$ m along its body axes and center of mass $\mathbf{c} = (0.1, -0.05, 0.03)^\mathsf{T}$ m relative to origin O. Its central inertia follows the box formula; the parallel-axis relation gives $\mathbf{I}_O$, while $\mathbf{h} = (0.2, -0.1, 0.06)^\mathsf{T}$ kg m. At one instant, in those body axes, specify

$$\begin{aligned} \mathbf{v} &= (0.4, -0.2, 1.1, 0.3, -0.1, 0.2)^\mathsf{T}, \\ \mathbf{a} &= (0.1, 0.2, -0.3, 0.5, 0.4, -0.2)^\mathsf{T}. \end{aligned} \tag{8.7}$$

The first three velocity entries are in rad/s and the last three in m/s; the corresponding acceleration units are rad/s^2 and m/s^2. Here $\mathbf{a}$ is physical spatial acceleration, whose lower block is not ordinary point acceleration. Substituting into the body balance gives

$$\begin{aligned} \mathbf{n}_O &\approx (-0.02492067, 0.05305867, 0.1679)^\mathsf{T}\ \mathrm{N\,m}, \\ \mathbf{F} &= (0.9064, 1.3418, -0.312)^\mathsf{T}\ \mathrm{N}. \end{aligned} \tag{8.8}$$

The companion obtains the same wrench from $\mathbf{Y}_b\boldsymbol{\pi}$ and from a separate center-of-mass Newton–Euler calculation. Projection onto an allowed relative rotation about O's positive z axis gives $\mathbf{S}^{\mathsf{T}}\mathbf{f} = 0.1679$ N m. Known external and descendant loads would need their own contributions before calling that projection an actuator demand. This single balance checks the mapping; it does not identify all ten body coefficients.

For known kinematics, recursive velocity/acceleration propagation does not depend on the unknown inertias. Backward wrench accumulation and joint projection are linear. Consequently the RNEA of Chapter 5 gives

$$\boldsymbol{\tau}_a + \boldsymbol{\tau}_{\text{ext}} = \mathbf{Y}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})\boldsymbol{\pi}_{\text{all}} + \boldsymbol{\tau}_{\text{loss}}. \tag{8.9}$$

External generalized loads use the sign of the applied load in each joint coordinate; assistance corresponds to positive power $\boldsymbol{\tau}_{\text{ext}}^{\mathsf{T}}\dot{\mathbf{q}}$. The actuators and external loads must together supply the rigid-body demand and the signed effort $\boldsymbol{\tau}_{\text{loss}}$ needed to balance resistance. The actual resistive generalized load is $-\boldsymbol{\tau}_{\text{loss}}$. Gravity is included once, using Chapter 5's RNEA convention. Unknown contact loads are not silently known terms. Linearity in inertial parameters is not linearity in configuration or velocity (Featherstone 2008; Wensing et al. 2018).

A single revolute joint illustrates the resulting model:

$$\tau_a = J\ddot{q} + b\dot{q} + \tau_c \operatorname{sgn}(\dot{q}) + \kappa \sin q + \tau_0. \tag{8.10}$$

Choose $q = 0$ vertically downward, with gravity requiring positive effort when $\sin q > 0$. Here J is effective inertia about the joint, b viscous drag, τ_c a direction-symmetric Coulomb term, κ a gravitational moment coefficient and τ_0 a constant offset. The Coulomb expression is used only during sliding, away from zero speed; it is not a stiction law. Hysteresis, compliance and temperature may require other terms.

These coefficients describe the declared arrangement. J can combine link, fixture, payload and reflected rotor inertia. τ_0 might represent a physical preload, an effort-sensor offset or an omitted load. A good fit alone does not decide which interpretation is correct. Available reference measurements and interventions determine which terms can be separated.

8.2 Design informative experiments

An experiment is informative relative to a model and a question. A motion useful for learning successful reaching may be poor for estimating inertia. Identification design therefore starts with the sensitivities and ambiguities of the chosen explanation, rather than selecting a waveform because it is commonly used.

8.2.1 Sensitivity, identifiability and parameter information

Let predicted observations stacked over an experiment be $\hat{\mathbf{y}}(\boldsymbol{\theta}; \mathcal{E})$, where $\mathcal{E}$ denotes the chosen input, initial conditions and configuration. The local sensitivity is

$$\mathbf{J}_\theta = \frac{\partial \hat{\mathbf{y}}}{\partial \boldsymbol{\theta}}. \tag{8.11}$$

A column near zero means that small changes in that parameter barely change this prediction to first order. Dependent columns indicate confounded first-order changes. For nonlinear models, local full rank does not guarantee a unique global solution. A symmetry can produce separate equivalent solutions elsewhere. Conversely, deficient rank at one point concerns first-order discrimination; it need not prove nonuniqueness of the nonlinear map.

For the linear regression $\mathbf{y} = \mathbf{Y}\boldsymbol{\theta} + \mathbf{e}$, a null direction $\mathbf{d}$ satisfies $\mathbf{Y}\mathbf{d} = \mathbf{0}$. Then $\boldsymbol{\theta}$ and $\boldsymbol{\theta} + \mathbf{d}$ predict the same record. A structural ambiguity persists over all admissible experiments; an experimental ambiguity can result from insufficient excitation. Robots can have identifiable combinations, or base parameters, without every link's ten coefficients being individually identifiable (Wensing et al. 2024).

When the observation covariance $\mathbf{R}$ is known, positive definite and independent of the parameters, a Gaussian mean model has information

$$\mathcal{I}(\boldsymbol{\theta}; \mathcal{E}) = \mathbf{J}_\theta^{\mathsf{T}}\mathbf{R}^{-1}\mathbf{J}_\theta. \tag{8.12}$$

For independent observations this becomes the sum of their contributions. Correlated errors require their joint covariance. If covariance also depends on parameters, the likelihood contains further information terms. The inverse information has

a covariance-bound interpretation only under the appropriate regularity and estimator assumptions; it is not an automatic certificate of achievable accuracy.

Information is directional. Repeating a rank-deficient experiment can strengthen already observed combinations while leaving another combination unidentified. Rank testing also needs numerical scales. If $\boldsymbol{\theta} = \mathbf{D}\boldsymbol{\eta}$ for chosen parameter scales, the regressor for dimensionless $\boldsymbol{\eta}$ is $\mathbf{YD}$. Report those scales when discussing singular values or conditioning. Changing units can change a raw condition number without changing the robot.

Worked example: steady motion cannot identify inertia. Suppose gravity and other known loads have been removed and the remaining effort is $\tau = Ja + bv$. At a constant velocity $v = 1$ rad/s, $a = 0$, and every row of the regressor is $[0 \;\; 1]$. A model with $J = 0.02$ kg m^2 and one with $J = 0.05$ kg m^2 both predict 0.1 N m when $b = 0.1$ N m s/rad. No number of those ideal observations separates their inertias. At $a = 2$ rad/s^2 and the same velocity, their predictions are 0.14 and 0.20 N m. The change in acceleration creates the distinction; a smaller training residual did not create it.

8.2.2 Excitation, interventions and confounded explanations

Excitation means varying the experiment so that the relevant model directions influence observations distinguishably. In a finite linear regression, full column rank is the basic algebraic requirement. Persistent excitation adds a sustained richness condition over time for a specified identification model. Neither phrase means that any large or complicated movement identifies every parameter.

A sum of sinusoids can vary rates and accelerations; reversals can contrast direction-dependent losses; changed poses can vary gravitational loading. Their usefulness depends on the model and the acquisition chain. If filtering removes the distinguishing frequencies, the commanded excitation is not the excitation represented in the retained observations.

Feedback can improve repeatability while suppressing visible disturbances. It can also keep the available regressors within a narrow regime. An external reference signal can provide variation independent of some disturbances, but the actual input, controller state and limiting events remain important. Opening the loop is not always necessary or appropriate.

An intervention changes a specified part of the arrangement to distinguish explanations. Changing a known payload inertia is different from merely observing that heavier trials respond differently: other changes in mounting, friction or controller settings must be controlled or recorded. The intervention supports a conclusion only through those assumptions.

For a joint driven by measured current i, consider

$$J\ddot{q} + b\dot{q} = \alpha i. \tag{8.13}$$

Without an independent effort or inertia reference, replacing (J, b, α) by $(sJ, sb, s\alpha)$ for $s > 0$ leaves the same motion. The observed response can identify ratios such as b/J and α/J, rather than their absolute values. A physical constraint such as positive inertia does not remove this common-scale ambiguity.

An independent known load can break that symmetry. It must be a known physical quantity with stated geometry and uncertainty, not another parameter freely rescaled by the estimator. The final joint investigation will develop this example explicitly.

8.2.3 Loads, trajectories and operating conditions

Choose variation for its physical effect. Acceleration emphasizes inertia; speed and direction help distinguish drag; pose changes gravity moments. A constant-speed sweep does not isolate friction if gravity changes through the sweep. A reversal changes acceleration and direction together, so a single reversal trace need not distinguish their contributions.

Worked example: paired directions and two speeds. Use the sliding model in Equation (8.10), at the same q and speed magnitude $v > 0$, with zero acceleration. Assume losses are direction-symmetric and that other conditions match. Then

$$\begin{aligned} \frac{\tau_+(q, v) + \tau_-(q, v)}{2} &= \kappa \sin q + \tau_0, \\ \frac{\tau_+(q, v) - \tau_-(q, v)}{2} &= bv + \tau_c. \end{aligned} \tag{8.14}$$

Take $b = 0.04$ N m s/rad, $\tau_c = 0.02$ N m, $\kappa = 0.3$ N m and $\tau_0 = 0.01$ N m. At $q = \pi/6$, the paired mean is 0.16 N m. The half-differences are 0.04 and 0.06 N m at $v = 0.5$ and 1 rad/s, respectively. Their slope and intercept recover b and τ_c. Repeating at $q = -\pi/6$ gives mean -0.14 N m; the two means recover κ and τ_0.

The arithmetic is exact because the generating model satisfies the pairing assumptions. During real sliding, samples at a pose are instants along a trajectory, not a stationary hold. Static friction, unequal heating, history-dependent losses or timing error can invalidate the cancellation. Data from both directions can reveal those failures rather than justify discarding them automatically.

Experiment design also has constraints. Available torque, speed, travel, temperature and loading limits restrict feasible excitation. An information objective can seek a large determinant, a strong weakest direction or small uncertainty in one target quantity. These are different choices. A trajectory that improves an unconstrained matrix criterion may be unrealizable or inappropriate for the actual apparatus (Tedrake, n.d.).

Specify initial conditions and recovery between trials. Randomized order can reduce confounding with slow drift, but cannot make an unrecorded temperature dependence disappear. Repeated baselines can expose drift. A successful run and a saturated run are different conditions, even if they use the same requested trajectory.

8.3 Estimate model parameters

Estimation turns a specified model and a record into a parameter value. It does not repair an undefined input, create missing excitation or establish that the model class is adequate. The objective, constraints and noise assumptions determine what the numerical answer means.

8.3.1 Constrained least squares, likelihood and regularization

Stack a linear model as $\mathbf{y} = \mathbf{Y}\boldsymbol{\theta} + \mathbf{e}$. For a known positive-definite observation covariance $\mathbf{R}$, generalized least squares solves

$$\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}\in\Theta}{\operatorname{argmin}}\,(\mathbf{y} - \mathbf{Y}\boldsymbol{\theta})^{\mathsf{T}}\mathbf{R}^{-1}(\mathbf{y} - \mathbf{Y}\boldsymbol{\theta}). \tag{8.15}$$

The feasible set Θ represents declared restrictions. For an unconstrained full-column-rank problem, differentiating gives

$$\mathbf{Y}^{\mathsf{T}}\mathbf{R}^{-1}\mathbf{Y}\,\hat{\boldsymbol{\theta}} = \mathbf{Y}^{\mathsf{T}}\mathbf{R}^{-1}\mathbf{y}. \tag{8.16}$$

These equations explain the solution. Numerically, whiten using a factorization of $\mathbf{R}$ and use QR or SVD rather than explicitly inverting the normal matrix. Forming normal equations squares the condition number of the whitened design in the usual two-norm analysis.

With fixed exact regressors, zero conditional error mean, correct covariance and no active constraints, the estimator is unbiased and

$$\operatorname{Cov}(\hat{\boldsymbol{\theta}}) = (\mathbf{Y}^{\mathsf{T}}\mathbf{R}^{-1}\mathbf{Y})^{-1}. \tag{8.17}$$

For independent Gaussian observation errors, the same estimate maximizes likelihood when $\mathbf{R}$ is parameter-independent. For unknown common variance, an ordinary regression can estimate it from residual sum of squares divided by $N - p$ under the usual independent, correct-model assumptions, with p fitted coefficients. Constraints, regularization, dependent samples or model error change this simple interpretation.

Worked example: separating inertia and viscous drag. Suppose other loads are known and removed, and collect four constructed instantaneous observations from independent trials:

$$\mathbf{Y} = \begin{bmatrix} 2 & 1 \\ -2 & 1 \\ 2 & -1 \\ -2 & -1 \end{bmatrix}, \qquad \boldsymbol{\theta} = \begin{bmatrix} 0.012 \\ 0.04 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 0.064 \\ 0.016 \\ -0.016 \\ -0.064 \end{bmatrix} \ \text{N m}. \tag{8.18}$$

The columns are acceleration in rad/s^2 and velocity in rad/s; the coefficients are J in kg m^2 and b in N m s/rad. Radians are treated as dimensionless in mechanical balances. The rows are admissible instantaneous conditions, not a claim that

four discontinuous states form one executable trajectory.

Here $\mathbf{Y}^{\mathsf{T}}\mathbf{Y} = \operatorname{diag}(16, 4)$ with the corresponding squared column units, and least squares recovers the generating coefficients. If each effort observation instead has an independent zero-mean error with known standard deviation 0.004 N m,

$$\sigma_J = 0.001 \text{ kg m}^2, \qquad \sigma_b = 0.002 \text{ N m s/rad}. \tag{8.19}$$

These uncertainties describe that noise model with exact kinematic regressors. The companion also verifies them through an exact finite error ensemble. They are not uncertainty estimates for differentiated encoder measurements on an actual joint.

Physical constraints need more than positive coefficients in an arbitrary parameterization. A positive mass and a positive-definite rotational inertia do not ensure a realizable nonnegative mass distribution. For the ten coordinates in Equation (8.5), define the second-moment matrix and pseudo-inertia

$$\boldsymbol{\Sigma}_m = \tfrac{1}{2}\operatorname{tr}(\mathbf{I}_O)\mathbf{1}_3 - \mathbf{I}_O, \qquad \mathbf{J}_m = \begin{bmatrix} \boldsymbol{\Sigma}_m & \mathbf{h} \\ \mathbf{h}^{\mathsf{T}} & m \end{bmatrix}. \tag{8.20}$$

Indeed, if ρ is mass density and $\mathbf{r}$ is measured from the declared body origin, then

$$\mathbf{J}_m = \int \begin{bmatrix} \mathbf{r} \\ 1 \end{bmatrix} \begin{bmatrix} \mathbf{r} \\ 1 \end{bmatrix}^{\mathsf{T}} \rho \, dV. \tag{8.21}$$

Nonnegative mass gives a positive-semidefinite matrix. With $m > 0$, its Schur complement is the mass-weighted central second moment, $\boldsymbol{\Sigma}_m - \mathbf{h}\mathbf{h}^{\mathsf{T}}/m$. This describes the distribution of material, not statistical uncertainty in an estimated parameter.

Strict positivity provides a useful nondegenerate physical-consistency constraint (Wensing et al. 2018). A semidefinite limit also permits ideal planar, line or point distributions; singular dynamics may then need separate attention. Positivity alone does not impose a particular known shape or manufacturing tolerance.

Worked example: a positive inertia can still be impossible. Take $m = 1$ kg, $\mathbf{h} = \mathbf{0}$ and $\mathbf{I}_O = \operatorname{diag}(1, 1, 3)$ kg m^2. The spatial inertia is positive definite. However, $\boldsymbol{\Sigma}_m = \operatorname{diag}(1.5, 1.5, -0.5)$ kg m^2. A negative second moment is impossible for nonnegative mass. Equivalently, the largest principal inertia exceeds the sum of the other two. This candidate should not be accepted as a rigid body merely because its fitted inverse dynamics is numerically convenient.

Regularization can encode a preference for a nominal model:

$$\min_{\boldsymbol{\theta} \in \Theta} \|\mathbf{R}^{-1/2}(\mathbf{y} - \mathbf{Y}\boldsymbol{\theta})\|^2 + \lambda \|\mathbf{L}(\boldsymbol{\theta} - \boldsymbol{\theta}_0)\|^2. \tag{8.22}$$

Choose $\mathbf{L}$ with appropriate parameter scales. The notation $\mathbf{R}^{-1/2}$ denotes a whitening factor, not a required explicit matrix inverse. A Gaussian prior gives a probabilistic interpretation to a corresponding quadratic penalty. Neither a prior nor physical feasibility creates new experimental information. Report sensitivity to these choices when the likelihood is weak.

Worked example: a noisy regressor changes the target. Let $y = \beta x_* + \epsilon$ and measured $x = x_* + \eta$, with centered, mutually uncorrelated x_*, η, ϵ. Then the population through-origin regression gives

$$\frac{\mathbb{E}[xy]}{\mathbb{E}[x^2]} = \beta \frac{\operatorname{Var}(x_*)}{\operatorname{Var}(x_*) + \operatorname{Var}(\eta)}. \tag{8.23}$$

For $\beta = 2$, $\operatorname{Var}(x_*) = 1$ and $\operatorname{Var}(\eta) = 0.25$, the result is 1.6. The companion uses equally weighted $x_* = \pm 1$, $\eta = \pm 0.5$ and $\epsilon = 0$ to verify it exactly.

In robot data, velocity and acceleration estimated from the same noisy positions can produce correlated errors in several regressor columns. Changing weights on the output residual does not automatically solve that errors-in-variables problem. Additional references, latent-state models, suitable instruments or a specified input/output error geometry may help, each under its own assumptions (Söderström 2007). A generic total-least-squares call is not a universal correction.

8.3.2 Output-error objectives, conditioning and numerical optimization

An equation-error objective substitutes measured or estimated motion into the differential balance. It can expose a linear parameter structure, but differentiated observations may be noisy. An output-error objective instead generates a predicted observation sequence from the model and compares it with recorded outputs. It avoids directly using measured acceleration as a regressor, while introducing other dependencies.

For a deterministic discrete rollout with known inputs,

$$\begin{aligned} \hat{\mathbf{x}}_{k+1} &= \mathbf{F}_d(\hat{\mathbf{x}}_k, \mathbf{u}_k, \boldsymbol{\theta}, h_k), \\ \hat{\mathbf{y}}_k &= \mathbf{H}(\hat{\mathbf{x}}_k, \boldsymbol{\theta}), \qquad \mathbf{r}_k = \mathbf{y}_k - \hat{\mathbf{y}}_k, \end{aligned} \tag{8.24}$$

minimize a stated function of the stacked residuals. Initial state, held-input meaning and timing are part of the prediction. An uncertain initial condition can be estimated as a nuisance parameter or marginalized in a probabilistic model; setting it to a convenient value does not make it known.

Unweighted squared output errors are a Gaussian likelihood only for the corresponding independent equal-variance observation-error model. Process disturbances generally produce correlated rollout errors. A linear input–output innovation model, for example, can write

$$y_k = G_{\boldsymbol{\theta}}(z)u_k + H_{\boldsymbol{\theta}}(z)\epsilon_k, \qquad \epsilon_k(\boldsymbol{\theta}) = H_{\boldsymbol{\theta}}^{-1}(z)[y_k - G_{\boldsymbol{\theta}}(z)u_k]. \tag{8.25}$$

Here z^{-1} denotes a one-sample delay, and the disturbance filter must support the stated causal innovation interpretation. Prediction error and free simulation error answer different questions; choosing a noise model matters especially under feedback (Forssell and Ljung 1999; Ljung 1999).

If a stacked Gaussian residual has parameter-dependent covariance $\mathbf{R}(\boldsymbol{\theta})$, negative log likelihood includes

$$\tfrac{1}{2}\mathbf{r}^{\mathsf{T}}\mathbf{R}(\boldsymbol{\theta})^{-1}\mathbf{r} + \tfrac{1}{2}\log\det\mathbf{R}(\boldsymbol{\theta}) \tag{8.26}$$

up to parameter-independent terms. Minimizing weighted squares alone would omit the covariance-volume term. This expression also assumes that the residual model itself is the stated joint Gaussian likelihood.

Worked example: an exact rollout still has a scale ambiguity. For Equation (8.13), take zero initial position and velocity, a constant $i_0 = 5$ A, and $J = 0.02$ kg m^2, $b = 0.1$ N m s/rad and $\alpha = 0.04$ N m/A. The delivered effort is 0.2 N m. Define $\beta = b/J$ and $\gamma = \alpha/J$. Integration gives

$$q(t) = \frac{\gamma i_0}{\beta}\left[t - \frac{1 - e^{-\beta t}}{\beta}\right], \qquad \beta = 5\ \mathrm{s}^{-1}. \tag{8.27}$$

The time constant is 0.2 s; $q(0.1) \approx 0.04261226$ rad and $q(0.2) \approx 0.14715178$ rad. This is an exact continuous-model solution, not a result of selecting a particular simulator integrator.

Doubling J, b, α preserves both ratios and every predicted position under the same current input. A perfect output fit does not resolve the absolute scale. The companion checks the zero output difference and the corresponding rank-two sensitivity for three parameters. An optimizer returning one triple has selected a representative; it has not supplied the missing reference.

For more general models, the objective can have multiple local minima, poorly scaled directions and nondifferentiable mode changes. Use parameter scales, feasible parameterizations, multiple justified initializations and derivative checks. State whether convergence means a small update, small gradient or another criterion. A numerical stopping rule is not evidence of a correct physical explanation.

Chapter 6 also matters here. An estimator can compensate for integration error by changing physical coefficients. Refine the numerical prediction enough for the inference being attempted, and distinguish temporal-discretization sensitivity from measurement uncertainty. Likewise, fitting a free time shift can hide an actuation delay unless the timing interpretation is independently justified.

8.4 Learn model components from data

Some relationships are too complicated or insufficiently known for a small chosen formula. Learning can represent such components while retaining mechanical structure around them. The resulting flexibility changes the questions of identifiability, availability of inputs and validation; it does not eliminate them.

8.4.1 Feature maps, neural approximators and automatic differentiation

A feature model represents an unknown component as

$$d(\boldsymbol{\xi}) = \boldsymbol{\phi}(\boldsymbol{\xi})^{\mathsf{T}}\mathbf{w}. \tag{8.28}$$

The features may be nonlinear in measured conditions while remaining linear in weights. A friction feature vector might contain speed, sliding direction and a temperature interaction. Its terms need declared units and domains. Highly correlated features can remain difficult to distinguish.

A neural approximation replaces the fixed feature construction with a parameterized function $d_{\boldsymbol{\psi}}(\boldsymbol{\xi})$. It can express more relationships, but a good fit on one trajectory does not determine how the function behaves elsewhere. Selecting history, coordinate representation and output quantity is part of model design.

Structure can constrain the component. For a scalar resisting effort, $d_\psi(v) = b_\psi(v)v$ with $b_\psi(v) \geq 0$ gives $vd_\psi(v) \geq 0$ and therefore dissipative power under that sign convention. The coefficient has drag units. This restriction would be inappropriate for a residual that also includes omitted conservative or active effects. A physically named residual is not automatically a physically constrained one (Tedrake, n.d.).

Automatic differentiation applies the chain rule to an executed computation (Baydin et al. 2018). For the smooth rollout in Equation (8.24), let $\mathbf{Z}_k = \partial\hat{\mathbf{x}}_k/\partial\boldsymbol{\theta}$. Forward sensitivity propagation gives

$$\begin{aligned}\mathbf{Z}_{k+1} &= \mathbf{F}_{x,k}\mathbf{Z}_k + \mathbf{F}_{\theta,k},\\ \frac{\partial\hat{\mathbf{y}}_k}{\partial\boldsymbol{\theta}} &= \mathbf{H}_{x,k}\mathbf{Z}_k + \mathbf{H}_{\theta,k}.\end{aligned} \tag{8.29}$$

These expressions assume the recorded inputs are held fixed with respect to the parameters. If a controller is re-executed within the rollout, its parameter/state dependence contributes additional chain-rule terms. The initial sensitivity depends on how initial conditions are specified.

The derivative is of the implemented prediction, including its integrator and selected branches. A derivative within one contact or clipping branch does not establish differentiability across its switching surface. Finite precision and finite-difference step size also affect derivative checks. Agreement of two derivative implementations tests that calculation, not the correctness of the physical model.

8.4.2 Residual models, available inputs and counterfactual limits

A structured residual model can write

$$\boldsymbol{\tau}_a + \boldsymbol{\tau}_{\text{ext}} = \mathbf{Y}\boldsymbol{\pi} + \mathbf{d}_{\boldsymbol{\psi}}(\boldsymbol{\xi}). \tag{8.30}$$

Declare what is included in the rigid baseline and what the residual may represent. Otherwise both parts can explain the same variation: an arbitrary residual can absorb a changed mass or gravity coefficient. Separate reference information, constrained function classes or deliberately varied experiments are needed to interpret that decomposition.

An appropriate input vector might include joint position, rate, current, temperature and a declared recent history. Availability depends on the intended use. Measured acceleration can be useful for offline inverse dynamics but may be unavailable when predicting the next acceleration. Future filtered state, a privileged simulator variable or a contact label derived using the outcome cannot become a deployable predictor merely because it was stored in the training table.

A residual learned from position references and output motion can describe an entire embedded closed loop. It should not automatically be interpreted as an actuator torque correction. The interface identified in the first section still determines the meaning of the learned relationship.

Counterfactual prediction asks what would happen under a changed input, controller or physical arrangement. Correlation in one collection does not alone answer that question. Suppose payload and commanded speed always increase together. A flexible function can attribute the extra effort to either variable on the observed data while predicting different outcomes when only one changes. Deliberately crossed conditions help distinguish those explanations within the declared model.

Retain a simple physical baseline and compare against it on separate conditions. If a residual improves prediction, report the quantity, regime and uncertainty affected. That improvement does not prove that the residual has identified one named physical effect, nor that its behavior is valid under arbitrary interventions.

8.5 Validate a model beyond fitting

Fitting asks how well a candidate can account for the records used to choose it. Validation asks how it behaves on evidence that can challenge its assumptions. A model can be useful within a stated regime without being a complete description of the apparatus. The aim is to establish that regime and the limits relevant to its intended use.

8.5.1 Held-out episodes, residual structure and uncertainty

Reserve complete trials or conditions that address the intended claim. Holding out time-adjacent rows from one filtered trace can retain shared state, calibration and processing information. A new speed tests another motion regime; a new load tests a changed dynamic arrangement; a new device tests unit variation. These are different levels of evidence.

The earlier steady-motion example provides a minimal validation challenge. Both candidate inertias fit $v = 1$ rad/s and $a = 0$. A held-out instantaneous condition with $a = 2$ rad/s^2 requires 0.14 N m under the constructed $J = 0.02$ model. The $J = 0.05$ candidate predicts 0.20 N m, an error of 0.06 N m. The comparison identifies which candidate is consistent with that generator. It is not a measured performance claim.

Evaluate the quantity the model is intended to support: predicted effort, free trajectory, one-step output or another declared quantity. These errors need not rank models equally. A predictor repeatedly corrected by observed state can remain accurate while an uncorrected rollout drifts. State which observations are supplied at evaluation time.

Inspect residuals in physical units and against time, pose, speed, direction, load and relevant operating conditions. Serial structure can suggest missing dynamics or filtering; direction structure can suggest an inadequate friction description; a rate-dependent discrepancy can suggest timing. These are diagnostic hypotheses, not unique diagnoses.

For an ordinary unconstrained linear least-squares fit, residuals are orthogonal to the fitted feature columns by construction. Generalized least squares instead gives $\mathbf{Y}^{\mathsf{T}}\mathbf{R}^{-1}\mathbf{r} = 0$. Neither training-set property is independent evidence that the model is correct. For a correctly specified innovation model, compare residuals with the appropriate causal past information. Under feedback, future inputs can depend on today's innovation, so demanding zero correlation at every positive and negative lag would be inappropriate (Ljung 2016).

With a correctly specified model, independent held-out observation error and parameter uncertainty, local predictive covariance can be approximated by

$$\mathbf{P}_{\text{pred}} \approx \mathbf{J}_{\text{test}}\mathbf{P}_{\theta}\mathbf{J}_{\text{test}}^{\mathsf{T}} + \mathbf{R}_{\text{test}}. \tag{8.31}$$

Add initial-state, process and shared-calibration terms when relevant; dependence requires cross terms. This expression does not cover arbitrary model inadequacy. A narrow interval is meaningful only relative to the assumptions used to construct it.

8.5.2 Measurement noise, parameter uncertainty and model inadequacy

Measurement noise concerns how observations vary around their measurement relationship. Parameter uncertainty concerns which coefficients remain plausible within the chosen model. Model inadequacy concerns relationships the model cannot express correctly. Device variation and changing conditions add further sources of response variation.

Repeated observations can help characterize some noise, while independent calibration can constrain shared measurement effects. Repeated trials under the same controller do not necessarily constrain a missing mode or a common

effort-scale ambiguity. Parameter covariance conditioned on a wrong model can be small even when its predictions are biased.

Use residual evidence to choose a specific next question. If error grows with speed, compare timing and drag explanations rather than immediately adding a high-order polynomial. If an added load changes apparent drag, check whether the intervention also changed bearing friction or temperature. If a neural residual hides all remaining errors, test whether the physical and learned parts are still separately interpretable.

Separate records used for fitting, tuning model complexity and final evaluation. Repeatedly selecting a model against the same supposed test set makes that set part of development. Chapter 9 develops general partitioning and provenance; here the identification claim must state what evidence remained independent.

Failure outside a regime can be informative. It may motivate a new model component, a revised boundary or a better measurement arrangement. A transparent local model with a documented failure can support more useful learning and design decisions than an apparently universal fit whose selection and assumptions are hidden.

8.6 Construct reusable identification data

A reusable identification record supports reconsidering the inference. It preserves the experiment, observations and transformations that made the fitted numbers possible. A parameter file without that context is a result to inspect, not a complete basis for reproducing the conclusion.

8.6.1 Repeated trials, device variation and experimental context

For a defined mechanism, retain the body and joint arrangement, fixture and load geometry, actuator interface, controller configuration and limits. Record which quantities were requested, measured or inferred, including their units, frames, origins and time interpretation. A delivered-effort estimate should retain its calibration rather than appear as an unexplained torque column.

Preserve raw indications alongside the transformations used to derive position, velocity, acceleration and loading. State filter windows, differentiation, alignment and missing-sample treatment. A different preprocessing choice can change the regressor and the identification result without changing the raw physical trial.

Repeated trials provide information only at the level actually repeated. Frames within a trial share initial conditions and processing; trials within one mounting share calibration; devices can differ in their physical coefficients. More samples from one unit do not establish population-wide parameter values. A collection aimed at device variation needs multiple identified units, not anonymous repetitions.

Retain attempted and limited trials as well as accepted fits. If an input is clipped or a run is interrupted, its actual conditions can differ from the planned design. A clean subset may be useful for one modeling objective, but selection must remain visible. General collection accounting is developed in Chapter 9.

The numerical companion uses small explicit records to preserve this correspondence. Its spatial-inertia routine and basis regressor implement Equations (8.4) and (8.6); an independent center-of-mass Newton–Euler calculation checks the resulting wrench. Its joint projection retains the same motion subspace and power convention as RNEA. Other routines verify rank, least-squares covariance, input-error attenuation, analytic output sensitivities and changed-load predictions.

These are checks on the constructed mathematics and its implementation. They do not certify that a real sensor has the assumed noise, that a particular servo delivers the modeled effort, or that a fitted simulator has been validated. A physical implementation should preserve the same semantic links while documenting what its measurements actually establish.

8.6.2 A joint investigation that distinguishes competing explanations

Return to the opening joint. Suppose measured current and recorded motion are compatible with the sliding, small-regime model $J\ddot{q} + b\dot{q} = \alpha i$, after separately justified compensation for other loads. The aim is to distinguish inertia, drag and effort-scale explanations. The assumptions are deliberately limited: constant coefficients, known timing, no saturation and no unmodeled load changes.

First, examine the input interface and timing. A reference current is not necessarily measured current, and current is not a calibrated joint effort by itself. Chapter 7's encoder/optical comparison can help challenge timing and geometry before mechanical coefficients are adjusted to absorb their errors.

Second, vary the kinematics. Constant-speed records constrain drag relative to effort scale but cannot identify inertia. Transients provide sensitivity to the ratios $\beta = b/J$ and $\gamma = \alpha/J$. Several rates, directions and initial conditions can test the assumed model and separate the sliding loss terms considered earlier. These variations still do not create an absolute mechanical scale.

Worked example: a known load supplies the missing scale. Attach a load with independently known added inertia $\Delta J = 0.01$ kg m^2 about the joint. Assume the intervention changes only J, leaving b, α and the measurement relationship unchanged. The two response rates satisfy

$$\beta_0 = \frac{b}{J}, \qquad \beta_1 = \frac{b}{J + \Delta J}. \tag{8.32}$$

Eliminating b gives

$$J = \frac{\beta_1 \Delta J}{\beta_0 - \beta_1}, \qquad b = \beta_0 J, \qquad \alpha = \gamma_0 J. \tag{8.33}$$

For $\beta_0 = 5$ s^{-1}, $\beta_1 = 10/3$ s^{-1} and $\gamma_0 = 2$ rad/(s^2 A), the recovered values are

$$J = 0.02\ \mathrm{kg\,m^2}, \qquad b = 0.1\ \mathrm{N\,m\,s/rad}, \qquad \alpha = 0.04\ \mathrm{N\,m/A}. \tag{8.34}$$

The unmodified and loaded time constants are 0.2 and 0.3 s. The companion confirms that doubling all three baseline coefficients preserves the original response but fails to preserve the response after this fixed added inertia. Combined sensitivity becomes full rank in this constructed experiment.

The known load has supplied information that a prior positive-mass constraint could not supply. If β_0 and β_1 are nearly equal, the denominator in Equation (8.33) makes recovery sensitive to rate uncertainty. A very small load can therefore be a poor reference even if its nominal value is known.

In a real apparatus, ΔJ follows from the load's mass, rotational inertia, orientation and lever distance. The parallel-axis relation from Chapter 4 is required; a mass value alone is insufficient. Mounting uncertainty contributes to the reference uncertainty. The intervention may also change friction, compliance or current limits. A shifted rate is evidence to interpret under those alternatives, not automatic proof of the assumed three-parameter model.

Third, challenge the selected model with held-out motion and loading. Predict the observation actually available in that test, rather than substituting a privileged acceleration estimate at every step. Compare both effort balance and free response when the intended use requires them. Preserve failure, residual structure and parameter sensitivity alongside the fitted coefficients.

The result is more than a fitted joint. The investigation identifies which observations support which conclusions, which ambiguities remain and which intervention changed the available information. For ExpGen PhySyst, this is the contribution of experimental design: generate experiences whose records can distinguish the explanations relevant to the next design or learning decision.

Exercises and further investigation. State units, system boundary, assumptions and which quantities are observed or inferred. Treat all supplied numerical records as constructed.

1. Derive the static feedback moments and the naive and instrumental coefficients. Change the reference variance and explain why instrument relevance matters. Distinguish this algebraic loop from a causal dynamic identification model.
2. Construct the ten spatial-inertia basis matrices in the declared order. Derive the body regressor and verify power-consistent joint projection against a direct Newton–Euler balance.
3. Show why parameterizing inertia by mass, center of mass and central inertia is not jointly linear. Explain how known kinematics preserves inverse-dynamics linearity and why forward dynamics generally does not.
4. Compare structural and experiment-specific null directions. For $\tau = J\ddot{q} + \kappa q$ in a small-angle approximation, show that a single sinusoidal frequency identifies a combination of J, κ. Explain what additional variation can separate them and when the small-angle approximation fails.
5. Reproduce the paired-direction example and examine its design matrix with one and two speeds. State why stationary holds with unknown static friction cannot replace the stated sliding observations.
6. Derive generalized least squares, its covariance and the conditions for its likelihood interpretation. Reproduce the four-row example. Rescale the parameter units and compare raw and scaled condition numbers.
7. Compute pseudo-inertia for a centered box and for the impossible inertia in the text. Distinguish strict physical consistency from degenerate mass distributions and from a parameter-covariance matrix.
8. Reproduce the errors-in-variables attenuation. Investigate correlated position, velocity and acceleration errors after differentiation. Identify what additional information a proposed correction needs.
9. Derive the exact current-step response and its parameter sensitivities. Compare analytic and finite-difference sensitivities. Verify the common-scale null direction and explain why a unique optimizer output need not imply unique physical parameters.
10. Define a residual-dynamics model with inputs available for its intended use. Propose a physically appropriate restriction and a held-out intervention that could expose an incorrect interpretation.
11. Derive the known-load recovery and propagate uncertainty in both fitted rates and added inertia. Examine the nearly equal-rate limit and a case in which drag also changes after mounting the load.
12. Design a complete joint identification investigation. Specify the intended model use, measured interfaces, excitation, independent references, repeated and held-out conditions, numerical checks and retained failures. Identify one conclusion the data would not justify.

Ljung (1999) provides a broader system-identification framework. Forssell and Ljung (1999) develops closed-loop estimation conditions, while Söderström (2007) examines noisy-regressor assumptions. The physical-consistency and identifiability studies by Wensing et al. (2018) and Wensing et al. (2024) connect inertial fitting to realizable bodies and identifiable combinations. These readings extend the examples without turning a successful fit into unqualified physical evidence.

Chapter 9 expands from a defined-system investigation to general collection and dataset design: deciding which experiences to generate, which populations and conditions they represent, and how their sources and transformations remain available for embodied learning.

Chapter 9 Data Collection and Dataset Design

A robot pushes an object toward a target. A camera records the motion, the controller stores requests, and a tracker estimates the object's pose. After several trials, one researcher keeps every technically valid record; another keeps only successful attempts. Both can describe their work as collecting manipulation data. Yet their datasets represent different populations of experience, and they support different conclusions.

A dataset begins before a file is written. It begins with decisions about which interactions are worth generating, which conditions to vary, which quantities to observe and which failures remain visible. Chapters 3 through 8 established physical and inferential meanings for those decisions. This chapter brings them together at the collection level.

For ExpGen PhySyst, designing data means designing the physical encounters and the evidential record that a learner can use. The objective is not maximal volume without qualification. It is a collection whose observations, selection and transformations are appropriate to an intended use. The running manipulation collection and all worked numbers below are constructed examples with a numerical companion, not a new hardware dataset or measured learning result.

9.1 Define the collection purpose and population

A population specifies the experiences to which a statement is intended to apply. It can be narrow: one apparatus, one object and one operating regime. It can also include variation across devices, operators, tasks or environments. A clear narrow population is more useful than an unqualified claim of generality.

9.1.1 Target conditions, units of collection and intended claims

Let C denote context and E an episode of interaction. Context can include mechanism configuration, object properties, surface, initial conditions and the operator or collection policy. A dataset samples from a collection law $P_{\mathrm{col}}(C, E)$, while the intended use concerns a target law $P_*(C, E)$. The two need not agree.

For a model or learner with parameters θ, define a relevant episode loss $L_\theta(E, C)$. Its target risk is

$$R_*(\theta) = \mathbb{E}_{(C,E)\sim P_*}[L_\theta(E, C)]. \tag{9.1}$$

This is a definition of the intended average, not a guarantee that a collected empirical average estimates it. The loss might concern pose prediction, effort reconstruction or a declared task outcome. Those choices require different observations and can value different experiences.

The collection law depends on the physical system and the actions that produce encounters. A mechanism's configuration, tangent velocity and actuator state affect subsequent observations; a human or policy selects actions from available information. More samples from the same narrow behavior do not automatically cover other behavior. Chapter 10 will develop how learned policies generate experience. Here we specify what the collection represents before choosing a learner.

Also define the unit receiving equal weight. A frame, action interval, episode, object and physical device are different units. The distinction is visible even in elementary averages.

Worked example: equal episodes and equal frames answer different questions. Suppose one episode has 10 frames with mean loss 0.2, and another has 100 frames with mean loss 0.8. The equal-episode mean is

$$\bar{L}_{\mathrm{episode}} = \frac{0.2 + 0.8}{2} = 0.5, \qquad \bar{L}_{\mathrm{frame}} = \frac{10(0.2) + 100(0.8)}{110} \approx 0.74545. \tag{9.2}$$

Neither arithmetic result is inherently wrong. They estimate different empirical averages. A frame-weighted objective gives longer episodes more influence; a device-weighted claim would require another grouping. A dataset description should make that choice explicit rather than let file length choose the intended population silently.

9.1.2 Coverage, uncertainty, missing information and collection budgets

Coverage concerns which relevant conditions and behaviors are represented. A count of recorded frames is not a coverage measure by itself. Ten thousand frames from one successful straight push may omit reversals, slip, occlusion, failed contact and recovery. Which omissions matter depends on what the learner will be asked to do.

For discrete context strata, the target average can be decomposed as

$$R_*(\theta) = \sum_c p_*(c)\, \mathbb{E}_*[L_\theta \mid C = c]. \tag{9.3}$$

This separates the frequency of a condition from behavior within it. Balancing counts across named conditions does not ensure representative behavior inside each condition, and an unrecorded variable can still create systematic differences.

Uncertainty is another design constraint. Extra images sharing one uncertain camera calibration do not independently reduce that calibration error. Chapter 7 showed why. Likewise, a dataset may lack delivered-effort measurements even when every command is stored. More complete sampling of requests does not reveal an unmeasured physical quantity without additional assumptions or observations.

A collection budget includes apparatus time, setup, resets, calibration, operator attention, annotation and recovery. A plan based only on successful motion duration can substantially understate the work needed. A useful budget states what counts as an attempt and which costs accrue when an attempt fails or its record is unusable.

Distinguish an observed absence from missing information. A tracker reporting no contact is different from a tracker failing during contact. A field absent because a sensor was never installed is different from a dropped packet. These cases can require different model assumptions and different next collection decisions. Dataset documentation should preserve such distinctions (Gebru et al. 2021).

9.1.3 A manipulation dataset: scope, collection effort and stopping criteria

Consider an instrumented pusher moving a rigid object toward a target region. Plan two device units, two surfaces and two starting regions: eight context cells. Use one declared collection method and six attempts per cell. This is a modest investigation of those conditions, not a claim of arbitrary-object manipulation.

Worked example: budget the attempted interaction. The plan has $2 \times 2 \times 2 \times 6 = 48$ attempts. Allocate 20 s to motion, 40 s to reset and 10 s to inspection per attempt, including attempts that fail. Then

$$T_{\text{attempts}} = 48(20 + 40 + 10) = 3360\ \text{s} = 56\ \text{min}. \tag{9.4}$$

Adding 20 min of setup and 15 min of calibration gives a nominal 91 min session. These are constructed allocations, not measured throughput. Surface changes, interruptions or longer recovery would need their own accounting.

A stopping rule defines when the plan is complete. A fixed number of attempts per condition, a fixed time budget and stopping after a chosen number of successes produce different collections. If collection stops after success and only that success is retained, the stored records conceal the effort required to obtain it.

Specify stopping criteria before reading the outcomes when they affect the intended comparison. Adaptive collection can be valuable, but its policy should be recorded: what information triggered another attempt, a changed condition or a decision to stop? An outcome-dependent stopping rule does not become a fixed-size sampling design because the final folder contains a convenient number of files.

Plan separate calibration and checking records. The calibration may support the measurement pipeline, while independent trials can challenge its behavior under changed conditions. A successful collection session does not by itself demonstrate that every retained label is accurate or that its future use is justified.

9.2 Design episodes and record their context

An episode is a chosen segment of interaction. Its boundaries organize observations, actions and outcomes, but should not erase the physical history that matters. Define what starts an episode, what ends it, and which context remains shared across several episodes.

9.2.1 Observations, actions, temporal alignment and episode boundaries

For a fixed-length schematic episode, distinguish the observation at the start of an action interval from the observation after that interval:

$$E = (C, o_0, a_0, o_1, a_1, \ldots, a_{T-1}, o_T). \tag{9.5}$$

The observation need not be the complete physical state. An action can be a request at a controller interface rather than the effort actually delivered during motion.

Each transition needs its actual temporal meaning. If a_k is issued at t_k and held until t_{k+1}, store that interval or the information needed to reconstruct it. A camera exposure can span part of the interval; a delayed pose estimate can describe an earlier instant. Placing them in the same row does not establish simultaneity.

For a simple constructed joint with delivered torque τ_k constant over $[t_k, t_{k+1}]$, constant inertia J and no other load, integration gives

$$v_{k+1} = v_k + \frac{\tau_k}{J} h_k, \qquad q_{k+1} = q_k + v_k h_k + \frac{\tau_k}{2J} h_k^2. \tag{9.6}$$

These are exact interval relations under the stated assumptions. They do not apply unchanged when the recorded action is a position reference or normalized input to an embedded servo. The data contract must identify which of those quantities is present.

A last observation can be valid without a subsequent action. An interrupted recording can have no trustworthy final observation. Do not manufacture a zero action, zero reward or repeated endpoint to make every row look identical. An episode can end because the task terminated, because a time budget expired, because an operator intervened or because logging failed. These meanings should remain distinct.

Episode-oriented representations such as RLDS make first/last and terminal semantics explicit (Google Research 2024; Ramos et al. 2021). In its canonical observation–action–reward alignment, the last step retains the final observation; its outgoing action, reward and discount are undefined. The flag `is_last` marks that last step, while `is_terminal` indicates an environment terminal state, independently of whether the task succeeded. A last but nonterminal step represents truncation. An episode-level `invalid` flag can instead identify an incomplete recording. These boundary rules depend on the declared alignment; RLDS also discusses a reward–observation–action alternative. A format can help preserve distinctions, but does not infer them from a file's length. Chapter 10 will explain their consequences for learning returns; the collection must first retain the actual end cause.

9.2.2 Device configuration, initial conditions, operator or policy identity

Context determines which physical system generated the episode. Nominally identical devices can differ in calibration, losses or limits. A changed tool or payload changes inertia and contact geometry. The controller configuration affects how requests become effort and motion.

Record the configuration relevant to the intended use: mechanism and object identity, mounting, actuator mode, calibration identity, control period, limits and relevant operating conditions. Context is not simply a serial number. It includes the relationships needed to interpret the signals.

Initial conditions also have several levels. A visible pose can be repeated while temperature, actuator history or contact preload differs. Chapter 6 showed the analogous issue in simulation resets. A collection should distinguish a measured initial quantity from a nominal reset target and from an unobserved state.

Operator and policy identity are part of the generation process. Demonstrations produced by a skilled operator, an assisted operator and a partially trained policy can have different coverage and failure modes. A human may use tactile, auditory or visual information that is not stored in the learner's observation stream. The resulting action is not necessarily reproducible from the recorded observation alone.

This issue is visible in systems that deliberately design the collection interface, such as Universal Manipulation Interface (Chi et al. 2024). The general lesson is that the interface affects what observations and actions can be aligned and reused. It is not a claim that every collection needs the same hardware or representation.

Retain revisions to context rather than quietly applying one final configuration to every earlier episode. A calibration

made after a session may not describe the original mounting. An independent fixed calibration can legitimately be shared across trials; its identity and uncertainty should remain visible.

9.2.3 Resets, interventions, failures and recovery

Resetting establishes another starting condition. It can require object replacement, unloading, homing, a controller restart or waiting for temperature to change. A reset target does not prove that the reset succeeded. Record a check appropriate to the task and preserve uncertainty about hidden state.

An intervention changes the interaction while it is underway. Examples include a human repositioning an object or a supervisory controller limiting a request. Such events can create useful recovery data, but they change the behavior being recorded. Do not attribute the result solely to the nominal autonomous policy.

Failure should identify a criterion and a stage. Failure to reach a goal, failure of pose tracking, command clipping and file corruption are different events. A technically valid episode may contain unsuccessful behavior; a successful physical motion may have an unusable observation record. These two axes should not be collapsed into one pass/fail flag.

Recovery can be part of the intended population. If a learner must recover from slip, keeping only the final successful approach removes relevant experience. Conversely, a collection intended to study one controlled contact regime may exclude some recovery intervals from that analysis. Preserve the original attempt and the exclusion rule so another use can make a different justified selection.

When segmenting one long interaction into several episodes, retain the parent interaction and boundary rule. Segments can share object state, operator decisions and recent history. Renaming each segment as an independent trial does not create an independent physical reset.

9.3 Represent observations and labels

Representations connect physical quantities to the numbers a learner sees. Consistent array dimensions are insufficient: two vectors can have the same length while describing different origins, units or times. Label design adds a second question: what judgment or inference gives the stored quantity its intended meaning?

9.3.1 Physical units, coordinate conventions and multimodal correspondence

Retain physical meaning before introducing a convenient learning encoding. A position in millimeters and one in meters need a scale conversion. A joint configuration and its tangent velocity can have different dimensions, particularly for floating joints. A quaternion sign change can leave the physical attitude unchanged. These are representation issues, not necessarily new physical events.

Featherstone's spatial language provides a stronger check for motion and loading records (Featherstone 2008). Let $\mathbf{E} = {}^{P}\mathbf{R}_{O}$ map O components into P, and let $\mathbf{r} = {}^{O}\mathbf{r}_{OP}$ point from O to P. The motion and wrench maps are

$$\mathbf{v}_P = \mathbf{X}_{PO}\mathbf{v}_O, \qquad \mathbf{X}_{PO} = \begin{bmatrix} \mathbf{E} & \mathbf{0} \\ -\mathbf{E}[\mathbf{r}]_\times & \mathbf{E} \end{bmatrix}, \qquad \mathbf{f}_O = \mathbf{X}_{PO}^{\mathsf{T}}\mathbf{f}_P. \tag{9.7}$$

Motion is angular-first and wrench is moment-first, each with its declared origin and axes. The wrench is the same applied load expressed about different reference points, not an action–reaction sign change.

The transformation preserves instantaneous power:

$$\mathbf{v}_P^{\mathsf{T}}\mathbf{f}_P = \mathbf{v}_O^{\mathsf{T}}\mathbf{f}_O. \tag{9.8}$$

A dataset transformation that changes this quantity without an intended physical change is inconsistent with the declared mechanics. Use compatible times as well as compatible spatial frames.

Worked example: a force record also needs its moment origin. Let $\mathbf{E}$ be the positive 90° rotation matrix about z, used as the component map from O to P, and $\mathbf{r} = (0.05, 0, 0)^{\mathsf{T}}$ m. Choose

$$\mathbf{v}_O = (0, 0, 2, 0.1, 0, 0)^{\mathsf{T}}, \qquad \mathbf{f}_P = (0, 0, 0, 1, 0, 0)^{\mathsf{T}}. \tag{9.9}$$

The angular and linear velocity blocks have units rad/s and m/s; the moment and force blocks have units N m and N. The transformed linear velocity at P is $(-0.1, 0.1, 0)^{\mathsf{T}}$ m/s. The same load about O is

$$\mathbf{f}_O = (0, 0, -0.05, 0, -1, 0)^{\mathsf{T}}. \tag{9.10}$$

Power is -0.1 W in either description. Rotating the force alone and discarding the translated moment would instead give zero when paired with $\mathbf{v}_O$. The negative power means this load removes mechanical energy at that instant. The companion checks the full transform, its inverse and the two power calculations.

Uncertainty must follow the same transformations. For an exact wrench map $\mathbf{A}$, covariance changes as $\mathbf{\Sigma}'_f = \mathbf{A}\mathbf{\Sigma}_f\mathbf{A}^{\mathsf{T}}$. Uncertain mounting adds calibration terms and potentially shared errors across episodes, as developed in Chapter 7. Transforming the mean without its uncertainty can misrepresent the label.

Multimodal correspondence needs an analogous declaration. An image, current indication and estimated pose might refer to different acquisition intervals. A geometric calibration establishes one kind of relationship; a clock calibration establishes another. Retain the source observations and those mappings so that a derived multimodal example can be inspected or regenerated.

9.3.2 Instructions, label definitions and annotator judgment

A label should identify an operational meaning. Successful pushing could mean that an object entered a region, remained there for a duration, stopped moving, or arrived without leaving a permitted area. These definitions can disagree on the same episode. A natural-language task instruction does not by itself define the measurement rule.

For a constructed goal test, one might require object-position error below a threshold, speed below a threshold and a specified dwell time. The pose reference, acquisition rate and treatment of occlusion are part of that rule. If the tracker disappears at the final instant, success is unknown under that rule unless another source supplies the evidence.

Instructions also affect the behavior generated. Asking an operator to finish quickly can produce a different trajectory population from asking for smooth motion. Asking annotators to infer intention rather than mark visible contact changes the label's evidence basis. Preserve the instruction and its revision, not just the final category.

Human disagreement can reveal ambiguous boundaries, competing interpretations or missing context. It can also arise from error or poor task execution. The annotation literature cautions against equating all disagreement with noise or all consensus with truth (Aroyo and Welty 2015). That argument does not establish the accuracy of a particular robotic label; it motivates examining why judgments differ.

Retain individual judgments and the adjudication rule when relevant. If majority voting produces a final label, it remains an aggregation of judgments. A vote fraction describes that panel's responses and is not automatically a calibrated probability of a unique physical truth. A revised instruction can change the labels without changing the underlying recorded interaction.

Good annotation design states which questions can be answered from the available evidence. A camera view that hides the contact cannot support an unqualified direct-contact label merely because an annotator must choose one of two buttons. Allowing an unknown or disputed status can preserve information that forced certainty would erase.

9.3.3 Reference measurements, inferred labels and annotation uncertainty

A reference label may be a calibrated instrument output, an estimate from several observations, a simulator state or a human judgment. Their errors and dependencies differ. The phrase *ground truth* should not replace a description of how a reference was obtained and what uncertainty remains.

A tracker-derived object pose inherits camera calibration, feature matching, visibility and timing assumptions. If the same image is both learner input and the source of its reference pose, errors in that reference can be systematically related to the input. More annotations from the same estimator do not create independent physical references.

Worked example: a threshold can be uncertain even when its value is precise. Let e be a one-dimensional signed position error, with success defined by $|e| \leq 0.02$ m. For illustration, assume the stated measurement model and prior

yield posterior $e \mid \mathcal{D} \sim \mathcal{N}(0.019, 0.003^2)$ in meters. Then

$$\Pr(|e| \le 0.02 \mid \mathcal{D}) = \Phi\left(\frac{0.02 - 0.019}{0.003}\right) - \Phi\left(\frac{-0.02 - 0.019}{0.003}\right) \approx 0.63056, \tag{9.11}$$

where Φ is the standard normal cumulative distribution. A central posterior interval with approximately 95% probability runs from 0.01312 m to 0.02488 m and crosses the threshold.

The probability follows from the explicitly assumed posterior. It is not implied by a sensor resolution specification or an annotator's confidence. The example concerns one coordinate and one criterion, not a full manipulation-success model. A hard label at the posterior mean would hide the modeled uncertainty near the boundary.

Possible representations include the estimated quantity with uncertainty, an explicit uncertain category or a probability under a declared model. Choose according to the learning objective and preserve how the representation was derived. Treating uncertain cases as rejected data also changes the retained population and therefore needs accounting.

Reference measurements can challenge rather than merely decorate labels. A separate geometric reference can reveal tracker bias; repeated independent annotations can reveal interpretation differences. Neither automatically removes shared calibration error or supplies unobserved physical events. Measurement uncertainty and a label rule are different parts of the record (Joint Committee for Guides in Metrology 2008).

9.4 Account for coverage and selection

A dataset is not only the files retained at the end. It is also the result of attempts, failures, missing observations and selection decisions. Those decisions determine which experiences remain available to the learner and which claims an average can support.

9.4.1 Attempted, retained, rejected and missing episodes

Use an attempt identity before knowing whether a useful file will result. An attempted interaction can produce a technically valid record, an invalid record or no retrievable record. Within valid records, task success and failure remain separate outcomes. Additional curation can select a subset for a particular learning objective.

Worked example: account for the same 48 attempts with different denominators. In the constructed pusher collection, suppose all 48 planned attempts occurred. Four have missing records and four have technically invalid records. The remaining 40 contain 30 known successes and 10 known failures:

$$48 = 4_{\text{missing}} + 4_{\text{invalid}} + 30_{\text{success}} + 10_{\text{failure}}. \tag{9.12}$$

Assume outcome evidence is unavailable for the eight missing or invalid cases. Success among valid records is $30/40 = 0.75$. The success fraction among all attempts is only bounded by

$$\frac{30}{48} \le p_{\text{attempted,empirical}} \le \frac{30+8}{48}, \qquad 0.625 \le p_{\text{attempted,empirical}} \le 0.79167. \tag{9.13}$$

These are logical bounds on this finite collection, not confidence intervals for a larger population.

Keeping only the 30 successes yields a useful success-conditioned subset for some purposes. Its $30/30$ success fraction is true by construction; it does not estimate collection or policy success. The original ledger allows those different statements to remain visible.

Missing data need not be random. Tracking can fail during occlusion, large motion or contact. A record that survives a quality check may therefore describe an easier subset of behavior. Likewise, an interrupted attempt can consume substantial resources even when it contributes no usable trajectory. Preserve the reason and known context where possible.

Counts should be available at relevant levels: attempted episodes per condition, valid observations per modality, selected episodes per analysis and unresolved outcomes. A total retained frame count cannot substitute for this accounting. Dataset lifecycle documentation explicitly asks about composition, missingness, collection and preprocessing (Gebru et al. 2021).

9.4.2 Acceptance rules, adaptive collection and selection bias

Let $A = 1$ mean that an attempted episode is retained. For any recorded episode/context pair with nonzero overall retention,

$$p_{\mathrm{ret}}(c, e) = \frac{\Pr(A = 1 \mid c, e)\, p_{\mathrm{attempt}}(c, e)}{\Pr(A = 1)}. \tag{9.14}$$

Retention can depend on outcome, visibility, duration or operator judgment. A successful-only subset and an all-attempt collection generally have different conditional behavior even within the same named context.

Sometimes a declared sampling imbalance can be adjusted. If context frequencies differ but the conditional loss law within each context remains appropriate for the target, and target-supported contexts have positive collection probability, then

$$R_*(\theta) = \mathbb{E}_{\mathrm{col}}\left[\frac{p_*(C)}{p_{\mathrm{col}}(C)} L_\theta(E, C)\right]. \tag{9.15}$$

This is a change-of-measure identity under those conditions. Weighting contexts alone cannot generally correct outcome-dependent retention that changes the conditional loss within each context.

Worked example: context balance changes the intended average. Take two context strata with collection frequencies $0.8, 0.2$, target frequencies $0.5, 0.5$, and conditional mean losses $0.1, 0.5$. The collection average is 0.18, whereas the target average is 0.30. Weights 0.625 and 2.5 recover the latter under the stated matching conditional laws.

For a constructed 100-record collection with those exact proportions, a common weight-concentration diagnostic is

$$N_{\mathrm{eff},w} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2} = 64. \tag{9.16}$$

This describes the dispersion of these weights. It is not a universal count of independent physical experiences or an exact learning-performance equivalent. Correlation and unequal conditional variability can matter too.

A context with no observed support cannot be recovered from retained examples by assigning a large weight. Nor can absent failure trajectories be reconstructed from successful trajectories alone. Additional collection or explicit modeling assumptions are needed.

Adaptive collection can improve coverage by seeking rare contacts, uncertain regions or failures. It also changes the generation law over time. Record the acquisition stage, collection policy and decision rule. Selecting the next condition after inspecting previous outcomes is different from a fixed randomized allocation. Its usefulness should be assessed against its intended learning objective, not hidden behind the final number of retained examples.

9.5 Plan comparisons and data partitions

A data partition defines what information is available during development and what remains for a stated assessment. It also specifies which variation is being challenged. Splitting files is not enough if their underlying physical sources or fitted transformations cross that boundary.

9.5.1 Matched collection conditions and declared resource budgets

A comparison needs a declared resource basis. Two methods can use equal numbers of retained demonstrations while requiring different numbers of attempts, operator hours or calibration procedures. Equal frame counts can also represent different amounts of independent physical experience.

For the constructed pusher collection, a method trained on the 30 retained successes has not used the same behavioral information as a method trained on all 40 valid records. Both may originate from the same 48 attempted interactions and the same apparatus session. Report those quantities separately rather than select whichever denominator makes one method appear more efficient.

Match conditions relevant to the question. To study a representation, hold the underlying episodes and information access fixed where possible. To study a collection interface, include its setup, sensing and operator requirements. To study

robustness across surfaces, retain the condition allocation and missingness rather than compare arbitrary pooled averages.

Dataset size can be expressed at several levels: devices, sessions, attempts, valid episodes, action intervals and observations. A useful learning curve states which level is varied and which remains fixed. Increasing windows from existing episodes is different from collecting new devices or new independent interactions.

A predeclared allocation also makes incomplete collection visible. If a difficult context produces few valid records, replenishing only that context changes the attempt budget and collection law. This can be a sensible decision, but the revision and its costs should remain part of the comparison.

These are foundations for interpreting a comparison, not a complete statistical test of one method against another. Chapter 16 develops broader claims and comparative evidence. Here the collection must retain the distinctions that such analysis needs.

9.5.2 Trial dependence, leakage and train-only preprocessing

Repeated observations are useful, but are not automatically independent replications. Frames share physical state, filtering and calibration; episodes can share a mounting or an operator strategy. The sampling unit should match the claim. The classical analysis of pseudoreplication makes the distinction between experimental units and subsamples explicit (Hurlbert 1984). That distinction does not imply that repeated measurements are useless.

Worked example: many correlated frames can leave uncertainty in an average. Take M independent episodes with m scalar observations each. Write $y_{ij} = \mu + b_i + \epsilon_{ij}$, with independent zero-mean episode effects of variance τ^2 and independent readout terms of variance σ^2. Then

$$\operatorname{Var}(\bar{y}) = \frac{\tau^2}{M} + \frac{\sigma^2}{Mm} = \frac{\tau^2 + \sigma^2}{Mm}[1 + (m-1)\rho], \qquad \rho = \frac{\tau^2}{\tau^2 + \sigma^2}. \tag{9.17}$$

For $M = 10$, $m = 20$ and $\rho = 0.8$, the variance is 0.081 times the marginal observation variance. The mean-equivalent independent count is

$$N_{\text{eff},\bar{y}} = \frac{Mm}{1 + (m-1)\rho} \approx 12.34568. \tag{9.18}$$

This result concerns the grand mean under a balanced compound-symmetry model. It is not a universal effective training-set size or a performance bound for a learned policy.

Dependence also affects partitions. Crops, temporal windows or alternative labels derived from the same physical episode should normally follow its assignment when testing generalization to new episodes. A random row split can place nearly identical or mutually dependent observations in both training and evaluation.

Transformations can broaden the raw support of one derived row. If its center is k, history length is h and future support is f, its source interval is $[k - h, k + f]$. For adjacent time-based partitions, disjoint supports require

$$k_{\text{test,min}} - h > k_{\text{train,max}} + f. \tag{9.19}$$

This prevents shared raw observations under the stated window rule; it does not establish statistical independence. Future support can be legitimate for an offline reference label, while remaining unavailable as an online policy input.

Fitted transformations need an information boundary too. In an inductive protocol, estimate normalization, feature selection and other learned preprocessing from the allowed training data and apply the frozen result to validation or test records. Model-assessment theory distinguishes fitting and selection from the evidence used to assess them (Hastie et al. 2009). An explicitly transductive protocol or a deployment calibration can allow different information, but should be named and reproduced accordingly.

Worked example: global normalization uses information outside training. Take a constructed scalar position feature in millimeters. Training values are $(0, 2)$ and held-out values $(100, 102)$. Training-only population mean and standard deviation are 1 mm and 1 mm. Pooling all four instead gives mean 51 mm and standard deviation $\sqrt{2501}$ mm. The held-out normalized values therefore change from $(99, 101)$ to approximately $(0.97980, 1.01980)$.

The second result used the held-out distribution to fit the transformation. That can be a different declared adaptation procedure, but is not the same inductive evaluation as freezing training-only normalization. Neither procedure resolves an

incorrectly defined physical frame; the feature must first have the intended geometric meaning.

A shared independently fixed calibration or software version is not automatically leakage. The question is whether information unavailable under the intended use entered fitting or evaluation, and whether the split genuinely challenges the stated novelty. A blanket rule that every shared dependency invalidates a split would incorrectly exclude ordinary shared instruments and common mathematics.

9.5.3 Held-out devices, tasks, configurations and embodiments

Name what is new in the held-out set. New episodes on familiar devices, new devices, new objects, new tasks and new embodiments ask different questions. Performance on one does not imply performance on all others. Embodiment changes can also alter action interfaces and feasible motion, not merely appearance.

Worked example: two partitions of the same collection test different claims. For the planned 48 pusher attempts, an episode-level plan allocates four training, one validation and one test attempt within each of the eight context cells: $32/8/8$ attempts. Both device units occur in every partition, so this plan does not test an unseen device.

An alternative holds device B out entirely. Device A has four cells; five training and one validation attempt per cell give $20/4$, while all 24 device-B attempts form the test allocation. The resulting claim concerns that held-out unit under the represented conditions, not every possible device in a population.

The companion assigns these allocations deterministically from attempt identities before filtering, without using outcomes. Applying the constructed validity ledger gives realized valid counts $27/7/6$ for the episode-level plan and $17/4/19$ for the held-out-device plan. Both sum to 40 valid records, but challenge different variation. The planned counts are not substituted for the realized counts.

Keep alternative analysis plans distinct. Choosing whichever split gives the best result after inspection turns the split into a development choice. The final assessment needs evidence appropriate to the declared selection process.

A device-specific calibration may be available during actual deployment. Using it in evaluation is appropriate only within that declared protocol; it changes the claim from transfer with no adaptation. Record what data supplied it, whether task outcomes were used, and what resources it required. Chapters 14 and 16 will develop transfer and broader evaluation in more detail.

9.6 Preserve sources and transformations

A reusable dataset should allow a reader to trace a stored example back to its sources and transformations. That lineage explains how an experience became a learning input or target. It also helps locate shared errors, reproduce partitions and revise an interpretation without losing the original evidence.

9.6.1 Measured, estimated, simulated and human-supplied information

Use evidence roles rather than treating every stored number as equivalent. A measured indication comes from an instrument and its acquisition chain. An estimate combines observations with a model. A simulated quantity is generated under computational assumptions. A human-supplied quantity can be an instruction, judgment or contextual description.

These roles can overlap in a lineage. A human can annotate a rendered image; a pose can be estimated from physical camera indications; a simulator can generate observations passed through another estimator. The role describes how the quantity was obtained, not a file extension or a guarantee of accuracy.

Retain direct observations separately from their interpretation when useful. A raw current indication and an inferred torque label are connected by calibration and actuation assumptions. A reconstructed pose and a task-success label can be connected by geometry, timing and an operational threshold. Replacing both with a single unexplained category prevents inspection of that chain.

Simulation can provide state unavailable on the physical system. That state can be useful for training or reference analysis, but should remain identified as privileged information if a deployed learner will not receive it. A simulated pose is exact with respect to a specified model state, not necessarily accurate with respect to the physical world.

Human context can be essential when apparatus changes are not visible in the sensors. A note that a payload or controller mode changed should be linked to the affected attempts, with uncertainty about its timing if necessary. Retrospective recollection is useful evidence, but should not be silently relabeled as a contemporaneous instrument observation.

9.6.2 Calibration, annotation, transformation and version lineage

Provenance represents how entities, activities and responsible agents are related. W3C's PROV-DM provides a general vocabulary for such relationships (Moreau and Missier 2013). A raw record and a derived label can be entities; a calibration application or annotation pass can be an activity; an operator or responsible software organization can be an agent. Merely recording that an activity used a file does not establish that every output was derived from every input.

A dataset lineage should retain the influences relevant to interpretation: source attempt, calibration, transformation method and parameters, annotation rule, responsible actor and revision. Revision produces a new identified version with a relationship to the earlier one. A filename reused for changing content loses that distinction.

Suppose one episode produces a tracked trajectory, several training windows and a goal label. All derive from one physical interaction. A split check can keep those observation-derived variants with the parent episode while allowing a fixed calibration and common software to be used across partitions. A fitted normalizer is different: its fitting activity must use only information allowed by the declared protocol.

The companion represents this distinction with typed dependencies. It accepts a shared fixed calibration, shared software and a normalizer fitted on training observations. It rejects a constructed example that places variants of one episode across an episode-held-out split, and another that fits normalization on test observations. Those checks enforce declared dependency rules; they do not infer every possible hidden relationship in an actual laboratory.

A content hash, compared with the recorded expected hash, can detect a changed byte sequence. It does not establish that units are correct, that timestamps are truthful or that the physical trial occurred as described. Likewise, a representable provenance graph is not automatically a complete, temporally consistent or scientifically valid history. PROV-DM is a data model, with additional constraints addressed separately.

Preserve transformation versions when labels are revised. If a new calibration shifts an estimated trajectory, regenerate the derived label under a new identity and retain the relationship to the old result. If that change alters acceptance, recompute the ledger and partition summary for the revised dataset. A silent overwrite would hide both the scientific revision and its effect on the represented population.

9.6.3 Access, permissions, licensing and conditions for reuse

Technical availability and permission to reuse are different questions. A reachable repository does not by itself describe the rights associated with its records, third-party material or accompanying software. Dataset documentation should state known access conditions, permitted uses, source terms and any unresolved restrictions (Gebru et al. 2021).

Physical collection can include people in images, speech, operator identities and private surroundings. Decide what information is necessary for the intended use and which information should remain restricted. A restricted source can still have a public description of its structure or an independently constructed example, provided those distinctions are accurately stated.

Rights information also has lineage. A derived dataset can combine sources with different access conditions; processing them does not automatically supply new permission to redistribute the originals. Record the relevant source and review decisions rather than treating a format conversion as a change of rights.

For scientific reuse, state more than an access address. Describe the intended use, target conditions, attempted and retained populations, modality meanings, label derivation, known missingness, partition protocol and processing versions. These details allow another reader to judge whether the data fit a new learning question. They do not guarantee that the data are suitable for every future purpose.

The numerical companion illustrates a compact part of this discipline. It links collection weights, attempt counts, physical transforms, uncertain labels, planned versus realized splits and typed source lineage. Its records are deliberately constructed, so their arithmetic can be checked without treating them as physical evidence. An actual collection needs the corresponding observations and documented processes, not merely the same field names.

For ExpGen PhySyst, a useful collection is an organized account of generated experience. Mechanics establishes what the quantities mean; acquisition and annotation establish what is known; sampling and selection establish what is represented; provenance preserves how those interpretations arose. Designing data connects all four to the embodied learning problem.

Exercises and further investigation. State the intended population, physical quantities, collection unit and information boundary. Distinguish constructed arithmetic from empirical evidence.

1. Define three uses for the same pusher record: motion prediction, successful-behavior learning and effort reconstruction. Specify which observations and population each requires, and one unsupported claim.
2. Derive the episode- and frame-weighted averages. Construct a case in which a longer failure changes a pooled average without changing the number of attempted episodes.
3. Design a collection budget that includes resets, calibration, annotation and unsuccessful attempts. Compare a fixed-attempt plan with a success-quota stopping rule.
4. Specify an episode's observation/action alignment, physical acquisition intervals and end causes. Explain why a terminal flag, task success and an incomplete recording are different.
5. Reproduce the spatial-transform and power example. Include an uncertain mounting and describe which errors are shared across episodes.
6. Define an operational manipulation-success rule with pose, speed and dwell criteria. Identify which sensor failures make the label unknown and what a human annotator can actually observe.
7. Reproduce the Gaussian threshold example. State which assumptions justify its posterior probability, and why a vote fraction or instrument resolution does not provide the same quantity.
8. Reconstruct the 48-attempt ledger and its outcome bounds. Propose different defensible subsets without deleting the original attempt identities or implying that unknown outcomes are failures.
9. Derive the retention law and context-weighting identity. Give a success-dependent selection for which context weights alone do not recover the intended outcome distribution.
10. Derive the balanced-cluster mean variance. Compare extra frames, extra episodes and extra devices; explain why neither effective-count formula in the chapter is a universal learning-data metric.
11. Reproduce both planned and realized partitions. Add history/future windows and identify their raw support. Design a held-out-configuration protocol and state what it does not test.
12. Construct a lineage for raw observations, calibration, pose estimates, success labels and normalized learning features. Include one legitimate shared dependency and two kinds of leakage. Explain what the graph and hashes cannot verify.

Gebru et al. (2021) provides lifecycle questions for dataset documentation. Hastie et al. (2009) supplies broader statistical learning and model-assessment foundations. Hurlbert (1984) clarifies replication and subsampling, while Aroyo and Welty (2015) examines information in annotation disagreement. The RLDS episode representation and PROV-DM lineage model offer useful vocabulary without replacing physical and statistical interpretation (Ramos et al. 2021; Moreau and Missier 2013).

Chapter 10 turns to learning feedback policies and generating experience through their actions. The collection foundations here determine what those policies observe, which experiences are retained and what can be concluded from the resulting data.

Chapter 10 Learning Feedback Policies

A hand can turn a cube repeatedly without ever holding a requested orientation. Another controller can bring a particular face upward and then stop. Both behaviors involve contact, motion and feedback, but they do not solve the same problem. The distinction begins with the objective, the observations and the meaning of an action. A learning algorithm does not supply those meanings automatically.

Chapters 3–8 established physical dynamics, actuation, simulation, measurement and identification. Chapter 9 asked which experiences a collection represents. Here the controller itself becomes a learned component of the arrangement that generates experience. Its actions change the states visited, the contacts encountered and the evidence available for its next update. Learning is therefore both a use of data and a cause of future data.

The chapter develops sequential objectives, value learning, policy gradients and a practical method for policy optimization. Throughout, the learned law remains connected to Featherstone mechanics through a declared observation and action interface. A small joint and a cube-orientation example make the equations inspectable. All numerical examples are constructed and checked by an arithmetic companion; they are not evidence of a trained robot policy or of hardware transfer.

10.1 From optimal control to sequential decisions

Feedback selects an action using information available during execution. An open-loop sequence chooses the requests in advance; a feedback policy can respond when the observed interaction changes. Learning a feedback law means adjusting this selection rule from experience, under a stated objective and information boundary.

10.1.1 Finite-horizon objectives, trajectories and dynamic programming

For an independent fixed-base joint coordinate set, retain the balance from Chapter 5:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{h}(\mathbf{q}, \dot{\mathbf{q}}) = \boldsymbol{\tau}_a + \boldsymbol{\tau}_p + \sum_i \mathbf{J}_i^{\mathsf{T}} \mathbf{f}_i^{\text{ext}}. \tag{10.1}$$

The rigid-body bias $\mathbf{h}$ includes the declared velocity and gravity terms; passive effects $\boldsymbol{\tau}_p$ are not counted twice. Body wrenches are moment-first, body motions angular-first, and their frames and origins must agree with the Jacobians. General joints use tangent velocity $\boldsymbol{\nu}$ with $\dot{\mathbf{q}} = \mathcal{N}(\mathbf{q})\boldsymbol{\nu}$ rather than treating quaternion entries as independent joint coordinates (Featherstone 2008).

The policy does not replace this balance. Let s_k collect the state needed to predict the modeled evolution: configuration, tangent velocity, actuator states and any other relevant dynamic variables. Let a_k be the policy action. An interface maps that action and the current physical state to actuator requests and delivered effort. Integration, contact and the interface together induce a discrete transition

$$s_{k+1} = F_\Delta(s_k, a_k), \qquad a_k = \pi_k(s_k), \tag{10.2}$$

for the deterministic fully observed case. Δ is the control interval, which may contain several engine steps. A neural network changes the choice of action, not the units or physical identity of the resulting effort. Contact constraints may be resolved implicitly inside F_Δ.

For a horizon of H intervals, a deterministic optimal-control problem is

$$\min_{\pi_0,\ldots,\pi_{H-1}} \left[\sum_{k=0}^{H-1} c_k(s_k, a_k) + g(s_H) \right], \qquad s_{k+1} = F_\Delta(s_k, a_k). \tag{10.3}$$

Costs may penalize tracking error, effort, excessive contact or task failure. Their scales must be defined. Adding a squared position error in meters to a squared effort in newton meters requires weights with the appropriate units, or explicit normalization by reference scales. An effort penalty is not necessarily energy: mechanical power is $\boldsymbol{\tau}_a^{\mathsf{T}}\boldsymbol{\nu}$ for compatible coordinates. Neither a reward penalty nor a sampled limit proves a hardware constraint.

Dynamic programming separates the best future cost from the current decision. Define the optimal cost-to-go W_k by

$$W_H(s) = g(s), \qquad W_k(s) = \min_{a \in \mathcal{A}_k(s)} \left[c_k(s,a) + W_{k+1}(F_\Delta(s,a)) \right]. \tag{10.4}$$

This follows by fixing the first action and optimizing the remaining decisions. It is an identity about the stated problem, not a claim that the minimization is easy. Nonlinear contacts, high-dimensional states and partial observation can make direct solution expensive. Approximate value functions and policies are ways to represent the result, with their own approximation and data requirements (Sutton and Barto 2018).

Worked example: mechanics determines the transition used by control. Consider one ideal revolute joint, with no gravity, friction or other external load and with constant delivered torque during each interval, inertia $J = 0.02$ kg m^2 and $\Delta = 0.1$ s. Write $\tau = 0.1a$ N m, and normalize position by 1 rad and velocity by 1 rad/s. Exact integration gives dimensionless state $\mathbf{x}$ with

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{B}a_k, \qquad \mathbf{A} = \begin{bmatrix} 1 & 0.1 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0.025 \\ 0.5 \end{bmatrix}. \tag{10.5}$$

The position coefficient includes $\Delta^2/(2J)$ and the velocity coefficient includes Δ/J, with the effort and state scales applied. They would differ for an embedded position servo or a different hold interval. This is the single-joint specialization of the mechanical balance, not an arbitrary data-driven transition matrix.

Use stage cost $\mathbf{x}^\mathsf{T}\mathbf{Q}\mathbf{x} + Ra^2$, $\mathbf{Q} = \operatorname{diag}(1, 0.1)$, $R = 0.05$, and terminal cost $\mathbf{x}^\mathsf{T}\mathbf{P}_H\mathbf{x}$ with $\mathbf{P}_H = \operatorname{diag}(4, 0.2)$. Substitution of a quadratic cost-to-go into Equation (10.4) and completion of the square give

$$\begin{aligned} \mathbf{K}_k &= (R + \mathbf{B}^\mathsf{T}\mathbf{P}_{k+1}\mathbf{B})^{-1}\mathbf{B}^\mathsf{T}\mathbf{P}_{k+1}\mathbf{A}, \\ \mathbf{P}_k &= \mathbf{Q} + \mathbf{A}^\mathsf{T}\mathbf{P}_{k+1}\mathbf{A} - \mathbf{A}^\mathsf{T}\mathbf{P}_{k+1}\mathbf{B}\mathbf{K}_k, \qquad a_k = -\mathbf{K}_k\mathbf{x}_k. \end{aligned} \tag{10.6}$$

Here the inverse is scalar; numerical implementations solve the corresponding linear system in the multiaction case. The arithmetic companion solves the two-interval problem, checks the rollout cost against $\mathbf{x}_0^\mathsf{T}\mathbf{P}_0\mathbf{x}_0$, and perturbs the chosen actions to check the minimum. For $H = 2$ and $\mathbf{x}_0 = (0.4, -0.2)^\mathsf{T}$, the gains are

$$\mathbf{K}_0 \approx (2.320747, 1.252993), \qquad \mathbf{K}_1 \approx (0.975610, 1.073171). \tag{10.7}$$

The actions are approximately $(-0.677700, 0.224076)$, with total cost 0.783086. The companion also solves the two-action quadratic problem directly, independently of the backward recursion, and checks its action gradient by finite differences. This is an unconstrained calculation; adding actuator limits changes the optimization and may invalidate this linear law. It is a transparent reference for what a learner is trying to approximate, rather than a prescription to use quadratic costs for every manipulation task.

10.1.2 Stochastic transitions and partial observability

Repeated requests need not give identical trajectories. Initial conditions, unmodeled interactions and sensor noise can vary. Write the transition kernel $p(ds' \mid s, a)$ and observation law $p(do \mid s)$ for a Markov model. Markov means that the declared state, together with the action, supplies the information needed for the next-state law. It does not mean that one camera frame or one joint-angle vector is automatically such a state.

Let r_{k+1} denote reward assigned to the transition from s_k under a_k to s_{k+1}. Its subscript makes the alignment explicit. For a policy π_θ and initial distribution ρ_0, define

$$\mathcal{J}(\theta) = \mathbb{E}_{\rho_0, p, \pi_\theta}\left[\sum_{k=0}^{H-1} \gamma^k r_{k+1} \right], \qquad 0 < \gamma \leq 1. \tag{10.8}$$

Reward maximization and cost minimization are compatible through a sign change, but terminal costs, discount and physical horizon must be matched. For finite-horizon problems the policy and value can depend on time or remaining duration. For an infinite discounted objective, convergence of returns requires suitable reward bounds or other integrability conditions.

A partially observed policy instead uses history $\mathcal{H}_k = (o_0, a_0, \ldots, a_{k-1}, o_k)$ or an internal memory derived from it. Two joints at the same angle but opposite velocity generally need different braking actions. A policy given angle alone

cannot distinguish them. A history may supply a velocity estimate, but cannot reveal every unmeasured contact or actuator state without sufficient information. Chapter 7 provides the estimation foundations.

Known dynamics under partial observation can be expressed on a belief distribution over state. A recurrent neural policy is a different, learned representation of history; it is not automatically a calibrated belief filter. A simulator may expose full state to a training value function while the actor receives deployable observations. This can help training, but the executed actor must retain its stated information boundary. Privileged simulator information cannot silently become a required physical sensor.

The data-design consequence is immediate. Record which observation the policy actually received, not just a richer state available elsewhere. Preserve observation delays, memory resets and randomization context. Otherwise the collected actions may appear inconsistent with their recorded inputs even when the policy behaved consistently with the information it really used.

10.2 Learning values and policies

Optimal control presumes a model or a way to evaluate consequences. Reinforcement learning develops value estimates or action-selection rules from interaction. Model-based and model-free methods differ in how they represent and use predictions; neither label removes the physical process that generated the transitions. This section introduces the relations needed to interpret common updates, rather than treating an optimizer as a complete experimental method.

10.2.1 Bellman relations and temporal-difference learning

For a fixed policy, the finite-horizon return from step k is $G_k = \sum_{j=k}^{H-1} \gamma^{j-k} r_{j+1}$. Define $V_k^\pi(s) = \mathbb{E}_\pi[G_k \mid s_k = s]$ and $Q_k^\pi(s,a) = \mathbb{E}_\pi[G_k \mid s_k = s, a_k = a]$. Conditioning on the next transition gives

$$\begin{aligned} Q_k^\pi(s,a) &= \mathbb{E}[r_{k+1} + \gamma V_{k+1}^\pi(s_{k+1}) \mid s, a], \\ V_k^\pi(s) &= \mathbb{E}_{a\sim\pi_k(\cdot|s)}[Q_k^\pi(s,a)], \qquad V_H^\pi = 0. \end{aligned} \tag{10.9}$$

A separately specified terminal payoff can be included in the final reward or in the boundary value, with its discount handled consistently. Optimal values replace the policy average with a maximizing choice. The same recursive structure appeared in optimal control; learning uses samples to estimate quantities that may not be available analytically.

For a continuing discounted task, a one-step temporal-difference residual is

$$\delta_k = r_{k+1} + \gamma b_k V_\phi(s_{k+1}) - V_\phi(s_k), \qquad b_k = 1 - \mathbf{1}\{\text{true terminal transition}\}. \tag{10.10}$$

The next value is omitted at a true terminal state. An external collection cutoff can still require it; the precise distinction is developed below. For a tabular on-policy estimate, a basic update adds $\alpha\delta_k$ to the current state's value. A fitted critic instead optimizes a loss against chosen targets. The bootstrapped value is an estimate, so temporal-difference learning is not ordinary regression against an independently measured return (Sutton and Barto 2018).

The policy that generated a transition matters. A record from another policy is not automatically a sample of the current policy's action law. Some methods use importance correction, some learn an action value from which a new policy is formed, and some impose coverage restrictions. The Bellman identity alone does not guarantee convergence of a particular function approximator, sampling process and update combination.

Reward must also preserve the intended physical question. A dense position reward can supply a learning signal before success, but may favor hovering near a target over completing a dwell-and-release criterion. An energy proxy based on squared action can change when action scaling changes even if mechanical work is unchanged. Keep task outcome, reward components and physical diagnostics separately observable. A high return cannot explain which behavior earned it when only the sum was stored.

One principled shaping construction adds a potential difference $F_k = \gamma\Phi(s_{k+1}) - \Phi(s_k)$ to reward (Ng et al. 1999). Over a finite trajectory the added return is

$$\sum_{k=0}^{H-1} \gamma^k F_k = -\Phi(s_0) + \gamma^H \Phi(s_H). \tag{10.11}$$

The intermediate terms cancel. With a fixed initial condition and appropriately treated terminal potential, such as zero terminal potential, the remaining change does not favor one action sequence over another. For a continuing problem with $0 < \gamma < 1$, a bounded potential makes the distant boundary vanish. An arbitrary dense reward need not have this form; leaving an action-dependent terminal potential changes the conclusion. The identity makes the assumptions inspectable rather than calling any helpful intermediate reward task-preserving.

10.2.2 Policy gradients, advantages and actor–critic updates

For a differentiable stochastic policy, trajectory probability contains factors for initial state, policy choices and physical transitions. If the initial law, transition law and reward definition do not depend directly on θ, the derivative of its log probability contains only policy-score terms. Using causality to discard rewards earned before an action gives

$$\nabla_\theta \mathcal{J} = \mathbb{E}\left[\sum_{k=0}^{H-1} \gamma^k \nabla_\theta \log \pi_\theta(a_k \mid s_k, k)\, G_k\right]. \tag{10.12}$$

This score-function derivation does not require differentiating a contact solver. It does require sampling and expectation conditions that justify the identity. Differentiable dynamics provide another route to derivatives, with the numerical and contact limitations discussed in Chapters 6 and 8.

An action-independent baseline $B_k(s_k)$ can be subtracted from G_k without changing the expected score term, because

$$\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}[\nabla_\theta \log \pi_\theta(a \mid s) B(s)] = B(s)\nabla_\theta \int \pi_\theta(a \mid s)\, da = 0. \tag{10.13}$$

Here the baseline is held fixed in the policy update; differentiating a learned baseline through this expression would add a different term. Subtracting a value baseline motivates the advantage $A_k^\pi = Q_k^\pi - V_k^\pi$. Positive advantage means better than the policy's expected continuation under that definition; it is not a separate measure of hardware competence or task success.

Worked example: a score derivative is not a derivative through a sampled action. Let $a \in \{0, 1\}$, reward equal a, and $p = \Pr(a = 1) = 1/(1 + e^{-\theta})$. Then $\mathcal{J} = p$ and $\nabla_\theta \mathcal{J} = p(1 - p)$. The log-policy score is $a - p$. At $p = 0.25$,

$$\mathbb{E}[(a - p)a] = 0.1875 = \mathbb{E}[(a - p)(a - 0.25)]. \tag{10.14}$$

The companion enumerates both actions and checks the derivative by finite differences. Differentiating the realized binary sample as if it were a smooth function of θ would miss this argument. A finite sample estimate has variability even though its expectation has the stated value.

Actor–critic methods use a learned value function to form lower-variance policy-update targets. Generalized advantage estimation, GAE, combines temporal-difference residuals with a trace parameter λ (Schulman et al. 2016). For a finite collected segment, one useful backward implementation is

$$\widehat{A}_k = \delta_k + \gamma\lambda c_k \widehat{A}_{k+1}, \qquad \widehat{A}_{\text{after segment}} = 0. \tag{10.15}$$

The bootstrap mask b_k in Equation (10.10) and the trace mask c_k have different roles. Set $c_k = 0$ when the next stored transition belongs to another episode, or when the collected segment ends. At an external cutoff the critic may still bootstrap from the final observation, even though the trace cannot continue into the reset episode. At a true terminal transition both continuation and cross-episode trace stop. For a genuine finite-horizon task, time and its terminal boundary belong to the task definition.

With $\lambda = 1$, summing the residuals telescopes into a bootstrapped multistep return minus the current value. Smaller λ changes the bias–variance tradeoff when the critic is approximate. An actor–critic update is therefore affected by reward, horizon, observation information, critic error and trace choices, not just the actor network.

Worked example: one segment, two possible end meanings. Consider three transitions, with rewards $(0.2, 0.4, 0.8)$. Let the four state-value estimates be $(1, 1.1, 1.3, 1.5)$, with $\gamma = 0.9$ and $\lambda = 0.8$. If the final state is a nonterminal external cutoff, the residuals are $(0.19, 0.47, 0.85)$ and the backward advantages are $(0.96904, 1.082, 0.85)$. If the same last reward instead leads to a true terminal state, the last residual is -0.5 and the advantages become $(0.2692, 0.11, -0.5)$.

The difference is caused by the task boundary, not an arithmetic preference. The companion also checks the $\lambda = 1$ forward-return identity. A reset observation must never replace the final pre-reset observation in this calculation.

The companion's `gae` function implements the same backward recursion. The following schematic makes the two boundaries explicit; `boundary[k]` is true at each episode or collected-segment end:

```
tail = 0.0
for k in reversed(range(segment_length)):
    next_v = 0.0 if terminal[k] else V(final_obs[k])
    delta = reward[k] + gamma * next_v - V(obs[k])
    tail = delta + gamma * lam * (not boundary[k]) * tail
    advantage[k] = tail
```

For a history-dependent critic, both value calls require the corresponding history, not just the displayed observation. These calculations consume the recorded end meanings; they cannot determine those meanings from the reward or buffer position alone.

10.3 Generating experience through exploration

The policy determines more than which action looks favorable at a visited state. It changes which states will be visited. Learning can improve a controller while narrowing the behavior represented in its newest data. Collection design must therefore describe both the desired behavior and how the system will encounter informative alternatives.

10.3.1 Policy-dependent coverage, exploration and resets

For a stationary Markov policy, let $d_k^\pi(s)$ denote the state law at step k. Its evolution is

$$d_{k+1}^\pi(s') = \int d_k^\pi(s) \int \pi(da \mid s)\, p(ds' \mid s, a)\, ds. \tag{10.16}$$

The notation is schematic for densities; sums replace integrals for discrete states. The occupancy law depends on dynamics, action interface, policy and initial distribution. A replay buffer full of successful motions need not cover recovery from slip or a distant initial pose. No update can directly learn an unobserved consequence solely by reweighting records with zero support for it.

Exploration can alter action choices, goals, starting states or apparatus conditions. Gaussian action noise is one choice, not a definition of informative exploration. After a saturated actuator map, many distinct latent samples may produce the same delivered request. During sustained contact, independent random joint requests may cause little useful object motion. More action entropy is therefore not automatically more physical coverage.

Reset design changes the experience population. Starting a cube in a stable grasp can make early learning feasible, while excluding grasp acquisition and recovery from a dropped object. Starting near a goal can teach local regulation without teaching how to reach the goal from a distant orientation. These can be legitimate component tasks if the boundary is explicit. They should not later be described as broader capabilities without additional evidence.

A useful investigation asks which missing experience would distinguish the plausible failure explanations. Is the policy missing contact-state information, unable to exert sufficient effort, poorly optimized, or rarely exposed to a difficult initial condition? More transitions under the same restricted policy may not answer that question. Record reset distributions, intervention rules and exploration settings alongside the trajectories they generate.

10.3.2 On-policy and off-policy records, termination and truncation

On-policy records are generated under the policy whose behavior is being estimated or improved, subject to the method's exact conventions. Off-policy methods use experience from other behavior policies. Reuse can improve data efficiency, but introduces a distribution mismatch. For one recorded action, the ratio

$$w_k = \frac{\pi_\theta(a_k \mid s_k)}{\mu(a_k \mid s_k)} \tag{10.17}$$

is defined only where the behavior law μ provides positive probability or density. Correcting a full trajectory generally involves more than this one ratio, and products can have high variance. Many algorithms use different approximate corrections or objectives. A replayed transition does not become on-policy merely because its reward was recomputed.

Preserve behavior-policy identity, action encoding and, where the method requires them, log probabilities at collection time. These probabilities refer to a defined random variable. The density of a latent Gaussian sample is not the ordinary density of a hard-clipped action, which has point masses at its limits. One can formulate the policy on the latent variable and include the fixed clipping map in the environment interface, but then the stored sample and update must use that formulation consistently. For an invertible squashing transform, densities of the transformed variable require its change-of-variables term.

End causes are equally important. A terminal state belongs to the task's specified continuation model. A rollout segment may end only because a collector needs to update its policy. A time limit may be either a true finite-horizon task boundary or an external limit on a continuing task. These cases need different targets (Pardo et al. 2018).

For a genuine finite-horizon task, remaining time generally belongs in the Markov state and value function, and continuation beyond the task horizon is absent. For an external cutoff of a continuing objective, preserve the final pre-reset observation and bootstrap its continuation value. Never carry a GAE trace into the next reset trajectory. An incomplete or corrupted recording is a separate data-validity problem; its final value cannot be inferred by assigning either flag blindly.

The dataset principles of Chapter 9 now become algorithmic requirements. A row needs its reward alignment, action meaning, next observation, actual interval, episode identity and end cause. When vectorized environments reset automatically, the returned next observation may already belong to a new episode. Collection code must preserve the transition's own final observation rather than pairing the reward with the reset state.

10.4 Learning in a physical simulator

Simulation offers repeated interaction and controllable conditions, but learning optimizes the implemented problem. The contact approximation, actuator model, observation timing, reset mechanism and reward computation all contribute to the experience. Faster computation increases the available budget; it does not establish that this experience matches a physical deployment.

10.4.1 PPO as a practical policy-optimization method

Proximal policy optimization, PPO, is a family of practical policy updates that reuse a recent rollout for several optimization passes (Schulman et al. 2017). For its clipped surrogate, let π_{old} be the policy that collected the batch and define $\varrho_k(\theta) = \pi_\theta(a_k \mid s_k)/\pi_{\text{old}}(a_k \mid s_k)$. With advantage estimates held fixed during an actor update,

$$L^{\text{clip}}(\theta) = \widehat{\mathbb{E}}_k\left[\min\left(\varrho_k \widehat{A}_k, \text{clip}(\varrho_k, 1-\epsilon, 1+\epsilon)\widehat{A}_k\right)\right]. \tag{10.18}$$

This objective is maximized. Its sign-dependent minimum removes an incentive to move the sampled ratio further in an already favorable direction. It does not constrain every new probability ratio to the clip interval, enforce a global trust region or guarantee improved return. Unsampled actions and parameter sharing can still change substantially.

Worked example: the negative-advantage case changes the clipping effect. For $\epsilon = 0.2$ and $\widehat{A} = 2$, a ratio 1.4 contributes $\min(2.8, 2.4) = 2.4$. For $\widehat{A} = -2$, a ratio 0.6 contributes $\min(-1.2, -1.6) = -1.6$. A ratio 1.4 with that negative advantage still contributes -2.8, retaining pressure against the unfavorable change. The companion checks both advantage signs and both ratios. Clipping the action itself would be a different operation with different physical and probability meanings.

A practical iteration collects transitions, constructs bootstrap targets and advantages, then optimizes actor and critic losses in minibatches. A value loss fits the critic to a declared return target; an entropy term may encourage exploration. Their coefficients, normalization conventions, optimizer settings and stopping rules affect the update. Advantage normalization can change the finite-batch surrogate, and the value loss need not share the actor's network parameters.

The original discounted policy-gradient expression includes the weighting appropriate to its stated objective. Common PPO implementations average surrogate terms over collected timesteps and use discounted advantage estimates. Their exact sampling and weighting conventions should be recorded; one should not silently identify every practical surrogate average with the exact finite-horizon gradient in Equation (10.12).

For a recurrent policy, minibatches also need compatible history, episode masks and hidden-state handling. Shuffling individual rows while forgetting their history changes the actor's input. For any policy, old log probabilities must correspond to the action variable, observation preprocessing and policy version used during collection. Changing normalization halfway through a ratio calculation can compare two different input representations rather than only two policies.

PPO supplies an update rule, not a complete reproducible training system. Reward, physical model, observation layout, action map, rollout length, batch size, update epochs and random seeds are part of the experiment. Matching the algorithm's name while changing these components is an adaptation, not an exact reproduction.

10.4.2 Randomization, curricula and computational budgets

Randomization varies conditions under a declared distribution: mass, friction, actuation, timing, observations or initial state. Some quantities are sampled once per episode; others evolve within an episode. Resampling a parameter each engine step can describe a different physical process from a fixed uncertain apparatus. Correlations matter: a payload changes both geometry and inertia, and actuator strength and delay may not vary independently.

An actor need not observe the randomization parameters. In that case, history may help infer their effect, but also makes the learning problem partially observed. Dynamics randomization has been used with relative position actions, low-level control and recurrent policies in manipulation (Peng et al. 2018). This is evidence for a concrete method, not a guarantee that any broad parameter range will produce transfer. The specified model family must still include relevant mechanisms, and training must make the needed behavior learnable.

A curriculum changes the experience distribution over training (Bengio et al. 2009). It may begin with supported objects, small goal changes or easier disturbances, then broaden the conditions. The schedule should state what changes, when it changes and which evidence triggers progression. A policy that masters an earlier stage has not thereby passed the final task. Replacing the task, controller or observation interface between stages may require more than continuing from the same network weights.

For ExpGen PhySyst, randomization and curricula are designs for generating experience. Their value lies in the information and behavior they make available, not simply the number of varied parameters. Retain the sampled context and stage identity in evaluation records, even when those fields are hidden from the actor. A fixed held-out evaluation distribution helps distinguish progress from an easier curriculum stage.

Worked example: transitions, updates and physical duration are different budgets. Suppose 16 parallel environments each collect 100 control transitions, with five engine steps of 0.004 s per action. The batch contains 1,600 transitions and 32 summed simulated seconds, although each environment has advanced only 2 s. Four optimizer epochs with minibatches of 400 produce 16 minibatch updates. They do not produce 6,400 new physical transitions. Real wall time depends on compilation, simulation, data movement, optimization and evaluation overhead; no speed result follows from these counts alone.

Report engine steps, control transitions, simulated duration, independent environment instances, optimizer passes and wall time separately. Changing action repeat changes the action interval and potentially the objective. A small-batch run with the same total transitions can have different gradient noise and update dynamics from a large-batch reference. The budget must be interpreted together with the experiment configuration.

10.5 Executing the learned feedback law

Training produces parameters for a particular implemented feedback law. Execution requires reconstructing that law's inputs, memory, timing and action map. A file with compatible tensor dimensions is not sufficient. The physical meaning of each channel must survive export and playback.

10.5.1 Observation history, normalization and action semantics

A deployable policy contract identifies observation components, order, units, coordinate frames, reference points, timestamps and preprocessing. It also identifies previous-action inputs, history length and hidden-state reset rules. A model that expects relative orientation cannot be given absolute quaternion components merely because both describe orientation. Quaternion ordering and the double representation of one rotation must be handled consistently.

For a normalized scalar input,

$$z_i = \frac{o_i - \mu_i}{\sqrt{\sigma_i^2 + \varepsilon_n}}, \tag{10.19}$$

the statistics, their fitting population, stabilizer and any clipping belong to the policy. Raw o_i, mean μ_i and standard deviation σ_i must share units; ε_n has squared-input units in this expression. A library may instead operate on already dimensionless features. State that convention rather than adding a unitless constant to an unexplained physical variance.

A bounded action can map to torque, velocity, absolute position, a relative target or a residual added to another controller. The reference point for a relative target matters: current measured position, previous target and a nominal pose give different recurrences. The actuator's own loop then changes requested effort and actual motion.

Worked example: the same action can deliver twice the effort. At $q = 0.3$ rad and $\dot{q} = 0.4$ rad/s, let policy output $a = 0.5$. One interface requests torque $0.12a$ N m, giving 0.06 N m. Another requests position $q^{\mathrm{ref}} = 0.3 + 0.2a = 0.4$ rad and uses $K_p = 2$ N m/rad, $K_d = 0.1$ N m s/rad. Its effort is

$$\tau^{\mathrm{req}} = K_p(q^{\mathrm{ref}} - q) - K_d\dot{q} = 0.16\,\mathrm{N\,m}, \qquad \tau_a = \mathrm{clip}(\tau^{\mathrm{req}}, -0.12, 0.12) = 0.12\,\mathrm{N\,m}. \tag{10.20}$$

For the ideal $J = 0.02$ kg m^2 joint with no gravity, friction or other external load, initial acceleration is 3 rad/s^2 in the first interface and 6 rad/s^2 in the second. Initial mechanical power is 0.024 W versus 0.048 W. These are ideal instant values, not servo performance measurements. The reference example uses a declared fixed anchor of 0.3 rad at this decision; recursively integrating target increments would be another law.

In MuJoCo, the meanings of `ctrl`, actuator transmission, gains, bias terms and force limits jointly determine generalized actuator effort. One control scalar does not by itself identify a joint torque. Likewise, position and tangent-velocity arrays need not have identical lengths for free or ball joints. The mappings established in Chapters 3, 5 and 6 remain part of learned-policy execution (DeepMind Technologies Limited, n.d., n.d.).

Useful playback records include raw observation, processed input, policy output, mapped request, delivered or estimated effort, limits, task state and timing. A physical effort estimate must retain its uncertainty and source. These channels distinguish an inappropriate action from a correct request that the apparatus could not realize.

10.5.2 Control frequency, limits and policy compatibility

Control frequency is part of the learned dynamics. Holding one torque for twice as long changes momentum by twice the impulse under the ideal constant-effort model. Holding a position target longer changes the interaction with an embedded loop. A different engine timestep with the same physical action interval can still alter numerical contact, as Chapter 6 explained.

Reward and discount also carry time semantics. If ℓ is a physical cost rate, an interval cost can approximate $\Delta\ell$. A per-step event bonus, by contrast, may intentionally have no duration factor. For continuous discount rate β in s^{-1},

$$\gamma(\Delta) = e^{-\beta\Delta}, \qquad \beta = -\frac{\log\gamma(\Delta)}{\Delta}. \tag{10.21}$$

Changing frequency while leaving every reward and discount unchanged can change the optimization problem, even before considering dynamics.

Worked example: a numerical discount has a physical timescale. With $\gamma = 0.99$ at $\Delta = 0.02$ s, the one-second discount is $0.99^{50} \approx 0.6050$. Keeping 0.99 at $\Delta = 0.05$ s instead gives $0.99^{20} \approx 0.8179$. Matching the original exponential timescale requires $\gamma' = 0.99^{2.5} \approx 0.97519$. This aligns one aspect of the objective; it does not make two differently sampled physical controllers equivalent.

Limits must identify where they act: actor output, target range, target rate, actuator force or mechanism constraint. A position request within a joint range can still generate excessive effort or contact load. A torque-limited controller can still accumulate dangerous momentum. Simulation tests can reveal discrepancies, but reward penalties and policy performance do not certify an actual mechanism's operation.

Compatibility testing should compare the same defined feedback law under declared initial conditions and observations. Begin with short deterministic traces of preprocessing and action mapping, then test physical rollouts and task outcomes. If a policy works in one engine implementation but not another, distinguish observation/interface differences, dynamics and numerical differences, and the policy's own limitations. Retraining is one possible response after the discrepancy is understood; it does not explain the discrepancy by itself.

10.6 Evaluating what the experience produced

Evaluation asks what behavior resulted under which conditions. A learning curve is evidence about an optimization experiment. A task-level test is evidence about an executed policy. An exported model and a video are useful artifacts, but neither substitutes for the relevant collection and evaluation records.

10.6.1 Learning curves, training data budgets and behavioral diagnostics

Plot or report return against a defined budget: environment transitions, physical interaction duration or wall time. The axis must distinguish new experience from optimizer reuse. Report separate evaluation episodes when using them to assess capability; training return can reflect exploration, reward normalization or a changing curriculum.

An episode's reward sum, discounted return and task success are different quantities. A policy can improve shaped return while dropping an object more often, or increase first-goal success while failing repeated goals. Inspect reward components, contacts, action saturation, stalls, resets, goal errors and episode end causes. State success and failure criteria before inspecting the results.

Random seeds change initialization, sampled conditions and optimization. Repeated runs can therefore reveal variation not visible in many evaluation episodes of one selected model. Conversely, repeating seeds under one narrow apparatus model does not establish transfer to another mechanism. Choose evaluation units and held-out conditions in accordance with the claims, using Chapter 9's collection principles. Chapter 16 develops broader comparative inference.

Keep model-selection evidence distinct from final evaluation. Selecting a checkpoint, reward weight or curriculum after repeated inspection of the same test conditions makes those conditions development information. A final held-out result should identify the selected policy, training budget, observation/action contract and evaluation protocol. A percentage without the number of trials and their task conditions is incomplete.

For data design, failures are diagnostic interactions. A rotation that stalls after contact, a drop at a particular pose and an action that remains saturated can suggest different next experiments. Their recordings can support recovery learning or reveal missing model mechanisms. Retaining only successful videos would remove the information needed to distinguish those explanations.

10.6.2 Continuous rotation and goal-conditioned orientation as distinct tasks

For an object orientation $\mathbf{R} \in SO(3)$ and desired orientation $\mathbf{R}_g$, a full-orientation error angle is

$$d_R = \cos^{-1}\left(\operatorname{clip}\left(\frac{\operatorname{tr}(\mathbf{R}_g^{\mathsf{T}}\mathbf{R}) - 1}{2}, -1, 1\right)\right), \qquad 0 \le d_R \le \pi. \tag{10.22}$$

The rotation matrices map the same object frame into the same world frame. Clipping protects against small numerical excursions of the cosine argument; it does not repair an inconsistent frame convention. Equivalent quaternion representations must yield the same physical error.

If the task only asks for one labeled face upward, let $\mathbf{n}_f$ be that face's unit normal in object coordinates and $\mathbf{z}_W$ the

upward world unit vector. Its directional error is

$$d_f = \cos^{-1}\big(\mathrm{clip}(\mathbf{z}_W^{\mathsf{T}}\mathbf{R}\mathbf{n}_f, -1, 1)\big)\,. \tag{10.23}$$

This does not constrain yaw about the upward normal. By contrast, an objective based on angular velocity about a declared axis can reward continued rotation. The angular velocity must be expressed in the same frame as that axis. Integrating its components is not generally a valid replacement for composing three-dimensional rotations.

Worked example: two face-up poses can have different full orientations. Take $\mathbf{R} = R_z(\pi/2)$, $\mathbf{R}_g = R_z(\pi)$ and $\mathbf{n}_f = \mathbf{z}_W = (0, 0, 1)^{\mathsf{T}}$. Here the body frame is distinguished, for example by face markings. The face-up error is zero, while the full-orientation error is $\pi/2$. An unmarked cube considered only up to geometric symmetries would require a different orientation criterion that minimizes over those symmetries. A positive angular velocity about world z may earn a continuous-spin reward even after passing the goal. A stationary successful orientation would earn no such velocity reward. Thus these objectives are not interchangeable descriptions of one benchmark.

A goal-conditioned policy receives an explicit representation of the goal, or an equivalent relative error, in addition to its other observations. If the goal changes while the actor's inputs do not, the actor cannot in general select a different action for the new goal. Adding letters to an otherwise physically matched cube can change the instruction and face mapping without changing rigid-body dynamics; a vision pipeline may still need to recognize the new appearance. A different cube's mass, friction or geometry changes the physical task as well.

For a holding task, define angular tolerance, velocity tolerance and dwell duration. For repeated reorientation, define how goals change and how drops or recovery are counted. For continuous rotation, define axis, direction, accumulated progress and allowed failures. State which capability an existing policy was trained and evaluated to perform before reusing it as an example.

ExpGen PhySyst makes these choices visible as a design of experience. Mechanics determines feasible interaction; sensing determines available information; the action interface determines how decisions reach the apparatus; reward and resets determine which experiences are favored; the learning update changes the generator. A useful policy experiment preserves these relationships well enough to explain both success and failure, and to design the next informative interaction.

Exercises and further investigation. State the task, physical units, information boundary and horizon in each answer. Distinguish constructed calculations from empirical evidence.

1. Derive the exact joint transition from constant delivered torque. Recompute it for another control interval and for a different torque scale. Explain why the same derivation does not describe a position servo.
2. Complete the two-interval Riccati calculation. Compare the optimal rollout cost with the value function, then impose an action bound and determine whether the unconstrained policy remains feasible.
3. Construct two physical states with the same policy observation but different appropriate actions. Propose a measurement or history that can distinguish them and identify a state that remains hidden.
4. Derive the finite-horizon value relation and the score-function policy gradient, including its discount weighting. State the assumptions on the transition law and justify an action-independent baseline.
5. Reproduce the binary-policy derivative analytically and by finite differences. Explain why one sampled gradient need not equal its mean.
6. Reproduce the TD/GAE example for both end meanings and for $\lambda = 1$. Show why using a reset observation or joining traces across episodes changes the target.
7. Evaluate the PPO surrogate for both advantage signs and several ratios. Explain what the clipping removes and why it does not constrain every probability change or guarantee higher return.
8. Design a record of behavior-policy identity, observations, action variables and end causes for a policy update. Distinguish the density of a latent sample from a clipped or squashed executed request.
9. Specify randomization and a curriculum for one manipulation task. Identify what is fixed per device, per episode and per timestep, and define a fixed final evaluation distribution.
10. Reproduce the rollout/optimizer budget and discount-timescale calculations. Explain which quantities change if action repeat doubles.

11. Reproduce the torque and position-interface example. Propose trace comparisons that separate an incorrect input map from insufficient actuator effort or different contact dynamics.
12. Specify continuous rotation, full-orientation regulation and a labeled-face-up task as three different problems. Define their goals, observations, success criteria, resets and useful failure recordings.

Sutton and Barto (2018) develops the broader reinforcement-learning foundations. The original GAE and PPO papers explain the specific estimators and update objectives introduced here (Schulman et al. 2016; Schulman et al. 2017). Pardo et al. (2018) clarifies time limits, while Peng et al. (2018) supplies a concrete manipulation example of dynamics randomization with a physical control interface. These sources support methods and distinctions, not a new reproduction claim for the constructed examples.

Chapter 11 turns to teleoperation: a human-guided feedback process whose interface also determines what actions and observations enter the dataset. The collection and physical semantics remain essential even when a person, rather than a learned actor, selects the requests.

Chapter 11 Teleoperation

A person moves a tracked controller to guide a robot's tool toward an object. The person sees the tool fall behind, moves farther, and then corrects when contact occurs. The recording contains hand poses, target poses, joint positions and images. Which of these is the demonstrated action? The answer depends on the interface between the person and the robot. The same hand movement can produce a different target after scaling, a different request after assistance, and a different physical motion after control and contact.

Chapter 10 studied a learned feedback policy as a generator of experience. Teleoperation places a human within that feedback process. The person selects actions using instructions, observations, previous experience and information that may never enter the stored dataset. The apparatus maps those decisions into feasible robot requests. Both contribute to the data subsequently available for embodied learning.

This chapter develops the physical and mathematical meaning of that mapping. Relative pose, retargeting, inverse kinematics, communication and control authority become parts of collection design. Featherstone's spatial language keeps frames, motion points and effort correspondence explicit. The numerical companion checks deliberately constructed examples; it does not establish the performance of a particular tracking device, network, robot or human operator.

11.1 Mapping human motion to a robot

A human and a robot have different bodies, workspaces and sensing. Teleoperation therefore requires a mapping, even when an interface feels direct. A joint-angle mapping chooses one kind of correspondence; a tool pose mapping chooses another. Neither becomes physically meaningful solely because its input and output arrays have matching dimensions.

11.1.1 Relative pose, scaling and clutching

Let a tracker report a hand-held frame's position $\mathbf{p}_H(t)$ and orientation $\mathbf{R}_H(t)$ in a tracking world. Let the robot tool's desired position and orientation be $\mathbf{p}_d$ and $\mathbf{R}_d$ in the robot world W. Each rotation maps its device or tool coordinates into the corresponding world coordinates. A fixed proper rotation $\mathbf{C}$ maps tracking-world vector components into robot-world axes. It is a calibrated axis correspondence, not a scale factor.

At an anchor time t_0, record the hand pose $(\mathbf{p}_{H0}, \mathbf{R}_{H0})$ and a robot target anchor $(\mathbf{p}_{E0}, \mathbf{R}_{E0})$. One spatial-increment mapping is

$$\begin{aligned} \mathbf{p}_d(t) &= \mathbf{p}_{E0} + s\mathbf{C}[\mathbf{p}_H(t) - \mathbf{p}_{H0}], \\ \Delta\mathbf{R}_H(t) &= \mathbf{R}_H(t)\mathbf{R}_{H0}^{\mathsf{T}}, \\ \mathbf{R}_d(t) &= \mathbf{C}\Delta\mathbf{R}_H(t)\mathbf{C}^{\mathsf{T}}\mathbf{R}_{E0}. \end{aligned} \tag{11.1}$$

Here $s > 0$ is a dimensionless translation scale after the position units have been reconciled. The displacement and rotation increment are expressed in world axes. The conjugation changes the rotation's axes; left multiplication applies that spatial increment to the tool anchor. Body-relative increments instead use $\mathbf{R}_{H0}^{\mathsf{T}}\mathbf{R}_H(t)$ and require a corresponding body-axis calibration and multiplication order. These alternatives agree only in special cases. Matrix order is part of the data contract.

The map deliberately leaves translation and orientation scales separate. An anisotropic translation gain can replace $s\mathbf{I}$, but then direction and distance are distorted according to that design. A scaled homogeneous matrix is not generally a rigid transformation. It must not be used as a Featherstone spatial transform for motion and wrench merely because it has a familiar matrix shape.

Worked example: axis mapping changes the direction of a request. Take $\mathbf{C} = R_z(\pi/2)$, $s = 0.5$ and hand displacement $(0.1, -0.04, 0.02)^{\mathsf{T}}$ m. The mapped displacement is $(0.02, 0.05, 0.01)^{\mathsf{T}}$ m. A tool anchor at $(0.3, 0.1, 0.2)^{\mathsf{T}}$ m therefore gives target $(0.32, 0.15, 0.21)^{\mathsf{T}}$ m. For hand rotation $\Delta\mathbf{R}_H = R_x(\pi/6)$ and tool anchor $R_z(\pi/4)$,

$$\mathbf{R}_d = R_y(\pi/6)R_z(\pi/4). \tag{11.2}$$

Reversing those last two factors generally gives a different target. The companion checks both translation and noncommuting rotation cases. Correcting a sign in code after seeing a surprising motion is not a substitute for defining the mapping and verifying it before collection.

A clutch allows the person to reposition the input device without moving the target. While the clutch is disengaged from motion control, retain the current target. Before resuming, replace both anchors by the current hand pose and the retained target pose. The relative displacement and rotation are then zero at re-engagement, so the target is continuous. Reusing an old hand anchor would instead interpret the repositioning as a robot request.

In the example, re-anchoring followed by another 0.02 m hand displacement along tracking x requests only 0.01 m along robot y. The previous target remains the starting point. Anchor the target or measured tool pose deliberately: choosing measured pose when the robot is lagging changes the reference and may cause a target jump. Store clutch events, anchor identities, scale and calibration versions, because they explain why one continuous hand trajectory can produce several different robot-target segments.

11.1.2 Embodiment differences and retargeting

A human finger angle is not automatically a robot finger angle. Joint axes, link lengths, coupling, ranges and contact surfaces differ. Likewise, a tracked device's reference point is not necessarily the robot's tool center. Retargeting chooses which relationships to preserve: fingertip locations, grasp aperture, tool orientation, joint posture or task-relative motion. These choices can conflict.

Suppose $\boldsymbol{\psi}_H$ describes selected human features and $\mathbf{g}(\mathbf{q})$ the corresponding robot features. A local posture retargeting problem can take the form

$$\min_{\mathbf{q}\in\mathcal{Q}} \|\mathbf{W}_g[\mathbf{g}(\mathbf{q}) - \mathbf{b}(\boldsymbol{\psi}_H)]\|^2 + \eta \left\|\mathbf{D}_q^{-1}(\mathbf{q} - \mathbf{q}_{\text{rest}})\right\|^2 . \tag{11.3}$$

The map $\mathbf{b}$ specifies the intended correspondence, $\mathbf{W}_g$ sets compatible feature scales, and $\mathbf{D}_q$ supplies joint-coordinate scales. The feasible set $\mathcal{Q}$ includes the modeled restrictions. The posture term resolves a preference; it does not prove that the desired contacts are achievable. This is a design formulation, not a uniquely correct human-to-robot mapping.

Featherstone motion variables make the reference-point issue concrete. Express all components in one world basis. If a rigid tool point E is offset by $\mathbf{r}_{OE}$ from origin O, then

$$\mathbf{v}_E = \begin{bmatrix} \boldsymbol{\omega} \\ \dot{\mathbf{p}}_E \end{bmatrix} = \mathbf{X}_{EO}\mathbf{v}_O, \qquad \mathbf{X}_{EO} = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ -[\mathbf{r}_{OE}]_\times & \mathbf{I} \end{bmatrix}. \tag{11.4}$$

Here $\dot{\mathbf{p}}_E = \dot{\mathbf{p}}_O + \boldsymbol{\omega} \times \mathbf{r}_{OE}$. The wrench dual is $\mathbf{f}_O = \mathbf{X}_{EO}^{\mathsf{T}}\mathbf{f}_E$, with moment-first ordering and moments taken about their stated points. Thus

$$\mathbf{v}_O^{\mathsf{T}}\mathbf{f}_O = \mathbf{v}_E^{\mathsf{T}}\mathbf{f}_E. \tag{11.5}$$

This is coordinate and point consistency, not a controller-stability result (Featherstone 2008).

Worked example: a tool offset contributes real velocity and moment. Let $\mathbf{r}_{OE} = (0.1, 0, 0)^{\mathsf{T}}$ m, $\boldsymbol{\omega} = (0, 0, 2)^{\mathsf{T}}$ rad/s and $\dot{\mathbf{p}}_O = (0.02, 0, 0)^{\mathsf{T}}$ m/s. Then $\dot{\mathbf{p}}_E = (0.02, 0.2, 0)^{\mathsf{T}}$ m/s. A force $(0, 3, 0)^{\mathsf{T}}$ N with zero moment about E has moment $(0, 0, 0.3)^{\mathsf{T}}$ N m about O. Both descriptions give power 0.6 W. Omitting either the offset velocity or the translated moment breaks the intended correspondence. The companion checks the transform against independent cross-product calculations.

The same reasoning applies to a camera, wrist tracker, fingertip or force sensor mounted away from the control point. Preserve the mounting transform and its uncertainty. A visually plausible target trajectory can still describe the wrong point on the mechanism.

11.2 Generating feasible robot motion

An intended tool pose is not yet a joint command. A robot's kinematics determines which instantaneous motions its joints can produce; limits and dynamics determine which requests can be realized over time. Teleoperation needs a policy for targets that cannot be reached, rather than assuming the human always supplies feasible ones.

11.2.1 Inverse kinematics and regularized least squares

For a fixed-base mechanism with ordinary joint coordinates, forward kinematics gives $(\mathbf{R}(\mathbf{q}), \mathbf{p}_E(\mathbf{q}))$. Define the angular-first endpoint Jacobian by

$$\begin{bmatrix} \boldsymbol{\omega}_W \\ \dot{\mathbf{p}}_E \end{bmatrix} = \mathbf{G}_E(\mathbf{q})\dot{\mathbf{q}}. \tag{11.6}$$

Its linear block is the velocity of point E. A world-origin spatial twist instead has linear block $\dot{\mathbf{p}}_E - \boldsymbol{\omega}_W \times \mathbf{p}_E$. A body twist has another basis and point convention. Their Jacobians must be transformed accordingly. A logarithm of a homogeneous pose error cannot be paired with any six-row Jacobian without checking these meanings (Lynch and Park 2017).

One local pose-feedback construction uses world-axis rotational error $\mathbf{e}_R = \operatorname{Log}(\mathbf{R}_d\mathbf{R}^\mathsf{T})^\vee$ and position error $\mathbf{e}_p = \mathbf{p}_d - \mathbf{p}_E$:

$$\mathbf{v}^* = \begin{bmatrix} \boldsymbol{\omega}_d + \mathbf{K}_R\mathbf{e}_R \\ \dot{\mathbf{p}}_d + \mathbf{K}_p\mathbf{e}_p \end{bmatrix}. \tag{11.7}$$

The logarithm selects a local rotation vector; near a rotation angle of π, branch ambiguity and continuity require care. The gains have units s^{-1} when errors and rates use the stated angular/linear units. For large rotations, the derivative of the logarithmic error includes rotation-Jacobian factors; it is not exactly minus angular velocity. Equation (11.7) is a chosen local feedback law, not a global convergence theorem. A stationary target has zero desired feedforward rates.

To match $\mathbf{G}_E\dot{\mathbf{q}}$ to $\mathbf{v}^*$, first reconcile angular and linear scales. Let $\mathbf{W}_v$ divide task-rate components by reference rates and let diagonal $\mathbf{D}_{\dot{q}}$ contain positive joint-rate scales. With dimensionless unknown $\boldsymbol{\xi}$, define

$$\widehat{\mathbf{G}} = \mathbf{W}_v\mathbf{G}_E\mathbf{D}_{\dot{q}}, \quad \widehat{\mathbf{v}} = \mathbf{W}_v\mathbf{v}^*, \quad \min_{\boldsymbol{\xi}} \|\widehat{\mathbf{G}}\boldsymbol{\xi} - \widehat{\mathbf{v}}\|^2 + \lambda^2\|\boldsymbol{\xi}\|^2. \tag{11.8}$$

For $\lambda > 0$, differentiating the quadratic gives the unique solution

$$(\widehat{\mathbf{G}}^\mathsf{T}\widehat{\mathbf{G}} + \lambda^2\mathbf{I})\boldsymbol{\xi} = \widehat{\mathbf{G}}^\mathsf{T}\widehat{\mathbf{v}}, \qquad \dot{\mathbf{q}}^{\mathrm{ref}} = \mathbf{D}_{\dot{q}}\boldsymbol{\xi}. \tag{11.9}$$

Solve this system numerically rather than forming an explicit inverse. Damped least squares limits the amplification associated with small singular values, at the cost of task residual (Wampler 1986). Using an SVD, a singular value σ contributes gain $\sigma/(\sigma^2+\lambda^2)$ instead of $1/\sigma$. The damping is dimensionless here because the variables have been scaled; its numerical value is not independent of those chosen scales.

Worked example: a planar arm supplies an inspectable Jacobian. For link lengths $l_1 = 0.3$ m and $l_2 = 0.2$ m,

$$\mathbf{p}_E = \begin{bmatrix} l_1\cos q_1 + l_2\cos(q_1+q_2) \\ l_1\sin q_1 + l_2\sin(q_1+q_2) \end{bmatrix},$$
$$\mathbf{J}_p = \begin{bmatrix} -l_1\sin q_1 - l_2\sin(q_1+q_2) & -l_2\sin(q_1+q_2) \\ l_1\cos q_1 + l_2\cos(q_1+q_2) & l_2\cos(q_1+q_2) \end{bmatrix}. \tag{11.10}$$

At $(q_1, q_2) = (\pi/6, -\pi/3)$, $\mathbf{p}_E \approx (0.433013, 0.05)^\mathsf{T}$ m and

$$\mathbf{J}_p \approx \begin{bmatrix} -0.05 & 0.1 \\ 0.433013 & 0.173205 \end{bmatrix} \quad \text{in m/rad.} \tag{11.11}$$

For requested endpoint velocity $(0.02, -0.01)^\mathsf{T}$ m/s, choose task-rate scales 0.1 m/s, joint-rate scales 1 rad/s and $\lambda = 0.1$. The companion checks the analytic Jacobian by finite differences. The damped joint rates are

$$\dot{\mathbf{q}}^{\mathrm{ref}} \approx (-0.0853814, 0.155727)^\mathsf{T} \quad \text{in rad/s.} \tag{11.12}$$

The Cartesian residual norm is approximately 0.00015824 m/s. Reporting that residual makes the tracking compromise visible; a finite joint-rate vector alone does not establish task accuracy. The two-row calculation concerns translation only; it does not claim independent control of the arm's complete spatial pose.

Repeated local solves can approach a pose target, with a declared step size, stopping criteria and treatment of infeasible targets. The linearized Jacobian is only local. Large integration steps can invalidate its prediction. A damped inverse also does not create an exact null-space projector: secondary posture commands can disturb the primary task unless the hierarchy or combined optimization is constructed accordingly.

11.2.2 Joint limits, singularities and reachable targets

At $q_1 = q_2 = 0$, the planar arm has

$$\mathbf{J}_p = \begin{bmatrix} 0 & 0 \\ 0.5 & 0.2 \end{bmatrix}. \tag{11.13}$$

No instantaneous joint velocity produces a nonzero horizontal endpoint velocity there. Damping returns a finite compromise, but cannot create a missing motion direction. Near the singularity, a direct inverse can request very large opposing joint speeds for modest horizontal motion. At $(q_1, q_2) = (0, 0.001)$ rad, the same horizontal request $(0.02, 0)^\mathsf{T}$ m/s gives inverse rates approximately $(66.6666, -166.667)^\mathsf{T}$ rad/s. With the same scales and damping as above, the rates are approximately $(0.008258, -0.020694)^\mathsf{T}$ rad/s, but most of the requested horizontal velocity remains unrealized. The companion compares these cases. A target inside a rough workspace boundary can still be locally difficult or unreachable under orientation, joint, collision or contact restrictions.

For ordinary joint coordinates and control interval h, one-step position and velocity bounds can be expressed as

$$\max\left(\dot{q}_i^{\min}, \frac{q_i^{\min} - q_i}{h}\right) \le \dot{q}_i^{\mathrm{ref}} \le \min\left(\dot{q}_i^{\max}, \frac{q_i^{\max} - q_i}{h}\right). \tag{11.14}$$

These use a local constant-rate step, not an exact prediction of actuator tracking. Add acceleration, collision and other restrictions only with their corresponding models. General configuration manifolds need a consistent integration rule. Solving the regularized objective subject to such bounds differs from clipping its unconstrained solution afterward.

Worked example: joint constraints should enter the solve. Consider a normalized one-dimensional task with Jacobian $(1, 1)$, target rate 1 and damping $\lambda = 0.1$. The unconstrained rates are both $1/2.01 \approx 0.497512$. If the first joint has upper admissible rate 0.1, the constrained optimum is

$$(\xi_1, \xi_2) = \left(0.1, \frac{0.9}{1.01}\right) \approx (0.1, 0.891089). \tag{11.15}$$

Simply clipping the first unconstrained component gives $(0.1, 0.497512)$ and a larger task error. A joint at 0.99 rad with upper position 1 rad and $h = 0.1$ s supplies the example's 0.1 rad/s bound when the reference-rate scale is 1 rad/s. The companion checks the objective and active constraint directly. This small quadratic problem illustrates the distinction; it is not a collision-avoidance or hardware guarantee.

Kinematic feasibility is also distinct from dynamic feasibility. Featherstone's articulated balance relates requested acceleration to required effort:

$$\mathbf{M}\ddot{\mathbf{q}}^{\mathrm{ref}} + \mathbf{h} = \boldsymbol{\tau}_a + \boldsymbol{\tau}_p + \sum_j \mathbf{J}_j^\mathsf{T} \mathbf{f}_j^{\mathrm{ext}}. \tag{11.16}$$

A feasible geometric path may require more actuator effort than available, especially under gravity, acceleration or contact. Position and rate limits alone do not account for that. Chapter 5 established these dynamics; Chapter 3 explained why requested quantities need not be measured delivered quantities.

The collection should preserve raw human target, feasible robot target, joint request and measured execution where available. Otherwise a projected or saturated action can be mistaken for the human's original command, and the learner may be trained to imitate an unexplained restriction.

11.3 Closing the human–robot feedback loop

The human acts on feedback, the robot responds through its controller, and the displayed observation returns later. Communication, sensing and processing therefore lie inside the effective loop. A smooth animation does not establish timely feedback or synchronized records.

11.3.1 Communication delay and asynchronous observations

Separate acquisition, transmission, reception, display, command issuance and physical application times. A camera exposure can span an interval; a pose estimate can describe an earlier image. A receive timestamp says when software

obtained a packet, not when the physical event occurred. Chapter 7 supplies clock-alignment and acquisition foundations. Use a common time basis with stated uncertainty when comparing streams.

At controller time t, define observation age by

$$a_o(t) = t - t_o^{\text{acquisition}}. \tag{11.17}$$

Its usefulness depends on which source sample was actually used. An out-of-order packet can arrive later but describe an older pose. Preserve sequence identity and acquisition time; a latest-received rule can reintroduce stale information. Even a latest-acquired rule requires a policy for dropout, clock uncertainty and discontinuous tracker resets.

Worked example: arrival order and physical order differ. Suppose a common clock has already been established. Packet 10 describes time 1.000 s and arrives at 1.080 s. Packet 11 describes time 1.020 s and arrives at 1.060 s. After processing both, a newest-acquisition rule retains packet 11. At time 1.100 s its age is 0.080 s. With a declared 0.120 s timeout, it is stale by time 1.200 s. The timeout expresses a chosen operating rule, not proof that every younger sample is adequate.

The action must also be associated with the feedback the human used. If a displayed image was acquired at 0.950 s, human input was recorded at 1.070 s and its command took effect at 1.100 s, the image is already 0.150 s old at application. Pairing the command with a newly acquired 1.100 s image for imitation would expose the learner to information the operator did not yet have. Such realignment can define a useful alternative learning target, but it must be identified as a transformation rather than the original observation–action pair.

Delay changes control behavior, not just record alignment. A simple constructed model shows the mechanism. Let an ideal scalar position integrator receive feedback velocity $u_k = -Kx_k$ each interval h. With no delay,

$$x_{k+1} = (1 - hK)x_k. \tag{11.18}$$

With one fixed interval of delay, $u_k = -Kx_{k-1}$ and

$$x_{k+1} = x_k - hKx_{k-1}, \qquad z^2 - z + hK = 0. \tag{11.19}$$

For $hK = 1.2$, the undelayed pole is -0.2, while the two delayed poles have magnitude $\sqrt{1.2} > 1$. The former decays and the latter is unstable in this ideal discrete model. Real humans, embedded loops and contact have richer dynamics; this is not a measured human-response model. It shows why adding a communication interval cannot be treated as a harmless file timestamp change.

Delay has a long history as a telemanipulation control problem. Niemeyer and Slotine (2004) discusses unknown constant transmission delays and the operation of wave-based transmission, including unwanted reflections and dynamics. Its setting and assumptions matter; it does not establish stability for arbitrary delayed position streaming, filtering or sampling. One-way motion input with visual feedback and bilateral force reflection are different control systems.

11.3.2 Filtering, hold behavior and operator authority

Tracking noise can cause unnecessary motion, but filtering also changes the response the person feels. For a scalar target sample p_k at a fixed interval h, a first-order filter can use

$$\bar{p}_k = (1 - \alpha)\bar{p}_{k-1} + \alpha p_k, \qquad \alpha = 1 - e^{-h/\tau_f}. \tag{11.20}$$

This coefficient matches the exact sampled response of a continuous first-order filter to a held input; τ_f is a time constant. It is not a universal fixed delay. Its frequency-dependent response attenuates and shifts different motion components differently. Componentwise averaging of rotation-matrix entries or quaternion coordinates can leave the intended rotation representation; orientation filtering requires a consistent manifold construction and branch/sign handling.

Worked example: smoothing and rate limiting alter different quantities. For $h = 0.02$ s, $\tau_f = 0.1$ s and a step from zero to 0.1 m, $\alpha \approx 0.181269$. The first filtered target is approximately 0.018127 m and the fifth is 0.063212 m. A separate rate limit of 0.2 m/s permits at most 0.004 m target change per interval. It changes the resulting path again. Recording only the raw 0.1 m target would conceal the commands actually presented to the robot during those intervals.

Between updates, a controller may hold a position target, hold a velocity request, interpolate targets or explicitly switch modes. Holding nonzero velocity continues motion; holding a position target can continue to apply effort against contact. Neither is equivalent to making the robot mechanically inactive. A dropout response must identify the actual controller behavior and its physical consequences.

An authority rule states when the human controls motion, when assistance modifies it, and what events stop, pause or reset the interface. A clutch, an autonomy takeover and a lost tracking signal are distinct events. Changing authority can require resetting filter state, reference anchors or integrators to prevent an artificial transient. A reference can be continuous while its derivative jumps, so continuity checks should match the quantity being controlled.

A schematic position-target update illustrates the stages:

```
if new_sample_is_valid_and_fresh:
    human_target = map_pose(sample, anchors, calibration)
    filtered_target = filter_target(human_target, filter_state)
    robot_target = enforce_declared_limits(filtered_target)
else:
    robot_target = apply_declared_dropout_rule(previous_target)
log_sample_and_all_transformed_targets()
```

The names describe application choices, not a complete safety or control implementation. Actual mode transitions and low-level actuator behavior must be defined as well. For learning, storing each stage explains whether a lag or restriction came from the person, the interface or execution.

11.4 Interpreting a demonstrated action

A demonstration is an interaction produced by a person and an apparatus. It is not a transparent measurement of intention. The record can contain proxies for a decision and the consequences of that decision, but these have different meanings and different causal relationships to the observations available to a learner.

11.4.1 Operator intent, transmitted targets, assistance and executed motion

Distinguish the task instruction, raw human input, mapped robot target, transmitted request, modified command, actuator response and measured motion. A joystick or tracker reading is an observed input. The intention to avoid an object or complete a grasp is an interpretation unless separately elicited. A final successful trajectory does not identify every intermediate choice the person intended.

For a shared-control example with compatible velocity requests, write

$$\mathbf{u}_{\text{mix}} = (1-\beta)\mathbf{u}_H + \beta\mathbf{u}_A, \qquad \mathbf{u}_{\text{cmd}} = \mathcal{P}(\mathbf{u}_{\text{mix}}; s), \quad 0 \le \beta \le 1. \tag{11.21}$$

Here $\mathbf{u}_H$ and $\mathbf{u}_A$ are human and assistance requests in the same frame, units and reference point. The map $\mathcal{P}$ can impose declared limits. Blending absolute poses or joint angles requires their own representation and topology; the velocity formula cannot simply be copied to quaternion coordinates.

Worked example: assistance changes the label before the robot moves. Let $\mathbf{u}_H = (0.08, 0.04)^{\mathsf{T}}$ m/s, $\mathbf{u}_A = (0.02, 0)^{\mathsf{T}}$ m/s and $\beta = 0.5$. Then $\mathbf{u}_{\text{mix}} = (0.05, 0.02)^{\mathsf{T}}$ m/s. Projecting onto a Euclidean speed ball of radius 0.03 m/s gives

$$\mathbf{u}_{\text{cmd}} = 0.03\frac{(0.05, 0.02)^{\mathsf{T}}}{\sqrt{0.05^2 + 0.02^2}} \approx (0.027854, 0.011142)^{\mathsf{T}}\ \text{m/s}. \tag{11.22}$$

These three vectors are different candidate labels. Only the first is the raw human request under this example's mapping; the last includes assistance and a speed constraint. None is yet a measured executed velocity. The companion checks the blend, projection and resulting differences without treating one label as universally preferable.

Even an unmodified command can differ from execution. For a constructed first-order velocity response with time constant τ_a, initial velocity zero and constant request u, the interval displacement is

$$p(h) - p(0) = u\left[h - \tau_a(1 - e^{-h/\tau_a})\right]. \tag{11.23}$$

At $u = 0.1$ m/s, $\tau_a = 0.05$ s and $h = 0.02$ s, the average interval velocity is about 0.01758 m/s. A finite difference of measured positions would estimate that average, not the 0.1 m/s request. This ideal lag model has no gravity, contact or measurement noise; actual discrepancies can have additional causes.

Choose the learning target according to the future interface. A policy that will send position targets can learn those targets, paired with the information available when they were chosen. A policy trained on executed motion and deployed as a torque controller needs another interpretation and a control layer. Predicting measured outcomes and imitating commands are different learning problems. Chapter 12 develops learning from demonstrations; it must inherit an explicit action contract rather than an unexplained column named action.

Store assistance mode and parameters, constraint activation and human input as separate fields when relevant. If assistance used a privileged object state unavailable to the learner, the final command may depend on unrecorded information. This can make an apparently simple supervised mapping ambiguous even when the physical behavior is repeatable.

11.4.2 Interventions and recovery as human-guided collection events

An intervention changes who or what controls the system. It may be a human takeover of a learned policy, a manual repositioning, a clutch event or assistance added to one component. These events should identify the authority before and after the change, the information available to each controller, and any physical state that was reset outside the ordinary action channel.

Human-gated interactive learning provides one example of explicitly separating the decision to intervene from the actions supplied under human control (Kelly et al. 2019). Its particular study does not imply that every human takeover is expert, correctly timed or fully observable. The person may react to sounds, contact sensations or a broader view not stored in the learner's inputs. Intervention timing can itself change with the interface and perceived response.

Recovery data is often valuable precisely because it visits states absent from clean successful demonstrations. An object that has slipped, a poor approach pose or a failed push can reveal what action restores progress. Deleting every such segment can remove the data needed for robust continuation. Conversely, joining a manual reset to an autonomous trajectory as if the robot caused the state jump creates a false transition.

A useful event record retains the parent interaction, the state or observation around the event, the new authority, its reason when known, and the transition's validity. Label a retrospective explanation as such; do not silently replace contemporaneous observations with later knowledge. A recovery segment and its earlier failure can share apparatus state and history, so they should remain linked when partitions test independence between episodes.

The resulting collection need not use every segment for every objective. One learner may use human-controlled recovery actions; another may use failure prediction labels; a third may exclude manual resets but retain their context. Preserve the original interaction and selection rule so that these derived datasets do not become competing undocumented versions of what happened.

11.5 Designing a collection interface

An interface is an experimental instrument. It selects what the person can perceive, what actions are convenient, which physical encounters are likely, and which quantities can be recorded. Designing it for embodied learning means balancing task access, physical correspondence, observation quality and collection effort rather than maximizing one signal count.

11.5.1 Tracked handheld and robot-mounted interfaces

A tracked handheld interface offers portable pose input and can support relative control without matching the robot's joint geometry. Its useful range depends on tracking, visibility, drift and clutching. Its pose alone does not measure object

contact force or guarantee robot reachability. Record what the tracker estimates, its coordinate convention and its failure indications. A handheld controller used to command a robot is also different from an instrumented handheld tool that acts on objects directly and is reconstructed later as a robot demonstration.

Universal Manipulation Interface illustrates the latter design: sensing and a handheld gripper organize human interaction for subsequent robot learning (Chi et al. 2024). Its interface and processing address representation, timing and embodiment constraints. It should not be described as ordinary live bilateral robot control, nor as proof that any handheld trajectory can be replayed by any robot.

Robot-mounted cameras and leader mechanisms provide other correspondences. A leader arm can supply joint or tool targets whose geometry resembles the follower. Its motion can still be scaled, filtered or constrained, and effort or contact at the follower may not be felt by the operator. ALOHA supplies a concrete example of bimanual leader–follower collection for robot learning (Zhao et al. 2023). Its non-bilateral position interface records leader positions as action targets and follower positions as observations. Their discrepancy matters through the low-level tracking controller; it is not itself a measured torque or a direct measure of human intention. Its collection arrangement and learning method are related but separable contributions; the interface does not by itself establish performance for another task or mechanism.

The choice affects observability. A robot-mounted view can preserve deployment camera geometry while hiding information the human used from outside that view. A handheld view can capture informative close motion but differ from the eventual robot camera. Force sensing, external views and annotations can add evidence, with calibration, timing and collection costs. The record should identify which information was available to the operator, to an assistant controller and to the future learner.

Use the physical contract to decide whether transfer is a coordinate conversion, a kinematic retargeting problem or a change in dynamics. Converting millimeters to meters does not resolve different contact geometry. Matching a tool path does not match required joint effort. Reconstructing a good camera trajectory does not prove successful object interaction. These are distinct questions that the collection can help investigate if its intermediate representations remain traceable.

11.5.2 Interface choices, operator variability and collection effort

Operators differ in familiarity, strategy, speed, perception and tolerance for the interface. An experienced operator can compensate for delay or a systematic mapping error, producing successful demonstrations that conceal the error from a later learner. A novice may reveal the same issue through frequent corrections. Neither observation alone determines a general ranking of skill or interface quality.

Separate practice sessions from retained collection when the distinction matters, and record instructions, familiarization and operator identity under the appropriate access conditions. To compare interfaces, declare the task conditions and what is matched: attempted interactions, operator time, successful examples or apparatus time. Learning and fatigue can confound a fixed ordering of interface trials. A comparison can alternate or balance conditions while retaining the actual order and dependencies. Chapter 16 develops comparative evidence more fully.

The collection budget includes setup, tracking calibration, robot checks, resets, inspection, unsuccessful attempts and recovery. A successful-motion duration omits much of the work that made those demonstrations possible. The same retained sample count can arise from very different physical effort and selection.

Worked example: useful demonstration duration is not collection cost. Suppose ten attempts each use 20 s of motion, 30 s of reset and 10 s of inspection. Setup takes 15 min and calibration 10 min. Total planned time is

$$T = 15 + 10 + \frac{10(20 + 30 + 10)}{60} = 35 \text{ min}. \tag{11.24}$$

If seven attempts succeed and each contributes its 20 s motion, the successful-motion subset contains 140 s. It does not represent the ten minutes of attempt work or the 35-minute session. These are constructed allocations; interruptions or repeated calibration would add cost. Store attempted, valid and selected counts separately, using the collection principles from Chapter 9.

Useful diagnostics include target-to-execution error, constraint activation, stale observations, clutch frequency, recoveries and unknown outcomes. Their interpretation is task-dependent. Frequent clutching may indicate a small input

workspace or simply a deliberate precision strategy. A low target error can occur while the wrong tool point is controlled. Choose diagnostics that distinguish relevant explanations, rather than treating a smooth trace as a sufficient quality label.

For ExpGen PhySyst, teleoperation designs a route from human judgment to physical interaction and then to reusable evidence. Its value is not only that it yields demonstrations. A well-specified interface makes clear which information generated each action, what the apparatus changed and which consequences were observed. Those distinctions allow later learning to build on human experience without mistaking an interface artifact for a general physical capability.

Exercises and further investigation. State frames, reference points, units, timing and information availability. Use the numerical companion to check constructed calculations, without treating them as device validation.

1. Derive the relative translation and spatial-rotation mapping. Reproduce the axis-map example and construct a non-commuting rotation case that exposes the wrong multiplication order.
2. Design a clutch state transition that preserves target continuity. Compare anchoring at the last target and at measured tool pose when the robot lags; identify the state that must be recorded.
3. Reproduce the tool-offset velocity, wrench and power example. Derive how its endpoint Jacobian relates to a Jacobian at another point.
4. Derive the planar-arm Jacobian and verify it by finite differences. Solve the scaled damped least-squares problem and report the task residual as well as the joint rates.
5. Compare an exact singularity with a nearby configuration. Explain what damping changes and why it cannot create a missing motion direction.
6. Reproduce the constrained-rate example. Compare its objective with a clipped unconstrained solution, then explain which dynamic and collision restrictions remain outside this model.
7. Reconstruct the packet and feedback timing example. Specify a valid common-clock assumption, a stale-data rule and the observation that should accompany an action for the chosen learning objective.
8. Derive both scalar feedback characteristic equations. Identify why their stability conclusions cannot be applied directly to every human, actuator or communication channel.
9. Reproduce the filter step response and rate bound. Describe position-hold, velocity-hold and mode-switch dropout behavior separately.
10. Reproduce the shared-control label and actuator-lag examples. Choose a learning target for a future position-controlled robot and list the transformations that must remain available.
11. Design an intervention/recovery record with parent interaction, authority, timing, observations and manual state changes. Distinguish unknown intention from an observed request.
12. Compare a tracked handheld input, a leader mechanism and an instrumented handheld collection tool for one task. Specify information available to operator and learner, retargeting limits and a collection budget that includes failures and resets.

Lynch and Park (2017) provides broader kinematic and inverse-kinematic foundations. The damped least-squares and classical delayed-teleoperation sources motivate particular mathematical and control choices (Wampler 1986; Niemeyer and Slotine 2004). UMI, ALOHA and human-gated interactive learning provide concrete interface and authority examples (Chi et al. 2024; Zhao et al. 2023; Kelly et al. 2019). Their published results are not new reproduction results of this chapter.

Chapter 12 develops learning from demonstrations. The mathematical learner will depend on the action, observation and episode meanings established here, even when the physical interface is no longer present during execution.

Chapter 12 Learning from Demonstrations

Two people guide a pusher around an obstacle. One goes left, the other right, and both reach the target. A learner predicts the average of their commands and moves straight toward the obstacle. Its prediction can have smaller squared error than either successful strategy. The difficulty is not necessarily too little data or too small a network. It can arise from the meaning of the prediction problem and the information supplied to it.

Chapter 11 traced a demonstration from human input through the interface to physical execution. This chapter asks how those records become a feedback policy. Behavioral cloning, action chunking, latent-variable models and diffusion policies make different assumptions about actions and their conditional distribution. All must eventually send a defined request through an actuator interface into the mechanics developed in Chapters 3–6.

For ExpGen PhySyst, a demonstration dataset is a designed source of experience. Its conditions, hidden information, corrections and label transformations determine what a learner can infer. A sophisticated generative model does not remove these requirements. The chapter combines statistical derivations with small physical and numerical examples; their companion checks arithmetic and stated mechanisms, not a newly trained manipulation system.

12.1 Behavioral cloning as statistical learning

Behavioral cloning fits an action-selection rule to demonstrations. The basic question is conditional: given information that will be available at execution, which demonstrated actions should the policy assign high probability or predict accurately? The resulting fit is supervised learning, but executing it changes future observations. That second fact distinguishes imitation of a robot controller from prediction on a fixed table.

12.1.1 Conditional likelihood, empirical risk and regularization

Let c_k denote the policy's conditioning information at decision k: for example, current and recent images, joint observations, previous requests and a task specification. It is not automatically the complete physical state. Let a_k be a declared action label, such as a joint position target, an endpoint velocity or delivered torque. Its meaning, frame, units and timestamp alignment come from the collection contract.

For dataset $\mathcal{D} = \{(c_i, a_i)\}_{i=1}^{N}$, regularized conditional likelihood minimizes

$$\widehat{\mathcal{L}}(\theta) = -\frac{1}{N}\sum_{i=1}^{N} \log p_\theta(a_i \mid c_i) + \lambda_r \Omega(\theta). \tag{12.1}$$

The empirical distribution is the one actually retained. Equal rows, equal episodes and equal operators give different weights, as Chapter 9 showed. Regularization Ω expresses a preference or constraint on the fitted model; it does not supply missing physical observations or correct an undefined action label.

For continuous actions, it is useful to define fixed reference scales before a numerical loss. With mean or reference $\mathbf{m}_a$ and positive diagonal scale matrix $\mathbf{D}_a$, write

$$\boldsymbol{\xi} = \mathbf{D}_a^{-1}(\mathbf{a} - \mathbf{m}_a), \qquad \mathbf{a} = \mathbf{m}_a + \mathbf{D}_a\boldsymbol{\xi}. \tag{12.2}$$

The network can model dimensionless $\boldsymbol{\xi}$. A density expressed in physical action coordinates has the Jacobian factor $|\det \mathbf{D}_a|^{-1}$. This factor is constant in a parameter update when the scales are fixed, but units and likelihood values are still representation-dependent. Fit data-dependent scales on the allowed training population and retain them for execution.

A Gaussian conditional model with fixed covariance $\sigma^2\mathbf{I}$ in normalized coordinates gives

$$-\log p_\theta(\boldsymbol{\xi} \mid c) = \frac{\|\boldsymbol{\xi} - \boldsymbol{\mu}_\theta(c)\|^2}{2\sigma^2} + \frac{d_a}{2}\log(2\pi\sigma^2). \tag{12.3}$$

Thus squared-error regression corresponds to a particular likelihood assumption, up to constants. If covariance is learned, its scale and log-determinant terms cannot be discarded. An absolute-error loss instead has a connection to a Laplace observation model under its own scale assumptions. Neither choice makes the label a measurement of intention.

Worked example: regularization changes a fitted rule. Take normalized scalar inputs $x = (-1, 0, 1)$ and labels $y = (-0.8, 0.2, 1.2)$. Fit $\hat{y} = b + wx$ by mean squared error plus $\lambda_r w^2$, leaving the intercept unpenalized. Since the inputs are centered, differentiation gives

$$b = 0.2, \qquad w = \frac{\sum_i x_i(y_i - b)}{\sum_i x_i^2 + N\lambda_r}. \tag{12.4}$$

At $\lambda_r = 0.1$, $w = 2/2.3 \approx 0.869565$, rather than the unregularized value 1. The companion checks the solution and objective gradient. The fit is deliberately elementary: more expressive models still need an explicit weighting, scale and regularization convention.

12.1.2 Representations, neural approximators and optimization

A policy can factor into an observation representation and an action predictor, $\hat{a} = g_\theta(f_\theta(c))$. Images may pass through a visual encoder; joint observations and task information can enter as additional features. A history can distinguish states that share one image but have different velocity or contact history. It does not guarantee that every relevant hidden state becomes identifiable.

Representation choices preserve or discard information. Absolute tool position, object-relative position and a pose increment referenced to the current tool frame answer different questions. A rotation representation must handle equivalent quaternions and the geometry of $SO(3)$. Subtracting raw quaternion components is not automatically a physical orientation error. A numerical feature can be useful without being a spatial motion vector; it must be decoded before entering mechanics.

For parameters shared across a representation and predictor, backpropagation uses the chain rule through the implemented computation. A minibatch step can be written

$$\theta^+ = \theta - \eta \nabla_\theta \left[\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \ell_\theta(c_i, a_i) + \lambda_r \Omega(\theta) \right]. \tag{12.5}$$

The derivative concerns that computation. It does not establish that features represent the intended physical quantity or that the chosen training examples cover deployment (Baydin et al. 2018).

Training and validation must preserve episode and transformation lineage. Adjacent windows share observations and actions. Splitting those windows randomly can expose nearly the same interaction during fitting and testing. A masked padded action at an episode end is not a demonstrated zero command. Preserve validity masks and normalize loss by the intended number of valid elements, rather than silently changing weights with padding length.

Optimization, model capacity and data quality are separate explanations for a poor result. A network can fit corrupted labels very accurately. Conversely, systematic ambiguity caused by missing task information may remain even after optimization converges. Inspect the observations and their label-generation chain before treating a lower training loss as evidence that the physical problem is solved.

12.1.3 Conditional means, multimodal behavior and loss choice

For squared loss, the optimal unconstrained deterministic predictor at fixed conditioning information is the conditional mean. Let $\mathbf{m} = \mathbb{E}[\mathbf{a} \mid c]$. Expanding the square gives

$$\mathbb{E}[\|\mathbf{a} - \mathbf{u}\|^2 \mid c] = \mathbb{E}[\|\mathbf{a} - \mathbf{m}\|^2 \mid c] + \|\mathbf{u} - \mathbf{m}\|^2. \tag{12.6}$$

The cross term vanishes by the definition of $\mathbf{m}$. This statistical optimum need not be a successful physical action. Feasible actions or trajectories can form disconnected sets, and their average can lie between useful alternatives.

Worked example: averaging two strategies can erase both. At the same recorded condition, suppose a normalized lateral request is $+1$ or -1, each with probability one half. Mean prediction zero has expected squared error 1; always predicting $+1$ has error 2. If the two labels describe going around opposite sides of an obstacle, the lower-loss mean can point into the obstacle. This is a constructed interpretation of the two modes, not a universal relation between loss and collision.

Temporal coherence matters as well. Suppose valid three-step strategies are $(1,1,1)$ and $(-1,-1,-1)$. Independent equiprobable mode choices at each step switch strategy within the three steps with probability $1-2/2^3 = 0.75$. Choosing one latent mode for the whole sequence produces no such switch in this toy construction. A joint action-sequence model can represent this dependence; independent per-step marginals cannot specify it by themselves.

Some apparent multimodality can be resolved by conditioning on a goal, history or observable context. Other variation represents genuinely different valid strategies. Mixture models, latent-variable decoders and diffusion models offer ways to represent distributions rather than one mean. The representation still needs enough data and information to learn the relevant alternatives. A broad distribution is not proof that its samples satisfy contact, actuator or collision restrictions.

12.2 From prediction to closed-loop behavior

An offline prediction is evaluated at an observation supplied by the dataset. An executed policy helps determine its next observation. Small errors can move the apparatus into conditions rarely demonstrated, where the learned action rule may have little support. This feedback changes the statistical question.

12.2.1 Deployment-induced covariate shift and compounding error

Retain the articulated balance

$$\mathbf{M}(\mathbf{q})\dot{\boldsymbol{\nu}} + \mathbf{h} = \boldsymbol{\tau}_a + \boldsymbol{\tau}_p + \sum_j \mathbf{J}_j^{\mathsf{T}} \mathbf{f}_j^{\text{ext}}. \tag{12.7}$$

Its tangent-coordinate bias is constructed consistently with the joint representation. Angular-first spatial motion and moment-first wrench use matching axes and reference points. Delivered actuator effort depends on the decoded policy request and controller interface (Featherstone 2008).

For a small interval h in one fixed smooth mode, a delivered-effort perturbation has local velocity effect

$$\delta\boldsymbol{\nu}_{k+1} \approx h\mathbf{M}^{-1}\delta\boldsymbol{\tau}_a. \tag{12.8}$$

This freezes the local inertia and other terms; it is not a complete multi-step sensitivity across contact changes. If the policy emits position targets, the low-level servo must first map target error into delivered effort. The same normalized prediction error can therefore have different physical consequences under different inertias, gains or action scales.

The policy induces a state distribution $d_k^{\pi_\theta}$, whereas offline behavioral cloning usually weights demonstrated conditions from an expert distribution $d_k^{\pi_E}$. Better prediction on the latter does not automatically bound loss on the former (Ross et al. 2011). This is deployment-induced covariate shift, not merely a different random train/test split of the same recorded population.

Worked example: a small effort bias persists through physical feedback. Take an ideal revolute joint with inertia $J = 0.02$ kg m^2, no gravity, friction or other load, and interval $h = 0.02$ s. Hold each delivered torque constant over its interval. An expert uses $\tau_E = -K_p q - K_d\dot{q}$, with $K_p = 0.8$ N m/rad and $K_d = 0.16$ N m s/rad. A learned law adds constant bias 0.01 N m. Exact held-torque integration uses

$$\begin{bmatrix} q_{k+1} \\ \dot{q}_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & h \\ 0 & 1 \end{bmatrix} \begin{bmatrix} q_k \\ \dot{q}_k \end{bmatrix} + \begin{bmatrix} h^2/(2J) \\ h/J \end{bmatrix} \tau_k. \tag{12.9}$$

The state components retain their different physical units; the matrix coefficients have the corresponding units. At initial state $(q, \dot{q}) = (0.2, 0)$, the bias changes the next position by 0.0001 rad and velocity by 0.01 rad/s. Under the stated stable sampled feedback, the biased law converges to $q = 0.01/0.8 = 0.0125$ rad and zero velocity, rather than the expert's zero equilibrium. The companion checks the closed-loop poles, trajectories and equilibrium. This example has no actuator saturation and does not describe a contact manipulation task.

A second constructed model illustrates a worst-case accumulation argument. Suppose a learner follows an expert path until an erroneous decision; afterward it enters an absorbing bad state. Its conditional error probability at each still-good

step is ε. For horizon H,

$$\Pr(\text{bad by step } H) = 1 - (1-\varepsilon)^H, \qquad \mathbb{E}[N_{\text{bad}}] = \sum_{t=1}^{H} [1 - (1-\varepsilon)^t]. \tag{12.10}$$

Since $1 - (1-\varepsilon)^t \le t\varepsilon$,

$$\mathbb{E}[N_{\text{bad}}] \le \min\left(H, \frac{\varepsilon H(H+1)}{2}\right). \tag{12.11}$$

This is a bounded bad-step count under a specified model. More general quadratic-horizon imitation bounds require their disagreement, cost and distribution assumptions. Neither this bound nor a classification error can be inferred directly from a joint-angle MSE. Stable feedback, recoverability and environment structure can yield very different behavior. For $H = 20$ and $\varepsilon = 0.02$ in the constructed absorbing process, the final bad-state probability is approximately 0.332392 and the expected bad-step count is 3.712791, below its 4.2 bound. These quantities count different aspects of the same process.

12.2.2 Coverage, interventions and recovery data

The next useful data can be a corrective action in a learner-visited state, rather than another nominal successful trajectory. Dataset Aggregation, DAgger, addresses this distinction by iteratively collecting states under the developing policy, querying expert actions at those states, and aggregating the labeled data (Ross et al. 2011).

At iteration j, write the accumulated set as

$$\mathcal{D}_j = \mathcal{D}_{j-1} \cup \{(c, a_E(c)) : c \text{ visited during iteration } j\}. \tag{12.12}$$

The expert must be able to provide a meaningful action for the queried condition. A policy mixture used during collection chooses which policy acts according to its defined procedure; it is not necessarily arithmetic averaging of two commands. Theoretical benefits depend on the learning, coverage, expert-query and cost assumptions. Adding arbitrary failure files to a folder is not by itself the DAgger procedure.

Human intervention provides one practical way to obtain corrective behavior, with the authority and information boundaries from Chapter 11. If the expert sees information absent from the learner's inputs, its action may not be identifiable from those inputs. If the physical state was manually reset, the transition cannot be treated as an ordinary robot action. Preserve the distinction between failure, recovery, intervention and invalid observation.

Coverage must also match the final task. Demonstrations from stable grasps do not establish recovery after a drop; successful pushes from one side do not define actions behind an obstacle. Recovery collection can broaden the experience, but retaining the original context and selection rules is necessary to understand what changed.

For data design, ask whether a missing behavior is caused by inadequate coverage, missing conditioning information, an unsuitable distribution model or a physically unrealizable request. These explanations call for different collection changes. A larger model cannot observe an unrecorded event, and another nominal trajectory may not teach the required recovery.

12.3 Learning sequences of action

Many physical tasks require coordinated action over time. A grasp must close while the hand approaches; a push needs a coherent direction; two arms can exchange an object. Predicting a sequence can represent this coordination and reduce how often a new high-level decision is required. It also creates a choice about how long to act before observing again.

12.3.1 Action chunking, attention and temporal ensembling

An action chunk predicted at decision k is

$$\mathbf{A}_k = (\mathbf{a}_{k|k}, \mathbf{a}_{k+1|k}, \ldots, \mathbf{a}_{k+H_p-1|k}). \tag{12.13}$$

The second subscript identifies when the prediction was made; the first identifies the physical slot it is intended to control. The prediction horizon H_p differs from the executed prefix $H_e \le H_p$. Replanning after H_e steps makes a feedback policy

even though each prefix is executed without a new high-level observation. The low-level actuator controller can still run feedback within that prefix.

Attention supplies one way to combine image features, proprioception and sequence queries. For row-wise queries $\mathbf{Q}$, keys $\mathbf{K}$ and values $\mathbf{V}$, a basic operation is

$$\mathrm{Attn}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\mathsf{T}}}{\sqrt{d_k}} + \mathbf{B}_{\mathrm{mask}}\right)\mathbf{V}. \tag{12.14}$$

The mask excludes unavailable or padded entries. Positional or time encodings identify ordering that a set of feature vectors alone does not provide (Vaswani et al. 2017). These learned arrays are not Featherstone spatial transforms: attention weights do not enforce rigid geometry, force balance or correct timestamps.

Worked example: attention combines evidence under learned weights. For two scalar scores $(0, \log 3)$, softmax gives weights $(0.25, 0.75)$. With values $(-1, 1)$, the output is 0.5. Masking the second entry instead gives output -1. The companion checks both cases. This arithmetic explains weighting and masking, not why particular learned keys will capture the relevant physical evidence.

An autoregressive decoder must mask future target tokens that would be unavailable during execution. A parallel chunk decoder can instead use queries for future action slots without leaking future information, provided those queries do not contain future measurements or target actions.

When a new chunk is predicted every step, several chunks contain a candidate for the same physical action slot. Temporal ensembling combines those aligned candidates. The original ACT description orders candidates from oldest to newest and uses

$$\bar{\mathbf{a}}_k = \frac{\sum_{i=0}^{n_k-1} e^{-mi}\mathbf{a}_k^{(i)}}{\sum_{i=0}^{n_k-1} e^{-mi}}, \qquad i = 0 \text{ is the oldest candidate.} \tag{12.15}$$

For positive m, this favors older predictions; a smaller m admits newer information more rapidly in that scheme (Zhao et al. 2023). The ordering must be explicit. Reversing array order while retaining the same weight formula changes the executed controller.

Worked example: ensemble only actions for the same time. At physical slot 3, suppose chunks generated at slots 0, 1, 2 and 3 predict values $(0.08, 0.12, 0.16, 0.20)$ for that slot. With $m = \log 2$, oldest-first normalized weights are $(8, 4, 2, 1)/15$, giving approximately 0.109333. Reversing the weighting gives approximately 0.170667. These are constructed normalized requests with a deliberately large decay for clarity, not ACT benchmark parameters. The relevant candidates are $a_{3|0}$, $a_{3|1}$, $a_{3|2}$ and $a_{3|3}$, not four consecutive future actions from one chunk.

Ensembling can smooth commands but also blends choices made from different observations. If those choices belong to incompatible task strategies, their average need not remain feasible. A smoother target is not automatically a better demonstrated action, and a sequence model must still preserve units, frames and the underlying action interface.

12.3.2 Latent-variable objectives, KL divergence and ACT

A conditional latent-variable model represents chunk variability through z:

$$p_\theta(\mathbf{A} \mid c) = \int p_\theta(\mathbf{A} \mid c, z) p(z \mid c)\, dz. \tag{12.16}$$

The latent can represent variation not specified by c, but its learned meaning is not guaranteed to correspond to a human intention or physical parameter. Introduce an approximate posterior $q_\phi(z \mid \mathbf{A}, c)$. Jensen's inequality gives

$$\begin{aligned}\log p_\theta(\mathbf{A} \mid c) \geq\ & \mathbb{E}_{q_\phi}[\log p_\theta(\mathbf{A} \mid c, z)] \\ & - D_{\mathrm{KL}}(q_\phi(z \mid \mathbf{A}, c) \,\|\, p(z \mid c)).\end{aligned} \tag{12.17}$$

The gap is the divergence from the approximate to the true posterior when the distributions and expectations are well-defined. The posterior may use the demonstrated future chunk during training because that chunk is the label. Execution must instead use an inference rule that does not require future demonstrated actions (Kingma and Welling 2014).

For a diagonal Gaussian posterior, training can use the reparameterization

$$z = \boldsymbol{\mu}_\phi(\mathbf{A}, c) + \boldsymbol{\sigma}_\phi(\mathbf{A}, c) \odot \eta, \qquad \eta \sim \mathcal{N}(0, \mathbf{I}). \tag{12.18}$$

The symbol $\odot$ denotes componentwise multiplication. Randomness now comes from a distribution independent of ϕ, while the sample is a differentiable function of the posterior parameters. Under the appropriate interchange conditions, backpropagation through that sample estimates the reconstruction gradient; the Gaussian KL below can be differentiated analytically. This pathwise derivative differs from treating a sampled value as a constant while attempting to train its sampling parameters.

For diagonal Gaussian posterior $\mathcal{N}(\boldsymbol{\mu}, \operatorname{diag} \boldsymbol{\sigma}^2)$ and standard-normal prior, the divergence is

$$D_{\mathrm{KL}}(q\|p) = \frac{1}{2} \sum_j \left(\sigma_j^2 + \mu_j^2 - 1 - \log \sigma_j^2\right). \tag{12.19}$$

It is measured in nats and is dimensionless. It penalizes a posterior that carries information inconsistent with the prior, while reconstruction encourages retaining information useful to predict the chunk. The balance depends on the chosen objective and its normalization.

ACT uses a conditional variational autoencoder with a Transformer-based action predictor (Zhao et al. 2023). In its training description, the posterior encoder receives the action sequence and proprioception, without the images used by the action decoder. At inference, that encoder is discarded and the latent is set deterministically to the standard normal prior mean, $z = 0$. This is a specific execution rule, not generic sampling from every conditional variational model.

The paper's implementation description uses an L1 reconstruction loss plus a weighted KL term:

$$\begin{aligned} \mathcal{L}_{\mathrm{ACT}} &= \mathbb{E}_{z\sim q_\phi}[\ell_{\mathrm{rec}}(\mathbf{A}, \widehat{\mathbf{A}}_\theta(c, z); \mathbf{m})] + \beta_{\mathrm{KL}} D_{\mathrm{KL}}(q_\phi \| \mathcal{N}(0, \mathbf{I})), \\ \ell_{\mathrm{rec}}(\mathbf{A}, \widehat{\mathbf{A}}; \mathbf{m}) &= \frac{\sum_j m_j |A_j - \widehat{A}_j|}{\sum_j m_j}. \end{aligned} \tag{12.20}$$

Here flattened-entry mask m_j is one for valid labels and zero for padding, with a nonzero valid count. The displayed convention uses mean valid-entry L1, matching the companion; other reduction conventions must be recorded with their coefficients. The action scale and KL reduction must also be specified. An arbitrary reconstruction norm and arbitrary KL weight are not automatically the exact negative bound in Equation (12.17). ACT's prose specifies L1 whereas its training-algorithm box prints MSE; an implementation or reproduction must resolve this source-level difference explicitly.

Worked example: a prior-mean latent is not a mean action. For $\boldsymbol{\mu} = (0.5, -0.25)$ and posterior standard deviations $(0.8, 1.2)$, the companion evaluates Equation (12.19) and obtains approximately 0.237072 nats. With reconstruction value 0.12 and $\beta_{\mathrm{KL}} = 0.1$, the weighted objective is approximately 0.143707. These are normalized constructed values. More generally, with decoder $g(z) = z^2$ and $z \sim \mathcal{N}(0, 1)$, $g(0) = 0$ while $\mathbb{E}[g(z)] = 1$. Thus deterministic decoding at prior mean need not equal the predictive mean, mode or a representative sampled strategy. ACT's inference choice should be evaluated as the feedback law actually executed.

12.4 Generative policies for demonstrations

A generative action model represents a conditional distribution from which actions or chunks can be produced. It can preserve alternatives that a single mean erases, but its sampling procedure introduces another computation between observation and physical command. Diffusion policies make that procedure explicit through iterative denoising.

12.4.1 Conditional denoising and diffusion-based action generation

Let $\mathbf{x}^{(0)}$ be a flattened, normalized action chunk. Superscript n now denotes diffusion level, not robot time. A Gaussian forward process uses schedule $0 < \beta_n < 1$, $\alpha_n = 1 - \beta_n$ and $\bar{\alpha}_n = \prod_{j=1}^{n} \alpha_j$:

$$q(\mathbf{x}^{(n)} \mid \mathbf{x}^{(0)}) = \mathcal{N}\left(\sqrt{\bar{\alpha}_n}\mathbf{x}^{(0)}, (1 - \bar{\alpha}_n)\mathbf{I}\right). \tag{12.21}$$

Repeated substitution of the Gaussian forward steps gives this marginal. Equivalently, sample $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ and form

$$\mathbf{x}^{(n)} = \sqrt{\bar{\alpha}_n}\mathbf{x}^{(0)} + \sqrt{1-\bar{\alpha}_n}\boldsymbol{\epsilon}. \tag{12.22}$$

This is artificial corruption of an action representation. It is not a model of motor noise, physical diffusion or sensor uncertainty.

A conditional denoiser predicts the added noise from the noisy chunk, level and policy conditioning. A common simplified objective is

$$\mathcal{L}_\epsilon(\theta) = \mathbb{E}_{\mathbf{x}^{(0)},c,n,\boldsymbol{\epsilon}}\left[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}^{(n)}, n, c)\|^2\right]. \tag{12.23}$$

The distribution over n, schedule, normalization and validity masks belong to the training definition. This simplified or reweighted variational objective is not the full exact negative log likelihood (Ho et al. 2020). A network minimizing it learns a statistical denoising relation from the supplied chunks and conditions.

For one canonical DDPM parameterization, the reverse mean is

$$\boldsymbol{\mu}_\theta(\mathbf{x}^{(n)}, n, c) = \frac{1}{\sqrt{\alpha_n}}\left[\mathbf{x}^{(n)} - \frac{\beta_n}{\sqrt{1-\bar{\alpha}_n}}\boldsymbol{\epsilon}_\theta(\mathbf{x}^{(n)}, n, c)\right]. \tag{12.24}$$

A sampling step draws around that mean using a declared reverse variance; the last step and alternative samplers need their own rules. One common variance choice is

$$\widetilde{\beta}_n = \beta_n\frac{1-\bar{\alpha}_{n-1}}{1-\bar{\alpha}_n}, \qquad \bar{\alpha}_0 = 1. \tag{12.25}$$

Sampling begins from an approximate noise prior when the final forward level sufficiently suppresses the data signal. An arbitrary very short schedule need not satisfy that approximation. Changing the sampler or number of steps changes the inference procedure, even with the same learned denoiser.

Worked example: forward corruption and reverse arithmetic are checkable. Use three illustrative noise levels with $\beta = (0.1, 0.2, 0.3)$, hence $\bar{\alpha} = (0.9, 0.72, 0.504)$. Let the clean normalized chunk be $(-1, 0, 1)$ and a chosen corruption vector $(0.5, -1, 0)$. The companion computes $\mathbf{x}^{(3)}$, recovers the clean chunk using the known corruption, and checks the reverse mean against the Gaussian posterior formula conditioned on that clean chunk. The resulting noisy chunk and reverse mean are

$$\begin{aligned}\mathbf{x}^{(3)} &\approx (-0.357793, -0.704273, 0.709930),\\ \boldsymbol{\mu}^{(3)}_{\text{oracle}} &\approx (-0.682211, -0.332634, 0.848528).\end{aligned} \tag{12.26}$$

The reverse mean concerns level 2, not the clean level 0. The third posterior variance is $0.3(1-0.72)/(1-0.504) \approx 0.169355$. If a noise prediction differs from the chosen vector by $(0.1, -0.2, 0.3)$, its mean squared noise error is $0.14/3$. Known corruption is an arithmetic fixture, not a trained denoiser. The three-level schedule retains substantial signal and is not presented as a complete noise-prior policy sampler.

Diffusion Policy applies conditional action-sequence generation and receding-horizon execution to robot learning (Chi et al. 2023). The image representation, observation horizon, predicted action horizon, executed prefix, noise schedule and sampling latency all affect the resulting controller. Its published method is a concrete design, not a guarantee that every conditional diffusion model will produce a feasible or successful robot action.

12.4.2 Conditioning information and multimodal action sequences

Conditioning should identify what the policy can know at the decision time: available images, state estimates, previous actions, goal and history. During offline learning, future data can define a label or a reference estimate, but it cannot silently enter the deployed actor's conditioning. A denoiser seeing future images during training solves a different prediction problem from one that must act before those images exist.

Generating an entire chunk can preserve correlations among its actions. It can represent one side-of-obstacle strategy for several steps rather than choosing a new mode independently at each step. That possibility does not prove the model learned the right dependencies. It may ignore a condition, collapse modes, or sample unsupported intermediate trajectories. Inspect sampled sequences as well as individual action histograms.

Different generative mechanisms should not be conflated. A conditional latent decoder transforms a sampled or fixed latent into actions; a diffusion policy iteratively transforms a noisy action sequence under conditioning. Neither is automatically a world model: its generated actions are not a prediction of the physical consequences of those actions. Chapter 13 treats generalist policies and predictive world models separately.

Physical restrictions remain outside a generic statistical action density. Collision, contact and dynamical restrictions can define nonconvex feasible trajectories; an effort limit does not ensure sufficient force to maintain a grasp. Projection or low-level control can modify generated actions, thereby changing the executed distribution. If those modifications matter, preserve them in evaluation records and train against the interface that will actually be used.

Data design can make multimodality more interpretable. Record task goals and observable context that distinguish strategies. Keep coherent trajectory segments, their timing and outcome evidence. Preserve valid recovery alternatives rather than replacing every varied behavior with one hand-selected template. A generative objective can use these records only if the correspondence between condition and action sequence remains physically meaningful.

12.5 Testing what a demonstration dataset supports

Evaluation needs both the statistical prediction problem and the executed feedback law. Offline errors can expose mislabeled or poorly modeled actions; physical rollouts can expose consequences absent from the demonstrated state distribution. They answer complementary questions.

12.5.1 Predicted chunks, execution horizons and feedback updates

Prediction horizon, execution horizon and observation horizon are distinct. With control interval h, a chunk of H_p actions describes $H_p h$ of requests under the declared action-hold convention. Executing only H_e of them gives a nominal high-level observation/update interval $H_e h$. Predicting a long chunk does not require committing to all of it, and frequent replanning does not eliminate low-level lag.

Worked example: chunk indices must follow the physical clock. Let $h = 0.02$ s, $H_p = 4$ and $H_e = 2$. Prediction spans 0.08 s and the executed prefix spans 0.04 s. A chunk generated for slots 10–13 that takes 0.035 s to compute is only available after almost two control intervals. If application is restricted to the next control boundary, the first available slot is 12. Using the slot-10 target then would introduce a two-slot timing mismatch.

Discarding elapsed predictions and using the appropriately indexed remainder is one alignment choice; it does not prove that an old observation still supports those future requests. Prediction, buffering or asynchronous inference can be designed differently, but need a stated policy for delay and stale plans. A four-action chunk may provide too little useful remainder for a chosen execution prefix once latency is accounted for.

A schematic receding-horizon controller makes the indexing visible:

```
condition = build_available_history(observations, timestamps)
plan = predict_chunk(condition)
plan = attach_intended_control_slots(plan, decision_time)
prefix = select_unexpired_prefix(plan, application_time)
execute_with_declared_low_level_controller(prefix)
record_plan_indices_requests_and_measured_execution()
```

Temporal ensembling adds aligned candidates from several plans; it does not justify averaging entries with different intended times. Relative pose chunks also need their anchor identity. A trajectory relative to one initial tool pose is not the same as accumulating pairwise deltas. Each update must reconstruct the representation the policy was trained to produce.

Compare controllers at a declared observation/update schedule and report inference time, action interval and discarded or stale targets. Fewer denoising steps may reduce computation but alter sample quality. A longer committed prefix may reduce update demand but weaken response to an unexpected contact. These are empirical tradeoffs to investigate, not quantities that the offline reconstruction loss determines alone.

12.5.2 Offline prediction and closed-loop diagnostics for imitation

Offline assessment should specify held-out episodes, conditions and label meaning. Report the scale and reduction of the loss, and inspect errors by task phase rather than only their pooled average. Rare contact changes or recovery actions can matter more to execution than common stationary frames. Likelihood, per-action error and sequence consistency measure different properties.

Worked example: identical offline fit can conceal opposite feedback. For an ideal scalar system $x_{k+1} = x_k + 0.1a_k$, suppose the only demonstrated condition is $x = 0$ with expert action zero. Policies $a = -x$ and $a = +x$ both have zero error on that record. Their closed-loop poles are 0.9 and 1.1 respectively. From a nonzero initial condition, one decays while the other grows in this unconstrained model. The companion checks both rollouts. The example establishes a lack of identification from that narrow dataset, not a general theorem that offline evaluation is uninformative.

Closed-loop tests should identify task success, intermediate failures, object motion, contact, saturation, intervention and termination causes. A policy can achieve low action error while failing a dwell criterion, or show imperfect imitation while completing the physical task by another valid strategy. Retain original observations, predicted chunk, selected actions, low-level requests and measured behavior to distinguish these cases.

Repeated sampling from one fixed observation measures variability of the generator. Repeated robot episodes measure behavior under interactions and observations that evolve. Repeated training seeds add another source of variation. They are not interchangeable counts. Declare how checkpoints and inference settings were chosen before interpreting a final held-out test; Chapter 16 develops the comparative-inference framework.

For ExpGen PhySyst, imitation learning closes a loop between designed experience and a new generator of experience. Mechanics constrains how predictions become interaction; observations and labels determine what is learnable; model and loss determine which statistical structure is represented; execution determines which states appear next. Useful evaluation preserves enough of this chain to choose the next informative collection or correction, rather than reporting one number detached from its physical meaning.

Exercises and further investigation. State action semantics, conditioning information, physical units and evaluation population. Separate constructed calculations from trained model or robot evidence.

1. Derive the fixed-covariance Gaussian regression loss, including action normalization. Explain what changes when covariance is learned or when a different action scale is used.
2. Reproduce the ridge fit and its gradient. Compare equal-row and equal-episode weighting for a collection with unequal episode lengths.
3. Prove the conditional-mean identity and reproduce the two-strategy example. Propose an observable condition that could resolve the modes, and one kind of variation that should remain a distribution.
4. Derive the local effort-to-velocity perturbation and reproduce the biased-joint rollout. Identify which assumptions fail across a contact change or actuator saturation.
5. Reproduce the absorbing-error example. State why its bounded bad-step count cannot be inferred from a generic physical-action MSE.
6. Specify one dataset-aggregation iteration, including expert information, authority and learner-visited states. Distinguish querying a label from physically resetting the apparatus.
7. Reproduce the attention and temporal-ensemble calculations. Show why time-slot alignment and oldest-first ordering matter.
8. Derive the variational bound and diagonal-Gaussian KL. Compare a true reconstruction likelihood with a weighted L1-plus-KL objective, and explain the training-versus-inference information boundary in ACT.
9. Reproduce the diffusion corruption, reverse mean and posterior variance example. Explain why oracle noise recovery is not a trained sampler and why diffusion level is not robot time.
10. Compare coherent sequence sampling with independent marginal sampling. Identify physical restrictions neither approach guarantees.

11. Reproduce the chunk-latency example and define an execution rule for expired predictions. State which observations and anchors remain associated with the executed requests.
12. Design offline and closed-loop tests for one demonstration dataset. Include task outcomes, failure diagnostics and a proposed next collection that would distinguish two competing explanations for a poor result.

The dataset-aggregation study develops the distribution-shift motivation and its specific learning assumptions (Ross et al. 2011). The original variational-autoencoder, attention and diffusion formulations provide broader model foundations (Kingma and Welling 2014; Vaswani et al. 2017; Ho et al. 2020). ACT/ALOHA and Diffusion Policy connect these ideas to concrete robot action interfaces (Zhao et al. 2023; Chi et al. 2023). Their reported experiments are not new reproduction results of this chapter.

Chapter 13 considers learning across heterogeneous experience and using predictive world models. The distinctions established here between an action distribution, a physical transition and a deployed feedback law remain essential as the scope of the data grows.

Chapter 13 Generalist Policies and World Models

Two arms receive the instruction to move a cube to the right. Their software returns the same seven numbers. One arm moves as intended; the other moves too far, rotates about a different axis or closes its gripper. The numbers had a common shape but different meanings. Now suppose a model generates a convincing video of the intended motion. The video may still omit the force needed to start sliding, the actuator delay or the contact that prevents the motion. Neither a shared representation nor a plausible prediction completes the physical argument.

Chapter 12 developed policies learned from demonstrations. This chapter broadens the source of experience and distinguishes three roles. A *generalist policy* is trained for a declared range of tasks, conditions or embodiments. A *world model*, as used here, predicts specified consequences from a history and proposed actions. A planner uses predictions and an objective to choose actions. These roles can be combined, but none follows automatically from the others. A large vision-language-action model can be a policy without an explicit predictive model of physical consequences.

For ExpGen PhySyst, the important question is what experience must be generated to support each role. Diverse demonstrations can broaden a policy's conditioning. Excited transitions can distinguish competing dynamics. Corrective observations can reveal where imagined behavior departs from interaction. We develop these connections using statistical objectives, Featherstone mechanics and small checked examples. The calculations explain mechanisms; they are not new training or deployment results.

13.1 Learning across heterogeneous experience

Heterogeneous experience differs in more than the appearance of a scene. Sources can use different robots, controllers, sensors, sampling intervals, task definitions and selection rules. Some contain demonstrations; others contain policy rollouts or measurements intended for identification. A useful shared dataset preserves these differences while specifying which relations a learner should share.

13.1.1 Shared representations, source mixtures and dataset weighting

Let e identify a source and its declared collection configuration. Let P_e describe the retained records from that source after documented transformations. A training record contains conditioning information c and an action label a, with source-specific validity masks when needed. A mixture objective is

$$R_{\mathbf{w}}(\theta) = \sum_{e=1}^{E} w_e \, \mathbb{E}_{(c,a)\sim P_e}[\ell_e(\theta; c, a)], \qquad w_e \geq 0, \quad \sum_e w_e = 1. \tag{13.1}$$

The sampling unit matters. Choosing a source, then an episode, then a valid timestep differs from choosing uniformly among all stored rows. Losses must also specify whether they average over valid coordinates, action slots or whole sequences. Doubling an episode's sampling frequency does not create another independent physical interaction.

Parameters shared across sources can encode useful visual, temporal or behavioral regularities. Source-specific encoders and output heads can handle some differences. Such sharing is a modeling choice rather than a proof that all sources measure the same variables. Missing camera channels require an explicit availability mask: a padded black image is not evidence that a camera observed darkness. Similarly, an absent force channel is not a measured zero force.

Worked example: one model, two mixture risks. Suppose a fixed model has dimensionless mean losses 0.1 and 0.9 on two defined sources. If their retained row counts are 90 and 10, uniform-row sampling gives $0.9(0.1) + 0.1(0.9) = 0.18$. Equal-source sampling gives 0.5. The model has not changed. The question posed by the aggregate has changed. Neither weighting is intrinsically correct: a target dominated by the first source and a requirement to serve both equally are different purposes.

If a target source mixture has weights v_e, importance factors v_e/w_e recover its expectation only when $w_e > 0$ wherever $v_e > 0$ and the within-source distributions are the intended ones. They cannot supply an unseen contact regime or undo

the state distribution induced by deploying a different controller. Chapter 9 established the more general population and selection argument.

Octo provides a concrete example of deliberate sharing. Its original architecture combines modality-specific tokenizers, a transformer with availability and temporal masks, and action readout heads. Its curated pretraining mixture restricts the action conventions; adaptation to new inputs or outputs can add tokenizers or heads and fine-tune the policy (Octo Model Team et al. 2024). This is useful flexibility, with a specified adaptation procedure, rather than an automatic conversion between every robot interface.

13.1.2 Embodiment, observation and action compatibility

For embodiment e, distinguish a raw observation history H_k^e, a representation encoder E_e, a model output u_k and an interface decoder D_e. A schematic execution chain is

$$c_k = E_e(H_k^e, g), \qquad u_k \sim \pi_\theta(\cdot \mid c_k), \qquad a_k^e = D_e(u_k, H_k^e). \tag{13.2}$$

Here g specifies the task. The decoder may restore units, select a robot-specific head, transform a frame or turn a relative request into a target anchored at the current pose. Its inputs and timing are part of the deployed policy. A decoder cannot generally recover information that the representation discarded.

Even an affine normalization changes the metric. If physical coordinates satisfy $a = b_e + D_e^{\text{scale}} u$, with an invertible diagonal scale, then

$$\|\hat{u} - u\|^2 = (\hat{a} - a)^\mathsf{T} (D_e^{\text{scale}})^{-\mathsf{T}} (D_e^{\text{scale}})^{-1} (\hat{a} - a). \tag{13.3}$$

Fit data-dependent scales only on allowed training data and retain them for decoding. Scaling a displacement does not turn it into a velocity. For a constant velocity request, displacement also depends on the hold interval. Joint ordering, sign, gripper convention, pose anchor and controller mode remain separate concerns.

Worked example: the same normalized request. Two ideal joint-position interfaces use zero reference and scales $0.2\,\text{rad}$ and $0.4\,\text{rad}$. Model output $u = 0.5$ requests $0.1\,\text{rad}$ and $0.2\,\text{rad}$ respectively. These are targets, not measured motion. If the second robot defines positive rotation in the opposite physical direction, its sign convention must also be reconciled. Likewise, $0.02\,\text{m}$ of displacement and $0.02\,\text{m/s}$ held for $0.1\,\text{s}$ request motions differing by a factor of ten.

An observation adapter must be equally explicit. A camera image is not automatically calibrated into a robot frame. A joint measurement and an estimated object pose can have different timestamps and uncertainty. Embodiment context can include kinematics, inertias, tool geometry and controller settings, but simply appending metadata does not ensure the network uses it correctly. Preserve the original record and the adapter version so that a failed behavior can be traced to representation, decoding, control or physical interaction.

13.2 Conditioning behavior on tasks

Tasks distinguish what should happen from what is currently observed. The same scene can support several valid goals. Without an informative condition, those goals become conflicting labels rather than a single intended behavior.

13.2.1 Language, goals and policy inputs

Language, a goal image, a target pose and a task identifier encode different kinds of information. A target pose names a geometric relation once its frame and object are defined. A sentence may identify an object or a relation but leave the approach path, force and termination unspecified. A goal image includes appearance and spatial information but may also contain irrelevant background cues. The collection protocol determines which distinctions can be learned.

For task condition g and available history H_k, a policy models $p_\theta(a_k \mid H_k, g, e)$. In a token-based action model with token sequence $b_{k,1:m}$, one possible factorization is

$$p_\theta(b_{k,1:m} \mid H_k, g, e) = \prod_{j=1}^{m} p_\theta(b_{k,j} \mid b_{k,<j}, H_k, g, e). \tag{13.4}$$

The index j orders output tokens; it is not the physical control clock. The policy still needs a rule converting the decoded sequence into a request at a specified physical time. Parallel or diffusion action heads use other factorizations, as Chapter 12 explained.

Worked example: a condition resolves a label ambiguity. Imagine a fixed scene and two equally frequent requests, left and right. For this constructed label model, the correct action is deterministically -1 or $+1$ given the request. Ignoring the request, the best single-choice classifier is correct half the time. The action-label entropy is $\log 2$ nats; conditioning on the request reduces it to zero. This proves that the condition is informative in the construction. It does not show that a robot can perceive the object, interpret a new sentence or execute the action.

Collection should include matched scenes with different goals and equivalent goals expressed in different ways. Otherwise the network may use scene appearance as a shortcut for the instruction. Store who supplied the condition, when it became available and how success was judged. A future final image may be a training label or a supplied goal; it must not silently enter the deployment history as if already observed.

13.2.2 Vision-language-action models and motor-control interfaces

A vision-language-action model combines visual information, a task description and an action output. The original OpenVLA turns continuous action coordinates into discrete tokens, trains next-token prediction on action tokens, and uses pretrained visual and language components. Its training curation includes restrictions on observations and single-arm end-effector control (Kim et al. 2024). Those restrictions are part of what makes its shared input and output spaces meaningful. The original Octo instead uses a diffusion action head and supports language or goal-image conditioning (Octo Model Team et al. 2024). These are different parameterizations of policies; neither architecture by itself asserts a model of how forces produce motion.

The physical boundary is the same as in earlier chapters. In Featherstone's angular-first convention, body motion and wrench are $\mathbf{v}_i = (\boldsymbol{\omega}_i, \mathbf{v}_{O_i})$ and $\mathbf{f}_i = (\mathbf{n}_{O_i}, \mathbf{F}_i)$, at declared body origins and in declared frames. Their pairing is power, and body balance is

$$\mathbf{f}_i^{\text{net}} = \mathbf{I}_i \mathbf{a}_i + \mathbf{v}_i \times^* \mathbf{I}_i \mathbf{v}_i. \tag{13.5}$$

Joint projection and propagation give, with compatible tangent velocity $\boldsymbol{\nu}$,

$$\mathbf{M}(\mathbf{q})\dot{\boldsymbol{\nu}} + \mathbf{h}(\mathbf{q}, \boldsymbol{\nu}) = \boldsymbol{\tau}_a + \boldsymbol{\tau}_p + \sum_j \mathbf{J}_j^{\mathsf{T}} \mathbf{f}_j^{\text{ext}}, \qquad \dot{\mathbf{q}} = \mathcal{N}(\mathbf{q})\boldsymbol{\nu}. \tag{13.6}$$

Here $\mathbf{h}$ has the declared rigid-body velocity and gravity terms, $\boldsymbol{\tau}_p$ contains separately modeled passive effects, and each external wrench is paired with its matching Jacobian. External forces must not also be counted inside another term. The model output is not automatically the delivered effort $\boldsymbol{\tau}_a$ (Featherstone 2008).

Worked example: equal torque, unequal dynamics. Two ideal horizontal joints have inertias 0.02 and $0.04\,\mathrm{kg\,m^2}$, no friction or other load, and direct delivered torque $0.01\,\mathrm{N\,m}$. Their accelerations are 0.5 and $0.25\,\mathrm{rad/s^2}$. A shared torque label therefore does not imply a shared state transition. If the output is instead a position target, the servo gains, saturation and internal state belong in the transition description too.

In a native MuJoCo implementation, the corresponding distinction is between configuration `qpos`, tangent velocity `qvel`, actuator request `ctrl` and resulting motion or sensed output. The actuator model determines how `ctrl` enters generalized effort; its array length need not equal the number of velocity coordinates (DeepMind Technologies Limited, n.d., n.d.). A VLA adapter must preserve that declared interface. A language embedding is not a spatial motion vector, and a token projection is not a rigid coordinate transform.

13.3 Learning predictive models from experience

An action policy answers what to request. A predictive model answers what may follow if a request is made. For embodied learning, useful predictions may concern joint motion, contact, task progress, observations or rewards. The

predicted quantities, intervention assumptions and time interval must be stated before calling a model a world model.

13.3.1 Latent state inference, transition prediction and observation models

Let o_k denote an observation at decision boundary k, a_k the request held or otherwise executed over the next declared interval, and r_{k+1} the reward attributed to that transition. Hidden physical state can include joint configuration, velocity, object state and controller memory. A learned latent representation need not expose these coordinates directly.

A recurrent stochastic model can use deterministic memory d_k and a stochastic latent z_k, with model state $s_k = (d_k, z_k)$:

$$\begin{aligned} d_{k+1} &= f_\theta(d_k, z_k, a_k, e), \\ z_{k+1} &\sim p_\theta(\cdot \mid d_{k+1}, e), \\ o_k &\sim p_\theta(\cdot \mid s_k, e), \\ r_{k+1} &\sim p_\theta(\cdot \mid s_k, a_k, s_{k+1}, g, e). \end{aligned} \tag{13.7}$$

The reward arguments here deliberately match our transition timing. A simpler reward head can be a special case if the chosen state contains the relevant information. Embodiment context e can be observed, inferred or absent, but those are different model assumptions.

When a real observation arrives, an inference model forms

$$q_\phi(z_k \mid d_k, o_k, e). \tag{13.8}$$

This is a filtering update using currently available information. An imagined rollout instead propagates the transition prior in Eq. (13.7) without supplying future observations. Providing each recorded future observation to the inference model tests reconstruction and filtering, not autonomous multi-step prediction. PlaNet develops this distinction with a recurrent state-space model and latent-space planning (Hafner et al. 2019).

Why introduce memory? A single image of a joint at $q = 0$ can be consistent with opposite velocities. Two noiseless states with velocities $\pm 0.2\,\mathrm{rad/s}$ and no acceleration reach $\pm 0.02\,\mathrm{rad}$ after $0.1\,\mathrm{s}$. The current image alone cannot choose the future. History, another sensor or a distribution over hidden states is needed. More pixels from the same instant need not resolve this temporal ambiguity.

Worked example: latent coordinates are not physical coordinates. Consider a fixed-axis joint with $J = 0.02\,\mathrm{kg\,m^2}$, directly delivered held torque, and no gravity, friction or other load. Over $\Delta = 0.1\,\mathrm{s}$, exact integration gives

$$\begin{bmatrix} q_{k+1} \\ v_{k+1} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0.1 \\ 0 & 1 \end{bmatrix}}_{A} \begin{bmatrix} q_k \\ v_k \end{bmatrix} + \underbrace{\begin{bmatrix} 0.25 \\ 5 \end{bmatrix}}_{B} \tau_k. \tag{13.9}$$

The coefficients carry the units implied by q in radians, v in radians per second and τ in newton metres. They follow from $q_{k+1} = q_k + \Delta v_k + \Delta^2 \tau_k/(2J)$ and $v_{k+1} = v_k + \Delta \tau_k / J$.

Now encode $z = Tx$, where $x = (q, v)^\mathsf{T}$ in these fixed units and $T = \mathrm{diag}(2, 0.5)$. Then

$$z_{k+1} = TAT^{-1} z_k + TB\tau_k, \qquad TAT^{-1} = \begin{bmatrix} 1 & 0.4 \\ 0 & 1 \end{bmatrix}, \quad TB = \begin{bmatrix} 0.5 \\ 2.5 \end{bmatrix}. \tag{13.10}$$

Decoding with T^{-1} produces exactly the same physical trajectory. The changed numbers do not indicate changed mechanics. This invertible feature rescaling is not a Featherstone spatial transform: it neither rotates a rigid frame nor changes a wrench reference point. Unconstrained latent coordinates therefore need not have unique physical meanings. Physical interpretation requires additional structure or measurements.

13.3.2 Prediction objectives, uncertainty and multi-step error

A model is fitted to a declared generative distribution, not to the word "world." For a sequence of length H, condition on actions, task and embodiment and take fixed initial memory. One factorization is

$$\begin{aligned} p_\theta(o_{0:H}, r_{1:H}, z_{0:H} \mid a_{0:H-1}, g, e) = p_\theta(z_0 \mid e) p_\theta(o_0 \mid s_0, e) \\ \times \prod_{k=0}^{H-1} p_\theta(z_{k+1} \mid d_{k+1}, e) p_\theta(o_{k+1} \mid s_{k+1}, e) \\ \times p_\theta(r_{k+1} \mid s_k, a_k, s_{k+1}, g, e). \end{aligned} \tag{13.11}$$

The deterministic memory recursion is implicit. An approximate posterior can factor as the initial inference update followed by the filtering updates in Eq. (13.8). Although it may ignore rewards in its inputs, it remains a valid approximate distribution for a variational bound if the required support conditions hold.

Writing $Y = (o_{0:H}, r_{1:H})$ and suppressing fixed conditioning for readability, Jensen's inequality gives

$$\log p_\theta(Y) = \log \mathbb{E}_{q_\phi}\left[\frac{p_\theta(Y, z_{0:H})}{q_\phi(z_{0:H})}\right] \geq \mathbb{E}_{q_\phi}[\log p_\theta(Y, z_{0:H}) - \log q_\phi(z_{0:H})]. \tag{13.12}$$

Expanding the log product yields

$$\begin{aligned} \mathcal{B} = \mathbb{E}_q \left[\sum_{k=0}^{H} \log p_\theta(o_k \mid s_k, e) + \sum_{k=0}^{H-1} \log p_\theta(r_{k+1} \mid s_k, a_k, s_{k+1}, g, e)\right] \\ - D_{\mathrm{KL}}(q_\phi(z_0 \mid d_0, o_0, e) \| p_\theta(z_0 \mid e)) \\ - \sum_{k=0}^{H-1} \mathbb{E}_q \left[D_{\mathrm{KL}}\big(q_\phi(z_{k+1} \mid d_{k+1}, o_{k+1}, e) \| p_\theta(z_{k+1} \mid d_{k+1}, e)\big)\right]. \end{aligned} \tag{13.13}$$

The outer expectation in each transition KL averages over the preceding latent history. Reconstruction and reward terms encourage useful predictions; the posterior-to-prior KL trains consistency between inferred and imagined states. Arbitrary loss weights, free-nats thresholds or auxiliary multi-step penalties change the objective. They are not automatically this exact lower bound. Neither a valid bound nor a low training loss proves that the fitted state is physically identifiable (Kingma and Welling 2014; Hafner et al. 2019).

For measured physical state, a Gaussian transition likelihood gives a residual weighted by inverse covariance and a log-determinant term. If different coordinates have different units, normalize or define a physical metric explicitly. Predicting images may devote most loss to background pixels while missing a small but decisive contact. Reward prediction can emphasize task relevance but can also discard distinctions needed for another task. Multi-step objectives should say which future observations are withheld, which actions are supplied and how resets or termination are handled.

PETS combines probabilistic dynamics with bootstrapped ensembles and propagates trajectory particles when planning (Chua et al. 2018). For an equally weighted ensemble of B_m conditional predictions with means μ_b and covariances Σ_b, total covariance is

$$\bar{\mu} = \frac{1}{B_m}\sum_b \mu_b, \qquad \Sigma_{\mathrm{mix}} = \frac{1}{B_m}\sum_b \Sigma_b + \frac{1}{B_m}\sum_b (\mu_b - \bar{\mu})(\mu_b - \bar{\mu})^{\mathsf{T}}. \tag{13.14}$$

This follows by writing each error as its within-component deviation plus its component-mean deviation; the cross terms have zero conditional mean. The denominator is B_m, because this is a specified mixture's covariance, not an unbiased sample-variance estimate. Interpreting the terms as aleatoric and epistemic uncertainty requires assumptions about noise, model family and fitting. Shared model bias can escape both.

Worked example: disagreement is only one uncertainty. Three scalar predictors have means $(0.8, 1.0, 1.2)$ in a declared normalized coordinate and conditional variance 0.04 each. Their mean is 1, between-model variance is 0.0266667 and total variance is 0.0666667. Reporting only disagreement omits the conditional variance. Conversely, three identical predictors can have zero disagreement while sharing a large error. An uncertainty score needs held-out calibration evidence before it becomes a quantitative reliability statement.

Multi-step error makes the distinction more consequential. Compare true transition F with a learned deterministic mean $\widehat{F}$, starting from states $x_0, \hat{x}_0$ and using the same action sequence. Suppose $\widehat{F}$ is L-Lipschitz in the chosen state

norm and $\|\widehat{F}(x,a) - F(x,a)\| \le \epsilon$ throughout a domain containing all relevant true and predicted states. Adding and subtracting $\widehat{F}(x_k, a_k)$ gives

$$e_{k+1} \le Le_k + \epsilon, \qquad e_H \le L^H e_0 + \epsilon \sum_{j=0}^{H-1} L^j, \quad e_k = \|\hat{x}_k - x_k\|. \tag{13.15}$$

The sum is $(1 - L^H)/(1 - L)$ for $L \neq 1$ and H for $L = 1$. This is a conditional deterministic bound, not a guarantee derived from mean validation error. It does not cover arbitrary contact switches, out-of-domain states or differing feedback actions. A latent error also needs a suitable decoder bound before it implies a physical error.

Worked example: a small transition discrepancy accumulates. In a normalized scalar system, let $F(x) = 0.9x$ and $\widehat{F}(x) = 0.95x$, with $x_0 = \hat{x}_0 = 1$. At step ten the true value is 0.3486784401 and the model predicts 0.5987369392. Both rollouts stay in $[-1, 1]$, where $\epsilon = 0.05$ and $L = 0.95$ are valid. Equation (13.15) gives an upper bound 0.4012630608 on an actual error of 0.2500584991. An average one-step score would not itself establish that uniform bound.

13.4 Using predictions to choose actions

A predictor becomes part of a controller only after an objective, search or learning procedure, information pattern and execution rule are added. Predicted trajectories are conditional on proposed actions; physical observations reveal the outcome of the actions actually applied.

13.4.1 Selecting actions through predicted consequences and planning objectives

At a decision boundary, infer a belief b_k from available observations. Plan a sequence $A_k = (a_k, \ldots, a_{k+H-1})$ over H intervals. A finite-horizon objective can be

$$A_k^\star \in \arg\min_{A_k \in \mathcal{A}_k} \mathbb{E}_{b_k, p_\theta} \left[\sum_{j=0}^{H-1} \gamma^j \ell(s_{k+j}, a_{k+j}, g) + \gamma^H V_f(s_{k+H}, g) \right]. \tag{13.16}$$

Here $\gamma \in [0, 1]$ is the discount factor and $\mathcal{A}_k$ expresses declared request constraints. The constraint set must not be mistaken for a guarantee that predicted contact or unobserved physical constraints hold. Costs can penalize task error, velocity, effort and other outcomes, with compatible units or explicit scales. Risk-sensitive objectives may add restrictions on predicted distributions, but their meaning depends on model calibration.

The cross-entropy method (CEM) searches a distribution over candidate action sequences. Sample candidates, predict their consequences, rank their scores, retain an elite set $\mathcal{E}$ and refit the distribution. For an unsmoothed diagonal Gaussian update obtained by maximum likelihood,

$$m_j^{\text{new}} = \frac{1}{|\mathcal{E}|} \sum_{n \in \mathcal{E}} A_j^{(n)}, \qquad (s_j^2)^{\text{new}} = \frac{1}{|\mathcal{E}|} \sum_{n \in \mathcal{E}} (A_j^{(n)} - m_j^{\text{new}})^2. \tag{13.17}$$

The index j runs over the sequence's coordinates and slots. Practical implementations may smooth updates, impose variance floors, truncate samples or warm-start search. These choices must be stated. CEM is a finite stochastic search, not a proof of global optimality. Re-evaluate any final mean or selected candidate using the intended constraints. PlaNet and PETS use CEM within model-predictive control (Hafner et al. 2019; Chua et al. 2018).

Worked example: acceleration and braking in a two-step plan. Use the ideal joint in Eq. (13.9), initially at rest at $q = 0$, with $|\tau_k| \le 0.04\,\mathrm{N\,m}$. The goal is $q_2 = 0.02\,\mathrm{rad}$ with low terminal velocity. Define dimensionless cost

$$C = 100 \left(\frac{q_2 - 0.02\,\mathrm{rad}}{1\,\mathrm{rad}} \right)^2 + \left(\frac{v_2}{1\,\mathrm{rad/s}} \right)^2 + 0.1 \sum_{k=0}^{1} \left(\frac{\tau_k}{1\,\mathrm{N\,m}} \right)^2. \tag{13.18}$$

The scales make the trade-off explicit; they are not universal tuning values. Applying the transition twice gives, in the stated units,

$$q_2 = 0.75\tau_0 + 0.25\tau_1, \qquad v_2 = 5\tau_0 + 5\tau_1. \tag{13.19}$$

Four candidate sequences illustrate the search:

(τ_0, τ_1) in N m	q_2 in rad	v_2 in rad/s	C
$(0, 0)$	0	0	0.04000
$(0.02, 0.02)$	0.020	0.200	0.04008
$(0.04, -0.04)$	0.020	0	0.00032
$(0.02, 0)$	0.015	0.100	0.01254

The best sampled sequence accelerates and then brakes. Position error alone would miss the difference from continued acceleration. The best two candidates give the CEM mean $(0.03, -0.02)\,\mathrm{N\,m}$. Its population standard deviations are $(0.01, 0.02)\,\mathrm{N\,m}$. Re-evaluating the mean gives $C = 0.003255$, worse than the best sampled sequence. Refitting a search distribution is not the same as monotonically improving its mean action.

In MPC only a declared prefix is applied, often the first action. The next observation updates the state belief and starts another plan. For probabilistic ensembles, preserve how model identity is propagated: sampling one model for a whole imagined trajectory differs from resampling at every step. PETS explicitly studies such alternatives (Chua et al. 2018). Temporal correlation in uncertain dynamics is lost if uncertainty is treated as independent fresh noise at every step.

13.4.2 Model-based learning and correction from physical observations

Online planning and learning a policy in imagination use a model differently. A planner searches a new sequence for the current belief. An imagined-policy method learns parameters that map model states to actions across many imagined trajectories. A schematic finite-horizon actor objective is

$$J_{\mathrm{imag}}(\psi) = \mathbb{E}_{s_0 \sim \mathcal{B},\, a_j \sim \pi_\psi,\, s_{j+1} \sim p_\theta} \left[\sum_{j=0}^{H-1} \gamma^j \bar{r}_\theta(s_j, a_j, s_{j+1}, g) + \gamma^H V_\eta(s_H, g) \right]. \tag{13.20}$$

Here $\mathcal{B}$ contains inferred starting states from real records, $\bar{r}_\theta$ is a predicted reward mean, and V_η supplies a learned terminal value. This expression is explanatory, not a claim that all world-model methods use the same return estimator.

Dreamer learns action and value models through latent imagined trajectories, then executes the learned action model while incorporating real observations. Its original formulation differentiates return estimates through the learned dynamics, with the world model fixed during behavior updates (Hafner et al. 2020). Ha and Schmidhuber's earlier recurrent world-model study is another influential example of learning behavior with a compressed predictive model (Ha and Schmidhuber 2018). These works concern defined systems and evaluations; the name does not confer universal physical understanding.

For a differentiable deterministic transition mean $\hat{s}_{j+1} = F_\theta(\hat{s}_j, a_j)$ and actor $a_j = \pi_\psi(\hat{s}_j)$, define transition Jacobians $F_s = \partial F_\theta / \partial s$, $F_a = \partial F_\theta / \partial a$ and actor Jacobians $\Pi_s = \partial \pi_\psi / \partial s$, $\Pi_\psi = \partial \pi_\psi / \partial \psi$, with other arguments fixed in each partial derivative. A pathwise sensitivity satisfies

$$\frac{d\hat{s}_{j+1}}{d\psi} = (F_s + F_a \Pi_s) \frac{d\hat{s}_j}{d\psi} + F_a \Pi_\psi. \tag{13.21}$$

Jacobians are evaluated along the imagined trajectory, with model parameters held fixed. Stochastic reparameterized rollouts add their declared noise variables. Automatic differentiation computes sensitivities of this computation, not automatically of the true robot. A planner or actor can exploit an inaccurate transition or reward model, especially outside the collected support (Baydin et al. 2018).

Worked example: replan after a measured discrepancy. The first torque $0.04\,\mathrm{N\,m}$ in the preceding example predicts $(q_1, v_1) = (0.01, 0.2)$ in the stated units. Suppose the next observation instead establishes $(0.011, 0.18)$. Use that state, without claiming to have identified the cause of the discrepancy. For one remaining step, the same model predicts $q_2 = 0.029 + 0.25\tau_1$ and $v_2 = 0.18 + 5\tau_1$. Differentiating the remaining terminal and effort cost gives

$$\tau_1^\star = -\frac{100(0.25)(0.009) + 5(0.18)}{100(0.25)^2 + 5^2 + 0.1} = -0.0358851675\,\mathrm{N\,m}. \tag{13.22}$$

All numerical coefficients use the normalization in Eq. (13.18). This value respects the declared torque bound. The revised prediction is approximately $(q_2, v_2) = (0.0200287081, 0.0005741627)$. This illustrates state correction and replanning, not a measured control result or evidence that the transition model has become accurate.

A discrepancy can come from state inference, wrong action decoding, delay, unmodeled forces or dynamics. Store planned, decoded, limited and applied requests separately when they differ. A model trained on a position target must include the servo response; relabeling the target as delivered torque changes the problem. Likewise, a predicted contact wrench must use the frame and reference point dual to its matching angular-first spatial Jacobian in Eq. (13.6). The wrench itself is moment-first.

For an ideal direct-torque simulation companion, the physical rollout maps q and v to configuration and tangent-velocity state, while the candidate torque enters through the declared actuator mapping before integration. A learned model can replace the rollout computation for planning, but its actions must retain that meaning. In code, batch and candidate axes are additional array indices; they do not change physical units. The accompanying `check_chapter13.py` checks these small calculations without requiring a simulator or a trained network.

13.5 Data requirements for broader capability

Broader capability requires more than a larger collection. It requires the right distinctions to be observable, actions to be interpretable and evaluation conditions to test the claimed extension. A model's apparent coverage can be broad in appearance and narrow in mechanics.

13.5.1 Coverage, adaptation and distribution shift

A policy dataset needs observations and task conditions that distinguish appropriate actions. A transition dataset needs enough state and input variation to distinguish relevant consequences. Video recorded under one controller may predict that controller's usual futures well while supporting poor predictions for alternative actions. Conditioning on an action does not by itself establish a causal intervention model: hidden state, operator decisions and selection can confound an observational comparison. Designed excitation and matched interventions make a stronger physical argument, subject to the limits of Chapter 8.

Worked example: unexcited data cannot choose the inertia. Suppose every recorded ideal-joint episode begins at $q = 0$, $v = 0$ and applies zero torque, with no other load. Inertias $J = 0.02$ and $J = 0.04\,\mathrm{kg\,m^2}$ predict the same stationary records exactly. Under a new held torque $0.01\,\mathrm{N\,m}$ for $0.1\,\mathrm{s}$, the first model predicts $(q, v) = (0.0025, 0.05)$ and the second predicts $(0.00125, 0.025)$ in radians and radians per second. Collecting more stationary records does not resolve the difference. A controlled probe can, provided torque, state and other loads are sufficiently known. This is a constructed identification problem, not an instruction to apply a particular hardware command.

Data design therefore spans task conditions, physical states, inputs, contacts, objects, viewpoints and embodiments. A useful plan states which axes vary, which remain fixed and which combinations are reserved. Corrective collection targets a diagnosed ambiguity or failure mode, rather than merely repeating easy successes. Include transitions near relevant boundaries, with justified limits and explicit reset conditions. Failures can be informative if their causes and observation quality are retained; a failure label alone does not identify its mechanism.

Adaptation also changes the evidence budget. Zero-shot execution, interface calibration, a new action head, full fine-tuning and online model correction are distinct procedures. Count target demonstrations, calibration trials, normalization changes, checkpoint selection and instruction tuning as applicable. Holding out test episodes while repeatedly choosing adapters from their outcomes does not preserve a held-out evaluation.

Source mixtures can encourage useful sharing and can also create negative transfer. Incompatible labels, a dominant repetitive source or missing distinctions may hurt a particular condition. Inspect source-wise behavior and controlled ablations before concluding that the model needs either more capacity or more undifferentiated data. Octo and OpenVLA make concrete curation and interface choices; their choices motivate investigation rather than a universal dataset recipe (Octo Model Team et al. 2024; Kim et al. 2024).

13.5.2 Method-specific tests of physical validity and task generalization

Different mechanisms need different tests. A generalist policy should be tested on named held-out factors with its adaptation budget stated. A world model should be tested for observation-conditioned inference, prior-only prediction and the particular quantities used for decisions. A model-based controller additionally needs evidence that actions chosen through prediction work when executed.

For predictive evaluation, initialize from a declared observation prefix, supply the recorded future action sequence, and withhold future observations for the prediction horizon. Compare appropriate quantities: joint and object motion in physical units, contact events with declared timing tolerance, task progress or reward, and image predictions where relevant. Reconstructing every frame with its own posterior answers a different question. Test uncertainty against held-out residuals and empirical coverage under named conditions; disagreement alone is not a confidence interval.

For physical-coordinate outputs, a dynamic residual can be written

$$r_{\mathrm{dyn}} = \mathbf{M}(\mathbf{q})\widehat{\dot{\boldsymbol{\nu}}} + \mathbf{h} - \boldsymbol{\tau}_a - \boldsymbol{\tau}_p - \sum_j \mathbf{J}_j^{\mathsf{T}} \mathbf{f}_j^{\mathrm{ext}}. \tag{13.23}$$

A small residual means consistency with the specified model, force accounting and derivative estimate. It does not establish that the model or unmeasured forces are correct. A latent model without a physical decoder cannot be tested by pretending its coordinates are joint accelerations. Impact intervals may need impulse balance instead of a smooth acceleration residual. Kinematic feasibility, friction, energy and task success are related but different checks.

Worked example: a broad average hides a weak condition. Consider constructed successes $90/100$, $8/10$ and $2/10$ for three declared conditions. The pooled rate is $100/120 = 0.833333$; the equal-condition average is $(0.9+0.8+0.2)/3 = 0.633333$. The last condition remains at 0.2 under either summary. The choice of summary changes the target population, not the individual outcomes. These counts are arithmetic fixtures, not a result for any cited model. Chapter 16 will develop comparative uncertainty and inference.

When testing language conditioning, change the requested goal while keeping the scene comparable; separately vary wording without changing the goal. For embodiment adaptation, distinguish familiar dynamics with a new appearance from a new actuator path or action space. For planning, compare model-predicted outcomes with executed outcomes, track clipping and timing, and examine whether optimization selects implausible transitions. Better offline prediction does not necessarily improve the ranking of candidate actions, and a good ranking on one task need not support another task's costs.

An ExpGen PhySyst is especially useful when its next interaction is chosen to make one of these distinctions observable. The system produces more than an action label: it produces an interpretable relation among conditions, requests, consequences and uncertainty. Generalist policies broaden the range over which such relations can inform behavior. World models make some consequences available for imagined comparison. Mechanics, measurement and designed collection determine when those comparisons can be trusted.

Exercises and further investigation. State source populations, physical interfaces, available information and the status of every result as constructed, source-reported or measured.

1. Derive the mixture-risk example. Compare uniform-row, equal-episode and equal-source sampling, and state the target population each is intended to represent.
2. Derive the normalization metric. Construct two identically shaped action arrays with incompatible physical meanings and specify the information required to reconcile them.
3. Reproduce the task-conditioning entropy calculation. Design matched-scene tests for goal sensitivity and wording invariance, and identify a shortcut that an unmatched collection could permit.
4. Trace an original OpenVLA or Octo action output through decoding, scaling, frame/anchor conventions, limiting and low-level control. Distinguish what the inspected paper specifies from what a particular implementation still needs to define.
5. Derive the ideal-joint transition from delivered torque and show that the two latent-coordinate systems decode to the same trajectories. Explain why this transformation is not a rigid spatial transform.
6. Derive the sequential variational bound from its factorization, including the initial-state KL. Explain why future-observation-assisted reconstruction is not an open-loop prediction test.
7. Derive the ensemble covariance decomposition and reproduce the scalar example. Give a shared-bias counterexample to interpreting disagreement as a calibrated error bound.
8. Prove the rollout-error recurrence and reproduce the ten-step example. Identify the assumptions that would fail at a contact switch, under a different feedback action sequence or outside the tested domain.
9. Reproduce the four MPC candidates and the elite-distribution update. Re-evaluate the updated mean and explain why it need not be the best sampled sequence.
10. Reproduce the one-step replanning optimum and check the effort bound. Contrast this online sequence optimization with training an actor through imagined trajectories.
11. Reproduce the unexcited-inertia ambiguity and propose a collection that can distinguish the models. State which measured quantities and nuisance effects control the strength of the conclusion.
12. Design a held-out evaluation for a policy and a predictive model using the same physical system. Include named variation axes, adaptation data, prior-only rollout tests, physical residual limits and executed task outcomes. Reproduce pooled and equal-condition summaries.

Octo and the original OpenVLA illustrate two generalist-policy designs with different architectures and explicit data curation (Octo Model Team et al. 2024; Kim et al. 2024). PETS and PlaNet develop uncertainty-aware or latent planning; Dreamer develops behavior learning through latent imagination (Chua et al. 2018; Hafner et al. 2019; Hafner et al. 2020). Read these as specified methods with evidence and limitations, rather than interchangeable demonstrations of a single universal capability.

Chapter 14 turns to transfer across simulation and reality. The question is no longer only whether a model predicts or a policy acts in its training setting, but which physical, perceptual and temporal relations survive a change in setting, and which data can establish or repair that correspondence.

Chapter 14 Transfer across Simulation and Reality

Imagine a joint controller that settles smoothly in simulation but oscillates when connected to a physical actuator. Increasing the simulated inertia makes one recorded trajectory look closer to the measured one. Yet the modified controller still oscillates. A later timestamp check reveals that physical commands are applied one control interval late. The inertia change compensated for part of a trace without repairing the feedback loop that generated it.

This example expresses the central difficulty of transfer. A discrepancy has several possible causes, and a successful fit is not necessarily a successful explanation. The physical dynamics, sensor interpretation, action interface and timing together determine what a learned policy actually does. Chapter 13 separated policies, predictive models and planning. Here we ask which relations between a simulated system and a physical one must hold for those components to remain useful.

For ExpGen PhySyst, transfer is also a problem of designing data. A measurement can calibrate an observation, distinguish actuator gain from inertia, reveal delayed execution or expose a contact regime absent from training. Simulated experience can explore alternatives at scale, while physical experience tests the correspondence those alternatives assume. The aim is a justified transfer decision for a declared task and operating range, rather than a single universal measure of simulator realism.

14.1 What transfers, and under which conditions

A model, a visual representation, a policy, a controller parameter and a dataset are different transferable objects. A policy may work on the physical system despite imperfect state prediction. A calibrated model may reproduce recorded trajectories while a new policy visits conditions where it is inaccurate. State what is retained, what is changed and what behavior is expected before evaluating a transfer.

14.1.1 Model correspondence and task-level transfer

Let F_r^{Δ} and F_s^{Δ} denote physical and simulated transitions over the same declared time interval Δ. Their states x_r, x_s include relevant robot and environment state, actuator and controller memory, and any timing buffers needed for the chosen boundary. Let Φ map physical state into a comparable simulated representation, and let D convert a simulated request a_s into a physical request. A local correspondence error can be written

$$d_F(x_r, a_s) = \Phi\big(F_r^{\Delta}(x_r, D(a_s, x_r))\big) - F_s^{\Delta}(\Phi(x_r), a_s). \tag{14.1}$$

The subtraction presumes compatible coordinates in a common local chart. For rotations, use a declared relative-rotation error rather than subtracting quaternion entries. If no defensible full-state map exists, compare specified observable quantities instead. An interface with memory must retain that memory in the augmented state or history.

Observation correspondence is separate: two states mapped by Φ need not produce equivalent images, encoder readings or inferred poses. Likewise, D can restore units or implement a servo target without making the underlying actuation identical. Matching state and action array dimensions establishes neither correspondence.

For a policy π and a task score S whose higher values are better, define expected performance in domain $d \in \{s, r\}$ by

$$J_d(\pi; \rho_d) = \mathbb{E}_{\xi \sim \rho_d,\, \zeta \sim P_d^{\pi}(\cdot|\xi)}[S(\zeta; \xi)], \qquad G_{\pi} = J_s(\pi; \rho_s) - J_r(\pi; \rho_r). \tag{14.2}$$

The condition ξ includes the declared initial state, object, task and other operating variables. The trajectory ζ follows the domain's closed-loop distribution P_d^{π}, including stochastic transitions and observations. A positive G_{π} means simulation is optimistic under these definitions. Different condition populations can change the gap without changing either implementation. Compare matched condition distributions when asking about transfer, and retain absolute physical performance: a gap of zero also occurs when both systems fail.

The recent reality-gap review distinguishes discrepancies in system models from differences in policy performance and organizes strategies that reduce or tolerate them (Aljalbout et al. 2026). That distinction guides the chapter. We do

not require every physical detail to be reproduced, but we do require evidence that omitted or mismatched details do not invalidate the claimed behavior.

14.1.2 Physical, visual, temporal and interface discrepancies

Physical discrepancies include geometry, mass properties, friction, compliance, backlash, contact and actuation. Visual and sensing discrepancies include calibration, appearance, exposure, noise, occlusion and estimator behavior. Temporal discrepancies include capture delay, clock offset, filtering, inference time and command application. Interface discrepancies include units, signs, ordering, reference frames, target anchors, saturation and reset logic. These categories interact: a changed camera can alter control commands, which changes contact and subsequent observations.

Featherstone's spatial notation makes one correspondence precise. Let X_{BA} map angular-first motion at origin O_A in frame A to motion at O_B in frame B. Power and kinetic energy require

$$\begin{aligned} v_B &= X_{BA} v_A, & f_B &= X_{BA}^{-\mathsf{T}} f_A, \\ I_B &= X_{BA}^{-\mathsf{T}} I_A X_{BA}^{-1}, & f_B^{\mathsf{T}} v_B &= f_A^{\mathsf{T}} v_A. \end{aligned} \tag{14.3}$$

Wrenches are moment-first and must use the declared origin. If $v_A = J_A \nu$, then $J_B = X_{BA} J_A$ and $J_B^{\mathsf{T}} f_B = J_A^{\mathsf{T}} f_A$. A frame conversion preserves generalized effort only when the motion and force transformations are dual (Featherstone 2008).

Worked example: a force at the wrong origin. Keep the frame axes aligned and place O_B at $r = (0.1, 0, 0)\,\mathrm{m}$ relative to O_A. With $[r]_\times$ the cross product matrix,

$$X_{BA} = \begin{bmatrix} I_3 & 0 \\ -[r]_\times & I_3 \end{bmatrix}. \tag{14.4}$$

For $v_A = (0, 0, 2, 0, 0, 0)^{\mathsf{T}}$, the angular speed is $2\,\mathrm{rad/s}$ and O_B moves at $(0, 0.2, 0)\,\mathrm{m/s}$. A $5\,\mathrm{N}$ force in the positive y direction at O_B, with zero moment there, corresponds to a $0.5\,\mathrm{N\,m}$ positive z moment at O_A. Power is $1\,\mathrm{W}$ in either representation. Copying the same six wrench numbers between origins loses the moment and breaks the equality. An apparent physical-model discrepancy can therefore be a representation error before any parameter has changed.

Visual randomization addresses another part of the correspondence. The early domain-randomization study varied rendered appearance, scene content and camera properties for object localization, then used the detector within a robot grasping pipeline (Tobin et al. 2017). Its setting and geometry assumptions matter. Appearance variation does not establish accurate force response, and correct joint dynamics does not establish that a vision policy receives equivalent inputs.

14.2 Collecting evidence of a discrepancy

The first useful comparison isolates what is being compared. Agreement under one command sequence is a statement about that sequence and its conditions. Agreement of task scores under feedback is a statement about the resulting closed loops. Both are useful, but they answer different questions.

14.2.1 Matched initial conditions, commands and aligned observations

For an offline replay, initialize the simulator from a defensible physical state estimate, replay the logged decoded requests with their actual hold durations, and compare observations at aligned times. Record the controller mode and state, actuator limits, object state and reset procedure. Replaying a policy's unscaled network output through a different decoder does not reproduce its physical requests. Starting at the same joint angle while ignoring velocity or servo memory does not reproduce the same initial state.

Let y_i^r be a calibrated physical observation and $\hat{y}_i^s$ its simulated counterpart at a matched observation time. One diagnostic is

$$r_i^y = y_i^r - \hat{y}_i^s, \qquad E_y = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \| W r_i^y \|^2}. \tag{14.5}$$

W defines units or scaling, and may account for specified measurement uncertainty. Report physical-coordinate errors alongside normalized scores. An orientation error needs its own geometric definition. A low replay error is evidence within the visited region, with the given initialization and input path; it is not a test of every future policy.

After observations diverge, two feedback policies generally issue different requests even with identical parameters. In a common chart, add and subtract transitions at matched states and inputs to separate three effects:

$$\begin{aligned} F_r(x_r, a_r) - F_s(x_s, a_s) = {} & [F_r(x_r, a_r) - F_s(x_r, a_r)] \\ & + [F_s(x_r, a_r) - F_s(x_s, a_r)] \\ & + [F_s(x_s, a_r) - F_s(x_s, a_s)]. \end{aligned} \tag{14.6}$$

Here the maps have already been expressed through the correspondence adapters and use one common interval. The terms describe model discrepancy, state divergence and request divergence. Under stated local Lipschitz bounds,

$$e_{k+1} \le \epsilon_F + L_x e_k + L_a \|a_k^r - a_k^s\|. \tag{14.7}$$

This is conditional on staying in the domain of those bounds. It does not convert an average replay score into a global feedback guarantee.

Worked example: delay resembles a tracking error. Suppose an otherwise matched joint signal is $q(t) = A\sin(\omega t)$, with $A = 0.1\,\mathrm{rad}$ and frequency $2\,\mathrm{Hz}$. Comparing $q(t)$ with $q(t-d)$ for $d = 0.02\,\mathrm{s}$ gives residual amplitude and full-period RMS

$$A_r = 2A\left|\sin\frac{\omega d}{2}\right|, \qquad E_r = \frac{A_r}{\sqrt{2}}, \qquad \omega = 4\pi\,\mathrm{rad/s}. \tag{14.8}$$

The amplitude is approximately $0.02506665\,\mathrm{rad}$ and RMS $0.01772480\,\mathrm{rad}$. Correct time alignment removes this constructed residual without changing inertia or friction. A phase lag at one frequency does not uniquely identify a pure delay: filters, actuator dynamics and clock errors can produce related symptoms. Capture and application timestamps, plus additional excitation, help distinguish them.

14.2.2 Residual structure and competing explanations

Return to the articulated balance in compatible tangent coordinates:

$$M(q)\dot{\nu} + h(q,\nu) = \tau_a + \tau_p + \sum_j J_j^{\mathsf{T}} f_j^{\mathrm{ext}}, \qquad \dot{q} = \mathcal{N}(q)\nu. \tag{14.9}$$

The rigid-body bias h contains the declared gravity and velocity terms; passive effects and external wrenches remain separately accounted. Featherstone recursions provide the physical structure. They do not measure unknown loads or convert a vendor current reading into a calibrated joint torque automatically (Featherstone 2008).

For a fixed-axis joint, consider the declared local model $J\ddot{q} + b\dot{q} = gu + \tau_{\mathrm{ext}}$, where u is a nominal torque request in newton metres and g is a dimensionless delivered-effort gain. Let subscript 0 denote nominal parameters, and δ mean true minus nominal. With acceleration estimate $\widehat{\ddot{q}} = \ddot{q} + e_a$, the nominal dynamic residual satisfies

$$\begin{aligned} r_0 &= J_0\widehat{\ddot{q}} + b_0\dot{q} - g_0 u - \tau_{\mathrm{ext},0} \\ &= \delta g\, u - \delta J\, \ddot{q} - \delta b\, \dot{q} + \delta\tau_{\mathrm{ext}} + J_0 e_a. \end{aligned} \tag{14.10}$$

This expansion assumes the displayed measurements and timing other than e_a are exact. Additional sensor and alignment errors add terms. Correlations with acceleration, velocity, command or contact suggest tests; they are not unique diagnoses. A flexible residual model can reduce r_0 while concealing a wrong interpretation of u.

Worked example: inertia and actuator gain explain the same trace. Let $u = 0.1\,\mathrm{N\,m}$ be held for $0.1\,\mathrm{s}$, starting from $q = 0$, $\dot{q} = 0$, with no friction, gravity or other load. The nominal model $J = 0.02\,\mathrm{kg\,m^2}$, $g = 1$ predicts acceleration $5\,\mathrm{rad/s^2}$ and final angle $0.025\,\mathrm{rad}$. A system with the same inertia but $g = 0.8$ instead has acceleration $4\,\mathrm{rad/s^2}$ and angle $0.020\,\mathrm{rad}$. Changing the simulated inertia to $0.025\,\mathrm{kg\,m^2}$ while keeping $g = 1$ reproduces that angle and acceleration exactly.

More generally, all pairs (cJ, cg) for $c > 0$ give the same ratio g/J and the same motion for any request history in

this ideal model. More command variation cannot separate the factors. A delivered-torque measurement or a known load intervention provides another relation. The problem is structural ambiguity, not merely insufficient optimizer iterations. Chapter 8 developed the general identification framework; here it prevents choosing a transfer repair from the wrong explanation.

Worked example: a numerical discrepancy before a physical one. For direct delivered torque $0.01\,\mathrm{N\,m}$ and $J = 0.02\,\mathrm{kg\,m^2}$, start at zero angle and velocity and integrate for $0.1\,\mathrm{s}$, with no other loads. The exact angle is $0.0025\,\mathrm{rad}$. With step $h_n = 0.01\,\mathrm{s}$, explicit Euler gives $0.00225\,\mathrm{rad}$ and velocity-first symplectic Euler gives $0.00275\,\mathrm{rad}$. Their difference is numerical, despite identical physical parameters. Halving the step halves each angle error in this constant-acceleration construction. A timestep study therefore precedes interpreting such disagreement as evidence about the hardware.

14.3 Choosing a transfer strategy

A useful strategy follows an identified uncertainty or discrepancy. Repair a unit or timing error at its interface. Calibrate a measurement using a reference. Identify parameters when the model structure and data support them. Revise a structure when a parameter adjustment cannot express the missing mechanism. Design robustness or adaptation for remaining variation that cannot be eliminated economically.

14.3.1 Calibration, identification and model revision

Calibration relates an instrument output to a reference quantity. Identification estimates a system model from input–output behavior. Model revision changes assumptions, such as adding actuator state, friction, compliance or a contact mode. These operations can all reduce disagreement, but they change different parts of the explanation.

Worked example: calibrate the observation. Take trusted reference angles -0.2, 0 and $0.2\,\mathrm{rad}$. Suppose a constructed sensor reports $y = 1.02q + 0.01\,\mathrm{rad}$. The uncorrected errors are $(0.006, 0.010, 0.014)\,\mathrm{rad}$, with RMS approximately $0.010520\,\mathrm{rad}$. Fitting the affine relation gives gain 1.02 and offset $0.01\,\mathrm{rad}$; using $\hat{q} = (y - 0.01\,\mathrm{rad})/1.02$ removes the errors exactly in this noiseless fixture. No physical joint dynamics have changed. A real calibration would retain reference uncertainty and test other angles, loads and temperatures (Joint Committee for Guides in Metrology 2008).

Parameter fitting should preserve physical admissibility and distinguish identified combinations from assumed values. Mass, center of mass and rotational inertia cannot be varied arbitrarily. If a parameter estimate changes drastically with timestep, filter or command convention, examine whether it is compensating for an implementation mismatch. A nominally accurate fit can fail outside its training excitation or contact mode.

One way to represent a missing generalized effort is a hybrid model

$$M(q)\dot{\nu} + h(q,\nu) = \tau_a + \tau_p + \sum_j J_j^{\mathsf{T}} f_j^{\mathrm{ext}} + r_\eta(H_k, a_k). \tag{14.11}$$

The residual r_η has generalized-effort units and ordering; its history arguments are declared. Adding a correction to the next state instead defines another model. A learned force correction, action transformation and observation correction are not interchangeable. Examine whether a learned term violates conservation, dissipative assumptions or known limits, and test whether it generalizes to held-out interactions. Low residual loss alone does not establish a physical cause.

Real-to-sim construction may recover geometry and appearance from physical observations, then estimate properties needed for interaction. A visually convincing asset can still have wrong collision geometry or mass properties. Conversely, a simple model may be adequate for a declared task. The required fidelity depends on which discrepancies affect its decisions, not on visual detail alone.

14.3.2 Physically justified randomization, adaptation and robust control

Domain randomization trains over a declared distribution $\rho(\theta)$ of simulated conditions. Let $C_\theta(\pi)$ be a lower-is-better expected cost for a policy in condition θ. Average and robust objectives are different:

$$\min_\pi \mathbb{E}_{\theta\sim\rho}[C_\theta(\pi)] \qquad \text{versus} \qquad \min_\pi \sup_{\theta\in\Theta} C_\theta(\pi). \tag{14.12}$$

An average objective weights rare conditions lightly; a worst-case objective depends on the chosen uncertainty set Θ. Neither makes unrepresented physical conditions disappear. Dynamics randomization has been used to train recurrent robot control under varied physical parameters and timing (Peng et al. 2018). That is evidence for a specified training and deployment design, not a theorem that any wide randomization range will transfer.

Worked example: a high average can hide a failed regime. For three constructed regimes, policies A and B have success probabilities $(1, 1, 0)$ and $(0.8, 0.8, 0.8)$. Under training weights $(0.49, 0.49, 0.02)$, their mean successes are 0.98 and 0.8. Their worst-regime successes are 0 and 0.8. Under target weights $(0.25, 0.25, 0.5)$, their means become 0.5 and 0.8. The apparent preference depends on the operating population and the criterion. These are supplied probabilities in an arithmetic example, not estimated outcomes for a trained policy.

Randomize coherent physical variables, their correlations and their time scales. A fixed unknown mass is normally drawn once per episode, whereas measurement noise may evolve every sample. Redrawing inertia every step represents a changing system and may require additional momentum accounting. A log-normal scale for a positive property differs from an unbounded additive Gaussian. Contact and material variation may be coupled to geometry, temperature or velocity; independent marginal draws can generate implausible combinations.

Worked example: positive inertia is not sufficient. For a central rotational inertia I_C, the second-moment matrix of a nonnegative mass distribution is

$$\Sigma_C = \tfrac{1}{2}\operatorname{tr}(I_C) I_3 - I_C \succeq 0. \tag{14.13}$$

The principal moments of I_C must satisfy the triangle inequalities. Consider the positive-definite but inadmissible example

$$\begin{aligned} I_C^{\text{bad}} &= \operatorname{diag}(0.01, 0.01, 0.05)\,\mathrm{kg\,m^2}, \\ \Sigma_C^{\text{bad}} &= \operatorname{diag}(0.025, 0.025, -0.015)\,\mathrm{kg\,m^2}. \end{aligned} \tag{14.14}$$

The second-moment matrix fails the test. The diagonal inertia $(0.03, 0.04, 0.05)\,\mathrm{kg\,m^2}$ instead gives $(0.03, 0.02, 0.01)$ for the second moments and passes this test. Positive mass, center-of-mass conventions and compatibility with the claimed geometry still need attention (Wensing et al. 2018).

Adaptation uses available interaction history to change a controller's context, parameters or decisions. A schematic context-conditioned policy is

$$\hat{z}_k = A_\phi(o_{k-L:k}, a_{k-L:k-1}), \qquad a_k = \pi_\psi(o_k, \hat{z}_k). \tag{14.15}$$

The context can be useful without identifying a unique physical parameter. The ideal-joint history above can reveal g/J while leaving g and J ambiguous. Report what is updated, which observations it uses, its reset rule and its behavior during the adaptation transient.

Rapid Motor Adaptation trains a base policy with an encoded environment context, then trains a history-based module to estimate that context. Its deployed actions are joint-position targets converted through a PD controller, rather than direct torque predictions (Kumar et al. 2021). This illustrates why learned context and the actuator interface both belong in the transfer description. Changing context online is different from updating network weights or estimating a physically named inertia.

Worked example: robustness must include execution delay. Take an ideal fixed-axis joint with no other load, measured velocity v_k, direct held torque $\tau_k = -K_v v_k$ and sampling interval h_c. Since $J\dot{v} = \tau$, exact held-input sampling gives

$$v_{k+1} = (1 - c)v_k, \qquad c = \frac{h_c K_v}{J}. \tag{14.16}$$

Strict decay of this sampled velocity subsystem requires $0 < c < 2$. If the same request is applied exactly one interval late, then

$$v_{k+1} = v_k - cv_{k-1}, \qquad \lambda^2 - \lambda + c = 0. \tag{14.17}$$

For real c, the second-order Schur conditions reduce to $c > 0$ and $c < 1$. Thus a delay halves the allowable velocity-feedback gain range in this construction. Position retains an integrator: damping velocity alone does not asymptotically regulate a desired angle.

Let the control interval be $h_c = 0.02\,\mathrm{s}$. The uncertain inertia ranges from 0.01 to $0.03\,\mathrm{kg\,m^2}$. Compare gains $K_v = 0.8$ and $K_v = 0.4$, both in $\mathrm{N\,m/(rad/s)}$:

J in $\mathrm{kg\,m^2}$	Undelayed pole $K_v = 0.8$	Delayed radius $K_v = 0.8$	Delayed radius $K_v = 0.4$
0.01	−0.600000	1.264911	0.894427
0.02	0.200000	0.894427	0.632456
0.03	0.466667	0.730297	0.516398

All undelayed poles are inside the unit circle. With delay and the higher gain, the smallest inertia is unstable. The lower gain ensures strict velocity decay throughout the stated interval, by $0 < c < 1$. This proof is specific to the ideal linear velocity loop, exact observations and one-slot delay. It does not establish stability of a contact-rich learned controller with saturation.

14.4 Collecting data to improve transfer

The next collection should reduce an uncertainty that matters for the transfer decision. Repeating a failed behavior may estimate its frequency more precisely without explaining it. A changed sensor, load or command protocol can be more informative if it distinguishes the leading causes.

14.4.1 Targeted measurements, interventions and coverage gaps

Begin with competing explanations and ask what observation would differ between them. A trusted encoder reference can distinguish scale/offset from true motion. A delivered-effort measurement can distinguish actuator gain from inertia. A known load change can add an independent relation. Additional capture and application timestamps can distinguish sensor alignment from delayed physical execution. Each measurement needs an uncertainty model and a record of what else changed.

Worked example: a known added inertia resolves a scale ambiguity. For the gain/inertia example, add independently known inertia $\Delta J = 0.01\,\mathrm{kg\,m^2}$ without changing gain, friction or other loads. The system $(J, g) = (0.02, 0.8)$ now accelerates at $0.08/0.03 = 2.666667\,\mathrm{rad/s^2}$. The alternative $(J, g) = (0.025, 1)$ predicts $0.1/0.035 = 2.857143\,\mathrm{rad/s^2}$. The models that agreed before the intervention no longer agree.

If a_0 and a_1 denote exact accelerations before and after the load change, at the same nonzero request u, then

$$a_0 = \frac{gu}{J}, \qquad a_1 = \frac{gu}{J + \Delta J}, \qquad J = \frac{\Delta J\, a_1}{a_0 - a_1}, \quad g = \frac{Ja_0}{u}. \tag{14.18}$$

Substituting $a_0 = 4$, $a_1 = 8/3$ in the stated units recovers $J = 0.02$ and $g = 0.8$. If $a_0 - a_1$ is small relative to measurement uncertainty, the inverse is poorly conditioned. Mounting a load can introduce friction, gravity or compliance; the inference fails if those changes are ignored. The fixture demonstrates a designed intervention, not a completed hardware identification.

Local information analysis can help choose excitation. For independent Gaussian observations with parameter-independent covariance R_i and mean $\mu_i(\theta)$, define sensitivity $S_i = \partial\mu_i/\partial\theta$. The Fisher information is

$$\mathcal{I}(\theta) = \sum_i S_i^{\mathsf{T}} R_i^{-1} S_i. \tag{14.19}$$

Derivatives must include state propagation and the controller when these depend on θ. Correlated errors require a joint covariance; parameter-dependent covariance contributes additional terms. Information criteria apply within those assumptions and chosen parameter scales. They are not a universal numerical quantity of experience.

Worked example: separate inertia and viscous friction locally. Suppose delivered torque is measured, acceleration and velocity are known, and the model is $\tau = J\ddot{q} + b\dot{q}$ with no other loads. Normalize acceleration, velocity and torque by reference scales $a_* = 1\,\mathrm{rad/s^2}$, $v_* = 1\,\mathrm{rad/s}$ and $\tau_* = 1\,\mathrm{N\,m}$. The dimensionless parameters are $\beta = (Ja_*/\tau_*, bv_*/\tau_*)^\mathsf{T}$, and the design rows are $(\ddot{q}/a_*, \dot{q}/v_*)$. Rows $(1,2), (2,4), (3,6)$ all point in one direction; their Gram matrix is

$$\Phi^\mathsf{T}\Phi = \begin{bmatrix} 14 & 28 \\ 28 & 56 \end{bmatrix}, \qquad \det(\Phi^\mathsf{T}\Phi) = 0. \tag{14.20}$$

Repeated measurements of this relation cannot separate the two normalized parameters. In contrast, rows $(1,0), (0,1)$ give the identity Gram matrix. With independent normalized torque-noise standard deviation 0.01, their information matrix is $10^4 I_2$, with determinant 10^8. This is a local construction with known regressors, not evidence that noisy acceleration estimates or a closed-loop hardware trial have the same information. Identifying J, b here assumes delivered torque is known, unlike the earlier unknown-gain problem.

After choosing a correction, retain separate data for fitting, choosing the correction and evaluating it. Physical adaptation trials and calibration measurements are part of the information budget. Newly generated simulated data should record the model revision and parameter distribution; the same policy name does not identify the experience generator. When a revised controller changes visited contacts, collect evidence in that new region rather than treating old replay agreement as sufficient coverage.

14.4.2 Testing observation pipelines, actuator paths and execution timing

Test each path with quantities whose expected meaning is known before connecting them into a closed loop. For observations, verify units, sign, frame, availability mask, capture time and preprocessing. For actions, verify output decoding, normalization, anchoring, limiting, queueing and application. For state, include controller integrators, filters, previous requests and estimator memory where they affect the next transition. A reset that clears physical state but preserves an old command queue does not reproduce an independently initialized episode.

At least three time intervals can matter:

$$d_{\mathrm{sense}} = t_{\mathrm{available}} - t_{\mathrm{capture}}, \qquad d_{\mathrm{infer}} = t_{\mathrm{ready}} - t_{\mathrm{start}}, \qquad d_{\mathrm{apply}} = t_{\mathrm{applied}} - t_{\mathrm{ready}}. \tag{14.21}$$

All timestamps require aligned clocks and explicit event definitions. End-to-end information age may also include scheduling waits between these events. A fixed delay, jitter and a clock offset are different models. Simulate a causal queue or filter when those mechanisms matter; adding independent observation noise is not equivalent to delaying a correlated signal.

The connection to native MuJoCo is explicit. Configuration `qpos` and tangent velocity `qvel` represent different spaces for some joints. `ctrl` enters an actuator model; it is not universally the delivered generalized torque. Sensor processing can include transformations, filters and delays outside the engine (DeepMind Technologies Limited, n.d., n.d.). A portable policy executor should preserve the same request meaning, observation construction and control interval when it replaces simulated observations with physical ones.

An implementation-level record can be organized as

```
record = (capture_time, available_time, observation,
          policy_time, model_output, decoded_request,
          limited_request, applied_time, measured_response)
```

This is a semantic schema, not an instruction to assume all fields are directly observable. Mark estimated or unavailable values. Link each record to model, controller, normalization and calibration revisions. The numerical companion independently recalculates the chapter's idealized examples. It is provided as `check_chapter14.py` and does not stand in for this complete physical logging pipeline.

Known-answer tests should include joint order and sign, a force at a known origin, a request at a limit, a missing observation and a reset with buffered history. Replay and closed-loop tests then expose interactions that isolated checks cannot reveal. Keep fitting and diagnosis separate: an interface repair can improve transfer immediately without any retraining, while retraining under a broken interface may hide the cause.

14.5 Evaluating transferred behavior

Transfer evaluation combines correspondence evidence with task behavior. Neither a small model residual nor a high simulated success rate is an adequate substitute for the physical outcome being claimed. Equally, a failed physical trial need not invalidate every useful relation in the simulator; it should narrow the question and guide another test.

14.5.1 Nominal and shifted operating conditions

Define nominal conditions before testing. Then name shifts in mechanics, appearance, sensing, timing or interface, rather than merging all variation into one difficult test set. Evaluate a fixed policy and adapter when estimating their transfer. If adaptation is part of the method, state its allowed information, update schedule, warm-up and reset rules. Include the transient and unsuccessful adaptation attempts.

Use compatible task definitions across domains. Success may require position/orientation tolerance, a dwell interval, limited effort and no drop or collision. The physical observation pipeline must be able to support that judgment. A simulator's exact object pose can define a training reward without becoming an available physical deployment measurement. Timeouts, resets and unavailable measurements affect the denominator and interpretation of reported outcomes.

Worked example: paired outcomes expose the transfer direction. Consider twenty matched condition pairs: twelve succeed in both domains, six only in simulation, one only physically and one in neither. The simulated success fraction is $18/20 = 0.9$, the physical fraction is $13/20 = 0.65$, and the gap in Eq. (14.2) is 0.25. Seven pairs are discordant. The marginal fractions alone do not reveal that structure. Matching does not make repeated physical outcomes deterministic or independent; these are constructed counts illustrating what to retain, not a significance test or hardware result.

Report physical task performance beside diagnostic quantities such as replay error, timing discrepancy and effort saturation. Separate a good trajectory match from accurate ranking of alternative policies. A simulator can preserve policy ranking while systematically overestimating all success rates, or match one policy's score while ranking others incorrectly. Ranking validity requires several actual matched comparisons, not a conclusion from visual similarity alone.

Retain failures with context: slip after contact, a missed target due to vision, a stale command, a timeout and a protective stop are different outcomes. Predefine whether assisted or interrupted trials count toward the task score and report interventions separately. Chapter 16 will develop experimental units, uncertainty intervals and comparisons; this chapter identifies the transfer relations those statistics should describe.

14.5.2 What simulation evidence establishes before physical testing

Simulation can establish that an implementation behaves in a specified executable model, explore sensitivity to declared parameters and expose failure mechanisms within its assumptions. Timestep and solver studies can test numerical dependence. Known-answer interface checks can test units, transforms and timing logic. Randomized evaluation can describe performance under the sampled simulated distribution.

Each result has a boundary. Agreement between two simulators is useful but does not identify which matches the physical system. A learned world model's plausible video does not establish contact forces or actuator response. A successful policy in a broad randomized set does not prove the physical system lies in that set, that its observations are comparable or that the deployed executor implements the same loop. Conversely, an inaccurate simulator may still support useful transfer when the policy is insensitive to the relevant discrepancies. That insensitivity is a claim to test under stated conditions, not an assumption granted by the model's complexity.

For ExpGen PhySyst, a productive transfer investigation links a diagnosis, a changed collection and a revised decision. The new data might repair a measurement scale, resolve a physical ambiguity, identify timing or add a missing contact regime. Its value is the distinction it enables and the behavior subsequently tested. Simulated and physical experience then complement each other: simulation explores a declared possibility space, and physical observation checks where that possibility space supports the intended use.

Exercises and further investigation. State the model, interface, time interval, uncertainty and evidence status before interpreting each calculation.

1. Define the transferable object and construct state, action and observation correspondence maps for one robotic task. Identify controller or buffer state that a simple joint-state map would omit.
2. Reproduce the spatial force/origin example and verify power and generalized-effort invariance. Explain why copying wrench numbers or applying an action normalization is not the same transformation.
3. Derive the closed-loop discrepancy decomposition. Compare an offline matched-input replay with a feedback test and state what each can establish about physical dynamics.
4. Derive the sinusoidal delay residual and calculate its amplitude and RMS. Propose another measurement that distinguishes delay from an actuator or observation filter.
5. Expand the scalar dynamic residual, including acceleration error. Reproduce the gain/inertia ambiguity and prove why more varied commands alone do not identify the two parameters in that model.
6. Reproduce the explicit and symplectic Euler examples at both timesteps. Distinguish numerical convergence from physical fidelity.
7. Reproduce the sensor calibration and describe a reference and held-out test that would be needed for a noisy physical instrument.
8. Compare randomized-mean and worst-regime performance in the supplied example. Test the two inertia matrices for physical consistency and specify which randomized parameters remain fixed within an episode.
9. Derive the undelayed and delayed stability conditions. Check the uncertain-inertia family for both gains, and identify assumptions that saturation, contact or jitter would invalidate.
10. Reproduce the known-added-inertia intervention. Study conditioning as the two accelerations approach each other, and explain what happens if mounting the load also changes friction.
11. Derive the information matrices for the two excitation designs. Explain why uncertain regressors, correlated errors and unknown actuator gain change the inference problem.
12. Design a nominal/shifted transfer evaluation with a fixed policy, a declared adaptation budget, matched replay, task outcomes and failure categories. Reproduce the paired-outcome example and identify the next physical observation most useful for a stated competing explanation.

The Annual Review provides a broad reading map of reality-gap causes, transfer strategies and evaluation questions (Aljalbout et al. 2026). The visual and dynamics randomization studies provide concrete examples with different transferable objects (Tobin et al. 2017; Peng et al. 2018); Rapid Motor Adaptation illustrates history-conditioned deployment (Kumar et al. 2021). Their reported experiments are source evidence, not new results of this chapter. Featherstone mechanics, measurement uncertainty and physical parameter consistency support the diagnostic calculations (Featherstone 2008; Joint Committee for Guides in Metrology 2008; Wensing et al. 2018).

Chapter 15 broadens the design question. Instead of taking the body, sensors, environment and collection protocol as fixed and repairing a transfer, we ask how to co-design them so that the resulting physical interactions generate useful, interpretable data for learning.

Chapter 15 Co-designing Physical Systems for Learning

A joint experiment has produced a thousand nearly identical traces, yet two explanations still fit. A manipulation dataset contains many successful demonstrations, yet its camera hides the contact that determines the next action. More repetitions may make the same patterns clearer without resolving either difficulty. The next useful change may be a reference measurement, a different sensor location, a new fixture or a different interaction, rather than another training run.

The preceding chapters developed ways to model, measure, operate and learn from physical systems. Here those methods become design alternatives. Instead of accepting the robot, environment and collection process as fixed, we ask which arrangement would generate records useful for a declared purpose. A mechanism changes the possible contacts; sensing changes the controller's information and actions; an interface changes both what a person demonstrates and what a learner can execute. These are coupled choices.

For ExpGen PhySyst, co-design concerns this complete arrangement and the evidence it makes possible. It does not require every component to be optimized simultaneously or imply that one objective measures experience. A careful comparison of feasible alternatives can be more useful than an elaborate optimization of an inadequate model. All numerical investigations below are constructed and independently recalculated; they are not new hardware trials or trained-policy results.

15.1 Formulating the collection problem

15.1.1 Capability goals, information needs and intended users

Begin with the decision the records should support. Estimating inertia, predicting a contact transition and learning a recovery policy are different purposes. Their useful observations, interventions and operating conditions need not coincide. A deployment user may need a reliable controller; a mechanism designer may need a defensible physical parameter; another researcher may need records that can be reanalyzed. These users can share an apparatus without sharing the same criterion.

Three distinctions from earlier chapters are essential. Observability concerns distinguishing changing states through a specified model and observation process. Identifiability concerns distinguishing unknown model properties through admissible experiments. Task sufficiency concerns whether retained information supports an intended decision or behavior. Identifying every mechanical parameter is unnecessary for some tasks, while accurate prediction on a narrow task may leave parameters unidentified.

Let a candidate arrangement be

$$d = (d_b, d_e, d_s, d_c, d_p). \tag{15.1}$$

The entries describe body and actuation, environment and task, sensing and acquisition, controller and interface, and collection/processing procedure. They may contain discrete choices and continuous parameters. Reset rules, operator involvement, timing and rejected-trial treatment belong here. Unknown properties $\boldsymbol{\theta}$ remain distinct from design choices: selecting nominal friction in a simulator does not set a physical surface's actual friction.

An arrangement induces encounters and a rule for recording them. With a sufficiently augmented Markov state x_k, fixed context c, observation y_k, collection action a_k and available history H_k, one possible factorization is

$$\begin{aligned} p_d(x_{0:T}, y_{0:T}, a_{0:T-1} \mid c) = {} & p_d(x_0 \mid c) O_d(y_0 \mid x_0, c) \\ & \times \prod_{k=0}^{T-1} \Big[\kappa_d(a_k \mid H_k, c) F_d(x_{k+1} \mid x_k, a_k, c) \\ & \qquad \times O_d(y_{k+1} \mid x_{k+1}, c) \Big]. \end{aligned} \tag{15.2}$$

The collection controller κ_d may involve a person. The state includes relevant actuator, controller, filter and timing memory; disturbances may require further augmentation. Terminal events require an absorbing state or a declared stopping rule. This is a modeling choice, not a claim that recorded joint angles are a complete state.

Changing sensing can alter O_d, the history available to κ_d, later actions and the contacts visited. It therefore changes more than the saved columns. Logging and a versioned transformation $T_{d,v}$ then map encounter records into a dataset $\mathcal{D}_v$. Capture, selection and interpretation remain separate operations.

15.1.2 Requirements, alternatives and physical constraints

A requirement states what an acceptable arrangement must achieve and how to judge it. For a contact controller, specify operating conditions, action meaning, available observations, tolerated delay, force range and evidence distinguishing successful contact from tracking failure. For identification, specify the parameter combination and required uncertainty within a stated model. A general request for high-quality data leaves these decisions unresolved.

Separate requirements from proposed solutions. Endpoint force might be measured by a load cell, inferred from joint effort and a contact model, or estimated from calibrated deformation. Each route has different bandwidth, uncertainty, mechanical burden and validity limits. Specifying one sensor before the information need can conceal alternatives.

Check feasibility before ranking convenience: positive mass, realizable inertia, reachability, effort and speed, stiffness, sensing range, timing, reset burden and resources constrain a design. Some constraints follow from a model; others require measurement. A model's feasible boundary is conditional on assumptions, not a tested physical operating limit.

Worked example: reduction changes several requirements. Consider an ideal rigid, lossless reducer with positive ratio n, motor angle $\theta_m = nq$, load inertia J_L and rotor inertia J_m. Exclude gravity, friction, compliance, voltage and thermal effects:

$$T = \tfrac{1}{2}(J_L + n^2 J_m)\dot{q}^2,$$
$$(J_L + n^2 J_m)\ddot{q} = n\tau_m, \qquad \ddot{q}(n) = \frac{n\tau_m}{J_L + n^2 J_m}. \tag{15.3}$$

Reduction multiplies effort and reflects rotor inertia by n^2. For constant positive motor effort τ_m,

$$\frac{d\ddot{q}}{dn} = \frac{\tau_m(J_L - n^2 J_m)}{(J_L + n^2 J_m)^2}, \qquad n_{\text{opt}} = \sqrt{J_L/J_m}. \tag{15.4}$$

This maximizes initial no-load acceleration in the ideal model, not whole-task performance.

Take $J_L = 0.02\ \text{kg m}^2$, $J_m = 0.0002\ \text{kg m}^2$, available motor effort $\tau_m = 0.02\ \text{N m}$, a motor speed bound of 100 rad/s and 4096 motor encoder increments per revolution:

n	effective inertia kg m^2	acceleration rad/s^2	speed bound rad/s	encoder increment rad
5	0.025	4	20	0.00030680
10	0.040	5	10	0.00015340
20	0.100	4	5	0.00007670

Only $n = 10$ meets separate requirements of at least $4.5\ \text{rad/s}^2$ initial acceleration and a kinematic speed bound of at least 8 rad/s. Finer encoder increments alone favor another choice. An increment is not total position accuracy, and a speed bound does not establish that the assumed torque is available at that speed. A physical choice must include the omitted electrical, mechanical and thermal behavior.

15.2 Designing the physical and computational arrangement

15.2.1 Body, environment, task, sensing and controller choices

Featherstone mechanics makes design changes physically explicit. Geometry and joint arrangement affect spatial transforms and motion subspaces; shape and material affect inertia; actuation affects delivered effort and internal states.

For a tree mechanism, using angular-first motion and moment-first force,

$$\begin{aligned} v_i &= X_{i\lambda}(q;d)v_{\lambda(i)} + S_i(q;d)\nu_i, \\ f_i^{\text{loc}} &= I_i(d)a_i + v_i \times^* I_i(d)v_i - f_i^{\text{ext}}, \\ f_i &= f_i^{\text{loc}} + \sum_{j\in\text{children}(i)} X_{ji}^{\mathsf{T}} f_j, \qquad \tau_i = S_i^{\mathsf{T}} f_i. \end{aligned} \tag{15.5}$$

Body acceleration a_i uses the gravity initialization in Chapter 5. The symbol a_k in Eq. (15.2) instead denotes a collection request. The wrench f_i^{ext} excludes gravity, which is already accounted for by that convention. The local required contribution f_i^{loc} is accumulated with child wrenches; it is not the body's net physical wrench. Inertias and wrenches use declared body origins and axes. Required generalized effort is not automatically an actuator command register (Featherstone 2008).

The reducer maps to Chapter 3's ideal native MuJoCo model. Load-body mass properties supply J_L, and joint armature can supply the separate $n^2 J_m$. A unit-gear actuator receiving joint-effort requests uses $u = n\tau_m$. An actuator receiving motor-effort requests through a scalar transmission factor n instead supplies that mapping inside the model. Applying both conversions multiplies effort twice; adding reflected inertia to both body and armature counts it twice. These are representation choices, not different physical predictions (DeepMind Technologies Limited, n.d., n.d.).

A change to n must therefore update each affected quantity and its meaning. Changing an actuator gear alone does not reconstruct rotor inertia, encoder calibration, electrical limits or a physical gearbox. Sensor mass belongs in the moving body's model; feature normalization belongs in the observation pipeline. The companion checks ideal relations, not a simulator implementation of these omitted mechanisms.

A camera bracket changes both inertia and the image relation. Fingertip shape changes contact geometry and available forces. Strong feedback may stabilize observable motion while obscuring the uncontrolled mechanism. A reset fixture improves repeatability but may narrow the collected population. These effects belong in the design argument.

The environment can also be an instrument. A known plane, reference stop, characterized load or visible marker supplies a relation against which to interpret observations. Its tolerances and effects on the interaction matter. A fixture that changes a task must be declared if the eventual behavior is expected to work without it.

Worked example: geometry changes force sensitivity. A planar revolute joint has a radial lever of length ℓ along x, with a tangential force F along y. At this instant, $\omega_z = \dot{q}$, endpoint speed is $\ell\dot{q}$, and external generalized effort is $\tau_{\text{ext}} = \ell F$. For this known force direction,

$$\tau_{\text{ext}}\dot{q} = F(\ell\dot{q}), \qquad \sigma_F = \frac{\sigma_\tau}{|\ell|}, \qquad \Delta J = m_s\ell^2. \tag{15.6}$$

The uncertainty relation assumes exact geometry and an unbiased scalar effort observation of standard deviation σ_τ, excluding uncertain other loads. The last relation approximates an endpoint sensor of mass m_s as a point mass.

For $F = 3$ N, $\dot{q} = 2$ rad/s and $\sigma_\tau = 0.01$ N m, changing ℓ from 0.1 to 0.2 m increases effort from 0.3 to 0.6 N m and endpoint speed from 0.2 to 0.4 m/s. Inferred-force standard deviation decreases from 0.1 to 0.05 N. A 0.05 kg sensor adds 0.0005 versus $0.002\ \text{kg}\,\text{m}^2$ to joint inertia. Better force sensitivity therefore accompanies a changed plant and effort requirement. Powers are 0.6 and 1.2 W: equal joint speed does not mean equal endpoint interaction.

In general, $v_C = J_C\nu$ and $\tau_C = J_C^{\mathsf{T}} f_C$. A frame or origin change requires $v_B = X_{BA}v_A$ and $f_B = X_{BA}^{-\mathsf{T}} f_A$, preserving $f_B^{\mathsf{T}} v_B = f_A^{\mathsf{T}} v_A$. Joint effort does not usually identify an arbitrary six-component wrench. Jacobian rank and contact assumptions determine which combinations can be inferred. Dividing joint efforts by link lengths does not generalize the scalar lever example.

15.2.2 Interactions that distinguish explanations or expand coverage

An interaction can separate explanations that previously agreed or expose a situation missing from a collection. Changing acceleration independently of velocity can separate inertia and viscous friction. New objects, contact modes and recovery states can broaden policy experience. The first aim concerns inference under a model; the second concerns situations available to a learner. Neither automatically implies the other.

Choose an intervention by asking what competing accounts predict will change. If they predict the same measured quantity, repetition cannot distinguish them even with negligible noise. A reference measurement can be more useful

than a larger command. Conversely, an identification fixture can clarify a mechanism while producing motions unlike the behavior to be learned.

Worked example: observations must distinguish the action. Suppose equally likely contact conditions require target displacements $a^* = +L$ and $a^* = -L$, where $L = 0.03$ m. The observation and its available history are identical in both cases. Under squared action error, the optimal deterministic prediction is zero, with risk $L^2 = 0.0009\ \mathrm{m}^2$. Perfectly recorded actions can still omit the condition explaining which action is appropriate.

Add a binary observation $z \in \{-1, +1\}$ reporting the condition correctly with probability $p = 0.9$, symmetrically. Then

$$\mathbb{E}[a^* \mid z] = (2p-1)Lz, \qquad \mathbb{E}[(a^* - \mathbb{E}[a^* \mid z])^2] = 4p(1-p)L^2. \tag{15.7}$$

Predictions become ± 0.024 m and risk becomes $0.000324\ \mathrm{m}^2$. These are exact results for the constructed probability model, not measured sensor accuracy or policy success. Another justified source of context could replace the added sensor. Earlier observations could resolve the ambiguity if they contain that context; this example assumes they do not.

A learner's capacity to represent several actions does not reveal which hidden physical condition is present. Chapters 12 and 13 provide the learning methods; collection design asks whether to reveal the condition, change the action task, retain an uncertainty-aware decision or accept a narrower claim.

15.3 Connecting collection choices to learning

15.3.1 Parameter information and downstream learning utility

Chapter 8 developed parameter-sensitive experiments. Co-design must also ask whether the resolved parameters matter to the intended use. An information criterion rewards a specified statistical distinction. It does not know which errors a task tolerates unless that relation enters the criterion.

Let $\mathcal{A}_\omega$ be a declared learning procedure, including randomness, resource budget and preprocessing conventions. Let $L_*(w; d_*)$ denote downstream loss under a fixed deployment arrangement d_* and target population. A collection-design objective is

$$\begin{aligned} w_d &= \mathcal{A}_\omega\big(T_{d,v}(\mathcal{R}_d)\big), \\ R(d) &= \mathbb{E}_{\mathcal{R}_d,\omega}[L_*(w_d; d_*)], \qquad \min_{d \in \mathcal{F}} R(d), \quad C_j(d) \le B_j. \end{aligned} \tag{15.8}$$

Here $\mathcal{R}_d$ is the collection of raw records, $\mathcal{F}$ the admissible arrangements, and C_j, B_j resource use and budget in the same declared units. The expectation includes collection outcomes and, inside L_*, downstream execution where applicable. This formulates a question; it does not assert that the expectation is analytically available or differentiable through hardware.

Fixing d_* compares collection alternatives for the same deployment question. If the deployed body also changes, compare the corresponding body–controller systems under a declared common task and resource account. That is broader than improving data for a fixed robot. Selection can also exclude information a learner needs: an informative raw stream does not guarantee an adequate training set.

Worked example: total precision can miss the target. Use a correctly specified linear model with two dimensionless parameters, known design rows and independent zero-mean, unit-variance Gaussian output errors. Two equal-size designs are

$$\Phi_A = \begin{bmatrix} 10 & 0 \\ 0 & 2 \end{bmatrix}, \qquad \Phi_B = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}, \qquad \mathcal{I}_d = \Phi_d^{\mathsf{T}} \Phi_d. \tag{15.9}$$

With parameter scales fixed, information determinants are 400 and 256; determinant maximization favors A. Unbiased least-squares covariances are

$$P_A = \mathrm{diag}(0.01, 0.25), \qquad P_B = \mathrm{diag}(0.0625, 0.0625), \qquad \mathrm{var}(r^{\mathsf{T}}\hat{\theta}) = r^{\mathsf{T}} P_d r. \tag{15.10}$$

For a prediction with $r = (0, 1)^\mathsf{T}$, B lowers variance from 0.25 to 0.0625. For $r = (1, 0)^\mathsf{T}$, A is better. These are estimation risks under the stated linear model, not demonstrated control benefits. Parameter information and usefulness for a chosen prediction are related but distinct objectives.

Choose meaningful reference scales before comparing information matrices with unlike physical parameter units. For a learned policy, test behavior through the specified training and execution pipeline rather than substituting an arbitrary determinant. Without a trusted utility model, use a pilot comparison and retain uncertainty about which collection change will help.

15.3.2 Collection effort, alternatives and comparable resource budgets

Purchase cost, setup time, operator attention, device hours, attempted interactions, resets, wear, calibration, processing and training are different resources. Report relevant quantities in their own units. A monetary or weighted scalar objective is legitimate when its conversion factors and preferences are explicit. An unexplained sum of seconds, joules and successful trials has no stable meaning.

Fewer demonstrations can require more specialist labor. More visual context can increase reconstruction and annotation burden. A compliant collection tool can change contact relative to deployment. Simulated trials can depend on costly physical calibration. Include the effort needed to make records meaningful, not merely robot motion time.

Worked example: count attempts before usable records. For a time budget B, setup time t_s and constant cycle time $t_c > 0$, an ideal deterministic schedule allows

$$N_{\text{attempt}} = \max\left(0, \left\lfloor \frac{B - t_s}{t_c} \right\rfloor\right), \qquad \mathbb{E}[N_{\text{usable}}] = p_u N_{\text{attempt}}. \tag{15.11}$$

The expectation assumes the same marginal usability probability p_u for each attempt; independence is unnecessary for this expectation, although relevant to variance. Usability means a declared recording or processing criterion, not automatically task success or learning value.

At $B = 3600$ s, arrangement A uses 300 s setup and 30 s per attempt, allowing 110 attempts. Stipulated $p_u = 0.6$ gives 66 expected usable records. B uses 900 s setup and 45 s per attempt, allowing 60 attempts; $p_u = 0.9$ gives 54 expected usable records. The greater count does not establish better coverage, labels or learning. B might observe a previously hidden contact. These are schedules and supplied probabilities, not measured collection yields.

Failed tasks and invalid recordings still consume resources. Preserve attempts and end causes. Resampling one episode from 30 to 60 Hz can double rows without creating a second physical encounter. Reuse can save collection effort, but its adaptation, calibration and validation belong in the account.

Design objectives can conflict, leaving several nondominated alternatives. One may be cheaper, another more observable and another better matched to deployment. Dominance is relative to declared objectives and constraints. Choosing among remaining alternatives requires priorities and evidence, not an assumption that every design has one naturally correct score.

15.4 From records to reusable experience

15.4.1 Context, interpretation and competing accounts

An encounter occurs under a particular arrangement. A record captures selected indications and context. An interpretation relates them to a physical or statistical account. A reusable result states where that account supports another decision. More detailed metadata can clarify an inference without making its physical assumptions true.

Retain the question, apparatus, conditions, observations, transformations, alternatives, result and limits. Dataset documentation and provenance make such relationships inspectable (Gebru et al. 2021; Moreau and Missier 2013). A lineage graph identifies computational inputs; it does not prove calibration, label validity or generalization. Replaying a record creates another computational use, not another occurrence of the original physical interaction.

Worked example: boundary changes the effort label. Add a tangential resisting force F at lever arm ℓ to the ideal reducer. With calibrated motor torque constant k_t,

$$(J_L + n^2 J_m)\ddot{q} = nk_t i - \ell F,$$
$$\tau_{\text{shaft}} = nk_t i - n^2 J_m \ddot{q}, \qquad J_L \ddot{q} = \tau_{\text{shaft}} - \ell F. \tag{15.12}$$

Electromagnetic effort referred to the joint includes acceleration of the reflected rotor. Output-side shaft effort acts on the load-only boundary. A current-derived label depends on that boundary and the other loads modeled.

In this separate fixture, $n = 10$, $J_m = 10^{-4}\ \text{kg m}^2$, $J_L = 0.02\ \text{kg m}^2$, $k_t = 0.1\ \text{N m/A}$, $i = 0.5$ A, $\ddot{q} = 2\ \text{rad/s}^2$ and $\ell = 0.2$ m. Then $nk_t i = 0.5$ N m, rotor acceleration requires a joint-referred 0.02 N m, $\tau_{\text{shaft}} = 0.48$ N m, and $F = 2.2$ N. The same current record supports several derived quantities but directly measures none of them. Their names, assumptions, uncertainties and reference points must survive reuse.

15.4.2 Human judgment, system updates and conditions for reuse

Human judgment enters task choice, demonstration, screening, annotation and explanation. Preserve the question and information available to the person judging. An acceptable motion does not establish its contact force; a successful task does not ensure a training-quality record. Keep ambiguity visible rather than silently converting judgment into certain physical labels (Aroyo and Welty 2015).

Version a changed calibration, synchronization, label rule or model separately from original indications. One episode can generate several derived datasets. Old and revised estimates are alternative interpretations, not independent samples. A retrospective smoother can use later observations; a deployment controller cannot be credited with information unavailable when it acted.

Before reuse, examine changes in mechanism, payload, friction, mounting, firmware, controller, operator, action meaning, task and target population. Some permit a known transform; others require calibration or new evidence. Matching array shapes or a successful model fit do not establish that a new arrangement lies inside an old claim.

The code representation follows the same logic. Preserve raw records separately from features and labels, with links to arrangement, calibration, transformation and learner versions. A semantic association is

```
episode -> arrangement + raw signals + event times
data -> episodes + calibration + transform + selection
result -> data + learner + execution + evaluation
```

These arrows express lineage, not executable software. Unavailable signals remain unavailable: a schema cannot manufacture a measured effort from an unnamed command. Appendix B collects data conventions and sample-access information; Appendix C connects investigations to optional tools. The reasoning needs no particular application or file format.

15.5 Organizing an investigation

An investigation must keep its design argument inspectable as observations and interpretations change. This book adopts six connected views: Claim, Experience, System, Grounding, Learning and Decision. They organize questions, not a validated six-stage method, a maturity scale or an exhaustive ontology. Revisit them in any order. A worksheet can preserve the argument, but completing a form does not establish correct reasoning.

15.5.1 Claim, Experience and System

Claim states a proposed capability, explanation or improvement, its conditions and relevant comparison. Saying a sensor improves learning leaves the sensor, learner and task unspecified. Saying a contact observation reduces ambiguity between two action labels under a stated distribution identifies something investigable. Separate a design goal from a conclusion supported by completed observations.

Experience identifies interactions that occurred or are proposed, their records and each record's role. Simulated trajectories, physical measurements, reconstructed demonstrations and human judgments have different assumptions. One

encounter can connect all these information sources through analysis; they are not independent physical repetitions. Explain what collection includes, including unsuccessful and missing attempts.

System describes the arrangement generating the interactions and records: body, environment, task, sensors, actuator interface, controller, operator, reset and selection. A load-only model and a rotor-inclusive model use different effort quantities. An inference about a complete servo cannot silently become a bare-motor property.

Together these views expose a consistency test. Does the system permit the proposed encounter, and do its records contain what the claim needs? If only position targets are recorded, a claim about measured joint torque needs another reference or an explicitly uncertain estimation model.

15.5.2 Grounding, Learning and Decision

Grounding explains why indications and transformations bear on the claim. Use calibration, physical balance, frames, units, timing, uncertainty, model assumptions and competing explanations. This is not one numerical evidence level. An unreported observation can leave one question unresolved while other aspects of a published project are well supported. Assess specific propositions rather than declaring an entire system invalid because one quantity was not reported.

Learning identifies the consumer of records and what changes: a parameter estimate, predictive model, policy, representation or human design decision. Specify input/output meaning and information available at execution. Reduced parameter uncertainty and improved behavior are different results; connect them by a test when claiming both.

Decision records what evidence warrants next: retain an arrangement, narrow a claim, revise calibration, change the system, collect another observation or stop an unproductive route. Explain alternatives, resources and remaining uncertainty. A qualified conclusion is useful if it rules out an inappropriate interpretation or identifies an informative change.

These views connect established mechanics, measurement, experimental design and learning; they do not replace those foundations. In paper analysis, distinguish reported evidence, present interpretation and questions requiring another experiment. In prospective design, distinguish proposed observations from completed evidence.

15.6 Comparative worked investigations

15.6.1 Redesigning a joint experiment after ambiguous observations

Worked investigation: calibrate before changing the mechanism. A horizontal fixed-axis joint receives delivered effort τ, has inertia J, no friction or other loads, and an encoder-related indication $y = sq + b$. Positive scale s is dimensionless and offset b is angular:

$$J\ddot{q} = \tau, \qquad \ddot{y} = (s/J)\tau, \qquad r = \ddot{y}/\tau = s/J. \tag{15.13}$$

Suppose $\tau = 0.1$ N m gives $\ddot{y} = 5$ rad/s^2. Thus $r = 50\ (\mathrm{kg\,m^2})^{-1}$. Both a scale of 1 with inertia $0.02\ \mathrm{kg\,m^2}$ and a scale of 1.2 with inertia $0.024\ \mathrm{kg\,m^2}$ explain it. More command variation cannot separate scale and inertia in this model. From rest, a 0.1 s hold gives the same indication change, 0.025 rad, despite different physical motions.

Compare three arrangements. A repeats the original motion. B adds a known-angle reference without changing the moving body. C replaces the actuator or adds a moving sensor before checking the encoder relation. A can reduce variability without resolving the ratio. B adds an independent geometric relation. C changes the system before explaining the ambiguity and can introduce inertia or calibration requirements. It may still be useful for another stated purpose.

In the constructed true system, $s = 1.2$, $b = 0.01$ rad and $J = 0.024\ \mathrm{kg\,m^2}$. A characterized reference establishes $q_0 = 0$ and $q_1 = 0.2$ rad, with $y_0 = 0.01$ and $y_1 = 0.25$ rad:

$$\hat{s} = \frac{y_1 - y_0}{q_1 - q_0} = 1.2, \qquad \hat{b} = y_0 - \hat{s}q_0 = 0.01\ \mathrm{rad},$$
$$\hat{J} = \hat{s}/r = 0.024\ \mathrm{kg\,m^2}. \tag{15.14}$$

Treating reference angles as exact and each indication as having independent standard uncertainty 0.001 rad, first-order

propagation gives

$$\sigma_s = \frac{\sqrt{2}\,(0.001)}{0.2} = 0.0070711, \qquad \sigma_J = \sigma_s / r = 0.00014142\ \mathrm{kg\,m^2}, \tag{15.15}$$

when r is exact. Reference-angle uncertainty, effort uncertainty, acceleration inference and correlation add terms (Joint Committee for Guides in Metrology 2008). Two points do not establish sensor linearity under every angle, load and temperature.

The interpretation changes a physical decision. If a controller treats y as q, an indicated target of 0.2 rad corresponds to $(0.2 - 0.01)/1.2 = 0.15833$ rad. Recalibration may be preferable to changing a mechanism blamed for the wrong motion. Raw indications stay unchanged; the calibrated trajectory is a derived version of the same episode. Subsequent physical trials are needed to test revised task execution.

Worked investigation: redesign excitation after calibration. After establishing angle and delivered-effort meaning, consider inertia, viscous friction and kinetic Coulomb friction:

$$\tau = J\ddot{q} + b_v\dot{q} + \tau_c\,\mathrm{sign}(\dot{q}), \qquad \dot{q} \neq 0. \tag{15.16}$$

This excludes sticking and static-friction transitions. Let $J = 0.024\ \mathrm{kg\,m^2}$, $b_v = 0.02\ \mathrm{N\,m\,s/rad}$ and $\tau_c = 0.01$ N m. Two exact sampled conditions $(\ddot{q}, \dot{q}) = (2, 1)$ and $(4, 2)$ in SI angular units require 0.078 and 0.146 N m. Their first two regression columns are proportional: $\ddot{q}/\dot{q} = 2\ \mathrm{s^{-1}}$. They identify $b_{\mathrm{eff}} = (2\ \mathrm{s^{-1}})J + b_v$ and τ_c, not J and b_v separately.

Add a condition with zero acceleration and velocity 1 rad/s:

$$\begin{bmatrix} 0.078 \\ 0.146 \\ 0.030 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0.024 \\ 0.020 \\ 0.010 \end{bmatrix}. \tag{15.17}$$

The display contains numerical SI coefficients; columns and parameter entries have their corresponding different physical units. The numerical design matrix has determinant 2 and full column rank. An alternative $J = 0.03$ and $b_v = 0.008$ with unchanged τ_c matches the first two efforts but predicts 0.018 N m at the third condition. Negative velocity can further check friction symmetry; it is unnecessary for these three columns to be independent.

The investigation required choices about references, input meaning, mechanics, excitation, estimation and intended use. An apparent need for a more complicated learner became a need for a reference followed by discriminating motion. Uncertain differentiation, changing friction, feedback and model inadequacy still need Chapter 8's treatment. Exact-regressor rank establishes neither low uncertainty nor valid extrapolation.

15.6.2 Revising a manipulation or demonstration collection system

Worked investigation: redesign a pusher collection interface. Suppose the target robot pushes a cube to a region under different initial poses and contact locations. A handheld tool collects demonstrations with one camera. Some successful-looking videos hide the contact; some reconstructed trajectories exceed robot reach; some action labels align with an earlier observation than is available in deployment. This is a constructed design problem, not an evaluation of a teaching package.

Compare A, more episodes with the existing arrangement; B, changed camera/tool geometry revealing contact with a calibrated relative-motion representation; and C, changed geometry plus execution-aware timing and robot-feasibility screening. They are different collection arrangements, not just file conversions. Equation (15.7) explains why context can matter, without predicting pusher success.

The target robot constrains demonstration use. At configuration q, require motion consistent with the Jacobian, limits and contact model:

$$v_C = J_C(q)\nu, \qquad \nu_{\min} \le \nu \le \nu_{\max}, \qquad \tau_{\mathrm{req}}(q, \nu, \dot{\nu}) \in \mathcal{T}(q, \nu). \tag{15.18}$$

Here $\mathcal{T}$ is the modeled available-effort set. Pose reachability does not establish execution at demonstrated speed or sustainable contact. A trajectory infeasible for one robot can still be useful elsewhere. Retain the source episode and its target-dependent screening decision.

Worked example: time alignment. At constant translational speed $v = 0.2$ m/s, an observation–action misalignment of 0.05 s gives a displacement discrepancy of 0.01 m. Keeping that contribution at or below 0.002 m requires

$$|e_x| = |v\,\delta t|, \qquad |\delta t| \leq 0.01 \text{ s}. \tag{15.19}$$

This is a local alignment requirement under constant velocity, not a sampling theorem, complete error budget or feedback-stability guarantee. It motivates capture/application timestamp measurement and compatible action horizons. More video frames do not remove fixed execution delay.

Research systems illustrate different choices. ALOHA combines a leader–follower apparatus, image and joint observations, target-action semantics and action-chunk learning (Zhao et al. 2023). UMI couples a handheld gripper and matching wrist-view geometry with visual–inertial trajectory reconstruction, a relative-pose policy interface, latency matching and robot-specific feasibility filtering (Chi et al. 2024). Its recovered actions are not robot joint-command measurements. Portability depends on tracking, observations and execution compatibility. These are source-reported designs, not evidence of universal superiority or claims that their authors adopt this book's framing.

Joint body–policy design has computational precedents too. Ha (2019) optimizes parameterized body dimensions and controller parameters in simulated locomotion environments. The study constrains variations of existing designs and discusses unrealistic results with relaxed constraints. It illustrates searching physical and control choices together, without establishing manufacturing feasibility, physical transfer or the usefulness of every generated record.

For the pusher comparison, fix the deployment task, learner and evaluation population. Retain attempts, processing failures, feasibility exclusions and annotation effort. Compare useful coverage and behavior under declared resource budgets, rather than counting successful videos alone. B and C share the geometry/context change; C adds timing and feasibility screening. Their comparison concerns those additions together, not an effect of camera placement. A complete-arrangement comparison can still be useful; explaining a component effect requires more controlled alternatives.

The eventual decision might adopt B, investigate C or narrow the task. None follows numerically from this chapter's illustrative counts. Physical feasibility, observation meaning, learning requirements and resources enter one argument. Chapter 16 develops the comparative evidence needed to justify its conclusions.

Exercises and further investigation. Separate calculations, proposed measurements and completed evidence. State units, boundaries, timing and assumptions.

1. Specify a capability goal and two intended users of its data. Identify different information needs; distinguish a requirement from a preferred implementation.
2. Construct a design-dependent collection law. Explain how changing sensing can change later contacts, and identify memory required by the assumed Markov state.
3. Derive reflected rotor inertia and the acceleration-optimal ratio. Check all reducer alternatives. What changes if available torque depends on speed or temperature?
4. Derive lever power and force sensitivity. Add geometry uncertainty, other loads and sensor mass; identify measurements needed for contact force inference.
5. Derive conditional predictions and risks for the binary sensor. Check $p = 0.5$ and $p = 1$. Why is action error not closed-loop performance?
6. Reproduce both information matrices and target variances. Change the target direction and explain the role of parameter scaling.
7. Compare collection alternatives under time and attempted-trial budgets, including setup, resets, calibration and processing. Identify coverage that expected usable counts miss.
8. Derive electromagnetic, output-shaft and endpoint-force quantities. List their required signals and explain why current cannot directly measure all three.
9. Reproduce the calibration/inertia investigation and uncertainty. Add uncertain reference angles and effort gain. Which ambiguity remains, and what design change would distinguish it?
10. Reproduce the three-parameter joint fit and alternative account. Propose a friction-symmetry test and explain why sticking requires a changed model.
11. Analyze a reported robot through the six organizing views. Separate source evidence, your interpretation and an

unanswered question. Do not treat unreported quantities as evidence of failure.

12. Compare pusher or gripper arrangements. Specify the deployment task, physical/timing constraints, learner, records, exclusions and a resource-matched evaluation. What result would warrant redesign?

Featherstone supplies the common mechanical language; uncertainty and identification support reference and excitation arguments (Featherstone 2008; Joint Committee for Guides in Metrology 2008; Ljung 1999). Dataset documentation and provenance support inspectable reuse (Gebru et al. 2021; Moreau and Missier 2013). ALOHA, UMI and joint morphology learning provide contrasting readings on apparatus, interfaces and learning (Zhao et al. 2023; Chi et al. 2024; Ha 2019). The standard-library companion `check_chapter15.py` recalculates constructed examples; it does not implement those research systems.

The outcome is a design argument: a specified purpose, feasible alternatives, physical and informational consequences, a declared consumer of records, and a decision bounded by evidence. This connects building a system with learning from its interactions. The final chapter asks what a comparison must establish before that decision becomes a supported result.

Chapter 16 Evaluating Data and Learned Capability

A redesigned collection system produces more usable recordings. A fitted model has smaller prediction error. A policy succeeds in more trials. These are three encouraging results, but they are not the same result. The new recordings may omit difficult contacts; the model may predict a quantity unavailable during deployment; the policy may have been tested on easier initial conditions. Evaluation must connect each conclusion to the object, population and comparison that support it.

Chapter 15 turned physical and computational choices into collection alternatives. This chapter asks what would justify retaining one of them. The answer combines mechanics, measurement and statistical inference. A physical diagnostic needs correct units, frames and timing. An average needs a defined population. An uncertainty interval needs an experimental unit and a repeated-sampling model. A learning claim needs evidence at the level at which learning or collection varies.

For ExpGen PhySyst, evaluation closes a design argument and often opens the next one. Its purpose is neither a universal data-quality score nor a ritual of obtaining a favorable number. It is to establish what changed, how much it mattered, under which conditions, and what remains uncertain. All numerical examples here are constructed and independently checked. They do not report newly trained policies, physical trials or benchmark reproductions.

16.1 Define the object and claim being evaluated

16.1.1 Collection systems, datasets, models and policies

Begin by naming the object. A collection system includes the apparatus, operator or collection controller, acquisition pipeline and procedure. Its evaluation can concern usable coverage, interpretability, effort or the benefit of its data to a specified learner. A dataset is the retained, transformed record of collection; its completeness and validity concern a declared purpose and population. A model predicts quantities under assumptions. A policy acts through an observation and actuator interface. Each object needs evidence appropriate to its role.

One investigation may evaluate several objects, but its conclusions must remain distinguishable. A valid file schema does not establish dataset representativeness. A smaller training loss does not establish prediction on new conditions. Accurate prediction of demonstration actions does not establish successful feedback execution. Conversely, a policy can solve a task with an imperfect physical model. These distinctions are reasons to choose complementary tests, not to dismiss any one kind of result.

Formulate the quantity to be estimated, or *estimand*, before choosing a statistic. For learned parameters w, operating conditions $c \sim \rho_*$ and an executed trajectory $\zeta \sim P^w(\cdot \mid c)$, define a higher-is-better score S:

$$\begin{aligned} \mu(w) &= \mathbb{E}_{c,\zeta}[S(\zeta; c)], \\ w_m = \mathcal{A}_m(\mathcal{D}, \omega), \qquad \mu_m &= \mathbb{E}_{\mathcal{D},\omega}[\mu(w_m)], \qquad \Delta = \mu_B - \mu_A. \end{aligned} \tag{16.1}$$

The learning procedure $\mathcal{A}_m$ consumes dataset $\mathcal{D}$ and training randomness ω. The trajectory law includes the complete executor, rather than only a network. Score units and direction must be stated: a time reduction can instead be defined explicitly as $\mu_A - \mu_B$ when smaller time is better.

For a fixed-checkpoint claim, hold w fixed and average execution conditions and outcomes. For a learning-procedure claim, average relevant training variation as well. For a collection-system claim, vary collection where the claim requires it; new training seeds on one fixed dataset do not independently repeat its collection. These are different experiments, even when their final tables all contain success percentages.

16.1.2 Task, embodiment and operating distribution

A task definition specifies goals, permissible actions, initial conditions, termination and outcome rules. Embodiment includes the mechanism, actuation, sensing and execution limits. The operating distribution describes which objects, con-

ditions, people or environments the result is intended to cover. Naming a task without these boundaries leaves considerable freedom to make a comparison easier or harder.

Suppose a pusher must place a cube inside a target region. A success rule might require a final pose tolerance, a dwell interval, no excluded contact and no intervention. Each term requires an observation or a declared judgment. A simulator's exact pose is not automatically a physical pose measurement. Timeout, protective stop, assisted completion and unavailable outcome are distinct events. Define how they enter the score before examining the results.

If conditions are stratified, use target weights rather than incidental test counts:

$$\mu_m = \sum_c \rho_*(c)\mu_{mc}, \qquad \hat{\mu}_m = \sum_c \rho_*(c)\bar{Y}_{mc}. \tag{16.2}$$

The sample averages $\bar{Y}_{mc}$ must estimate the intended conditional outcomes. The equation does not repair biased sampling within a stratum or supply evidence for an absent one.

Worked example: an easier test reverses a ranking. Let two conditions be easy and hard. Construct the following outcome ledger:

Method	easy successes/attempts	hard successes/attempts	pooled fraction
A	81/90	4/10	0.85
B	19/20	48/80	0.67

B has higher observed success within both conditions: 0.95 versus 0.90 and 0.60 versus 0.40. Yet pooling favors A because it received more easy tests. For an equally weighted target, A's standardized fraction is 0.65 and B's is 0.775. This calculation changes the mixture being estimated; it does not remove sampling uncertainty or establish that the two conditional populations were otherwise comparable.

Use separate nominal and shifted conditions when evaluating transfer. If a result concerns one device and one mounting, retain that restriction. Multiple objects on the same robot do not establish device-to-device variation. A fixed benchmark suite and a sampled population of future tasks likewise support different generalizations.

16.2 Quantify performance and uncertainty

16.2.1 Measurement quality, model residuals and task outcomes

Measurement uncertainty concerns how indications relate to quantities. Outcome variation describes differences among encounters. Sampling uncertainty concerns an estimate produced from a finite collection of units. None is a substitute for the others. Repeating a biased measurement can narrow the spread of its average while leaving the bias unchanged. Calibration and model inadequacy need their own treatment (Joint Committee for Guides in Metrology 2008).

Physical diagnostics should correspond to a declared balance or prediction. For smooth motion, one generalized dynamic residual is

$$r_\tau = M(q)\widehat{\dot{\nu}} + h(q,\nu) - \tau_a - \tau_p - \sum_j J_j^{\mathsf{T}} f_j^{\mathrm{ext}}. \tag{16.3}$$

Here h contains rigid-body gravity and velocity terms, τ_p contains separately accounted passive effects, and external wrenches use compatible points and frames. Quantities must be evaluated at matched times. Generalized coordinates can include translation and rotation, so residual components need declared scales or covariance weighting before forming a scalar norm. At an impact, use an appropriate impulse balance rather than forcing a smooth-acceleration interpretation onto an impulsive event.

Featherstone's spatial language makes the physical meaning precise. For motion $v = (\omega^{\mathsf{T}}, v_O^{\mathsf{T}})^{\mathsf{T}}$ and wrench $f = (n^{\mathsf{T}}, F^{\mathsf{T}})^{\mathsf{T}}$,

$$\begin{aligned} P &= f^{\mathsf{T}} v = n^{\mathsf{T}}\omega + F^{\mathsf{T}} v_O, \\ v_B &= X_{BA} v_A, \qquad f_B = X_{BA}^{-\mathsf{T}} f_A, \qquad f_B^{\mathsf{T}} v_B = f_A^{\mathsf{T}} v_A. \end{aligned} \tag{16.4}$$

Frame and origin transformations must act dually. Component magnitudes of a wrench need not remain unchanged. If a residual wrench transforms the same way and Q_A is its declared positive-definite metric, then

$$Q_B = X_{BA} Q_A X_{BA}^{\mathsf{T}}, \qquad r_B^{\mathsf{T}} Q_B r_B = r_A^{\mathsf{T}} Q_A r_A. \tag{16.5}$$

For an invertible wrench covariance, the covariance and its inverse metric therefore transform in different directions:

$$\Sigma_B = X_{BA}^{-\mathsf{T}} \Sigma_A X_{BA}^{-1},$$
$$Q_A = \Sigma_A^{-1}, \qquad Q_B = \Sigma_B^{-1}.$$

Copying one numeric weighting matrix to every origin generally changes the diagnostic (Featherstone 2008).

Worked example: retain the physical port when scoring. Let O_B lie 0.2 m along x from O_A, with aligned axes. A body has $\omega_z = 1$ rad/s and origin velocity $(0.3, 0, 0)$ m/s at O_A. The point velocity at O_B is $(0.3, 0.2, 0)$ m/s. A 10 N force along y with zero moment at O_B has a 2 N m positive z moment at O_A. Both descriptions give 2 W.

Now consider a residual of 1 N in that force component. At O_A its moment component is 0.2 N m. Applying fixed reference scales of 1 N and 1 N m and summing normalized component squares gives 1 at O_B but 1.04 at O_A. The frames describe the same residual; the discrepancy comes from keeping the metric fixed. Transporting the metric as in Eq. (16.5) preserves the value 1.04.

Integrating compatible power gives mechanical work, $W_a = \int \tau_a^{\mathsf{T}} \nu \, dt$. Signed work, positive work, absolute power integral and electrical energy $\int V I \, dt$ are different quantities. A holding actuator can perform zero mechanical work while consuming electrical energy. Low work alone establishes neither task completion nor acceptable contact.

Worked example: a circular residual can be perfect. An ideal joint with true $J = 0.02\ \mathrm{kg\,m^2}$, delivered effort 0.1 N m and no other loads accelerates at 5 rad/s^2. A model with $J_0 = 0.025\ \mathrm{kg\,m^2}$ predicts 4 rad/s^2. Using the model's own predicted acceleration in its residual gives $J_0(4) - 0.1 = 0$ N m. Using an independent exact acceleration observation instead gives $J_0(5) - 0.1 = 0.025$ N m. The first calculation checks internal algebra, not physical correspondence.

The same problem arises when unmeasured contact forces are chosen solely to minimize a residual. State which quantities are observed, inferred or imputed by the model under test. Report measurement errors, prediction residuals and task outcomes together where relevant, without treating a small diagnostic as proof of every other property.

16.2.2 Experimental units, repeated trials, dependence and confidence intervals

The experimental unit is the level at which a comparison is independently replicated or assigned. Frames, windows, episodes, sessions, devices, collection datasets, training runs and tasks can occupy different levels. Repeated observations are useful, but are not automatically independent replications (Hurlbert 1984). Preserve parent identities so the analysis can respect the dependence.

For independent, identically distributed scalar observations, let $\bar{Y}$ be the sample mean and s the sample standard deviation. The standard error $s/\sqrt{n}$ estimates the spread of the mean, not the spread of individual outcomes. Under an independent normal sampling model,

$$\frac{\bar{Y} - \mu}{s/\sqrt{n}} \sim t_{n-1}, \qquad \bar{Y} \pm t_{1-\alpha/2,n-1} \frac{s}{\sqrt{n}} \tag{16.6}$$

gives an exact Student confidence procedure. Outside that model, an appropriate approximation, dependence model or resampling procedure is needed. A small sample does not justify normality by itself (NIST/SEMATECH, n.d.).

A confidence level describes the long-run coverage of a procedure under its assumptions. After computing one frequentist interval, the parameter is fixed; the interval does not assign it a posterior probability. Coverage also does not include an unmodeled calibration bias or an unrepresented operating regime.

Worked example: no observed failures is not zero failure risk. For n fixed independent Bernoulli trials with common failure probability p, the probability of zero failures is $(1-p)^n$. Inverting this event gives the exact one-sided $1-\alpha$ upper bound after zero failures:

$$p_U = 1 - \alpha^{1/n}. \tag{16.7}$$

At $\alpha = 0.05$, zero failures in 30 trials gives approximately 0.0950; in 59 trials it gives approximately 0.0495. These are illustrative confidence calculations, not required robot-reliability targets. A shared session effect or adaptive stopping changes the sampling problem.

For a general success count k, a useful approximate binomial procedure is the Wilson score interval. With $\hat{p} = k/n$ and $z = z_{1-\alpha/2}$, invert $n(\hat{p} - p)^2 \leq z^2 p(1-p)$ to obtain

$$\frac{\hat{p} + z^2/(2n) \pm z\sqrt{\hat{p}(1-\hat{p})/n + z^2/(4n^2)}}{1 + z^2/n}. \tag{16.8}$$

For 18 successes in 20 trials and $z = 1.96$, the interval is approximately $[0.6990, 0.9721]$. Unlike an unqualified symmetric plug-in interval, it does not collapse to zero width at a boundary. Its coverage is approximate, not a cure for clustered trials or an unknown denominator. The handbook also describes exact binomial-tail inversion (NIST/SEMATECH, n.d.).

Learning adds another replication level. Suppose N independently generated training units each produce M conditionally independent evaluation scores. Let between-unit variance of true means be σ^2_{train}, and expected within-unit score variance be σ^2_{exec}. For the balanced grand mean, total variance gives

$$\text{var}(\hat{\mu}) = \frac{\sigma^2_{\text{train}}}{N} + \frac{\sigma^2_{\text{exec}}}{NM}. \tag{16.9}$$

Within a training unit, scores share its learned policy. Additional rollouts reduce only the second term. If the claim includes collection variability, training units must replicate collection or the analysis must model the shared dataset separately.

Worked example: more rollouts leave training uncertainty. Stipulate $\sigma^2_{\text{train}} = 0.0025$ and $\sigma^2_{\text{exec}} = 0.09$ in squared score units:

N	M	variance of grand mean	standard error
5	100	0.0006800	0.02608
5	10000	0.0005018	0.02240
20	25	0.0003050	0.01746

The first and third schedules use 500 evaluation episodes, but the third requires more independent training units. Their training/collection costs are not equal merely because rollout counts match. The second schedule uses vastly more rollouts while retaining the first term's uncertainty. These are supplied variance components, not estimates from new training.

Bootstrap intervals must resample the relevant units. For a fixed suite of tasks with independent training runs within each, stratified resampling can draw runs within each task while retaining the task weights (Agarwal et al. 2022). Resampling tasks addresses another population claim. Resampling frames from one episode cannot create device-level replication. No resampling scheme restores missing calibration, hidden outcomes or a fundamentally mismatched population.

16.3 Compare collection and learning alternatives

16.3.1 Resource-matched collection alternatives, baselines and ablations

A baseline answers a specific alternative explanation. A simple physical model tests whether a learned component adds useful prediction. An existing controller tests whether a learned policy improves behavior. A repeat of the original collection arrangement tests whether a redesign improves what its records support. No single baseline answers all three questions.

Declare resources that could explain an apparent advantage: physical attempts, operator time, setup/calibration, adaptation, training compute, tuning and evaluation. Equal retained demonstrations need not mean equal collection effort. Equal training iterations need not mean equal compute. An analysis can legitimately compare complete systems at their actual costs; it must then make that comparison explicit rather than imply a controlled component experiment.

Matching conditions can improve a comparison when it retains relevant covariance. For independent matched pairs, define

$$D_i = Y_{Bi} - Y_{Ai}, \qquad \hat{\Delta} = \bar{D}, \qquad \widehat{\mathrm{SE}}(\bar{D}) = s_D/\sqrt{n},$$
$$\mathrm{var}(Y_B - Y_A) = \mathrm{var}(Y_B) + \mathrm{var}(Y_A) - 2\,\mathrm{cov}(Y_B, Y_A). \tag{16.10}$$

Pair the complete condition, not merely a seed number. For physical experiments, order, resetting, wear, temperature and carryover can matter. Randomized or counterbalanced order can address some of these issues; pairing alone does not establish independent pairs or positive covariance.

Worked example: compare paired differences. Construct four higher-is-better scores for A, $(10, 20, 30, 40)$, and B, $(11, 22, 31, 42)$. Differences are $(1, 2, 1, 2)$, with mean 1.5, sample standard deviation approximately 0.57735 and standard error 0.288675 score units. Ignoring pairing and using an independence formula gives a standard error approximately 9.2241, because the variation among conditions overwhelms the small within-condition difference.

Under a hypothetical independent normal model for pair differences, Eq. (16.6) applies with three degrees of freedom. Using $t_{0.975,3} = 3.182446$ yields approximately $[0.5813, 2.4187]$ score units. This illustrates that procedure; the four constructed differences do not establish normality or a real improvement. For binary paired outcomes, a small-sample normal-difference justification would be inappropriate; analyze the paired outcome structure using a method suitable for that data.

An ablation changes a component while retaining a meaningful comparison. Removing a sensor and retraining the learner asks about an optimized system without that sensor. Masking the sensor only during testing asks about sensitivity of the existing policy to missing input. These are different interventions. Report retraining, retuning and replacement work instead of calling both simply a sensor ablation.

Worked example: sensing and learning interact. Let the first index indicate old/new sensing and the second old/new learning. Stipulate success probabilities

$$\mu_{00} = 0.60, \qquad \mu_{10} = 0.70, \qquad \mu_{01} = 0.65, \qquad \mu_{11} = 0.90. \tag{16.11}$$

The sensing improvement is 0.10 with the old learner but 0.25 with the new one. The interaction contrast is

$$I = (\mu_{11} - \mu_{01}) - (\mu_{10} - \mu_{00}) = 0.15. \tag{16.12}$$

The total improvement is 0.30, which cannot be uniquely assigned to sensing or learning as an independent contribution. These are supplied probabilities, not measured causal effects. Empirical attribution needs appropriate allocation, comparable conditions, declared resource budgets and uncertainty at the right replication level.

Learning curves make another resource choice visible. Let b count new physical collection episodes, and let $\mu_m(b)$ be a declared performance measure after using that budget. If a piecewise-linear interpolation is chosen, its normalized area is

$$A_m = \frac{1}{b_K - b_0} \sum_{k=0}^{K-1} (b_{k+1} - b_k) \frac{\mu_m(b_k) + \mu_m(b_{k+1})}{2}. \tag{16.13}$$

This summarizes one stated budget range; it is not a unit-free universal sample-efficiency score when horizontal axes or populations differ.

Worked example: endpoint and budget-average answer different questions. At budgets $(0, 10, 20)$, stipulate normalized scores $A = (0.2, 0.8, 0.8)$ and $B = (0.2, 0.5, 0.9)$. The trapezoidal summaries are 0.65 and 0.525. A has the larger area; B has the better endpoint. Neither summary should be substituted for the other. Interpolation between the three supplied points is an assumption, and the zero-budget score may use prior learning whose cost must be disclosed. Turning episodes into more windows changes neither the physical collection budget nor its lineage.

16.3.2 Learning benefit, failure cases and alternative explanations

A learning-benefit claim connects collection to a specified consumer and an outcome assessed independently of development choices. Better data coverage can improve a learner, have no effect on that learner, or expose an inadequacy in its representation. An absence of observed improvement can result from uninformative data, inadequate optimization, insensitive evaluation or high uncertainty. Distinguishing these possibilities requires useful diagnostics and comparisons rather than a single score.

Keep attempted outcomes visible. If s known successes, f known failures and u unknown outcomes exhaust a fixed ledger of n attempts,

$$\frac{s}{n} \le p_{\text{ledger}} \le \frac{s+u}{n}, \qquad n = s + f + u. \tag{16.14}$$

These are logical bounds on that ledger, not population confidence intervals. A separate analysis is needed to infer future performance. Complete-case analysis estimates another quantity unless the missingness mechanism justifies its use for the intended target.

Worked example: preserve the denominator and the timeout. Of 100 attempts, 70 are known successes, 20 known failures and ten have unknown outcomes. Complete-case success is $70/90 \approx 0.7778$; the realized all-attempt fraction lies between 0.70 and 0.80. Recording failure is not automatically task failure, and an assisted completion needs the declared assistance rule.

Now compare completion times. A succeeds once in ten attempts in 1 s and has nine 60 s timeouts. B succeeds in every attempt in 10 s. Time among successes favors A, but its charged duration per attempted trial is $(1+9\cdot60)/10 = 54.1$ s versus B's 10 s. Report completion probability and time with its censoring or timeout definition. Neither conditional speed nor one penalized scalar should silently replace the full outcome description.

Failures also help distinguish explanations: slip after contact, a stale command, loss of tracking, a limit, a timeout and an intervention are different mechanisms. Predeclare categories where feasible and retain unexpected ones. A small mean residual can coexist with rare damaging errors or a systematic failure in one regime.

Development selection can inflate results even when each estimate is unbiased before selection.

Worked example: choosing the maximum creates optimism. Let two independent estimates each equal $\mu - \delta$ or $\mu + \delta$ with probability one half, where $\delta > 0$. Of the four equally likely pairs, three have maximum $\mu + \delta$ and one has maximum $\mu - \delta$. Thus

$$\mathbb{E}[\max(\hat{\mu}_1, \hat{\mu}_2)] = \tfrac{1}{4}(\mu - \delta) + \tfrac{3}{4}(\mu + \delta) = \mu + \tfrac{1}{2}\delta. \tag{16.15}$$

For $\mu = 0.5$ and $\delta = 0.1$, the selected report averages 0.55. This exact toy does not model every learning experiment. It explains why checkpoint, hyperparameter, metric and split selection must be part of the assessment procedure, rather than hidden behind a single final value.

Use development evidence for choosing the system, and a protected final assessment for estimating its stated outcome. If final outcomes guide another revision, they have become development information. Repeated testing until an interval excludes zero also changes the sampling procedure. Either predefine the assessment or use an inference method that accounts for the adaptive procedure.

Across heterogeneous tasks, inspect per-condition results and distributions alongside aggregate scores. A weighted empirical score profile can report the fraction exceeding threshold t:

$$\hat{F}(t) = \sum_c \rho_*(c) \frac{1}{n_c} \sum_{i=1}^{n_c} \mathbf{1}\{Y_{ci} > t\}. \tag{16.16}$$

It differs from applying a threshold to each condition's sample mean. Robust summaries such as an interquartile mean trim tails and therefore change the functional being estimated. Pointwise confidence bands for a profile are not simultaneous coverage of the entire curve. Such tools are useful complements, with declared normalization and weights, rather than automatic remedies for selection or incompatible protocols (Agarwal et al. 2022).

16.4 From evidence to the next decision

16.4.1 Retain, qualify or revise a design, model or claim

A comparison should end with a decision proportional to the evidence. State a practically meaningful improvement or tolerated loss in task units when one is needed. A detectable difference may be too small to justify a costly change. An interval that includes zero does not prove equivalence; it may leave meaningful benefit and harm unresolved. Equivalence or noninferiority requires a declared margin and an appropriate comparison, not simply failure to reject a difference.

Return to the joint in Chapter 15. A new reference resolves an encoder-scale ambiguity. That supports a calibration and parameter interpretation under stated assumptions. It does not by itself establish that a new controller is more reliable. The next evaluation must execute the revised controller on the intended operating conditions and retain relevant failures. The inference and behavior claims can both be useful, but require different evidence.

For a manipulation system, improved visibility may reduce action-label ambiguity while adding mass or latency. If recording yield improves but policy performance does not, retain the proven improvement and qualify the downstream claim. Examine whether the learner used the added information, whether execution remained compatible, and whether the test had enough relevant variation. A null comparison can guide a useful redesign without establishing that sensing or data never matters.

Separate what is observed from what is concluded. An observed difference can be clear even when its cause is uncertain. An entire revised arrangement can be preferable under the declared budget without proving which component caused the gain. Conversely, a small controlled component effect need not make the complete system worthwhile. The decision should identify the tested alternative, unresolved explanation and next informative change.

16.4.2 Reusable knowledge and conditions for independent use

Reusable knowledge includes the conditions under which a relation or artifact supports a new decision. The same dataset can support geometric reconstruction, imitation or model identification to different degrees. An executable script establishes one computational route; it does not establish all intended scientific uses.

Distinguish three operational achievements. Another person may reproduce a numerical result from supplied artifacts. Another implementation or acquisition may recover a stated relation under comparable conditions. Another consumer may use the material successfully for a new purpose. Each needs its own evidence; completing the first does not establish the second or third.

A useful independent-use check asks whether a reader can reconstruct the physical/action meaning, calibration, timing, selection and reported denominator, then perform the stated computation or use. Missing information can make that check inconclusive while leaving the original result useful within its scope. Report the missing link rather than treating all uncertainty as either complete validation or total failure.

For example, raw current and encoder records may permit a revised motor-effort estimate if the torque constant, transmission, acceleration and boundary are known. They cannot establish contact force when the other loads needed by the balance remain unknown. A trajectory screened out for one robot can still support perception research, but its rejection and original context must remain visible. These are bounded reuse conditions, not generic claims that all recorded experience transfers.

16.5 Make the investigation reproducible

16.5.1 Linking derivation, collection, code, data and result

Reproducibility begins with a chain of meaning, not only a launch command. The derivation names quantities; the collection arrangement determines which are observed; transformations create estimates and labels; code implements the analysis; the result describes a specified population. Each link can be computationally correct while interpreting the wrong quantity.

Make the mechanics-to-code mapping explicit. Configuration coordinates and tangent velocities can have different dimensions. An actuator request is not automatically delivered generalized effort. Indices must name joints, actuators or velocity degrees of freedom correctly. A force sensor's origin and axes must agree with a Jacobian or spatial transform. Capture time, estimator availability and command application are different events. The native MuJoCo examples in Chapters 3 and 6 illustrate these distinctions; their evaluation should not silently replace measured acceleration with the engine's own prediction.

Retain an artifact identity for the model, collection arrangement, raw episodes, calibration, derived dataset, split, learner, checkpoint, executor and evaluation rules. A result row should have a defined parent at each relevant level. Preserve attempted trials and reasons for exclusion. A source hash verifies byte identity, not physical validity; a lineage graph makes a derivation inspectable, not automatically true (Gebru et al. 2021; Moreau and Missier 2013).

Simple independent checks can expose mistakes before a large experiment: verify a unit conversion, transform a known wrench and preserve power, recalculate a residual from separately supplied inputs, and reconstruct one outcome denominator. Test the pipeline's treatment of an interrupted trial and unavailable measurement. These checks complement physical validation rather than replacing it.

The chapter's standard-library companion, `check_chapter16.py`, independently recalculates its constructed examples. It does not collect data, train a policy or validate hardware. Appendix A collects mathematical conventions; Appendix B data conventions and sample access; and Appendix C optional computational tools. Their reference role does not make them substitutes for a complete evaluation argument.

16.5.2 Figures, videos and communication of bounded conclusions

A figure or video should preserve the claim being evaluated. A trace needs axes, units, frame or point definitions where relevant, time alignment and distinctions among measured, estimated and predicted quantities. State what uncertainty bands represent: across-trial spread, standard error, measurement uncertainty and confidence intervals are not interchangeable. A selected successful trace does not describe the distribution of attempts.

For a physical demonstration video, disclose playback speed, cuts, resets, interventions and whether the displayed state is measured, reconstructed or simulated. Link the clip to its trial and outcome rule. If only representative examples are shown, explain selection and retain the complete outcome ledger. A convincing video is valuable qualitative evidence, but its evidential role differs from an independently sampled performance estimate.

A bounded conclusion identifies the system, data, learner, executor, population, resource budget, effect and uncertainty. Explain failures and competing accounts, along with what another person would need to repeat the calculation or test a new use. Avoid making the claim broader than the experimental units: one robot, session or dataset remains one at that level, regardless of the number of frames.

The final report should also say what the evidence changed. Perhaps a sensor revision improved useful coverage, a model explained one regime, a policy improved under a specified budget, or a proposed mechanism was not distinguishable from another. These are different contributions. Presenting their limits clearly makes them more reusable.

Exercises and further investigation. State the estimand, unit, population, physical meaning and assumptions. Distinguish proposed tests from completed evidence.

1. Formulate separate claims about a collection system, dataset, model and fixed policy for one task. Identify the evidence each needs and a result that would not establish it.
2. Reproduce the target-mixture example. Propose a fairer allocation and state which within-condition differences still require control.
3. Derive power and residual-metric transport for a change of origin. Recalculate the 2 W example and the normalized residual costs.
4. Reproduce the circular dynamic residual. Propose independent observations and describe how acceleration uncertainty changes the test.
5. Derive Wilson's interval from the score inequality. Check zero and all successes; contrast its approximation with the exact one-sided zero-failure bound and its fixed-sampling assumptions.
6. Derive the two-level variance expression. Compare training and rollout allocation under a total resource budget, including collection variation if the claim concerns the collection system.
7. Reproduce paired means, covariance and standard errors. State when a Student interval is justified and design an order/reset protocol for a physical comparison.
8. Reproduce the factorial interaction and budget-area examples. Explain what each comparison can and cannot attribute to a component.
9. Reconstruct missing-outcome bounds and timeout accounting. Add an assisted outcome category and define how its interpretation affects the claimed task performance.
10. Derive maximum-selection optimism and propose a protected final assessment. Explain what changes if final outcomes guide another revision.
11. Design an independent-use test for an existing model or dataset. Separate computational replay, new acquisition and a different downstream use; identify the artifacts and physical context each requires.
12. Write a bounded conclusion for one constructed comparison and describe a suitable trace, table or video. Specify the denominator, uncertainty, exclusions and next warranted decision.

For further reading on mean and binomial intervals, consult the NIST/SEMATECH handbook (NIST/SEMATECH, n.d.). Replication and subsampling have a longer experimental-design history (Hurlbert 1984). Modern learned-policy evaluation highlights run variation, interval estimation and robust aggregate reporting (Agarwal et al. 2022). Featherstone mechanics and measurement uncertainty preserve the physical meaning of those comparisons (Featherstone 2008; Joint Committee for Guides in Metrology 2008). These sources support different parts of the argument, rather than one universal evaluation formula.

The book began with a joint whose changed motion admitted several explanations. Designing data meant deciding what to change and observe so that those explanations, or useful actions, could be distinguished. Evaluation now asks what that designed encounter actually justified. ExpGen PhySyst connects these steps: arrange physical interactions, preserve what their records mean, learn for a stated purpose, and use bounded evidence to decide what to build or investigate next.

Appendix A Mathematical and Implementation Reference

This appendix collects the conventions and computational identities used throughout the book. It is a reference to the arguments in the chapters, not a replacement for their assumptions. The mechanical notation follows Featherstone (2008): spatial motion is angular-first, and its dual wrench is moment-first. The implementation examples use native MuJoCo 3.16.0. A library's array names do not determine its physical conventions; its version, model and evaluation stage remain part of the interpretation.

For ExpGen PhySyst, these conventions connect a mathematical quantity to a record that another person can interpret. A useful conversion preserves body, origin, axes, units, sign, time and signal role. A shape check alone cannot establish that correspondence. The worked numbers below are constructed. Arithmetic checks and small simulator-interface tests do not establish physical parameter accuracy, trained capability or transfer.

A.1 Notation and physical identities

A.1.1 Symbols, units, frames and reference points

Bold symbols denote vectors or matrices where that distinction is useful. A dot is a physical-time derivative, a hat an estimate, and a superscript T a transpose. Subscripts and superscripts naming frames specify representation; algorithm names or labels such as "comp" specify roles. A superscript -1 denotes an inverse mathematically; code should normally apply a linear solve. $\mathbf{1}_d$ denotes the d-dimensional identity.

Symbol	Meaning and dimension	Physical interpretation
$\mathbf{q}, \boldsymbol{\nu}$	Configuration parameters n_q; tangent velocity n_v	$\dot{\mathbf{q}} = \mathcal{N}(\mathbf{q})\boldsymbol{\nu}$; n_q need not equal n_v.
$\mathbf{v}_O$	Spatial motion, 6×1	$[\boldsymbol{\omega}; \mathbf{v}_{O,\mathrm{lin}}]$; angular rad/s, linear m/s.
$\mathbf{a}_O$	Physical spatial acceleration, 6×1	Angular rad/s^2; linear block is not generally ordinary point acceleration.
$\mathbf{f}_O$	Wrench, 6×1	$[\mathbf{n}_O; \mathbf{F}]$; moment N m, force N, acting on a named body.
$\boldsymbol{\tau}$	Generalized effort, $n_v \times 1$	Power-conjugate to $\boldsymbol{\nu}$; revolute torque or prismatic force.
$\mathbf{I}_O, \mathbf{M}$	Spatial inertia 6×6; joint-space inertia $n_v \times n_v$	$T_i = \frac{1}{2}\mathbf{v}_i^{\mathsf{T}}\mathbf{I}_i\mathbf{v}_i$ for one body; $\sum_i T_i = \frac{1}{2}\boldsymbol{\nu}^{\mathsf{T}}\mathbf{M}\boldsymbol{\nu}$ for the system.
$\mathbf{J}_O$	Point-specific spatial Jacobian, $6 \times n_v$	$\mathbf{v}_O = \mathbf{J}_O\boldsymbol{\nu}$; angular and linear rows retain different units.
$\mathbf{u}, \mathbf{z}$	Control input; internal actuator state	Neither is generally delivered generalized effort. Sizes and units depend on the actuator and transmission.
$\mathbf{x}, \mathbf{y}$	Statistical state; observation	Each component has its own meaning and unit; these need not be complete mechanical state or direct measurement.

Local definitions remain authoritative. In particular, J can denote a scalar rotational inertia in Chapter 3 or a Jacobian elsewhere; h can denote timestep, bias or momentum; R can denote rotation, resistance or measurement covariance. A scalar action a_k is not spatial acceleration $\mathbf{a}_i$. Such reuse is harmless when the role and dimensions are stated, but unsafe when copied into an untyped data column.

Use SI internally unless another convention is explicit. Radians retain their interpretive role even though they are dimensionless in SI. Degrees-to-radians conversion is $q_{\mathrm{rad}} = \pi q_{\mathrm{deg}}/180$; rates use the same factor, while gains follow their own units. For example, a torque gain in N m/degree becomes $K_{\mathrm{rad}} = (180/\pi)K_{\mathrm{deg}}$ in N m/rad. A six-vector combines units: there is no universal physical norm obtained by adding its component squares. Declare reference scales or a covariance when forming a statistical loss.

Three accelerations and a sensor. For a body's absolute motion expressed in its own body-fixed frame B,

$$\mathbf{a}_O^B = \begin{bmatrix} \boldsymbol{\alpha}^B \\ \mathbf{a}_{O,\mathrm{class}}^B - \boldsymbol{\omega}^B \times \mathbf{v}_{O,\mathrm{lin}}^B \end{bmatrix}. \tag{A.1}$$

The lower block differs from the classical acceleration of O relative to an inertial world. A gravity-initialized inverse-dynamics recursion instead uses $\mathbf{a}_i^{\mathrm{alg}} = \mathbf{a}_i - \mathbf{a}_{g,i}$, where $\mathbf{a}_{g,i}$ is the spatial representation of uniform gravity. An ideal accelerometer at P, with sensor axes S, reports specific force ${}^S\mathbf{R}_W(\mathbf{a}_{P,\mathrm{class}}^W - \mathbf{g}^W)$ before its calibration, bias and noise terms. These three quantities are not interchangeable; Chapters 4, 5 and 7 derive their relationships.

Worked example: rotation separates acceleration meanings. At one instant take aligned body and world axes, constant angular speed $\omega_z = 2$ rad/s about a fixed origin, and P displaced by $(0.2, 0, 0)$ m. Then $\mathbf{v}_{P,\mathrm{lin}} = (0, 0.4, 0)$ m/s and $\mathbf{a}_{P,\mathrm{class}} = (-0.8, 0, 0)$ m/s^2. The spatial acceleration is zero because its linear block subtracts $\boldsymbol{\omega} \times \mathbf{v}_{P,\mathrm{lin}} = (-0.8, 0, 0)$ m/s^2. With $\mathbf{g}^W = (0, 0, -9.81)$ m/s^2, an ideal aligned accelerometer at P reads $(-0.8, 0, 9.81)$ m/s^2. A zero spatial acceleration therefore does not imply a zero sensor reading or zero applied net force.

A.1.2 Rotations, spatial transforms and inertia identities

Let $\mathbf{E} = {}^B\mathbf{R}_A$ map vector components from axes A to axes B, and let $\mathbf{r} = {}^A\mathbf{r}_{AB}$ point from O_A to O_B in A. Define $[\mathbf{r}]_\times \mathbf{b} = \mathbf{r} \times \mathbf{b}$. Then $\mathbf{E}^\mathsf{T}\mathbf{E} = \mathbf{1}_3$, $\det \mathbf{E} = 1$ and $[\mathbf{E}\mathbf{r}]_\times = \mathbf{E}[\mathbf{r}]_\times \mathbf{E}^\mathsf{T}$. For compatible instantaneous descriptions,

$$\begin{aligned} {}^B\mathbf{p}_P &= \mathbf{E}({}^A\mathbf{p}_P - \mathbf{r}), \\ \mathbf{X}_{BA} &= \begin{bmatrix} \mathbf{E} & \mathbf{0} \\ -\mathbf{E}[\mathbf{r}]_\times & \mathbf{E} \end{bmatrix}, & \mathbf{v}_B &= \mathbf{X}_{BA}\mathbf{v}_A, \\ \mathbf{f}_B &= \mathbf{X}_{BA}^{-\mathsf{T}}\mathbf{f}_A, & \mathbf{I}_B &= \mathbf{X}_{BA}^{-\mathsf{T}}\mathbf{I}_A\mathbf{X}_{BA}^{-1}. \end{aligned} \tag{A.2}$$

Destination-first labels fix the direction. The inverse uses rotation $\mathbf{E}^\mathsf{T}$ and displacement $-\mathbf{E}\mathbf{r}$; composition is $\mathbf{X}_{CA} = \mathbf{X}_{CB}\mathbf{X}_{BA}$. Changing description does not subtract an observer's motion. Form relative motion only after both absolute motions share a point and axes.

For $\mathbf{v} = [\boldsymbol{\omega}; \mathbf{v}_{\mathrm{lin}}]$, the spatial cross operators and rigid-body inertia are

$$\begin{aligned} \mathbf{v}\times &= \begin{bmatrix} [\boldsymbol{\omega}]_\times & \mathbf{0} \\ [\mathbf{v}_{\mathrm{lin}}]_\times & [\boldsymbol{\omega}]_\times \end{bmatrix}, & \mathbf{v}\times^* &= -(\mathbf{v}\times)^\mathsf{T}, \\ \mathbf{I}_O &= \begin{bmatrix} \mathbf{I}_C + m[\mathbf{c}]_\times[\mathbf{c}]_\times^\mathsf{T} & m[\mathbf{c}]_\times \\ m[\mathbf{c}]_\times^\mathsf{T} & m\mathbf{1}_3 \end{bmatrix}. \end{aligned} \tag{A.3}$$

Here $\mathbf{c}$ points from O to the center of mass C; $\mathbf{I}_C$ is rotational inertia about C in the same axes. A positive-mass physical body has nonnegative principal moments satisfying the triangle inequalities. Positive definiteness alone is not the complete mass-realizability condition. Inertia maps motion to momentum, not another motion vector.

The following invariants are useful implementation checks:

$$P = \mathbf{f}^\mathsf{T}\mathbf{v} = \boldsymbol{\tau}^\mathsf{T}\boldsymbol{\nu}, \qquad T = \tfrac{1}{2}\mathbf{v}^\mathsf{T}\mathbf{I}\mathbf{v}, \qquad \boldsymbol{\tau} = \mathbf{J}^\mathsf{T}\mathbf{f}. \tag{A.4}$$

All paired quantities must describe the same point, axes and physical interaction. For a wrench covariance $\boldsymbol{\Sigma}$ and its inverse metric $\mathbf{Q}$, the transformations go in different directions:

$$\boldsymbol{\Sigma}_B = \mathbf{X}_{BA}^{-\mathsf{T}}\boldsymbol{\Sigma}_A\mathbf{X}_{BA}^{-1}, \qquad \mathbf{Q}_B = \mathbf{X}_{BA}\mathbf{Q}_A\mathbf{X}_{BA}^\mathsf{T}. \tag{A.5}$$

The second formula assumes an invertible covariance. A dimensionless quadratic residual is invariant only with the corresponding metric.

Worked example: change both axes and origin. Take $\mathbf{E} = R_z(-\pi/2)$, $\mathbf{r} = (0.2, 0, 0)$ m, $\boldsymbol{\omega}_A = (0, 0, 2)$ rad/s, $\mathbf{v}_{O_A,\mathrm{lin}} = (0.1, 0.2, 0)$ m/s, $\mathbf{n}_A = (0, 0, 0.2)$ N m and $\mathbf{F}_A = (0, 3, 0)$ N. Equation (A.2) gives

$$\boldsymbol{\omega}_B = (0, 0, 2)\ \mathrm{rad/s}, \qquad \mathbf{v}_{O_B,\mathrm{lin}} = (0.6, -0.1, 0)\ \mathrm{m/s},$$
$$\mathbf{n}_B = (0, 0, -0.4)\ \mathrm{N\,m}, \qquad \mathbf{F}_B = (3, 0, 0)\ \mathrm{N}.$$

Both descriptions give 1 W. For a body with $m = 2$ kg, $\mathbf{c}_A = (0.2, 0, 0)$ m and $\mathbf{I}_C^A = \mathrm{diag}(0.01, 0.02, 0.03)$ kg m^2, O_B is its center of mass. The spatial-inertia quadratic gives $T = 0.43$ J in either frame: 0.06 J rotational and 0.37 J translational. Rotating the linear block without shifting its point fails these checks.

Orientation parameters and increments. Use scalar-first Hamilton quaternions $Q = (s, \mathbf{u})$ for the body-to-world rotation. With $[\mathbf{u}]_\times$ as above,

$$Q_1 \otimes Q_2 = (s_1 s_2 - \mathbf{u}_1^\mathsf{T}\mathbf{u}_2, s_1\mathbf{u}_2 + s_2\mathbf{u}_1 + \mathbf{u}_1 \times \mathbf{u}_2),$$
$$\dot{Q} = \tfrac{1}{2} Q \otimes (0, \boldsymbol{\omega}^B) = \tfrac{1}{2}(0, \boldsymbol{\omega}^W) \otimes Q. \tag{A.6}$$

For a rotation vector $\boldsymbol{\phi}$, define $\mathrm{Exp}_Q(\boldsymbol{\phi}) = (\cos(\|\boldsymbol{\phi}\|/2), \sin(\|\boldsymbol{\phi}\|/2)\boldsymbol{\phi}/\|\boldsymbol{\phi}\|)$ with its continuous zero limit. A constant body-angular increment uses $Q^+ = Q \otimes \mathrm{Exp}_Q(h\boldsymbol{\omega}^B)$; a world-angular increment multiplies on the left. Normalize as needed. Q and $-Q$ have identical orientation. Shortest-increment logarithms need sign handling and have an axis ambiguity at angle π. A pose-log error, a finite displacement and an instantaneous point velocity have different meanings even when all have six entries.

A.2 Inference, optimization and dynamics algorithms

A.2.1 Spatial recursions and numerical checks

The following RNEA summary assumes a fixed-base tree, scalar joints with constant local motion subspaces $\mathbf{S}_i$, parent $p(i)$ and transform $\mathbf{X}_i$ from parent to child. All local wrenches and inertias use child axes and origin. Initialize $\mathbf{v}_0 = 0$ and $\mathbf{a}_0^{\mathrm{alg}} = -[\mathbf{0}; \mathbf{g}^W]$ in world-aligned base axes. In an outward pass,

$$\mathbf{v}_{J_i} = \mathbf{S}_i \dot{q}_i, \qquad \mathbf{v}_i = \mathbf{X}_i \mathbf{v}_{p(i)} + \mathbf{v}_{J_i},$$
$$\mathbf{c}_i = \mathbf{v}_i \times \mathbf{v}_{J_i}, \qquad \mathbf{a}_i^{\mathrm{alg}} = \mathbf{X}_i \mathbf{a}_{p(i)}^{\mathrm{alg}} + \mathbf{S}_i \ddot{q}_i + \mathbf{c}_i,$$
$$\mathbf{f}_i^{\mathrm{loc}} = \mathbf{I}_i \mathbf{a}_i^{\mathrm{alg}} + \mathbf{v}_i \times^* \mathbf{I}_i \mathbf{v}_i - \mathbf{f}_i^{\mathrm{ext}}. \tag{A.7}$$

External wrenches exclude gravity already included through the base. The last line is a *required local contribution*, not the body's physical net external wrench. Accumulate inward from the leaves:

$$\mathbf{f}_i = \mathbf{f}_i^{\mathrm{loc}} + \sum_{j:p(j)=i} \mathbf{X}_j^\mathsf{T} \mathbf{f}_j, \qquad \tau_i = \mathbf{S}_i^\mathsf{T} \mathbf{f}_i. \tag{A.8}$$

General joints require their joint model and any intrinsic bias acceleration. Closed loops and contacts require a constraint treatment; they are not additional unconstrained tree joints.

With $\mathbf{h}$ containing gravity and rigid-body velocity bias, the corresponding tangent-space balance is

$$\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{h} = \boldsymbol{\tau}_a + \boldsymbol{\tau}_p + \sum_j \mathbf{J}_j^\mathsf{T} \mathbf{f}_j^{\mathrm{ext}} + \mathbf{J}_c^\mathsf{T} \boldsymbol{\lambda}. \tag{A.9}$$

Count each load once. Passive damping on the right is negative for positive velocity; contact cannot also appear as the same external load. In a smooth constrained interval, $\mathbf{J}_c\dot{\boldsymbol{\nu}} + \dot{\mathbf{J}}_c\boldsymbol{\nu} = \mathbf{b}_c$ specifies the chosen acceleration constraint. Impacts instead require impulse and velocity balance. A contact solver's regularization belongs to its model and cannot be removed by relabeling its output.

For a free tree without passive or external loads, $\mathbf{h} = \mathrm{ID}(\mathbf{q}, \boldsymbol{\nu}, \mathbf{0})$. Mass columns follow from zero velocity and gravity: $\mathbf{M}_{\cdot j} = \mathrm{ID}(\mathbf{q}, \mathbf{0}, \mathbf{e}_j; \mathbf{g} = 0)$. CRBA accumulates locked-subtree composite inertia, $\mathbf{I}_i^{\mathrm{comp}} = \mathbf{I}_i + \sum_j \mathbf{X}_j^\mathsf{T} \mathbf{I}_j^{\mathrm{comp}} \mathbf{X}_j$.

ABA instead eliminates movable-joint accelerations. For a scalar joint,

$$\mathbf{U}_i = \mathbf{I}_i^{\text{art}}\mathbf{S}_i, \qquad D_i = \mathbf{S}_i^{\mathsf{T}}\mathbf{U}_i, \qquad u_i = \tau_i - \mathbf{S}_i^{\mathsf{T}}\mathbf{p}_i^{\text{art}},$$
$$\ddot{q}_i = D_i^{-1}[u_i - \mathbf{U}_i^{\mathsf{T}}(\mathbf{X}_i\mathbf{a}_{p(i)}^{\text{alg}} + \mathbf{c}_i)]. \tag{A.10}$$

The articulated inertia and bias require the complete inward elimination in Chapter 5; they are not simply a body's inertia and net wrench. A nonzero scalar D_i, or a suitable joint-block solve, is needed. For a dense forward calculation, solve Eq. (A.9) for acceleration instead of explicitly inverting $\mathbf{M}$.

Worked example: reproduce the coupled calculation. The two-link example in Chapter 5 specifies lengths $(0.30, 0.20)$ m, masses $(0.40, 0.25)$ kg and uniform slender links. At $\mathbf{q} = (\pi/6, -\pi/3)$, rates $(0.50, -0.25)$ rad/s and desired accelerations $(1, -0.5)$ rad/s^2, with downward gravity and no passive or other external loads, analytic dynamics and spatial RNEA give $\boldsymbol{\tau} \approx (1.399887, 0.216186)$ N m. Reusing its arithmetic implementation checks both mass columns and the inverse/forward calculation. This is a consistency check on that fixture, not a new implementation or a validation of a physical arm's parameters.

Check	What it can expose
$\mathbf{X}^{-1}\mathbf{X} = \mathbf{1}$; power and energy invariance	Direction, point-shift, ordering and inertia transport errors.
$\mathbf{v}\times^* = -(\mathbf{v}\times)^{\mathsf{T}}$	Incorrect dual cross operator or transpose.
$\mathbf{M} = \mathbf{M}^{\mathsf{T}}$ and positive energy	Some algebra and mass-model errors; not all physical-realizability errors.
RNEA versus analytic fixture; forward recovery	Gravity, coupling, indexing and load-accounting discrepancies.
Independent point or force reference	Physical mismatch that a model's own imputed acceleration can conceal.
Timestep refinement at matched physical times	Numerical sensitivity; does not establish real-world fidelity.

A.2.2 Probability, estimation and optimization reference formulas

For a random vector $\mathbf{x}$, covariance is $\boldsymbol{\Sigma}_x = \mathbb{E}[(\mathbf{x}-\boldsymbol{\mu}_x)(\mathbf{x}-\boldsymbol{\mu}_x)^{\mathsf{T}}]$. Under a local linearization $\mathbf{y} \simeq g(\boldsymbol{\mu}_x) + \mathbf{G}\delta\mathbf{x} + \boldsymbol{\epsilon}$,

$$\boldsymbol{\Sigma}_y \simeq \mathbf{G}\boldsymbol{\Sigma}_x\mathbf{G}^{\mathsf{T}} + \boldsymbol{\Sigma}_\epsilon + \mathbf{G}\boldsymbol{\Sigma}_{x\epsilon} + \boldsymbol{\Sigma}_{x\epsilon}^{\mathsf{T}}\mathbf{G}^{\mathsf{T}}. \tag{A.11}$$

Cross terms vanish only with the appropriate zero cross-covariance assumption. Calibration parameters can be appended to $\mathbf{x}$; reusing a common calibration does not create independent errors. Orientation covariance belongs to a stated local perturbation convention, not automatically four quaternion entries or a generic spatial velocity (Joint Committee for Guides in Metrology 2008; Barfoot 2024).

Bayes' rule gives $p(\mathbf{x} \mid \mathbf{y}) \propto p(\mathbf{y} \mid \mathbf{x})p(\mathbf{x})$ with its normalization. For residual $\mathbf{r}(\boldsymbol{\theta})$ and positive-definite Gaussian error covariance $\mathbf{R}$, the negative log likelihood is, up to a constant,

$$L(\boldsymbol{\theta}) = \tfrac{1}{2}\mathbf{r}^{\mathsf{T}}\mathbf{R}^{-1}\mathbf{r} + \tfrac{1}{2}\log\det\mathbf{R}. \tag{A.12}$$

The log-determinant term matters if $\mathbf{R}$ is fitted. For exact regressors $\mathbf{y} = \boldsymbol{\Phi}\boldsymbol{\theta} + \boldsymbol{\epsilon}$ with zero conditional error mean and known covariance, generalized least squares satisfies

$$(\boldsymbol{\Phi}^{\mathsf{T}}\mathbf{R}^{-1}\boldsymbol{\Phi})\hat{\boldsymbol{\theta}} = \boldsymbol{\Phi}^{\mathsf{T}}\mathbf{R}^{-1}\mathbf{y}, \qquad \operatorname{cov}(\hat{\boldsymbol{\theta}}) = (\boldsymbol{\Phi}^{\mathsf{T}}\mathbf{R}^{-1}\boldsymbol{\Phi})^{-1}. \tag{A.13}$$

The covariance expression assumes full column rank and the stated model. Noisy regressors or correlated feedback disturbances need additional analysis. Whiten with a factor $\mathbf{L}$ satisfying $\mathbf{L}\mathbf{L}^{\mathsf{T}} = \mathbf{R}$, solve using $\mathbf{L}^{-1}\boldsymbol{\Phi}$ and $\mathbf{L}^{-1}\mathbf{y}$, and prefer QR or SVD over explicitly forming normal equations in ill-conditioned problems. Parameter scaling changes the numerical conditioning, not the physical experiment's identifiability.

Worked example: weighted observations and a common bias. Two independent position observations, 0.98 m and 1.04 m, have variances 0.0004 m^2 and 0.0001 m^2. Precision weighting gives 1.028 m and variance 0.00008 m^2. Add a shared zero-mean bias with variance 0.0001 m^2, independent of the individual errors. The off-diagonal covariances are now nonzero; GLS retains the same mean and gives variance 0.00018 m^2. Treating the bias as two independent errors changes the problem and exaggerates the benefit of repetition.

For sensitivity $\mathbf{H}_\theta = \partial g/\partial \boldsymbol{\theta}$ and parameter-independent Gaussian noise, a local information matrix is $\mathbf{H}_\theta^\mathsf{T}\mathbf{R}^{-1}\mathbf{H}_\theta$. Rank concerns locally distinguishable parameter combinations, not a universal score for useful experience. With fixed dimensionless scales, compare this matrix only for the intended parameterization and population. Chapter 8 develops excitation and identification limits. For instance, exact rows $(\ddot{q}, \dot{q}) = (1,0), (0,1), (1,1)$ and efforts $(0.02, 0.01, 0.03)$ recover $J = 0.02$ kg m^2 and $b = 0.01$ N m s/rad in $\tau = J\ddot{q} + b\dot{q}$. Rows with proportional acceleration and velocity cannot separate them.

For a linear state model, take zero-mean process and measurement noise independent across times and noise types and independent of initial error. With invertible innovation covariance $\mathbf{S}_y$, the Kalman equations are given below. An existing posterior $(\hat{\mathbf{x}}^+, \mathbf{P}^+)$ predicts the next prior; "new" marks the subsequent correction:

$$\begin{aligned} \hat{\mathbf{x}}^- &= \mathbf{F}\hat{\mathbf{x}}^+ + \mathbf{B}\mathbf{u}, & \mathbf{P}^- &= \mathbf{F}\mathbf{P}^+\mathbf{F}^\mathsf{T} + \mathbf{Q}, \\ \mathbf{S}_y &= \mathbf{H}\mathbf{P}^-\mathbf{H}^\mathsf{T} + \mathbf{R}, & \mathbf{K} &= \mathbf{P}^-\mathbf{H}^\mathsf{T}\mathbf{S}_y^{-1}, \\ \hat{\mathbf{x}}^+_{\text{new}} &= \hat{\mathbf{x}}^- + \mathbf{K}(\mathbf{y} - \mathbf{H}\hat{\mathbf{x}}^-), \\ \mathbf{P}^+_{\text{new}} &= (\mathbf{1} - \mathbf{K}\mathbf{H})\mathbf{P}^-(\mathbf{1} - \mathbf{K}\mathbf{H})^\mathsf{T} + \mathbf{K}\mathbf{R}\mathbf{K}^\mathsf{T}. \end{aligned} \tag{A.14}$$

Solve for the innovation gain rather than forming $\mathbf{S}_y^{-1}$. The last line is the Joseph form. Joint Gaussian assumptions give an exact conditional distribution; otherwise the stated covariance assumptions give the optimal linear estimate, not a general nonlinear posterior. For position/velocity and interval h, continuous white acceleration with spectral density q_c and independently held random acceleration with variance σ_a^2 instead imply different process covariances:

$$\mathbf{Q}_{\text{white}} = q_c \begin{bmatrix} h^3/3 & h^2/2 \\ h^2/2 & h \end{bmatrix}, \qquad \mathbf{Q}_{\text{held}} = \sigma_a^2 \begin{bmatrix} h^4/4 & h^3/2 \\ h^3/2 & h^2 \end{bmatrix}. \tag{A.15}$$

Here q_c has units m^2/s^3, while σ_a^2 has units m^2/s^4. Adjust both transition and process covariance when h changes.

Worked example: one scalar correction. For prior position 1 m with variance 0.0004 m^2 and observation 1.03 m with independent error variance 0.0001 m^2, $K = 0.8$. The posterior mean is 1.024 m and Joseph covariance is 0.00008 m^2. An unmodeled common bias is not removed by this update.

For local inverse kinematics, let $\mathbf{J}$ map joint rates to a point-specific task velocity. Choose task scaling $\mathbf{W}_v$ and positive diagonal joint-rate scaling $\mathbf{D}_\nu$, and set $\mathbf{G} = \mathbf{W}_v\mathbf{J}\mathbf{D}_\nu$, $\mathbf{b} = \mathbf{W}_v\mathbf{v}^*$. For dimensionless $\boldsymbol{\xi}$, damped least squares solves

$$\min_{\boldsymbol{\xi}} \|\mathbf{G}\boldsymbol{\xi} - \mathbf{b}\|^2 + \lambda^2\|\boldsymbol{\xi}\|^2, \qquad (\mathbf{G}^\mathsf{T}\mathbf{G} + \lambda^2\mathbf{1})\boldsymbol{\xi} = \mathbf{G}^\mathsf{T}\mathbf{b}, \quad \boldsymbol{\nu} = \mathbf{D}_\nu\boldsymbol{\xi}. \tag{A.16}$$

With $\lambda > 0$ this quadratic has a unique solution. Limits and contact feasibility need explicit constraints. A damped pseudoinverse does not create an exact null-space projector (Wampler 1986). For dimensionless $\mathbf{G} = \operatorname{diag}(1, 0.01)$, $\mathbf{b} = (1,1)$ and $\lambda = 0.1$, the solution is $(0.990099, 0.990099)$; the undamped solution is $(1, 100)$. Damping limits amplification while accepting a task residual. These are scaled coordinates, not unspecified rad/s applied directly to a robot.

Likelihoods, covariance and optimization answer different questions. A well-conditioned fit does not imply an adequate model, and a small training loss does not imply successful execution. Bellman objectives, policy gradients, demonstration likelihoods, latent-model objectives and sampling algorithms retain their definitions in Chapters 10 through 13. Confidence procedures, replication and protected comparisons remain in Chapter 16; no unqualified confidence formula should replace those assumptions.

A.3 Mathematics-to-implementation maps

A.3.1 Simulator state, force and control arrays

Table entries below refer to MuJoCo 3.16.0 and a completed evaluation at a stated configuration and time. The model defines sizes and addresses; never infer them from a drawing or assume identical object, joint, actuator and degree-of-freedom indices (DeepMind Technologies Limited, n.d., n.d.). Here $n_b =$ `model.nbody` and $n_j =$ `model.njnt`; n_u, n_a and n_{out} correspond to `nu`, `na` and `nout`. An actuator count is `model.nactuator`.

Field or function	Size	Meaning and caution
`qpos`	n_q	Configuration storage; includes quaternion parameters for ball/free joints.
`qvel`, `qacc`	n_v each	Generalized tangent velocity and evaluated acceleration. The opt-in `discrete` mode uses a step-acceleration convention; see Chapter 6. Neither array is the componentwise derivative of all `qpos` entries.
`ctrl`, `act`	n_u, n_a	Control scalars and internal activation states. Controls, actuator count and force-output count need not be equal.
`actuator_force`	n_{out}	Effort in actuation space; transmission maps it into generalized effort.
`qfrc_actuator`	n_v	Delivered generalized actuator effort in the simulated model, not independently measured hardware torque.
`qfrc_bias`	n_v	Rigid-body velocity/gravity bias on the left of the balance, not passive damping.
`qfrc_passive`, `qfrc_applied`	n_v each	Passive and user-applied generalized contributions, respectively.
`qfrc_constraint`	n_v	Constraint contribution; includes more than contact alone.
`xfrc_applied`	$n_b \times 6$	User-applied Cartesian force then torque, world axes, about each body's center of mass.
`jnt_qposadr`, `jnt_dofadr`	n_j each	Start addresses for a joint's configuration and velocity entries. Use its type to determine the block length.
`body_inertia`	$n_b \times 3$	Principal rotational moments about the center of mass in the inertial frame specified by `body_ipos`/`body_iquat`; not a spatial 6×6 matrix.
`mj_fullM`	$n_v \times n_v$ output	In 3.16.0, expands sparse `data.M` through the model/data/destination interface. Legacy `qM` snippets are not this interface.

In ordinary use the generalized balance combines actuator, passive, applied, Cartesian-applied and constraint effects against the inertia and bias. Applied Cartesian loads are projected into generalized coordinates; adding that projection again as `qfrc_applied` would double count it. `mj_inverse` computes required generalized effort under its specified dynamics, not a general allocator of realizable actuator controls.

Construct motion at the point actually observed. For an existing model, state and named site, a compatible forward evaluation and the site's Jacobians give motion in world axes at that site:

```
import numpy as np
import mujoco
mujoco.mj_forward(model, data)
nv = model.nv
Jp = np.zeros((3, nv))
Jr = np.zeros((3, nv))
mujoco.mj_jacSite(model, data, Jp, Jr, site_id)
v_site = np.concatenate((Jr @ data.qvel, Jp @ data.qvel))
tau_ext = Jr.T @ moment_at_site + Jp.T @ force_at_site
power = tau_ext @ data.qvel
```

The wrench components are world-expressed at the same site; `power` equals their pairing with `v_site`. This is not the world-origin space twist: its linear block would instead be $\mathbf{v}^W_{E,\mathrm{lin}} - \boldsymbol{\omega}^W \times \mathbf{p}^W_E$. Chapter 11 explains this distinction. `mj_objectVelocity` returns rotation then translation about the named object's center, with axes selected by its local flag. Internal `cvel` arrays use another center convention; do not treat them as body-origin measurements without the corresponding conversion.

Worked example: a control scalar is not a joint torque. For a hinge about z, put the body center of mass at the joint, set its principal moments to $(0.01, 0.015, 0.02)$ kg m^2, and add joint `armature="0.005"`. With gravity, passive loads and contacts absent, the total hinge inertia is 0.025 kg m^2. An ideal motor transmission `gear="3"`, with unity gain and no limits, maps control 0.1 to actuation-space effort 0.1 and generalized effort 0.3 N m. The modeled initial acceleration at rest is therefore 12 rad/s^2. Only a unit transmission would make those two effort numbers equal. The armature term must not also be added to the body's inertia.

For the same hinge at $q = 0.3$ rad and $\dot{q} = 2$ rad/s, a site 0.2 m along the body's x axis has

$$\begin{aligned} \mathbf{v}^W_E &= (-0.4 \sin 0.3, 0.4 \cos 0.3, 0)\ \mathrm{m/s}, \\ \boldsymbol{\omega}^W &= (0, 0, 2)\ \mathrm{rad/s}. \end{aligned}$$

A world force $(0, 3, 0)$ N and pure moment $(0, 0, 0.2)$ N m at that site project to $\tau_{\mathrm{ext}} = 0.2 + 0.6 \cos 0.3 \approx 0.773202$ N m. `mj_applyFT` and the Jacobian transpose give the same projection when their point, axes and signs agree. This check verifies a specified model/interface calculation, not the calibration of a real actuator.

`mj_forward` updates derived quantities without time integration. After directly changing state, recompute the required stages before reading derived arrays. `mj_step` advances the state; derived quantities need not all have been evaluated at the new state's time. Do not change a logging experiment's command schedule inadvertently when calling a controller callback again. States needed for exact continuation can include activation, history, warm starts, plugins, controller memory and random-generator state, beyond `qpos`/`qvel`. Chapter 6 gives the numerical and reset qualifications.

A.3.2 Permutations, representations, rates and signal dimensions

Changing component order is a permutation, not a frame transformation. For a linear-first motion or force-first wrench, use

$$\boldsymbol{\Pi} = \begin{bmatrix} \mathbf{0} & \mathbf{1}_3 \\ \mathbf{1}_3 & \mathbf{0} \end{bmatrix}, \qquad \mathbf{v}_{\mathrm{ang}} = \boldsymbol{\Pi}\mathbf{v}_{\mathrm{lin}}, \qquad \mathbf{f}_{\mathrm{mom}} = \boldsymbol{\Pi}\mathbf{f}_{\mathrm{force}}. \tag{A.17}$$

Applying the same permutation to the paired dual quantities preserves power because $\boldsymbol{\Pi}^{\mathsf{T}}\boldsymbol{\Pi} = \mathbf{1}_6$. It does not fix incompatible origins, axes, units or force directions. For example, `mj_contactForce` returns force then torque in the contact frame. Reorder, rotate and, when needed, shift its moment to the chosen point; identify which body's load and which sign are intended. A contact identifier is not a joint index.

A MuJoCo free joint stores world translational velocity and body-local angular velocity. Reordering those six entries alone does not produce one common-axes Featherstone vector. Recover the appropriate point velocity through the model's Jacobian or the documented object-velocity function. Similarly, read a joint's starting addresses rather than using its ordinal number to index `qpos` or `qvel`. A model with one free joint and one hinge has $n_q = 8$ and $n_v = 7$; its number of controls is a separate design choice.

For configuration increments use representation-aware operations:

```
q_next = data.qpos.copy()
mujoco.mj_integratePos(model, q_next, data.qvel, dt)
nu_interval = np.zeros(model.nv)
mujoco.mj_differentiatePos(model, nu_interval, dt,
                           data.qpos, q_next)
```

This constructs a constant tangent increment; it is not a complete dynamics step. The recovered interval velocity follows the function's representation and branch conventions, not arbitrary componentwise quaternion differences. For $Q_0 = (\cos\frac{\pi}{4}, 0, 0, \sin\frac{\pi}{4})$, body-local angular velocity $(1, 0, 0)$ rad/s corresponds initially to world angular velocity $(0, 1, 0)$ rad/s. Right multiplication by $\mathrm{Exp}_Q(0.1, 0, 0)$ for 0.1 s preserves that interpretation. Replacing Q by $-Q$ changes no orientation, although the raw Euclidean difference of those unit quaternions has norm 2.

Transmission and signal conversions. For the fixed-housing scalar model of Chapter 3, with an ideal rigid, lossless reducer, $\theta_m = Nq$ and $\dot{\theta}_m = N\dot{q}$. Its transmitted input-shaft effort τ_s and output effort τ_t obey $\tau_t = N\tau_s$, while $J_{\text{reflected}} = N^2 J_m$. These are gearbox-port efforts, not automatically electromagnetic torque $k_t i$. Without motor friction, the rotor balance gives

$$\tau_t = Nk_t i - N^2 J_m \ddot{q}. \tag{A.18}$$

With $N = 5$ and $J_m = 0.0002$ kg m^2, the joint-referred contribution is 0.005 kg m^2. If $k_t i = 0.1$ N m and $\ddot{q} = 2$ rad/s^2, joint-referred electromagnetic effort is 0.5 N m, but output-shaft effort is 0.49 N m even without gearbox losses. If $Nk_t i$ is used as generalized drive effort, include the reflected rotor inertia exactly once in the model. Do not apply the ratio a second time or label this effort as measured output torque. Other loads, losses and compliance add further distinctions between electromagnetic, shaft and contact effort. Store each conversion's boundary, scale, zero, direction and validity range, not only its output number.

Finally distinguish physics, control and observation clocks. A physics step of 0.002 s and a held control period of 0.020 s give ten physics steps per command, not ten independent decisions. At 0.4 m/s, a 0.030 s timing mismatch alone gives a local displacement discrepancy of 0.012 m. Capture, arrival, estimator availability and application times answer different questions; a clock correction does not itself remove transport delay. For a chosen first-order filter time constant τ_f, $\alpha = \exp(-T_s/\tau_f)$ weights the previous filtered state and $1 - \alpha$ the new input. It specifies a discrete pole, not an arbitrary universal smoothing coefficient.

An episode's $(o_k, a_k, r_{k+1}, o_{k+1})$ indexing must name the interval in which the action was applied. Discounting per decision, terminal outcomes and externally truncated records have different semantics. Action-prediction horizons, executed chunks, diffusion steps and token positions are not interchangeable robot timesteps. These meanings connect the physical reference to Chapters 9–13.

The companions `check_appendix_a.py` and `check_appendix_a_mujoco.py` check the constructed arithmetic and the stated native interface separately. A version mismatch should be resolved explicitly rather than silently importing an older signature. A reader designing data should be able to follow one quantity from its equation, through this conversion, to the actual stored signal and back. That inspectable connection is the purpose of the reference.

Appendix B Data Collection and Sample Reference

A useful record states what was observed, how it was obtained and which interpretations remain possible. A column called `torque` is not yet a torque measurement; an image file is not yet an observation available to a controller at its decision time. This appendix provides a compact reference for those distinctions. Chapters 7–9 develop the underlying measurement, identification and dataset-design arguments. Appendix A supplies the mathematical and implementation conventions; Appendix C explains the accompanying checks.

The sample registry is deliberately selective. It identifies available source records and their limitations rather than promising a complete dataset for every modality discussed in the book. Device and course records are examples of problems that arise in many ExpGen PhySyst investigations. They are not prerequisites for understanding the derivations.

B.1 Signal and representation reference

B.1.1 Electrical, proprioceptive and inertial measurements

Electrical signals and effort. At least four quantities can appear under the informal name current: supply current, individual phase current, a transformed current component used by the drive, and a requested current setpoint. Their electrical boundaries and averaging processes differ. For the fixed-housing motor model of Chapter 3, $\tau_e = k_t i$ requires the current variable for which the torque constant k_t is defined. A logged supply current cannot simply be substituted for that variable. Nor does motor electromagnetic torque equal transmitted output-shaft torque during rotor acceleration or in the presence of transmission losses.

A complete electrical channel description therefore states the measurement point, sign convention, gain and offset, bandwidth or averaging window, quantization, clipping, and the operating mode. Voltage should likewise identify whether it is a bus measurement, applied winding voltage or a requested duty ratio. Thermal state, supply limitations and protection modes can alter the relation between a command and the achieved motion. Record these when available; state explicitly when they are absent.

The selected XC330 physical demonstration illustrates this boundary. Its `Present Current` is documented in the accompanying record as input-supply telemetry. It is not an independent winding-current or output-torque measurement. Its usefulness includes examining a repeatable telemetry response under a declared controller and load condition; it does not supply a calibrated output-torque sensor.

Position, velocity and actuator state. Proprioceptive records concern the system's own configuration, motion and internal state. An encoder reading may refer to the motor rotor, gearbox output or another point in the transmission. For an ideal scalar encoder,

$$q = s\frac{2\pi}{C}(n - n_0), \tag{B.1}$$

where C is counts per revolution of the *specified shaft*, n_0 is the calibrated zero and $s \in \{-1, +1\}$ fixes direction. Applying a reduction ratio again to an output-shaft encoder is a conversion error. Wraps, multi-turn state and discontinuities need a declared treatment before numerical differentiation.

Example B.1A constructed encoder record For an output-shaft encoder with $C = 4096$, $n_0 = 1024$ and $s = +1$, two readings 1024 and 1034 separated by 0.020 s give

$$\Delta q = \frac{20\pi}{4096} = 0.01534 \text{ rad}, \qquad \bar{\dot{q}} = 0.76699 \text{ rad/s}. \tag{B.2}$$

This is an interval velocity calculated from two indications. It is not a separately measured instantaneous velocity, and two positions do not provide an independent acceleration measurement. The assumed scale and zero belong to this constructed example, not to a calibration of an XC330 or RS06.

A drive-reported velocity may be filtered, differentiated internally or estimated from a motor model. Retaining both it and a derived velocity is useful only if their origins remain distinguishable. Record firmware, operating mode and estimator

configuration when those affect the signal. Temperature, status flags and fault transitions also belong to the actuator record: removing them can make a protection event look like unexplained model error.

Inertial signals. A gyroscope measures angular rate in its sensor axes, subject to scale, bias, noise and saturation. An accelerometer measures specific force, not world-frame translational acceleration. Under the convention of Chapter 7, an ideal sensor at point P obeys

$$\boldsymbol{y}_a = {}^{S}R_W({}^{W}\boldsymbol{a}_P - {}^{W}\boldsymbol{g}) + \boldsymbol{b}_a + \boldsymbol{\epsilon}_a. \tag{B.3}$$

State the axes, gravity convention and sensor location. Lever-arm terms matter when using its reading to infer acceleration at a different point. A stationary accelerometer is not generally expected to report zero. Calibrated orientation is an estimate derived from signals and assumptions; it is not another raw inertial channel.

B.1.2 Geometric, visual, force and tactile representations

Geometry and pose. A position needs an origin and axes. A rigid pose needs a direction: in ${}^{A}T_B$, coordinates in B are mapped into A. Store length units, handedness and the point represented. A quaternion needs element order, active/passive meaning, source and target frames, and normalization policy. The equivalence of Q and $-Q$ prevents componentwise differences from being an orientation error. State whether an angular increment is local or world expressed, and use the tangent-space operations in Appendix A.

A matrix shape is insufficient to describe a pose record. Two 9-D vectors may both contain position and the first two rotation-matrix columns yet refer to different episode origins, action semantics and availability times. Such records are not interchangeable merely because the arrays can be concatenated.

Images and reconstructed geometry. An image record should identify the camera, intrinsics, distortion model, resolution, color/depth convention and relevant calibration version. Record exposure time or interval and any rolling-shutter convention, separately from host receipt time. Image compression and resizing are transformations; they should not silently replace the original camera model. Depth values need their physical interpretation, invalid-value encoding and scale.

A reconstructed pose or point cloud is an estimate. Preserve the original image references, reconstruction method and parameters, alignment frame, quality indicators and failure status. Missing tracking should remain missing rather than being represented as a plausible stationary pose. The registry does not claim that raw synchronized images or externally validated poses exist for every actuator record.

Forces and contact. Use Featherstone's moment-first ordering $\boldsymbol{f} = (\boldsymbol{n}^{\mathsf{T}}, \boldsymbol{F}^{\mathsf{T}})^{\mathsf{T}}$. State the point about which the moment is taken, the expression axes, the body receiving the load and the sign convention. Changing point changes moment even when the physical force is unchanged. The dual transformation in Appendix A preserves instantaneous power; copying six numbers into a differently ordered interface does not.

A load-cell indication, estimated external wrench, commanded effort and simulator contact force are different records. For a tactile array, add sensor geometry, taxel ordering, contact area assumptions and whether values represent pressure, normal force, deformation or an uncalibrated response. Integration over taxels requires the associated geometry and calibration. Absence of a logged contact signal does not prove absence of contact.

B.2 Acquisition and record conventions

B.2.1 Units, clocks, calibration and temporal uncertainty

Every channel needs a quantity, unit, reference, source and time meaning. Prefer SI units in derived records while preserving the original indication and its conversion. Degrees versus radians and milliseconds versus seconds are common, readily detectable errors. More subtle errors retain the unit while changing the point, direction or boundary of the quantity.

Distinguish at least acquisition, device stamp, host receipt, estimator availability and command application. A host receive-end timestamp says when a read transaction completed, not when every value in the packet was sampled. A device counter can wrap or reset; retain its raw value and the unwrapping rule. A fitted relation

$$t_{\text{reference}} = a\, t_{\text{device}} + b \tag{B.4}$$

changes the clock representation. It does not remove transport delay, exposure duration, sample-and-hold behavior or scheduling jitter.

For local position error induced by timing uncertainty, $\delta \boldsymbol{p} \simeq \boldsymbol{v}\, \delta t$. A constructed 0.030 s mismatch at 0.4 m/s gives 0.012 m even with perfect spatial calibration. When timing error is independent, zero mean and small, its covariance contribution is approximately $\boldsymbol{v}\boldsymbol{v}^{\mathsf{T}}\sigma_t^2$. A persistent offset is a bias; it should not be hidden in that variance term. Exposure intervals and correlated timing errors need a richer model.

Calibration records should identify the artifact or reference used, date, configuration, fitted parameters, residuals, validity range and uncertainty. For a conversion $\boldsymbol{y} = g(\boldsymbol{r}, \boldsymbol{\gamma})$, first-order uncertainty propagation includes both raw indication $\boldsymbol{r}$ and calibration $\boldsymbol{\gamma}$, including their cross-covariance when they are dependent. Reusing one uncertain calibration across many episodes creates shared error, not independent fresh noise.

B.2.2 Episode context, missing records and collection identity

An episode needs a collection identity before it needs a learning label. Record the apparatus and revisions, actuator/controller configuration, object/load condition, task and initial state, operator or generator, start/end reasons, and interventions. Separate the intended request, accepted controller target and any measured response. Preserve a reset or fault as an event rather than joining unrelated intervals into a smooth trajectory.

Useful missingness fields distinguish not requested, unavailable, invalid, out of range and lost during transport. A missing current value is not a measured zero current. A repeated last value is a held or imputed value, not a fresh independent observation. Resampling should retain the parent timestamps and describe interpolation, filtering and gap limits.

An audit ledger counts attempted, captured, retained and rejected episodes and names each rejection rule. The physical trial is often a more suitable experimental unit than a row or training window. Thirty simultaneous held-out events recorded on two motors yield sixty motor episodes, but not sixty independent collection events. Randomly splitting nearby frames can leak an encounter across training and testing (Chapter 9).

Load descriptions are part of identity. The selected public XC330 records use a *declared* 110 g fixture, while the population example uses a separately declared 113 g fixture. An older 98 g label is another case. Do not average these numbers or relabel a historical record to manufacture agreement. The current source descriptions do not establish independent weighing of every fixture. Parameter inference should state that uncertainty rather than silently treating a declared mass as an exact measurement.

B.3 Source and derivation provenance

B.3.1 Measured and simulated sources, estimates and human contributions

The book uses the following distinctions throughout.

Measured source: a recorded indication from a physical acquisition process, with the actual device, configuration and time meaning stated.

Simulated source: values produced by an executable model and its numerical, reset and observation procedures.

Estimate: a quantity inferred from other information, such as a calibrated joint angle, reconstructed pose, fitted inertia or policy value.

Human contribution: a demonstration, task choice, annotation, correction or assessment supplied by a person.

Constructed example: an explicitly assumed case for deriving or checking an argument, with no claim of physical collection or training.

These are provenance descriptions, not mutually exclusive modalities. A human demonstration can contain measured camera images and estimated poses; a simulated rollout can contain an image and a human-supplied success label. A simulated sensor can include noise without becoming a physical measurement.

Preserve the distinction between *source-reported* physical origin and independent verification. Inspecting a file, recomputing its summary and checking its hash do not rerun its collection, calibrate its sensors or validate its claimed hardware conditions. A saved success ledger documents a particular reported evaluation; it cannot establish that the matching policy checkpoint and executable environment have been supplied.

B.3.2 Annotation, transformation and version lineage

Treat a derived artifact as a new version with named parents. Keep the raw bytes, calibration version, selection criteria, transformation code and parameters, and the resulting hash. A cryptographic hash checks byte identity; it does not establish truth, licensing or physical validity. Commit identities freeze a repository snapshot; a branch name alone does not.

An annotation record should name its question, operational definition, annotator or procedure, time and uncertainty or disagreement where relevant. A success label and a fault flag answer different questions: a recording with no injected fault can still violate a reachability or quality criterion. A model-generated label needs its model version and inputs just as a fitted physical parameter needs its estimation procedure.

When preparing sequence-learning data, preserve the relation between observations and actions. The physical handheld collection route labels a next measured pose, whereas the selected physical arm-teleoperation sample records the accepted controller target. The source recorder confirms that distinction; the selected sample is not a handheld-recorder dataset. A simulated teleoperation record can also label an accepted target. A shared 9-D layout alone therefore does not establish the same conditional-learning problem. The episode frame, relative-height convention, action interval and allowed latency must also agree. An absent metadata field is not repaired by assuming a convenient default; if its meaning is supplied by a dataset card, retain that dependency.

Provenance supports useful revision. If a new calibration changes derived poses, regenerate the derived data and evaluation under a new identity while retaining the earlier version. If a rejection criterion changes, report its effect on the retained population. If train/test partitions change, previous reported errors belong to the previous split. These relationships make reusable experience possible without pretending that the original trace already contains its final interpretation.

B.4 Sample registry and access

B.4.1 Available samples, experiment identities and source status

The accompanying `companion/sample-registry.json` records selected public course artifacts at the immutable revision `b8b1bb4`, inspected on 19 October 2026. Each entry identifies a path, experiment, source kind, byte identity, representation and supported uses. The registry contains metadata and links, not another copy of the data or pretrained models. Its source revision is separate from the earlier course-coverage audit used to plan the book.

Sample family	Available information	Interpretation boundary
XC330 physical demonstration	2601 telemetry rows and 1301 command rows; declared mode, gains and no external payload.	Host receive-end time; supply-current telemetry; no independent shaft torque or synchronized camera validation.
XC330 identification	Selected physical/derived traces and 30 held-out episode metrics.	Declared 110 g fixture; fitted-model error is not an independent verification of the load mass.
RS06 records	A selected physical session and separate simulated position/torque records.	Physical position-mode signals and synthetic torque replay have different origins and usable conclusions.
XC330 population	A paired motor subset and event-level split information.	Declared 113 g case; two motor records per event do not double independent experimental replication.
Arm and pusher	A physical arm motion log; five physical pusher episodes with 2147 frames; a separate simulated pilot.	The physical sample is all training data; it supplies no held-out task success estimate.
Saved learning artifacts	A source neural-actuator artifact and a saved 50-episode ACT evaluation ledger.	Artifact identities are task specific; the ACT checkpoint referenced by the ledger is absent.

The ten-episode simulated pusher pilot contains 638 frames. Its acceptance notes establish six definite rejections; the other four are candidates, not four validated demonstrations. Image and contract review remain part of acceptance. The saved ACT ledger contains 32 successes in 50 trials and reports 0.64; it is a source-reported simulation evaluation, not a result reproduced by the book's arithmetic companions.

A cube policy with another observation or goal contract cannot replace the absent continuous-rotation deployment bundle. Checkpoints with different observations, goals and action contracts are not substitutes. Likewise, the registry does not promise ready ACT, diffusion or generalist checkpoints simply because those methods are explained in the book. Availability is a property of the identified distribution, not of a method's name or a figure showing its result.

B.4.2 Permissions, licensing and independent reconstruction

Public readability is not a redistribution license. An applicable license must cover the particular code, data, media or trained artifact and the intended use. A software license in a subdirectory does not automatically cover every dataset or supplier illustration in the repository. Where an applicable data license was not established, the registry says so; it does not grant permission. Personal identifiers, operator recordings and institutional constraints require their own review.

Readers can independently reconstruct the book's *constructed* calculations from the equations and companions without these source samples. Reconstructing a physical experiment requires further apparatus, calibration, operating conditions and permissions. Reconstructing a trained policy requires the correct environment, observation/action transformations, training/evaluation configuration and actual checkpoint. A saved metric alone is insufficient.

For further collection, write a small data card before starting: the intended question; physical boundary; measured and requested quantities; units and frames; clock and availability definitions; calibration; episode and rejection rules; split unit; provenance; and permitted sharing. Revise it after a pilot reveals missing information. Designing this record together with the apparatus is part of designing data for embodied learning, not paperwork added after learning has finished.

Appendix C Computational Companions and Investigations

The companions make selected derivations inspectable through short numerical calculations. They also support investigations in which a reader changes an assumption, predicts its consequence and checks that prediction. Neither a passing assertion nor an attractive simulator animation is evidence of a physical capability by itself. This appendix states what the supplied computations do, how they relate to the book, and how to carry an inquiry beyond a worked example.

C.1 Reproducible computation

C.1.1 Environments, numerical checks and result identity

The authoritative mathematical argument is in the chapter text. The `tools/check_chapterNN.py` companions for Chapters 3–16 implement selected constructed calculations using the Python standard library. Appendix A has a separate arithmetic checker. The Appendix B/C checker validates the registry structure and the constructed encoder, timing and calibration examples. Their inputs are explicit fixtures; their assertions check quantities, signs and declared tolerances. They emit JSON summaries, not experimental measurement files or benchmark scores.

From the book repository's root, the complete arithmetic suite is:

```
python3 tools/run_companions.py
```

A focused example can be inspected and run independently:

```
python3 tools/check_chapter04.py
python3 tools/check_appendix_a.py
python3 tools/check_appendix_bc.py
```

These checks require no network, credentials, robot, training run or external dataset. The review environment used Python 3.9.6 for this standard-library route. Record the interpreter version when repeating it; satisfying these checks is not a compatibility claim for every future Python release.

A separate native interface check. `tools/check_appendix_a_mujoco.py` uses an existing isolated environment with both the MuJoCo Python package and engine at version 3.16.0. It checks small constructed cases: array/tangent dimensions, quaternion increments, actuator transmission, point/angular velocity conventions and spatial-force ordering. It does not load a learned policy or run a hardware controller. After independently providing that exact environment, use its Python interpreter to run the script; no installer is launched by the arithmetic suite. Version checking is intentional because an interface claim belongs to a stated implementation, not to the name MuJoCo in general.

The remaining tools have different effects. `tools/build.py` produces the manuscript and outline PDFs with XeLaTeX and Biber. `tools/render_outline.py` regenerates the structural proposal from its JSON, not the authored chapters. Publication and credential-connection helpers are editorial tools for the private book project; they are not learning exercises and are not called by the companions. Do not run a synchronization helper to check an equation.

What should a reproducibility record contain? Retain the source commit, script identity, interpreter/dependency versions, input identity, command, output and tolerances. An output hash establishes byte identity under that run; floating-point comparisons generally need stated numerical tolerances rather than assumed bit-for-bit portability. Separate the script's input assumptions from properties it actually checks. If a tolerance is changed, explain the numerical reason before accepting a previously failing result. A negative test that rejects an inconsistent frame or source type is often as informative as a successful calculation.

C.1.2 Separating worked examples from hardware procedures

There are three distinct levels of checking:

1. *Constructed arithmetic*: do the stated equations give the stated result for the assumed inputs?
2. *Implementation correspondence*: does a specified software interface implement the declared representation and mapping in a controlled case?
3. *Empirical evaluation*: does an actual apparatus or learned system exhibit the claimed behavior under declared conditions?

The supplied arithmetic suite addresses the first. The optional native MuJoCo checker provides bounded evidence for parts of the second. Neither establishes the third. Hash inspection of an available checkpoint also does not execute or qualify that policy.

Consider the reflected-inertia example in Appendix A. Calculating $N^2 J_m$ checks an ideal rigid transmission with a fixed housing. It does not identify the inertia of a purchased actuator or verify a current telemetry channel. A simulator implementation must include that inertia exactly once and map generalized effort consistently. A hardware procedure would additionally require the real actuator's signal semantics, calibration, loading, limits and operating conditions. Those procedures belong to the particular apparatus documentation, not to a generic textbook command.

A useful investigation changes one defined element at a time. For example, first predict the effect of moving the wrench reference point; then compute the dual transformation and verify invariant power; finally identify which simulator fields represent those quantities. If a physical comparison is later performed, its data and assumptions receive a new record. Do not rename a constructed output as measured evidence.

C.2 Tools for developing an investigation

C.2.1 The six-view project record and worked use

Chapter 15 introduces six connected views: Claim, Experience, System, Grounding, Learning and Decision. They organize questions that can be revisited as evidence changes. They are neither a validated maturity score nor a required linear sequence. A record is useful when another person can reconstruct the argument and identify what would change its conclusion.

Example C.1A constructed joint-identification investigation A simplified fixed-housing joint has $J\ddot{q} = \tau$, while its reported position is $y = sq + b$. Suppose the experiment identifies only

$$\ddot{y} = \frac{s}{J}\tau, \qquad \frac{s}{J} = 50\ (\mathrm{kg\,m^2})^{-1}. \tag{C.1}$$

The pairs $(s, J) = (1, 0.020)$ and $(1.2, 0.024)$, with J in kg m^2, produce the same acceleration of the reported coordinate under the same input. This statement assumes a known applied generalized effort, constant scale/offset and no omitted gravity, friction or compliance. It does not hold merely because a drive has a field labelled current.

An independently known angle can distinguish the scale hypotheses. Under the additional constructed assumption $b = 0$, $q = 0.5$ rad produces $y = 0.5$ or 0.6, respectively. Once the scale is established independently, the same ratio can determine J within the stated model. Without that extra information the response identifies a combination, not two separate physical properties.

View	A useful entry for this investigation
Claim	Can the chosen observations distinguish output inertia from sensor scale, rather than merely fit their ratio?
Experience	Applied excitation, recorded indications and the independent known-angle encounter; include acquisition and reset conditions.
System	Joint, fixed housing, inertia, effort source, encoder and acquisition process; state excluded effects.
Grounding	Effort meaning, units, zero/scale reference, timing and the assumptions behind $J\ddot{q} = \tau$.
Learning	Fit the identifiable ratio; compare parameter explanations; update a model only to the extent supported by the additional reference.
Decision	With response alone, report the ambiguity. With an independent reference, revise the inference and retain both versions.

This example ties the views to the theory of Chapters 7, 8 and 15. Increasing the number of otherwise identical response samples can reduce noise without resolving a structural ambiguity. Changing the collection arrangement can provide the missing information. That is a precise sense in which an ExpGen PhySyst investigation designs data rather than only accumulates it.

A compact project record can use six headings with links to shared evidence: one question and competing explanations; one versioned system description; a collection manifest; a calibration and transformation record; model or policy artifacts; and a decision with its evidence and remaining limits. The files need not duplicate the data. Connect entries by stable identifiers, record changes and preserve the distinction between a proposal, an attempted experiment and an accepted conclusion. No particular form or software is required by the six views.

C.2.2 Optional asMagicBrain workflow

asMagicBrain can serve as an optional environment for reading, editing and organizing these records. Its role is documentary: bring the question, mathematics, source identities and interpretations into a navigable project. It does not supply evidence that the physical model is correct, that a sample is licensed or that a policy will transfer. The investigation remains usable with a text editor, version control and the same independent files.

Maintain a single authoritative source for each artifact. Link a dataset by its versioned registry entry and distinguish it from a generated summary. Keep private preparation, personal information, credentials and restricted sources outside a public distribution. Visibility in an explorer is not an access-control boundary. Before a deliberate synchronization, reconcile local and remote changes, inspect the intended destination and review the actual diff. Afterward, verify the destination content rather than relying only on a synchronization indicator.

No particular plugin version, automatic synchronization behavior, invitation mechanism or interactive viewer capability is assumed here. Interfaces can change without changing the mathematical or evidential requirements of the project. The book's computations and manuscript are ordinary versioned files; their use does not depend on adoption of asMagicBrain by a reader or team.

C.3 Independent study and course use

C.3.1 A many-to-many course reading map

Independent reading follows the argument: actuation and mechanics; simulation, measurement and identification; collection and learning; then transfer, co-design and evaluation. A course can revisit these foundations through apparatus-centered activities instead. The map below indicates connections, not a one-to-one correspondence or a requirement to reproduce semester-specific procedures.

Course investigation	Book reading	Central connection
Introduction	1–2	Physical AI, purposeful records and ExpGen PhySyst.
Actuator basics	3; 4, 7	Physical feedback, effort boundaries and measurement.
Digitization	4–6	Frames, inertia, topology and executable models.
Quantification	6–8; 14, 16	Identification, correspondence and held-out inference.
Multiple actuators	5, 8–9	Coupling, variation and related-trial records.
Learned manipulation	4–6, 10; 14, 16	Contact, policy interfaces and bounded evaluation.
Visual observation	4, 7; 10, 14	Pose, uncertainty and observation age.
Arm and teleoperation	3–7, 11	Kinematics, motion, relative mapping and timing.
Demonstration collection	7, 9, 11	Labels, acquisition and dataset construction.
Imitation and deployment	9–10, 12–14, 16	Learned actions, distribution shift and evidence.
Project investigation	1, 7–16; A–C	Connected views, informative experiments and reproducible decisions.

For a joint investigation, begin with Chapters 3–8, then select the collection and evaluation questions in Chapters 9, 15 and 16. For a teleoperated manipulation investigation, the geometry of Chapter 4, acquisition of Chapter 7 and mapping of Chapter 11 establish the meaning of demonstrations before Chapter 12 uses them for learning. The predictive and generalist methods in Chapter 13 broaden that argument; their presence does not imply that every method has an accompanying trained model.

C.3.2 Extensions, companion data and further investigations

The chapter exercises are the first extensions. A further investigation should state a question, prediction, evidence source and decision rule before producing a larger software system. The following directions remain useful at several levels of practical access.

1. *Representation*: move a force's reference point, rotate its axes and check power invariance. Identify the corresponding API fields before comparing implementations.
2. *Acquisition*: inject a declared timestamp offset or missing interval into a constructed trace. Compare the induced error with the first-order timing prediction and state where the approximation fails.
3. *Identification*: compare response-only fitting with an independent scale reference or known-load intervention. Explain which parameter combinations become identifiable under each assumption.
4. *Collection*: count encounters rather than overlapping windows; compare event-held-out and frame-random partitions without interpreting them as estimates of the same generalization question.

5. *Learning*: derive a loss for a next-measured-pose label and contrast it with one for an accepted controller target. State what must change in data collection or execution for the comparison to be meaningful.
6. *Transfer and design*: distinguish a model revision from a controller compensation; propose an encounter that separates the competing explanations and assess it using Chapters 14–16.

The registry in Appendix B offers source records for selected investigations. Begin by checking the entry's experiment, representation, rights and availability; then check the actual source bytes if acquired under appropriate permission. An unavailable checkpoint or missing calibration is a limit on an executable investigation, not a reason to invent the missing artifact. A reader can still examine a derivation or a saved record while stating that narrower scope.

The companions connect physical assumptions, representations and computations to inspectable conclusions. ExpGen PhySyst asks the reader to design encounters that answer a stated question and retain the context needed to judge the answer.

References

Agarwal, Rishabh, Max Schwarzer, Pablo Samuel Castro, Aaron Courville, and Marc G. Bellemare. 2022. "Deep Reinforcement Learning at the Edge of the Statistical Precipice." Inspected version 4 of the NeurIPS 2021 paper; first submitted August 30, 2021, January 5, 2022. Accessed October 19, 2026. https://doi.org/10.48550/arXiv.2108.13264. arXiv: 2108.13264. https://arxiv.org/abs/2108.13264v4.

Aljalbout, Elie, Jiaxu Xing, Angel Romero, Iretiayo Akinola, Caelan Reed Garrett, Eric Heiden, Abhishek Gupta, Tucker Hermans, Yashraj Narang, Dieter Fox, Davide Scaramuzza, and Fabio Ramos. 2026. "The Reality Gap in Robotics: Challenges, Solutions, and Best Practices." First published as a Review in Advance December 2, 2025; volume published May 2026, *Annual Review of Control, Robotics, and Autonomous Systems* 9:403–432. Accessed October 19, 2026. https://doi.org/10.1146/annurev-control-031924-100130. https://www.annualreviews.org/content/journals/10.1146/annurev-control-031924-100130.

Aroyo, Lora, and Christopher A. Welty. 2015. "Truth Is a Lie: Crowd Truth and the Seven Myths of Human Annotation." *AI Magazine* 36 (1): 15–24. https://doi.org/10.1609/aimag.v36i1.2564.

ATI Industrial Automation. 2020. *FAQ—Force/Torque Sensors.* ATI Industrial Automation, March 30, 2020. Accessed October 19, 2026. https://www.ati-ia.com/library/documents/FT_FAQ.pdf.

Barfoot, Timothy D. 2024. *State Estimation for Robotics.* 2nd ed. Cambridge University Press. ISBN: 9781009299893. https://doi.org/10.1017/9781009299909.

Baydin, Atılım Güneş, Barak A. Pearlmutter, Alexey Andreyevich Radul, and Jeffrey Mark Siskind. 2018. "Automatic Differentiation in Machine Learning: A Survey." *Journal of Machine Learning Research* 18 (153): 1–43. Accessed October 19, 2026. https://www.jmlr.org/papers/v18/17-468.html.

Bengio, Yoshua, Jérôme Louradour, Ronan Collobert, and Jason Weston. 2009. "Curriculum Learning." In *Proceedings of the 26th Annual International Conference on Machine Learning,* 41–48. Accessed October 19, 2026. https://ronan.collobert.com/pub/2009_curriculum_icml.pdf.

Boston Dynamics and Toyota Research Institute. 2024. "Boston Dynamics and Toyota Research Institute Announce Partnership to Advance Robotics Research." Joint research announcement, October 16, 2024. Accessed October 19, 2026. https://bostondynamics.com/news/boston-dynamics-toyota-research-institute-announce-partnership-to-advance-robotics-research/.

Brooks, Rodney A. 1991. "Intelligence without Representation." *Artificial Intelligence* 47 (1–3): 139–159. https://doi.org/10.1016/0004-3702(91)90053-M.

Chi, Cheng, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin Burchfiel, and Shuran Song. 2023. "Diffusion Policy: Visuomotor Policy Learning via Action Diffusion." In *Proceedings of Robotics: Science and Systems XIX.* https://doi.org/10.15607/RSS.2023.XIX.026.

Chi, Cheng, Zhenjia Xu, Chuer Pan, Eric Cousineau, Benjamin Burchfiel, Siyuan Feng, Russell L. Tedrake, and Shuran Song. 2024. "Universal Manipulation Interface: In-the-Wild Robot Teaching without In-the-Wild Robots." In *Proceedings of Robotics: Science and Systems XX.* https://doi.org/10.15607/RSS.2024.XX.045.

Chua, Kurtland, Roberto Calandra, Rowan McAllister, and Sergey Levine. 2018. "Deep Reinforcement Learning in a Handful of Trials Using Probabilistic Dynamics Models." Version 2, November 2, 2018; NIPS 2018 paper. https://doi.org/10.48550/arXiv.1805.12114. arXiv: 1805.12114 [cs.LG]. https://arxiv.org/abs/1805.12114v2.

Cyber-Physical Systems Public Working Group. 2017. *Framework for Cyber-Physical Systems: Volume 3, Timing Annex.* NIST Special Publication 1500-203. Version 1.0; public working group report. National Institute of Standards and Technology, September. https://doi.org/10.6028/NIST.SP.1500-203.

DeepMind Technologies Limited. n.d. “MuJoCo: API Reference, Functions.” Site Jacobians, position differentiation and integration; stable documentation for version 3.16.0 inspected on the access date. Accessed October 19, 2026. https://mujoco.readthedocs.io/en/stable/APIreference/APIfunctions.html.

———. n.d. “MuJoCo: API Reference, Types.” Version 3.16.0; selected mjModel and mjData fields inspected. Accessed October 19, 2026. https://mujoco.readthedocs.io/en/3.16.0/APIreference/APItypes.html.

———. n.d. “MuJoCo: Computation.” Actuation and soft contact models; online stable documentation inspected on the access date. Accessed October 19, 2026. https://mujoco.readthedocs.io/en/stable/computation/index.html.

———. n.d. “MuJoCo: Overview.” Floating-object configuration and velocity conventions; stable documentation for version 3.16.0 inspected on the access date. Accessed October 19, 2026. https://mujoco.readthedocs.io/en/stable/overview.html#floating-objects.

———. n.d. “MuJoCo: Simulation.” Simulation loop, state, reset and sleeping; stable documentation for version 3.16.0 inspected on the access date. Accessed October 19, 2026. https://mujoco.readthedocs.io/en/stable/programming/simulation.html.

———. n.d. “MuJoCo: XML Reference.” Motor actuator and joint armature; online stable documentation inspected on the access date. Accessed October 19, 2026. https://mujoco.readthedocs.io/en/stable/XMLreference.html#actuator-motor.

———. n.d. “MuJoCo: XML Reference, Sensors.” Sensor frames, internal loading, histories and delays; stable documentation for version 3.16.0 inspected on the access date. Accessed October 19, 2026. https://mujoco.readthedocs.io/en/stable/XMLreference.html#sensor.

Driscoll, Tobin A., and Richard J. Braun. n.d. “Fundamentals of Numerical Computation.” Online edition; sections 6.2, Euler’s method, and 11.3, Absolute stability. Accessed October 19, 2026. https://fncbook.com/.

Featherstone, Roy. 2008. *Rigid Body Dynamics Algorithms*. New York: Springer. ISBN: 978-0-387-74314-1. https://doi.org/10.1007/978-1-4899-7560-7.

Forssell, Urban, and Lennart Ljung. 1999. “Closed-loop Identification Revisited.” *Automatica* 35 (7): 1215–1241. https://doi.org/10.1016/S0005-1098(99)00022-9.

Gebru, Timnit, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. 2021. “Datasheets for Datasets.” *Communications of the ACM* 64 (12): 86–92. https://doi.org/10.1145/3458723.

Google DeepMind. n.d. “MuJoCo DC Motor Model.” Accessed October 19, 2026. https://mujoco.readthedocs.io/en/stable/_static/dcmotor.pdf.

———. n.d. “MuJoCo Source Code.” Version 3.16.0; engine dynamics, support functions and data structures. Accessed October 19, 2026. https://github.com/google-deepmind/mujoco/tree/3.16.0.

———. n.d. “MuJoCo Source Code: engine_core_util.c.” Version 3.16.0; Jacobian shortcuts and object-frame definitions. Accessed October 19, 2026. https://github.com/google-deepmind/mujoco/blob/3.16.0/src/engine/engine_core_util.c.

Google Research. 2024. “RLDS: Dataset Format.” Official README at commit b35dac3; repository archived November 29, 2025, September 26, 2024. Accessed October 19, 2026. https://github.com/google-research/rlds/blob/b35dac3a6b73396b0cb8773095999c4b5d70947c/README.md.

Ha, David. 2019. “Reinforcement Learning for Improving Agent Design.” Inspected author preprint, version 3, December 2, 2019, *Artificial Life* 25 (4): 352–365. Accessed October 19, 2026. https://doi.org/10.1162/artl_a_00301. https://arxiv.org/abs/1810.03779v3.

Ha, David, and Jürgen Schmidhuber. 2018. “Recurrent World Models Facilitate Policy Evolution.” In *Advances in Neural Information Processing Systems 31,* 2451–2463. Curran Associates. https://papers.nips.cc/paper/7512-recurrent-world-models-facilitate-policy-evolution.

Hafner, Danijar, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. 2020. “Dream to Control: Learning Behaviors by Latent Imagination.” Version 3, March 17, 2020; first submitted December 3, 2019. https://doi.org/10.48550/arXiv.1912.01603. arXiv: 1912.01603 [cs.LG]. https://arxiv.org/abs/1912.01603v3.

Hafner, Danijar, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. 2019. “Learning Latent Dynamics for Planning from Pixels.” In *Proceedings of the 36th International Conference on Machine Learning,* 97:2555–2565. Proceedings of Machine Learning Research. PMLR. https://proceedings.mlr.press/v97/hafner19a.html.

Hairer, Ernst, Christian Lubich, and Gerhard Wanner. 2006. *Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations.* 2nd ed. Springer Series in Computational Mathematics 31. Berlin and Heidelberg: Springer. ISBN: 978-3-540-30663-4. https://doi.org/10.1007/3-540-30666-8.

Hastie, Trevor John, Robert Tibshirani, and Jerome Friedman. 2009. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction.* 2nd ed. New York: Springer. ISBN: 9780387848570. https://doi.org/10.1007/978-0-387-84858-7.

Ho, Jonathan, Ajay Jain, and Pieter Abbeel. 2020. “Denoising Diffusion Probabilistic Models.” In *Advances in Neural Information Processing Systems,* vol. 33. Inspected arXiv version 2, December 16, 2020. https://doi.org/10.48550/arXiv.2006.11239. arXiv: 2006.11239.

Hogan, Neville. 1985. “Impedance Control: An Approach to Manipulation: Part I—Theory.” *Journal of Dynamic Systems, Measurement, and Control* 107, no. 1 (March): 1–7. https://doi.org/10.1115/1.3140702.

Hurlbert, Stuart H. 1984. “Pseudoreplication and the Design of Ecological Field Experiments.” *Ecological Monographs* 54 (2): 187–211. https://doi.org/10.2307/1942661.

Jackson, John. n.d. “5: British Tabulating Machine Company.” The National Museum of Computing. Accessed October 19, 2026. https://www.tnmoc.org/bh-5-british-tabulating-machine-company.

Joint Committee for Guides in Metrology. 2008. *Evaluation of Measurement Data—Guide to the Expression of Uncertainty in Measurement.* JCGM 100:2008. GUM 1995 with minor corrections. Bureau International des Poids et Mesures. https://doi.org/10.59161/JCGM100-2008E.

———. 2026. *Evaluation of Measurement Data—Guide to the Expression of Uncertainty in Measurement: Amendment 1: Nonlinearity in Measurement Models.* JCGM 100:2008/Amd 1:2026. Bureau International des Poids et Mesures. https://doi.org/10.59161/PPDI3267.

Kalman, Rudolf Emil. 1960. “A New Approach to Linear Filtering and Prediction Problems.” *Journal of Basic Engineering* 82 (1): 35–45. https://doi.org/10.1115/1.3662552.

Kelly, Michael, Chelsea Sidrane, Katherine Rose Driggs-Campbell, and Mykel John Kochenderfer. 2019. “HG-DAgger: Interactive Imitation Learning with Human Experts.” In *IEEE International Conference on Robotics and Automation,* 8077–8083. IEEE. https://doi.org/10.1109/ICRA.2019.8793698.

Kester, Walt. 2008. *What the Nyquist Criterion Means to Your Sampled Data System Design.* Tutorial MT-002. Revision A. Analog Devices, October. Accessed October 19, 2026. https://www.analog.com/media/en/training-seminars/tutorials/MT-002.pdf.

Kim, Moo Jin, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russell L. Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. 2024. "OpenVLA: An Open-Source Vision-Language-Action Model." Version 3, September 5, 2024. https://doi.org/10.48550/arXiv.2406.09246. arXiv: 2406.09246 [cs.RO]. https://arxiv.org/abs/2406.09246v3.

Kingma, Diederik Pieter, and Max Welling. 2014. "Auto-Encoding Variational Bayes." In *International Conference on Learning Representations.* Inspected arXiv version 11, December 10, 2022. https://doi.org/10.48550/arXiv.1312.6114. arXiv: 1312.6114.

Kumar, Ashish, Zipeng Fu, Deepak Pathak, and Jitendra Malik. 2021. "RMA: Rapid Motor Adaptation for Legged Robots." Version 1, July 8, 2021; RSS 2021 paper. https://doi.org/10.48550/arXiv.2107.04034. arXiv: 2107.04034 [cs.LG]. https://arxiv.org/abs/2107.04034v1.

Ljung, Lennart. 1999. *System Identification: Theory for the User.* 2nd ed. Upper Saddle River, NJ: PTR Prentice Hall. ISBN: 9780136566953.

———. 2016. "Sysid Course VT1 2016: An Overview." Linköping University. Graduate course slides. Accessed October 19, 2026. https://rt.isy.liu.se/student/graduate/idkurs/Id1_4.pdf.

Lynch, Kevin M., and Frank C. Park. 2017. *Modern Robotics: Mechanics, Planning, and Control.* Cambridge University Press. ISBN: 9781107156302. https://doi.org/10.1017/9781316661239.

Miriyev, Aslan, and Mirko Kovač. 2020. "Skills for Physical Artificial Intelligence." *Nature Machine Intelligence* 2 (11): 658–660. https://doi.org/10.1038/s42256-020-00258-y.

Moreau, Luc, and Paolo Missier, eds. 2013. *PROV-DM: The PROV Data Model.* W3C Recommendation. World Wide Web Consortium, April 30, 2013. Accessed October 19, 2026. https://www.w3.org/TR/2013/REC-prov-dm-20130430/.

Murray, Richard M., Zexiang Li, and S. Shankar Sastry. 1994. *A Mathematical Introduction to Robotic Manipulation.* CRC Press. ISBN: 9780849379819.

Ng, Andrew Y., Daishi Harada, and Stuart Russell. 1999. "Policy Invariance under Reward Transformations: Theory and Application to Reward Shaping." In *Proceedings of the 16th International Conference on Machine Learning,* 278–287. Accessed October 19, 2026. https://people.eecs.berkeley.edu/~russell/papers/icml99-shaping.pdf.

Niemeyer, Günter Dieter, and Jean-Jacques E. Slotine. 2004. "Telemanipulation with Time Delays." *The International Journal of Robotics Research* 23, no. 9 (September): 873–890. https://doi.org/10.1177/0278364904045563.

NIST/SEMATECH. n.d. "NIST/SEMATECH e-Handbook of Statistical Methods." Sections 1.3.5.2, Confidence Limits for the Mean, and 7.2.4.1, Confidence Intervals, inspected online. Accessed October 19, 2026. https://www.itl.nist.gov/div898/handbook/.

Octo Model Team, Dibya Ghosh, Homer Walke, Karl Pertsch, Kevin Black, Oier Mees, Sudeep Dasari, Joey Hejna, Tobias Kreiman, Charles Xu, Jianlan Luo, You Liang Tan, Lawrence Yunliang Chen, Pannag Sanketi, Quan Vuong, Ted Xiao, Dorsa Sadigh, Chelsea Finn, and Sergey Levine. 2024. "Octo: An Open-Source Generalist Robot Policy." Version 2, May 26, 2024. https://doi.org/10.48550/arXiv.2405.12213. arXiv: 2405.12213 [cs.RO]. https://arxiv.org/abs/2405.12213v2.

OpenCV. n.d. "Camera Calibration and 3D Reconstruction." OpenCV 4.13.0 documentation. Accessed October 19, 2026. https://docs.opencv.org/4.13.0/d9/d0c/group__calib3d.html.

Pardo, Fabio, Arash Tavakoli, Vitaly Levdik, and Petar Kormushev. 2018. "Time Limits in Reinforcement Learning." In *Proceedings of the 35th International Conference on Machine Learning,* 80:4045–4054. Proceedings of Machine Learning Research. PMLR. Accessed October 19, 2026. https://proceedings.mlr.press/v80/pardo18a.html.

Peng, Xue Bin, Marcin Andrychowicz, Wojciech Zaremba, and Pieter Abbeel. 2018. “Sim-to-Real Transfer of Robotic Control with Dynamics Randomization.” In *IEEE International Conference on Robotics and Automation,* 3803–3810. IEEE. https://doi.org/10.1109/ICRA.2018.8460528.

Ramos, Sabela, Sertan Girgin, Léonard Hussenot, Damien Vincent, Hanna Yakubovich, Daniel Toyama, Anita Gergely, Piotr Stanczyk, Raphael Marinier, Jeremiah Harmsen, Olivier Pietquin, and Nikola Momchev. 2021. “RLDS: An Ecosystem to Generate, Share and Use Datasets in Reinforcement Learning.” Version 1, November 4, 2021. Accessed October 19, 2026. https://doi.org/10.48550/arXiv.2111.02767. arXiv: 2111.02767. https://arxiv.org/abs/2111.02767.

ROBOTIS. n.d. “XM430-W350-T/R.” ROBOTIS e-Manual. Accessed October 19, 2026. https://emanual.robotis.com/docs/en/dxl/x/xm430-w350/.

Ross, Stéphane, Geoffrey J. Gordon, and James Andrew Bagnell. 2011. “A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning.” In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics,* 15:627–635. Proceedings of Machine Learning Research. PMLR. Accessed October 19, 2026. https://proceedings.mlr.press/v15/ross11a.html.

Schaal, Stefan. 1997. “Learning from Demonstration.” In *Advances in Neural Information Processing Systems 9,* 1040–1046. Proceedings of NIPS 1996. Cambridge, MA: MIT Press. https://proceedings.neurips.cc/paper/1996/hash/68d13cf26c4b4f4f932e3eff990093ba-Abstract.html.

Schulman, John, Philipp Moritz, Sergey Levine, Michael I. Jordan, and Pieter Abbeel. 2016. “High-Dimensional Continuous Control Using Generalized Advantage Estimation.” In *International Conference on Learning Representations.* Inspected arXiv version 6, October 20, 2018. https://doi.org/10.48550/arXiv.1506.02438. arXiv: 1506.02438.

Schulman, John, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. “Proximal Policy Optimization Algorithms.” Version 2, August 28, 2017. Accessed October 19, 2026. https://doi.org/10.48550/arXiv.1707.06347. arXiv: 1707.06347. https://arxiv.org/abs/1707.06347.

Shannon, Claude Elwood. 1949. “Communication in the Presence of Noise.” *Proceedings of the IRE* 37 (1): 10–21. https://doi.org/10.1109/JRPROC.1949.232969.

Shaw, Kenneth, Ananye Agarwal, and Deepak Pathak. 2023. “LEAP Hand: Low-Cost, Efficient, and Anthropomorphic Hand for Robot Learning.” In *Proceedings of Robotics: Science and Systems XIX.* https://doi.org/10.15607/RSS.2023.XIX.089.

Söderström, Torsten. 2007. “Errors-in-variables Methods in System Identification.” *Automatica* 43 (6): 939–958. https://doi.org/10.1016/j.automatica.2006.11.025.

Sutton, Richard S., and Andrew G. Barto. 2018. *Reinforcement Learning: An Introduction.* 2nd ed. Cambridge, MA: The MIT Press. ISBN: 9780262039246.

Tedrake, Russell L. n.d. “Underactuated Robotics: Multi-Body Dynamics.” Massachusetts Institute of Technology. Accessed October 19, 2026. https://underactuated.mit.edu/multibody.html#impulsive-collision-dynamics.

———. n.d. “Underactuated Robotics: System Identification.” Massachusetts Institute of Technology. Online working chapter; relevant sections inspected on the access date. Accessed October 19, 2026. https://underactuated.mit.edu/sysid.html.

The National Museum of Computing. n.d. “The Turing–Welchman Bombe.” Accessed October 19, 2026. https://www.tnmoc.org/bombe.

Tobin, Joshua, Rachel Fong, Alex Ray, Jonas Schneider, Wojciech Zaremba, and Pieter Abbeel. 2017. "Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World." Version 1, March 20, 2017; author byline uses Josh Tobin. https://doi.org/10.48550/arXiv.1703.06907. arXiv: 1703.06907 [cs.RO]. https://arxiv.org/abs/1703.06907v1.

Todorov, Emanuel, Tom Erez, and Yuval Tassa. 2012. "MuJoCo: A Physics Engine for Model-Based Control." In *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems,* 5026–5033. IEEE. https://doi.org/10.1109/IROS.2012.6386109.

Turing, Alan Mathison. 1950. "Computing Machinery and Intelligence." *Mind* 59 (236): 433–460. https://doi.org/10.1093/mind/LIX.236.433.

Vaswani, Ashish, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. "Attention Is All You Need." In *Advances in Neural Information Processing Systems,* 30:5998–6008. Accessed October 19, 2026. https://papers.nips.cc/paper_files/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html.

Wampler, Charles W., II. 1986. "Manipulator Inverse Kinematic Solutions Based on Vector Formulations and Damped Least-Squares Methods." *IEEE Transactions on Systems, Man, and Cybernetics* 16 (1): 93–101. https://doi.org/10.1109/TSMC.1986.289285.

Wensing, Patrick Michael, Sangbae Kim, and Jean-Jacques E. Slotine. 2018. "Linear Matrix Inequalities for Physically Consistent Inertial Parameter Identification: A Statistical Perspective on the Mass Distribution." *IEEE Robotics and Automation Letters* 3 (1): 60–67. https://doi.org/10.1109/LRA.2017.2729659.

Wensing, Patrick Michael, Günter Niemeyer, and Jean-Jacques E. Slotine. 2024. "A Geometric Characterization of Observability in Inertial Parameter Identification." *The International Journal of Robotics Research* 43 (14): 2274–2302. https://doi.org/10.1177/02783649241258215.

Wensing, Patrick Michael, Albert Wang, Sangok Seok, David Otten, Jeffrey Lang, and Sangbae Kim. 2017. "Proprioceptive Actuator Design in the MIT Cheetah: Impact Mitigation and High-Bandwidth Physical Interaction for Dynamic Legged Robots." *IEEE Transactions on Robotics* 33, no. 3 (June): 509–522. https://doi.org/10.1109/TRO.2016.2640183.

Wiener, Norbert. 2019. *Cybernetics, or Control and Communication in the Animal and the Machine.* 2nd ed. Reissue of the 1961 second edition; first edition published in 1948. The MIT Press. https://doi.org/10.7551/mitpress/11810.001.0001.

Zhao, Tony Z., Vikash Kumar, Sergey Levine, and Chelsea Finn. 2023. "Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware." In *Proceedings of Robotics: Science and Systems XIX.* https://doi.org/10.15607/RSS.2023.XIX.016.